

# ENGINEERING DRAWING

[PLANE AND SOLID GEOMETRY]

[IN FIRST-ANGLE PROJECTION METHOD]



by

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### FOREWORD

It gives me pleasure to introduce this text-book on Engineering Drawing by Prof. N. D. Bhatt of the Birla Vishvakarma Mahavidyalaya to students of Engineering. Prof. Bhatt has been teaching this subject for over twenty years and has deservedly earned the reputation of being one of the best teachers in the subject.

This book covers the syllabus usually prescribed for the Pre-engineering and First Year of the Degree and Diploma courses in Engineering and deals with the fundamental principles of this basic subject which have been treated by Prof. Bhatt with his characteristic lucidity.

I am sure the book will prove a boon to students and help them to acquire a sound knowledge of the subject without which a really satisfactory progress cannot be achieved in any branch of Engineering.

June 6, 1958

**S. B. JUNNARKAR**

M.B.E., B.A., B.Sc. Hons. (Eng.), London  
POONA

# ELEMENTARY ENGINEERING DRAWING

## PREFACE

From the very early days, man realized that if he had to construct any structure or machine correctly and methodically, he must first record his ideas before starting construction work. These recorded ideas become more vivid and forceful if they are shown on paper in form of a drawing of the structure or machine. Such a drawing will be of very great help to the man who looks after the construction of this structure or machine.

Indeed, "technical drawing is the language of engineering". Without the good knowledge of drawing, an engineer is nowhere and he could not have constructed the various magnificent structures or intricate machines. Evidently, any one connected in any way, with engineering construction must understand this language of engineers. Technical drawing is, therefore, indispensable today and shall continue to be so as long as engineering and technology continue to be of use in the activities of man.

By means of drawing, the shape, size, finish, colour and construction of any object (no matter how complex) can be described accurately and clearly. The engineer should develop his skill, in two phases of technical drawing; first, he must be able to draw clearly and rapidly, the freehand technical sketches; secondly, he must be proficient in drawing to scale the instrumental drawing. The purpose of the present volume is to give the basic principles of the instrumental drawing only.

The book covers the syllabi in Engineering Drawing of many University Colleges and Polytechnics in India and has been written keeping in view the difficulties of a beginner in the subject of Engineering Drawing. I am quite hopeful that this book will serve its purpose very well for young and budding engineers.

I am highly indebted to Principal S. B. Junnarkar for his valuable guidance and for his kindness to write a suitable foreword for the book. I am also thankful to Prof. V. B. Priyani of Birla Vishvakarma Mahavidyalaya for going through the initial manuscript and for offering constructive suggestions. Finally, I feel grateful to the following:

(i) The authorities of the Universities of Bombay, Poona and Gujarat, and the Department of Technical Education, Bombay, for their kind permission to include a few questions set at their examinations. (ii) Mr. N. M. Panchal and Mr. M. D. Bhatt for their help in preparing pencil sketches. (iii) Mr. L. D. Bhatt for preparing the excellent typed manuscript. (iv) Mr. Ramanbhai C. Patel of Charotar Book Stall for careful proof-reading and for his efforts to see the book out in proper time. (v) The Anand Press authorities for the care and interest shown in the printing and get-up of this book. (vi) The Prabhat Process Studio for the promptness and good work of block-making.

Any suggestion to improve the value of this book will be gratefully received and will be incorporated in subsequent editions after due scrutiny.

June 30, 1958

**N. D. BHATT**

## EIGHTH EDITION

Upto six editions, the book hardly underwent any fundamental change. Under the inspiration of the Indian Standards Institution, the industries in a large measure have switched on to the third-angle projection system. So have a vast majority of Technical Institutions. "Elementary Engineering Drawing", too, must keep step with the changes. There was a phased change-over beginning with the seventh edition. This edition has completed the change. Nevertheless, the first-angle projection system has not been altogether discarded. The fundamentals of both the systems are juxtaposed and elucidated.

The chapters on (i) Development of surfaces, (ii) Isometric projection and (iii) Conversion of pictorial views into orthographic views have been treated in some greater depth. This and the increase of the illustrative problems and practice exercises have evidently enlarged the size of the book and inevitably led to a little rise in price.

I should not miss this opportunity to record my gratitude to the numerous teachers for their very useful suggestions and the students for their excellent response to the book, without which this edition should not have come into being.

January 26, 1970

**N. D. BHATT**

## FORTY-SECOND EDITION

The first edition was originally published in 1958 and now has entered in its 42nd Edition during 42 years, proves its popularity and utility among the teachers as well as the students of Engineering Institutions of our country.

This edition has been entirely revised and enlarged by adding the following *four* chapters:

- (1) Screw Threads (2) Screwed Fastenings (3) Riveted Joints and Welded Joints (4) Computer Aided Drafting.

It was thought desirable to include this fourth chapter on 'Computer Aided Drafting' which has now acquired an important place in this subject. We are thankful to Dr. S. S. Khandare of Y. C. College of Engineering, Nagpur for contributing this chapter. We are also highly obliged to Shri R. C. Patel of Charotar Publishing House, Anand for correcting the Proofs.

This book provides a wealth of information and describes an exciting new paradigm and is also a valuable tool for all the Engineering students, who wish to deepen and acquire a sound knowledge of this important subject (Language of Engineers) without which a really satisfactory progress cannot be achieved in any branch of Engineering.

We shall feel obliged to receive comments, suggestions and opinions from the readers to enhance the utility of the book.

August 15, 2000

**N. D. BHATT**  
**V. M. PANCHAL**

## FIFTIETH EDITION

We are inundated with joy to present before you the FIFTIETH EDITION of this most standard text-book. At the outset, we would like to mention that splendid response to earlier editions is prodigious. Despite such favourable responses, it was felt that the utility of the book could be further enhanced. This is one of the most comprehensive revisions since the book was first published. As a result, all the drawings have been redrawn with utmost intelligibility. Many new examples, drawings are incorporated along with some new text matter. This text-book is thoroughly revised, extensively enlarged, completely updated. Chapter on Computer Aided Drafting (CADr) is entirely rewritten with inclusion of *50 self-interactive and self-learning practice modules*. This book accompanied by a computer CD as a novel pedagogical concept, containing *51 selected audiovisual animation modules* presented for better visualization and understanding of the subject of Engineering Drawing.

We take this opportunity to thank Prof. V. M. Panchal, Former Professor in Mechanical Engineering, Faculty of Technology and Engineering, M. S. University of Baroda for revising the entire book and adding many new typical examples. We express our hearty gratitude to Prof. Pramod R. Ingle, Lecturer in Mechanical Engineering, B and B Institute of Technology, Vallabh Vidyanagar for redraw all the drawings with precision, rewriting the chapter on Computer Aided Drafting (CADr) and revising the entire book.

We are also indebted to Prof. R. S. Bhatt, Associate Professor in Mechanical Engineering, Birla Vishvakarma Mahavidyalaya (Engineering College), Vallabh Vidyanagar, Anand; Prof. Mukesh A. Bulsara, Assistant Professor in Mechanical Engineering, G. H. Patel College of Engineering and Technology (GCET), Vallabh Vidyanagar, Prof. N. V. Patel, Ex-Professor, L. D. Engineering College, Ahmedabad who were kind enough to send their suggestions to us for the improvement of the book. We are also grateful to Prof. S. H. Makwana, Lecturer in Mechanical Engineering, B & B Institute of Technology, Vallabh Vidyanagar, who spared no efforts in the tedious task of diligently correcting the final proofs. We are sincerely thankful to Cognifront, Nasik for preparing a unique CD and Repro India Ltd., Mumbai for their hearty co-operation and excellent printing of the book.

In spite of all the pains taken, it is possible that some errors may have escaped our attention. We shall be grateful if they are brought to our notice by sending e-mail at [charotar@cphbooks.com](mailto:charotar@cphbooks.com), so that they can be corrected in subsequent edition. We strongly urge the readers to send their comments, suggestions and opinions to enhance the utility of the book.

August 24, 2010

**PUBLISHERS**

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## CONTENTS OF CD-ROM



### NOTE ABOUT THE CD ACCOMPANYING THE BOOK

This book is accompanied by a CD, which contains audiovisual modules for better visualization and understanding of the subject.

It has been found in our research done over hundreds of students and dozens of colleges and universities that visualization IQ is lacking in different quantities among male and female students. It is therefore necessary to aid learning process by use of high quality computer animations as a novel pedagogical concept.

This CD contains 51 modules and CD is provided FREE with 50th Edition of the book.

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## Chapter

# 1



# DRAWING INSTRUMENTS AND THEIR USES

## 1-1. INTRODUCTION



*Drawing instruments* are used to prepare drawings easily and accurately. The accuracy of the drawings depends largely on the quality of instruments. With instruments of good quality, desirable accuracy can be attained with ease. It is, therefore, essential to procure instruments of as superior quality as possible.

Below is the list of minimum drawing instruments and other drawing materials which every student must possess:

1. Drawing board
2. T-square
3. Set-squares —  $45^\circ$  and  $30^\circ$ - $60^\circ$
4. Drawing instrument box, containing:
  - (i) Large-size compass with inter-changeable pencil and pen legs
  - (ii) Lengthening bar
  - (iii) Small bow compass
  - (iv) Large-size divider
  - (v) Small bow divider
  - (vi) Small bow ink-pen
  - (vii) Inking pen
5. Scales
6. Protractor
7. French curves
8. Drawing papers
9. Drawing pencils
10. Sand-paper block
11. Eraser (Rubber)
12. Drawing pins, clips or adhesive tapes
13. Duster
14. Drafting machine
15. Roll-n-draw.

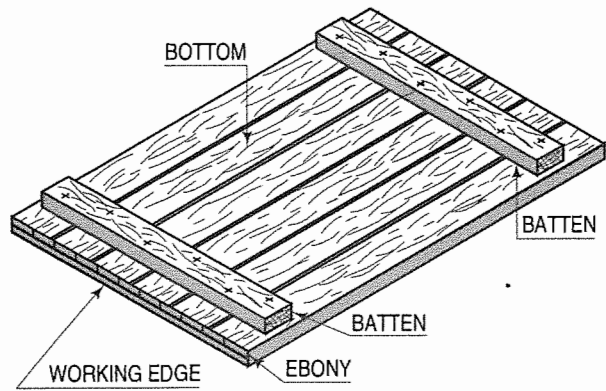
We shall now describe each of the above in details with their uses:



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 1 for Introduction of the subject and various drawing instruments.

**1-2. DRAWING BOARD** (FIG. 1-1) 

Drawing board is rectangular in shape and is made of strips of well-seasoned soft wood about 25 mm thick. It is cleated at the back by two battens to prevent warping. One of the edges of the board is used as the *working edge*, on which the T-square is made to slide. It should, therefore, be perfectly straight. In some boards, this edge is grooved throughout its length and a perfectly straight ebony edge is fitted inside this groove. This provides a true and more durable guide for the T-square to slide on.



| TABLE 1-1<br>SIZES OF DRAWING BOARDS |             |
|--------------------------------------|-------------|
| Designation                          | Size (mm)   |
| B0                                   | 1000 × 1500 |
| B1                                   | 700 × 1000  |
| B2                                   | 500 × 700   |
| B3                                   | 350 × 500   |

FIG. 1-1

Drawing board is made in various sizes. Its selection depends upon the size of the drawing paper to be used. The sizes of drawing boards recommended by the Bureau of Indian Standards (IS:1444-1989) are tabulated in table 1-1.

For use in schools and colleges, the last two sizes of the drawing boards are more convenient. Large-size boards are used in drawing offices of engineers and engineering firms. The drawing board is placed on the table in front of the student, with its working edge on his left side. It is more convenient if the table-top is sloping downwards towards the student. If such a table is not available, the necessary slope can be obtained by placing a suitable block of wood under the distant longer edge of the board.

**1-3. T-SQUARE** (FIG. 1-2) 

A T-square is made up of hard-quality wood. It consists of two parts — the stock and the blade — joined together at right angles to each other by means of screws and pins. The stock is placed adjoining the working edge of the board and is made to slide on it as and when required. The blade lies on the surface of the board. Its distant edge which is generally bevelled, is used as the working edge and hence, it should be perfectly straight. The nearer edge of the blade is never used. The length of the blade is selected so as to suit the size of the drawing board. Now-a-days T-square is also available of celluloid or plastic with engraved scale.

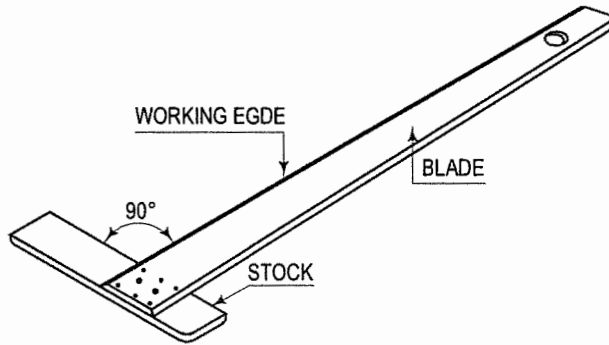


FIG. 1-2

**Uses:**

- (i) The T-square is used for drawing horizontal lines. The stock of the T-square is held firmly with the left hand against the working edge of the board, and the line is drawn from left to right as shown in fig. 1-3. The pencil should be held slightly inclined in the direction of the line (i.e. to the right) while the pencil point should be as close as possible to the working edge of the blade. Horizontal parallel lines are drawn by sliding the stock to the desired positions.

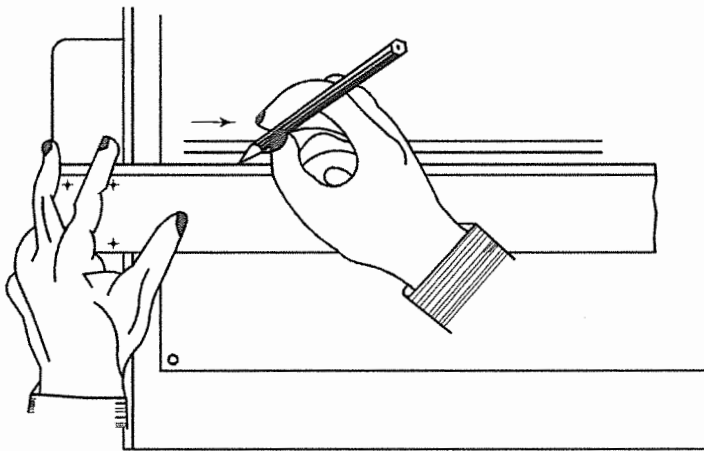


FIG. 1-3

- (ii) The working edge of the T-square is also used as a base for set-squares to draw vertical, inclined or mutually parallel lines. A pencil must be rotated while drawing lines for uniform wear of lead. The T-square should never be used on edge other than the working edge of the board. It should always be kept on the board even when not in use.
- (iii) Testing the straightness of the working edge of the T-square: Mark any two points A and B (fig. 1-4) spaced wide apart and through them, carefully draw a line with the working edge. Turn the T-square upside down as shown by dashed lines and with the same edge, draw another line passing through the same two points. If the edge is defective the lines will not coincide. The error should be rectified by planing or sand-papering the defective edge.

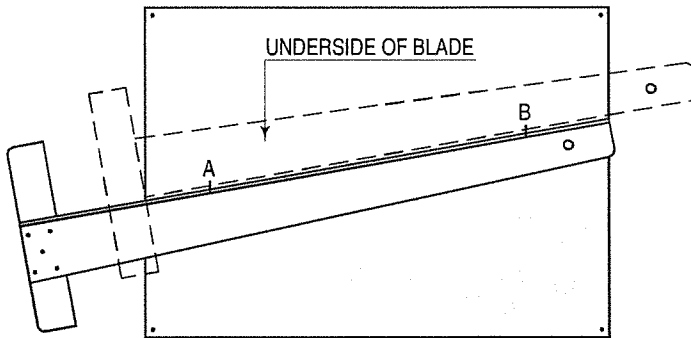


FIG. 1-4

### 1-4. SET-SQUARES (FIG. 1-5)

The set-squares are made of wood, tin, celluloid or plastic. Those made of transparent celluloid or plastic are commonly used as they retain their shape and accuracy for a longer time. Two forms of set-squares are in general use. A set-square is triangular in shape with one of the angle as right angle. The  $30^\circ$ - $60^\circ$  set-square of 250 mm length and  $45^\circ$  set-square of 200 mm length are convenient sizes for use in schools and colleges.

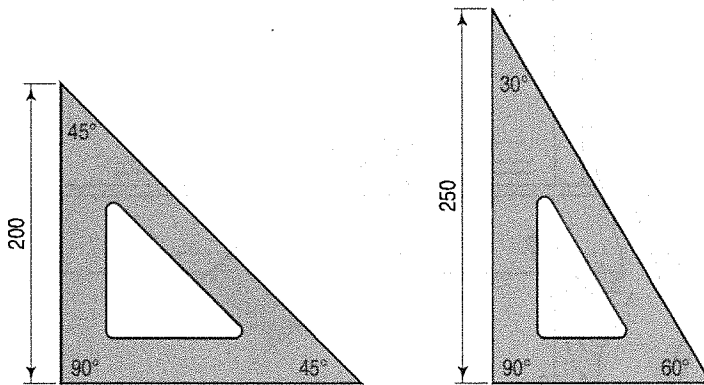


FIG. 1-5

#### Uses:

- (i) Set-squares are used for drawing all straight lines except the horizontal lines which are usually drawn with the T-square. Vertical lines can be drawn with the T-square and the set-square.
- (ii) In combination with the T-square, lines at  $30^\circ$  or  $60^\circ$  angle with vertical or horizontal lines can be drawn with  $30^\circ$ - $60^\circ$  set-square and  $45^\circ$  angle with  $45^\circ$  set-square. The two set-squares used simultaneously along with the T-square will produce lines making angles of  $15^\circ$ ,  $75^\circ$ ,  $105^\circ$  etc.
- (iii) Parallel straight lines in any position, not very far apart, as well as lines perpendicular to any line from any given point within or outside it, can also be drawn with the two set-squares.
- (iv) A circle can be divided in six, eight, twelve and twenty four equal parts by using set-squares and T-square.

**Problem 1-1.** To draw a line perpendicular to a given horizontal line from a given point within it.

- (i) Place the T-square a little below the given line (fig. 1-6).
- (ii) Arrange any one set-square with one of the edges containing the right angle touching the working edge of the T-square, and the other edge passing through the given point.
- (iii) Hold the T-square and the set-square in this position firmly with the left hand.
- (iv) With the right hand, draw the required line through the given point in the upward direction as shown by the arrow. The pencil point should always be in contact with the edge of the set-square. A perpendicular from any given point outside the line can also be drawn in the same manner. Vertical parallel lines may be drawn by sliding the set-square along the edge of the T-square to the required positions.

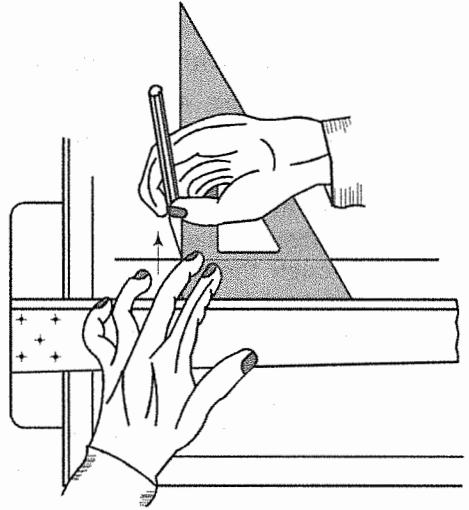


FIG. 1-6

**Problem 1-2.** To draw a line inclined at  $45^\circ$ ,  $30^\circ$  or  $60^\circ$  to a given horizontal line from a given point.

- (i) Place the edge containing the right angle of the  $45^\circ$  set-square on the edge of the T-square (fig. 1-7).
- (ii) Slide it so that its longest edge (hypotenuse) passes through the given point and then draw the required line. The same line will make  $45^\circ$  angle with the vertical line passing through that point.
- (iii) By turning the set-square upside down, the line making  $45^\circ$  angle in the other direction will be drawn. The lines can also be drawn by placing the set-square so that its longest edge coincides with the edge of the T-square and the other edge passes through the given point. A circle can similarly be divided into eight equal parts by lines passing through its centre (fig. 1-8).

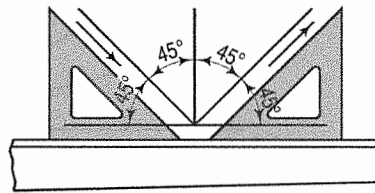


FIG. 1-7

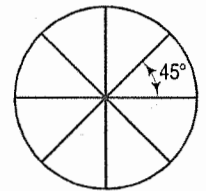


FIG. 1-8

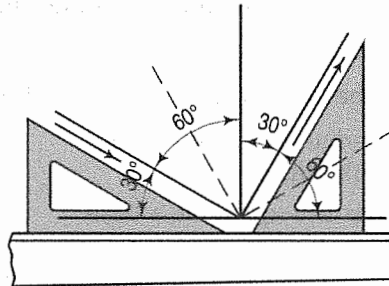


FIG. 1-9

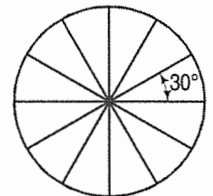


FIG. 1-10

Lines inclined at  $30^\circ$  or  $60^\circ$  to a given horizontal line can similarly be drawn with the aid of a  $30^\circ$ - $60^\circ$  set-square (fig. 1-9). A circle may be divided into twelve equal divisions in the same manner (fig. 1-10).

**Problem 1-3.** To draw a line inclined at  $15^\circ$  to a given horizontal line from a given point.

- (i) Place the  $30^\circ$ - $60^\circ$  set-square with its longer edge containing the right angle, coinciding with the edge of the T-square (fig. 1-11).
- (ii) Arrange the  $45^\circ$  set-square with its longest edge on the longest edge of the  $30^\circ$ - $60^\circ$  set-square.
- (iii) Slide the  $45^\circ$  set-square so that one of its edges containing the right angle passes through the given point, and draw the required line. The line drawn with the other edge will make  $15^\circ$  angle with the vertical line and  $105^\circ$  or  $75^\circ$  angles with the horizontal line. A circle may thus be divided into 24 equal parts with the aid of the set-squares (fig. 1-12).

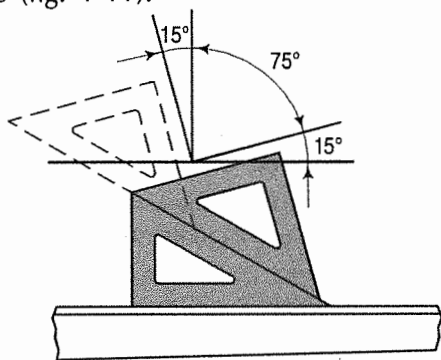


FIG. 1-11

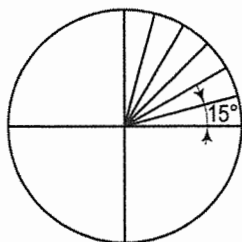


FIG. 1-12

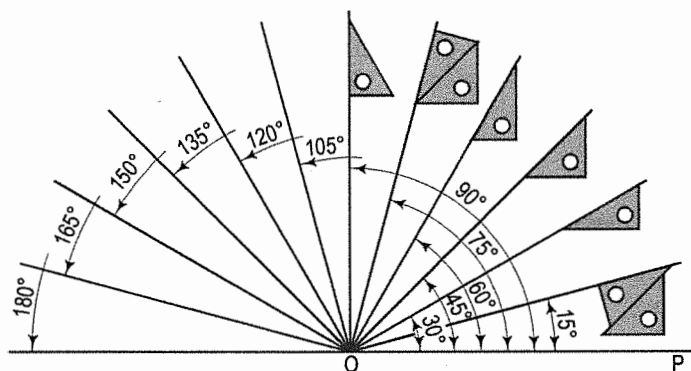


FIG. 1-13

Fig. 1-13 shows methods of drawing lines (with the aid of the T-square and set-squares) making angles with the horizontal line in multiples of  $15^\circ$  upto  $180^\circ$ .

**Problem 1-4.** To draw a line parallel to a given straight line through a given point. The line  $AB$  and the point  $P$  are given (fig. 1-14).

- (i) Arrange an edge of a set-square coinciding with  $AB$ .
- (ii) Place the other set-square as a base for the first.
- (iii) Hold the second set-square firmly and slide the first, till its arranged edge is along the point  $P$ .
- (iv) Draw the line  $CD$  through  $P$ .  $CD$  is the required parallel line.

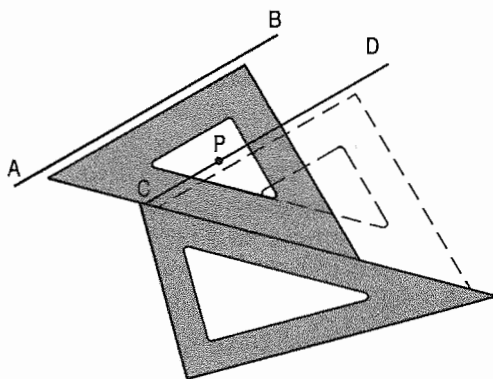


FIG. 1-14

By keeping the edge of the T-square as base for the set-square, parallel lines, long distances apart, can be drawn.

**Problem 1-5.** To draw a line perpendicular to a given line through a point within or outside it.

The line  $PQ$  and the point  $O$  are given (fig. 1-15).

**Method I:**

- (i) Arrange the longest edge of one set-square along  $PQ$ .
- (ii) Place the second set-square or T-square as base along one of the edges containing the right angle.
- (iii) Holding the base set-square firmly, rotate the first set-square so that its other edge containing the right angle coincides with the edge of the base set-square.
- (iv) Slide the first set-square till its longest edge is on the point  $O$  and draw the required line  $AB$ .

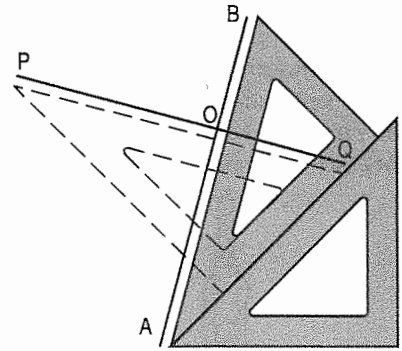


FIG. 1-15

**Method II:**

- (i) Arrange one set-square with an edge containing the right angle along the line  $PQ$  (fig. 1-16).
- (ii) Place the second set-square or T-square as a base under the longest edge.
- (iii) Slide the first set-square on the second till the other edge containing the right angle is on the point  $O$  and draw the required line  $AB$ .

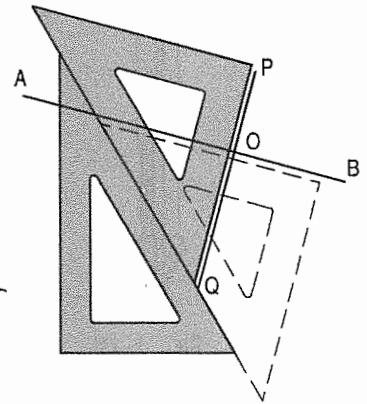


FIG. 1-16

**Problem 1-6.** To draw a line parallel to a given straight line at a given distance, say 20 mm from it (fig. 1-17).

Let  $AB$  be the given line.

- (i) From any point  $P$  in  $AB$ , draw a line  $PQ$  perpendicular to  $AB$  (Problem 1-5).
- (ii) Mark a point  $R$  such that  $PR = 20$  mm.
- (iii) Through  $R$ , draw the required line  $CD$  parallel to  $AB$  (Problem 1-4).

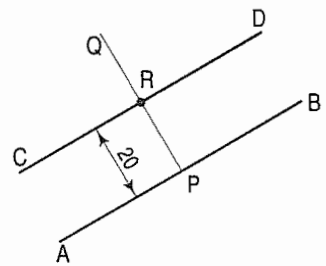


FIG. 1-17

## 1-5. DRAWING INSTRUMENT BOX



The drawing instrument box contains the following as mentioned earlier:

- |                                                                 |                       |
|-----------------------------------------------------------------|-----------------------|
| (1) Large-size compass with interchangeable pencil and pen legs | (5) Small bow divider |
| (2) Lengthening bar                                             | (6) Small bow ink-pen |
| (3) Small bow compass                                           | (7) Inking pen.       |
| (4) Large-size divider                                          |                       |

**(1) Large-size compass with interchangeable pencil and pen legs (fig. 1-18):**

The compass is used for drawing circles and arcs of circles. It consists of two legs hinged together at its upper end. A pointed needle is fitted at the lower end of one leg, while a pencil lead is inserted at the end of the other leg. The lower part of the pencil leg is detachable and it can be interchanged with a similar piece containing an inking pen. Both the legs are provided with knee joints. Circles upto about 120 mm diameter can be drawn with the legs of the compass kept straight. For drawing larger circles, both the legs should be bent at the knee joints so that they are perpendicular to the surface of the paper (fig. 1-19).



FIG. 1-18

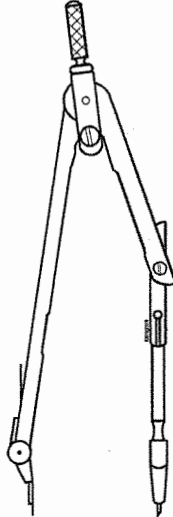


FIG. 1-19

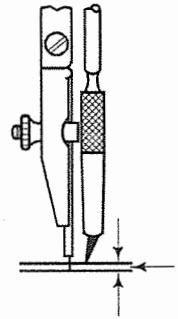


FIG. 1-20

As the needle is required to be inserted slightly inside the paper, it is kept longer than the lead point. *The setting of the pencil-lead relative to the needle, and the shape to which the lead should be ground are shown in fig. 1-20.*

To draw a circle, adjust the opening of the legs of the compass to the required radius. Hold the compass with the thumb and the first two fingers of the right hand and place the needle point lightly on the centre, with the help of the left hand. Bring the pencil point down on the paper and swing the compass about the needle-leg with a twist of the thumb and the two fingers, in clockwise ↻ direction, until the circle is completed. The compass should be kept slightly inclined in the direction of its rotation. *While drawing concentric circles, beginning should be made with the smallest circle.*

**(2) Lengthening bar:** Circles of more than 150 mm radius are drawn with the aid of the lengthening bar. The lower part of the pencil leg is detached and the lengthening bar is inserted in its place. The detached part is then fitted at the end of the lengthening bar, thus increasing the length of the pencil leg (fig. 1-21).

It is often necessary to guide the pencil leg with the other hand, while drawing large circles.

**(3) Small bow compass:** For drawing small circles and arcs of less than 25 mm radius and particularly, when a large number of small circles of the same diameter are to be drawn, small bow compass is used (fig. 1-22).

Curves drawn with the compass should be of the same darkness as that of the straight lines. It is difficult to exert the same amount of pressure on the lead in the compass as on a pencil.

*It is, therefore, desirable to use slightly softer variety of lead (about one grade lower, HB or H) in the compass than the pencil used for drawing straight lines, to maintain uniform darkness in all the lines.*

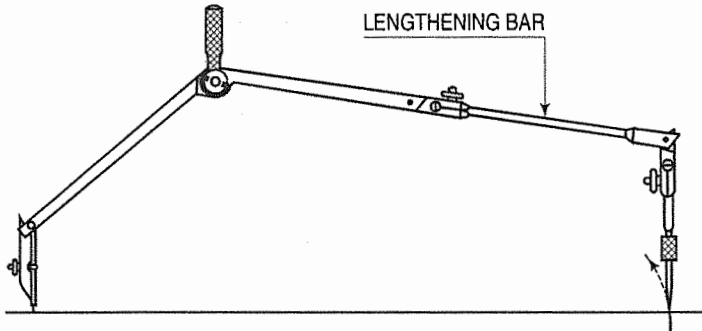


FIG. 1-21

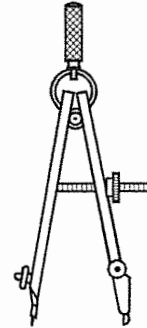


FIG. 1-22



FIG. 1-23

(4) **Large-size divider:** The divider has two legs hinged at the upper end and is provided with steel pins at both the lower ends, but it does not have the knee joints (fig. 1-23).

In most of the instrument boxes, a needle attachment is also provided which can be interchanged with the pencil part of the compass, thus converting it into a divider.

The dividers are used:

- (i) to divide curved or straight lines into desired number of equal parts,
- (ii) to transfer dimensions from one part of the drawing to another part, and
- (iii) to set-off given distances from the scale to the drawing.

They are very convenient for setting-off points at equal distances around a given point or along a given line.

(5) **Small bow divider:** The small bow divider is adjusted by a nut and is very convenient for marking minute divisions and large number of short equal distances.

**Problem 1-7.** To divide a straight line into a number of equal parts — say 3.

The straight line AB is given (fig. 1-24).

- (i) Set the legs of the divider so that the steel pins are approximately  $\frac{1}{3}$  of the length of the line apart.

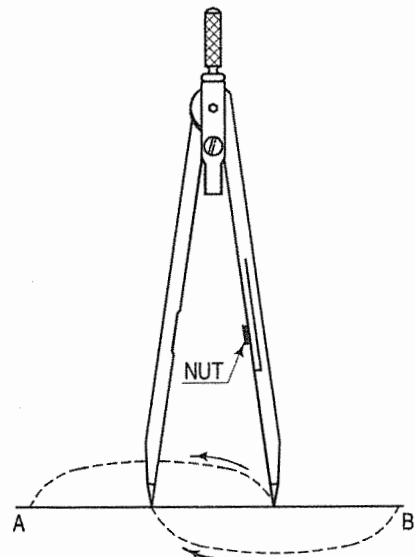


FIG. 1-24

- (ii) Step this distance lightly from one end of the line, say *B*, turning the divider first in one direction and then in the other. If the last division falls short, increase the set distance by approximately  $\frac{1}{3}$  of the difference by means of the nut, keeping the other point of the divider on the paper. If the last division goes beyond the end of the line, decrease the set distance by  $\frac{1}{3}$  of the difference.
- (iii) Re-space the line, beginning from the starting point, and adjusting the divider until the required setting is obtained.

With some practice, it will be possible to obtain the desired result with less trials and in short time. The trial divisions should be set-off as lightly as possible so that the paper is not pricked with large and unnecessary holes.

Any arc or a circle can similarly be divided into any number of equal divisions.

(6) **Small bow ink-pen:** It is used for drawing small circles and arcs in ink.

(7) **Inking pen (fig. 1-25):** This is used for drawing straight lines and non-circular arcs in ink. It consists of a pair of steel nibs fitted to a holder made of metal or ivory. Ink is filled between the two nibs to about 6 mm length by means of a quill which is usually fitted to the cork of the ink bottle. The gap between the nibs through which the ink flows and upon which the thickness of the line depends is adjusted by means of the screw *S*.

The pen should be kept sloping at about  $60^\circ$  with the paper in the direction of drawing the line and the ends of the nibs should be slightly away from the edge of the T-square or set-square. The screw should be on the side, farther from the T-square.

As the ink dries rapidly, the pen should be used immediately after it is filled. The inside faces of the nibs should be frequently cleaned for the ink to flow freely and to maintain uniformity in thickness of lines. Ink should never be allowed to dry within the pen. There should be no ink on the outside of the nibs and hence, the pen should never be dipped in ink.

For drawing large circles and circular arcs, inking attachment should be fitted in place of the pencil leg in the compass.

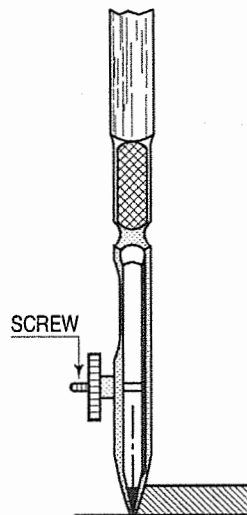


FIG. 1-25

## 1-6. SCALES (FIG. 1-26)

Scales are made of wood, steel, celluloid or plastic or card board. Stainless-steel scales are more durable. Scales may be flat or of triangular cross-section. 15 cm long and 2 cm wide or 30 cm long and 3 cm wide flat scales are in common use. They are usually about 1 mm thick. Scales of greater thickness have their longer edges bevelled. This helps in marking measurements from the scale to the drawing paper accurately. Both the longer edges of the scales are marked with divisions of centimetres, which are sub-divided into millimetres.

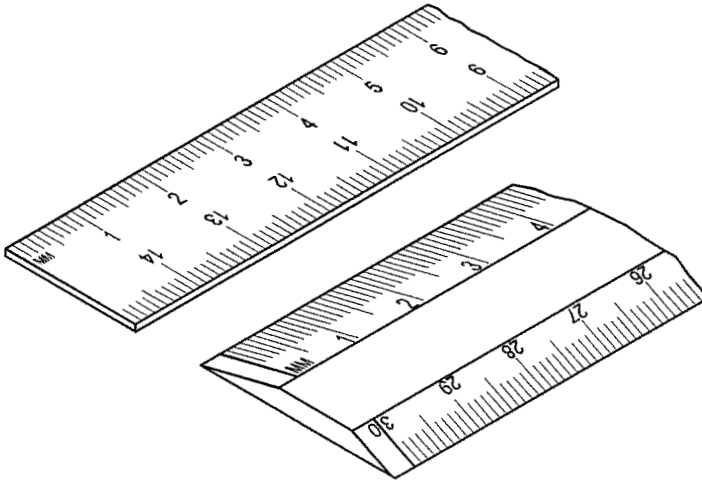


FIG. 1-26

Various other types of scales are described in chapter 4.

The scale is used to transfer the true or relative dimensions of an object to the drawing. It is placed with its edge on the line on which measurements are to be marked and, looking from exactly above the required division, the marking is done with a fine pencil point. *The scale should never be used as a straight-edge for drawing lines.* The card-board scales are available in a set of eight scales. They are designated from M1 to M8 as shown in table 1-2.

TABLE 1-2  
STANDARD SCALES

| Designation | Description |               | Scale  |
|-------------|-------------|---------------|--------|
| M1          | Full size   |               | 1:1    |
|             | 50          | cm to a metre | 1:2    |
| M2          | 40          | cm to a metre | 1:2.5  |
|             | 20          | cm to a metre | 1:5    |
| M3          | 10          | cm to a metre | 1:10   |
|             | 5           | cm to a metre | 1:20   |
| M4          | 2           | cm to a metre | 1:50   |
|             | 1           | cm to a metre | 1:100  |
| M5          | 5           | mm to a metre | 1:200  |
|             | 2           | mm to a metre | 1:500  |
| M6          | 3.3         | mm to a metre | 1:300  |
|             | 1.66        | mm to a metre | 1:600  |
| M7          | 2.5         | mm to a metre | 1:400  |
|             | 1.25        | mm to a metre | 1:800  |
| M8          | 1           | mm to a metre | 1:1000 |
|             | 0.5         | mm to a metre | 1:2000 |

### 1-7. PROTRACTOR (FIG. 1-27)

Protractor is made of wood, tin or celluloid. Protractors of transparent celluloid are in common use. They are flat and circular or semi-circular in shape. The commonest type of protractor is semi-circular and of about 100 mm diameter. Its circumferential edge is graduated to  $1^\circ$  divisions, is numbered at every  $10^\circ$  interval and is readable from both the ends. The diameter of the semi-circle (viz. straight line 0- $180^\circ$ ) is called the base of the protractor and its centre  $O$  is marked by a line perpendicular to it.

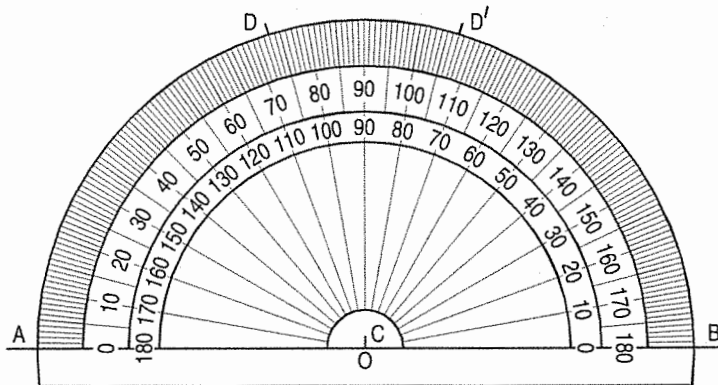


FIG. 1-27

The protractor is used to draw or measure such angles as cannot be drawn with the set-squares. A circle can be divided into any number of equal parts by means of the protractor.

**Problem 1-8.** To draw a line making an angle of  $73^\circ$  with a given line through a given point in it.

Let  $AB$  be the line and  $C$  the point in it.

- (i) Set the protractor with its base coinciding with  $AB$  (fig. 1-28) and its centre exactly on the point  $C$ .
- (ii) Mark a point  $D$  opposite to the  $73^\circ$  division and join  $C$  with  $D$ . Then  $\angle ACD = 73^\circ$  (fig. 1-28). Another point  $D'$  can be marked against the reading from the other side. In this case  $\angle BCD' = 73^\circ$  while  $\angle ACD' = 107^\circ$ .

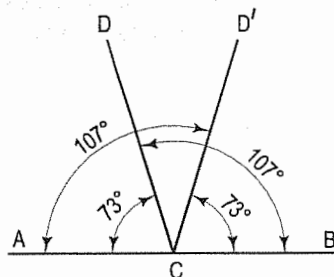


FIG. 1-28

### 1-8. FRENCH CURVES

French curves are made of wood, plastic or celluloid. They are made in various shapes, one of which is shown in fig. 1-29. Some set-squares also have these curves cut in their middle.

French curves are used for drawing curves which cannot be drawn with a compass. Faint freehand curve is first drawn through the

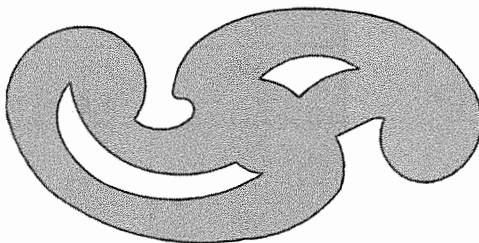


FIG. 1-29

known points. Longest possible curves exactly coinciding with the freehand curve are then found out from the french curve. Finally, neat continuous curve is drawn with the aid of the french curve. Care should be taken to see that no corner is formed anywhere within the drawn curve.

## 1-9. DRAWING PAPERS

Drawing papers are available in many varieties. For ordinary pencil-drawings, the paper selected should be tough and strong. It should be uniform in thickness and as white as possible. When the rubber eraser is used on it, its fibres should not disintegrate. Good quality of paper with smooth surface should be selected for drawings which are to be inked and preserved for a long time. It should be such that the ink does not spread. Thin and cheap quality paper may be used for drawings from which tracings are to be prepared. The standard sizes of drawing papers recommended by the Bureau of Indian Standards (B.I.S.). are given in table 2-1.

Surface area of A0 size is one square metre. Successive format sizes (from A0 to A5) are obtained by halving along the length or doubling along the width. The areas of the two subsequent sizes are in the ratio 1:2. See fig. 1-30.

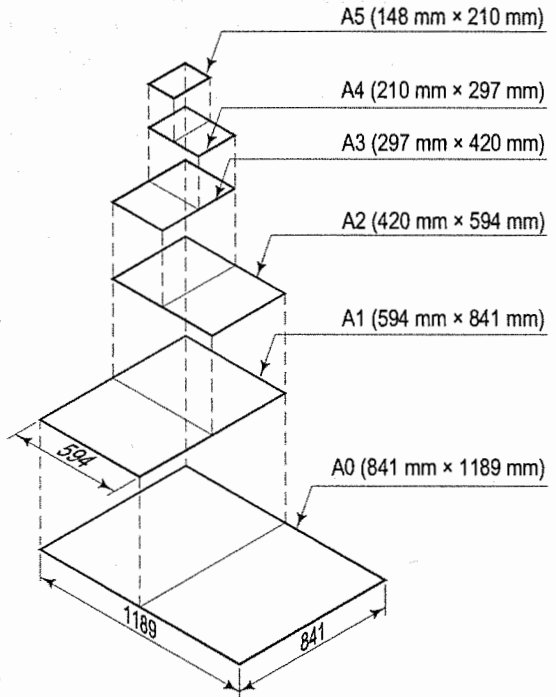


FIG. 1-30

## 1-10. DRAWING PENCILS

The accuracy and appearance of a drawing depend very largely on the quality of the pencils used. With cheap and low-quality pencils, it is very difficult to draw *lines of uniform shade and thickness*. The grade of a pencil lead is usually shown by figures and letters marked at one of its ends. Letters HB denote the medium grade. The increase in hardness is shown by the value of the figure put in front of the letter H, viz. 2H, 3H, 4H etc. Similarly, the grade becomes softer according to the figure placed in front of the letter B, viz. 2B, 3B, 4B etc.

Beginning of a drawing should be made with H or 2H pencil using it very lightly, so that the lines are faint, and unnecessary or extra lines can be easily erased. The final fair work may be done with *harder pencils, e.g. 3H and upwards*. Lines of uniform thickness and darkness can be more easily drawn with hard-grade pencils.

H and HB pencils are more suitable for lettering and dimensioning. For freehand sketching, where considerable erasing is required to be done, soft-grade pencils such as HB should be used.

Great care should be taken in mending the pencil and sharpening the lead, as the uniformity in thickness of lines depends largely on this. The lead may be sharpened to two different forms:

- (i) Conical point and
- (ii) Chisel edge.

The conical point is used in sketch work and for lettering etc. With the chisel edge, long thin lines of uniform thickness can be easily drawn and hence, it is suitable for drawing work.

To prepare the pencil lead for drawing work, the wood around the lead from the end, other than that on which the grade is marked, is removed with a pen-knife, leaving about 10 mm of lead projecting out, as shown in fig. 1-31(a).

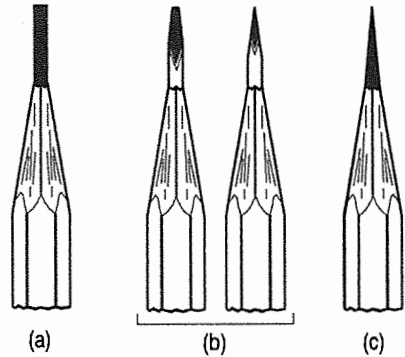


FIG. 1-31

The chisel edge [fig. 1-31(b)] is prepared by rubbing the lead on a sand-paper block, making it flat, first on one side and then on the other by turning the pencil through a half circle. For making the conical end [fig. 1-31(c)] the pencil should be rotated between the thumb and fingers, while rubbing the lead.

The pencil lead should occasionally be rubbed on the sand-paper block (while doing the drawing work) to maintain the sharpness of the chisel edge or the pointed end.

Instead of wooden pencils, Mechanical clutch pencils with a different lead size and grade like 5 mm, 4 mm and H, 2H, HB etc., are also available. Sharpening is not required in such pencils.

### 1-11. ERASER (RUBBER)

Soft India-rubber is the most suitable kind of eraser for pencil drawings. It should be such as not to spoil the surface of the paper. Frequent use of rubber should be avoided by careful planning.

### 1-12. DRAWING PINS, CLIPS OR ADHESIVE TAPES

These are used to fix the drawing paper on the drawing board. The needle part of the pin is generally made of steel, while the head may be of plated mild steel or brass. Pins of about 15 mm to 20 mm diameter and about 1 mm thick flat heads made of brass are quite convenient, as they do not rust. Pins should be so inserted that the heads sit on the surface of the paper. Clips or adhesive tapes are often used instead of the pins. (Refer fig. 1-32).

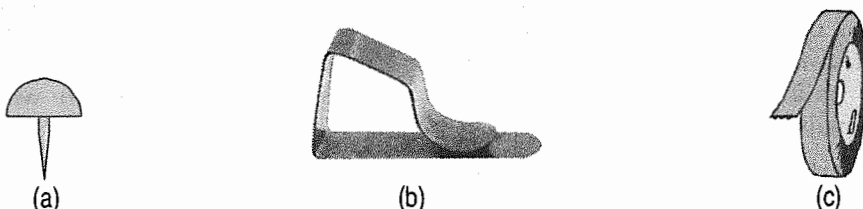


FIG. 1-32

### 1-13. SAND-PAPER BLOCK

It consists of a wooden block about 150 mm  $\times$  50 mm  $\times$  12 mm thick with a piece of sand-paper pasted or nailed on about half of its length, as shown in fig. 1-33.

The sand-paper, should be replaced by another, when it becomes dirty or worn out. This block should always be kept within easy reach for sharpening the pencil lead every few minutes.

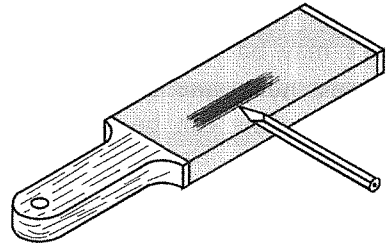


FIG. 1-33

### 1-14. DUSTER

Duster should preferably be of towel cloth of convenient size. Before starting work, all the instruments and materials should be thoroughly cleaned with the duster. The rubber crumbs formed after the use of the rubber should be swept away by the duster and not by hand. The underside of the T-square and the set-squares or the drafting machine which continuously rub against the paper should be frequently cleaned.

### 1-15. DRAFTING MACHINE (FIG. 1-34)

The uses and advantages of the T-square, set-squares, scales and the protractor are combined in the drafting machine. Its one end is clamped by means of a screw, to the distant longer edge of the drawing board. At its other end, an adjustable head having protractor markings is fitted. Two blades of transparent celluloid accurately set at right angles to each other are attached to the head.

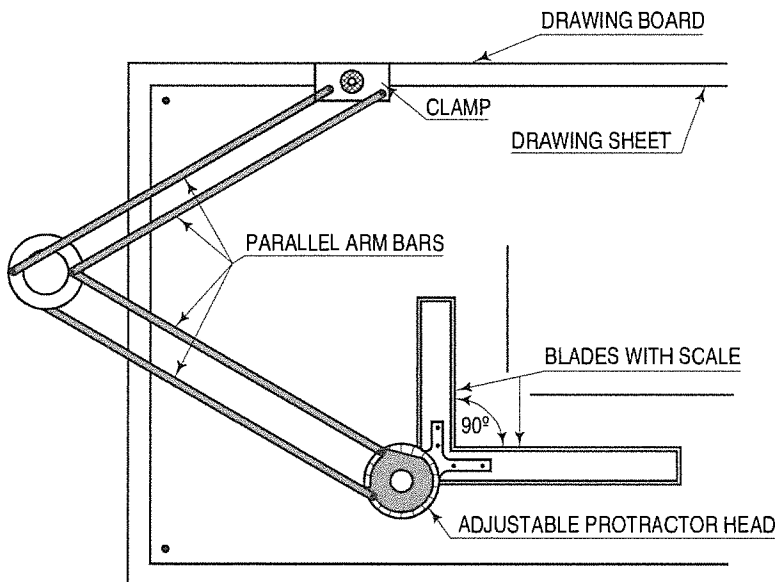


FIG. 1-34

The machine has a mechanism which keeps the two blades always parallel to their respective original position, wherever they may be moved on the board. The blades have scales marked on them and are used as straight edges. In some machines, the blades are removable and hence a variety of scales can be used. The blades may be set at any desired angle with the help of the protractor markings.

Thus, by means of this machine, horizontal, vertical or inclined parallel lines of desired lengths can be drawn anywhere on the sheet with considerable ease and saving of time. Drafting machines are common among the college students and draughtsmen.

### 1-16. ROLL-N-DRAW (FIG. 1-35)



It consists of graduated roller, scale of 16 centimeter and protactor. It is ideal for drawing vertical lines, horizontal lines, parallel lines, angles and circles.

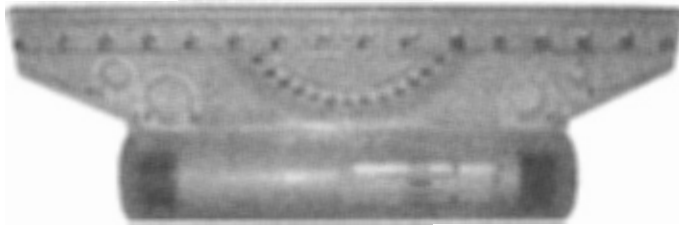


FIG. 1-35

### 1-17. GENERAL SUGGESTIONS FOR DRAWING A SHEET



(1) **Cleaning the instruments:** Clean the drawing board and the T-square and place them on the table, with the working edge of the board on your left-hand side and the stock of the T-square attached to that working edge. Clean all other instruments and materials and place them on a neat piece of paper by the side of the board. When a drafting machine is used, clean the drafting machine before fixing on drawing board.

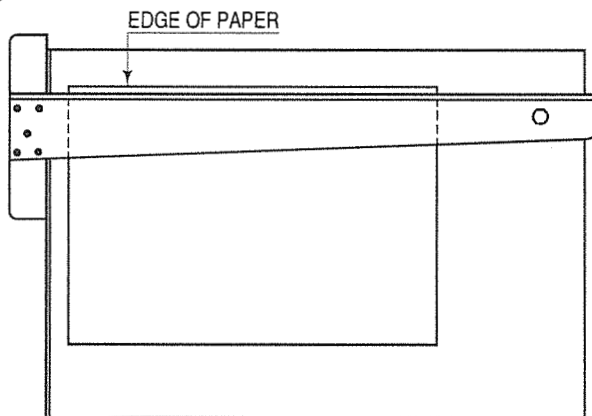


FIG. 1-36

(2) **Pinning the paper to the drawing board:** Place the paper at about equal distances from the top and bottom edges of the board and one of its shorter edges at about 25 mm from the working edge of the board. When the paper is

of a much smaller size than that of the board, it may be placed with its lower edge at about 50 mm from the bottom edge of the board. Insert a pin in the left-hand top corner of the paper and at about 10 mm from its edges. Adjust the paper with the right hand, bringing its upper edge in line with the working edge of the T-square (fig. 1-36).

Stretch the paper gently to make it perfectly flat and insert the second pin at the right-hand bottom corner. In the same manner, fix two more pins at the remaining corners. Push the pins down firmly till their heads touch the surface of the paper.

**(3) Border lines:** Perfectly rectangular working space is determined by drawing the border lines. These may be drawn at *equal distances of about 20 mm to 25 mm from the top, bottom and right-hand edges of the paper and at about 25 mm to 40 mm from the left-hand edge.* More space on the left-hand side is provided to facilitate binding of the drawing sheets in a book-form, if so desired.

**To draw the border lines** (fig. 1-37): [Dimensions shown are in mm.]

- (i) Mark points along the left-hand edge of the paper at required distances from the top and bottom edges and through them, draw horizontal lines with the T-square or by mini-drafter.
- (ii) Along the upper horizontal line, mark two points at required distances from the left-hand and right-hand edges, and draw vertical lines through them by mini-drafter.

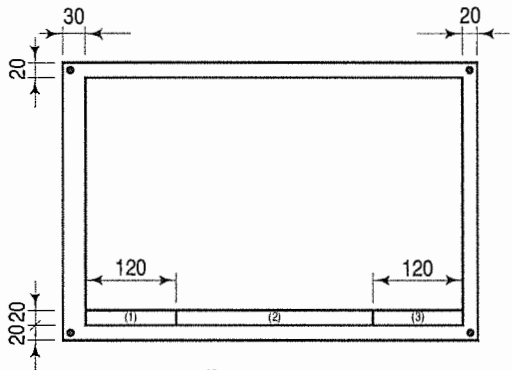


FIG. 1-37

- (iii) Erase the extra lengths of lines beyond the points of intersection.
- (iv) One more horizontal line at about 10 mm to 20 mm from the bottom border line may also be drawn and the space divided into three blocks. A title block as shown in fig. 2-2 must be drawn in left-hand bottom corner above block-3; in which
  - (a) name of the institution,
  - (b) title of the drawing, and
  - (c) name, class etc. of the student may be written.

**(4) Spacing of drawings:** When only one drawing or figure is to be drawn on a sheet, it should be drawn in the centre of the working space. For more than one figure, the space should be divided into suitable blocks and each figure should be drawn in the centre of its respective block.

*Important: The subject of Engineering Drawing cannot be learnt only by reading the book. The student must have practice in drawing. With more practice he can attain not only the knowledge of the subject but also the speed. He may possess drawing instruments of the best quality. But to gain proficiency in the subject he should pay a lot of attention to accuracy, draftsmanship i.e. uniformity in thickness and shade of lines according to its type, nice lettering and above all general neatness.*

## EXERCISES 1

1. In an A2 size sheet, copy fig. 1-38(a) to fig. 1-38(g) as per layout shown in fig. 1-39. [All dimensions are in mm.]

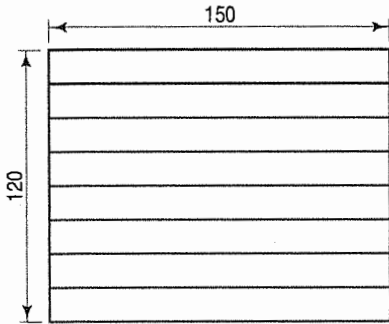


FIG. 1-38(a)

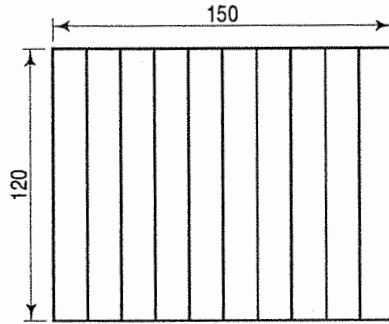


FIG. 1-38(b)

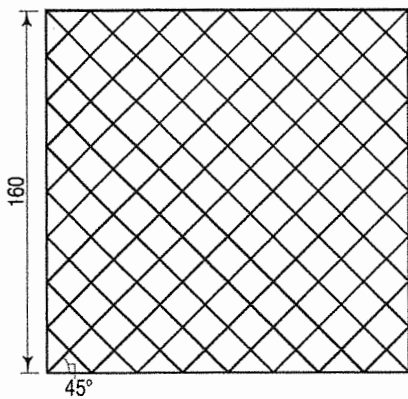


FIG. 1-38(c)

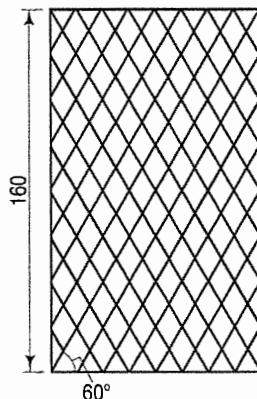


FIG. 1-38(d)

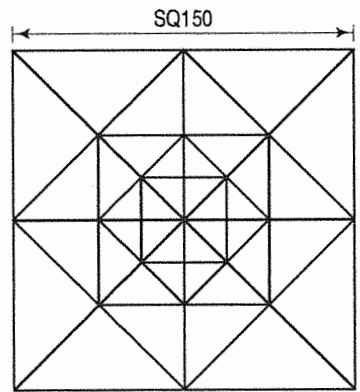


FIG. 1-38(e)

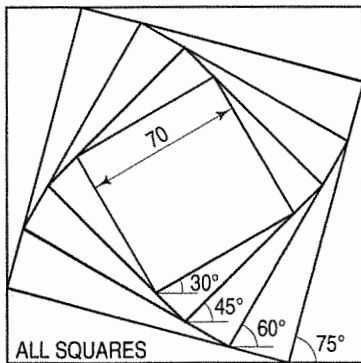


FIG. 1-38(f)

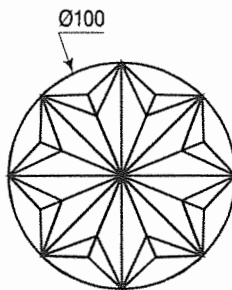


FIG. 1-38(g)

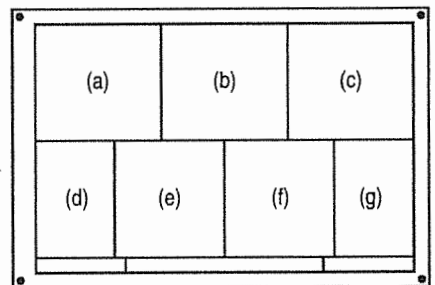


FIG. 1-39

2. Draw a circle of 75 mm diameter. With the aid of T-square and set-squares only, draw lines passing through its centre, dividing it into  
(i) eight, (ii) twelve and (iii) twenty four equal parts.

3. Without using a protractor, draw triangles having following base angles on a 75 mm long line as base:
  - (i)  $75^\circ$  and  $15^\circ$ , (ii)  $60^\circ$  and  $75^\circ$ , (iii)  $135^\circ$  and  $15^\circ$ , (iv)  $105^\circ$  and  $45^\circ$ .
4. Draw a line 125 mm long and divide it into seven equal parts by means of a divider.
5. Copy fig. 1-40(a) to fig. 1-40(d) as per layout shown in fig. 1-41.  
[All dimensions are in mm.]

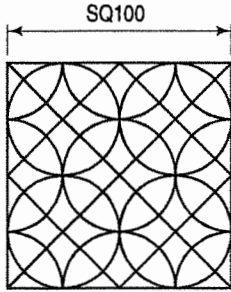


FIG. 1-40(a)

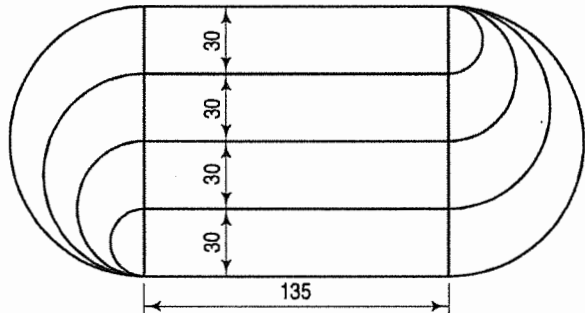


FIG. 1-40(b)

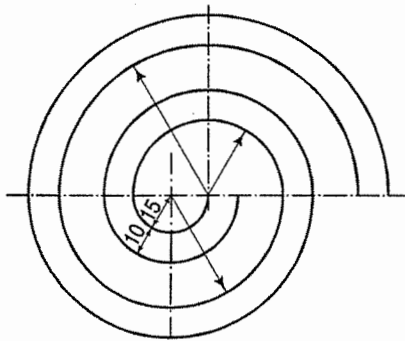


FIG. 1-40(c)

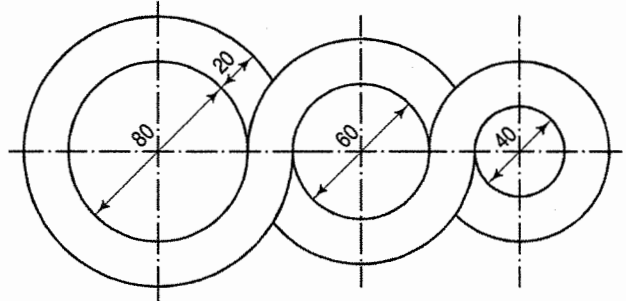


FIG. 1-40(d)

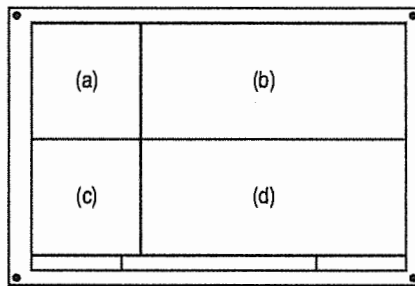


FIG. 1-41

6. Draw lines (using the protractor) meeting at the end A of a line AB and making with it the angles: (1)  $27^\circ$ , (2)  $49^\circ$ , (3)  $115^\circ$ , (4)  $151^\circ$ .
7. Draw a straight line AB of any length. Mark any point O on it. Through O, draw all possible angles in multiples of  $15^\circ$ , with the aid of the T-square and set-squares.
8. Fill up the blanks with appropriate words selected from the list given:
  - (a) The edge of the board on which T-square is made to slide is called its \_\_\_\_\_.

(1) working edge (2) straight edge (3) chisel edge.

- (b) To prevent warping of the board \_\_\_\_\_ are cleated at its back.  
(1) packings (2) wooden blocks (3) battens.
- (c) The two parts of the T-square are \_\_\_\_\_ and \_\_\_\_\_.  
(1) vertical and horizontal edge (2) straight edge (3) stock (4) blade.
- (d) The T-square is used for drawing \_\_\_\_\_ lines.  
(1) vertical (2) curve (3) horizontal.
- (e) Angles in multiples of  $15^\circ$  are constructed by the combined use of \_\_\_\_\_ and \_\_\_\_\_.  
(1) T-square (2) set-squares (3) protractor.
- (f) To draw or measure angles, \_\_\_\_\_ is used.  
(1) set-squares (2) T-square (3) protractor.
- (g) For drawing large-size circles, \_\_\_\_\_ is attached to the compass.  
(1) straight bar (2) bow compass (3) lengthening bar.
- (h) Circles of small radii are drawn by means of a \_\_\_\_\_.  
(1) lengthening bar (2) bow divider (3) bow compass.
- (i) Measurements from the scale to the drawing are transferred with the aid of a \_\_\_\_\_.  
(1) scale (2) compass (3) divider.
- (j) The scale should never be used as a \_\_\_\_\_ for drawing straight lines.  
(1) set-squares (2) working edge (3) straight edge.
- (k) \_\_\_\_\_ is used for setting-off short equal distances.  
(1) compass (2) bow divider (3) scale.
- (l) For drawing thin lines of uniform thickness the pencil should be sharpened in the form of \_\_\_\_\_.  
(1) chisel edge (2) conical (3) pointed.
- (m) Pencil of \_\_\_\_\_ grade sharpened in the form of \_\_\_\_\_ is used for sketching and lettering.  
(1) soft (2) low (3) conical point (4) chisel.
- (n) \_\_\_\_\_ are used for drawing curves which cannot be drawn by a compass.  
(1) bow compass (2) protractor (3) French curves.
- (o) To remove unnecessary lines \_\_\_\_\_ is used.  
(1) duster (2) chalk (3) sand box (4) eraser.
- (p) Uses of the T-square, set-squares, scale and protractor are combined in the \_\_\_\_\_.  
(1) set-squares (2) drafting machine (3) compass.
- (q) Circles and arcs of circles are drawn by means of a \_\_\_\_\_.  
(1) lengthening bar (2) divider (3) compass.
- (r) Inking pen is used for drawing \_\_\_\_\_ in ink.  
(1) writing (2) curves (3) straight lines.
- (s) Set-squares are used for drawing \_\_\_\_\_, \_\_\_\_\_ and \_\_\_\_\_ lines.  
(1) horizontal (2) vertical (3) inclined (4) parallel.

Answer to Ex. 8:

a-1, b-3, c-3 and 4, d-3, e-1 and 2, f-3, g-3, h-3, i-3, j-3, k-2, l-1, m-1 and 3, n-3, o-4, p-2, q-3, r-3, s-2, 3 and 4.

# Chapter 2

## SHEET LAYOUT AND FREE-HAND SKETCHING



### 2-1. SHEET LAYOUT

(1) **Sheet sizes:** The preferred sizes of the drawing sheets recommended by the Bureau of Indian Standards (B.I.S.) are given below as per SP : 46 (2003). Refer fig. 1-30:

TABLE 2-1

| Sheet designation | Trimmed size (mm) | Untrimmed size (mm) |
|-------------------|-------------------|---------------------|
| A0                | 841 × 1189        | 880 × 1230          |
| A1                | 594 × 841         | 625 × 880           |
| A2                | 420 × 594         | 450 × 625           |
| A3                | 297 × 420         | 330 × 450           |
| A4                | 210 × 297         | 240 × 330           |
| A5                | 148 × 210         | 165 × 240           |

The layout of the drawing on a drawing sheet should be done in such a manner as to make its reading easy and speedy. Fig. 2-1(a) and fig. 2-1(b) shows an A1 size sheet layout. All dimensions are in millimetres.

(2) **Margin:** Margin is provided in the drawing sheet by drawing margin lines [fig. 2-1(a)]. Prints are trimmed along these lines. After trimming, the prints would be of the recommended trimmed sizes of the trimmed sheets.

(3) **Border lines:** Clear working space is obtained by drawing border lines as shown in [fig. 2-1(a)]. More space is kept on the left-hand side for the purpose of filing or binding if necessary. When prints are to be preserved or stored in a cabinet without filing, equal space may be provided on all sides (fig. 2-3).

(4) **Borders and frames:** SP : 46 (2003) recommends the borders of 20 mm width for the sheet sizes A0 and A1, and 10 mm for the sizes A2, A3, A4 and A5. Frame shows the clear space available for the drawing purpose.

(5) **Orientation mark:** Four centring marks are drawn as shown in fig. 2-1(b) to facilitate positioning of the drawing for the reproduction purpose. The *orientation mark* will coincide with one of centring marks which can be used for the orientation of drawing sheet on the drawing board.



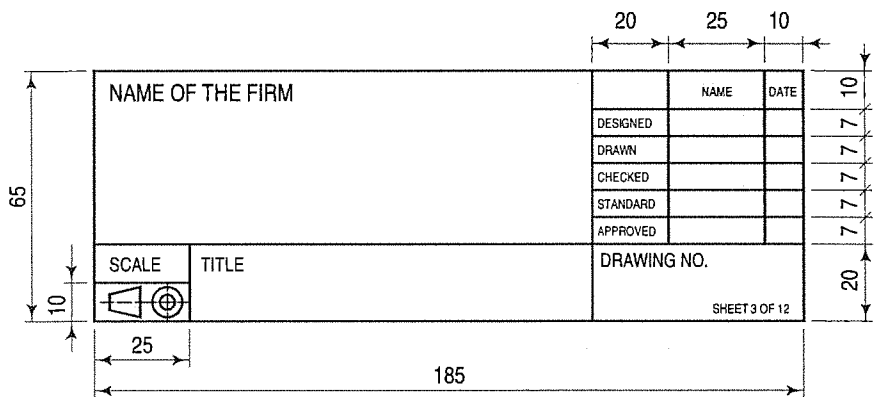


FIG. 2-2

TABLE 2-2  
PARTICULARS OF TITLE BLOCK

|    |                                                                                           |
|----|-------------------------------------------------------------------------------------------|
| 1. | Name of the firm.                                                                         |
| 2. | Title of the drawing.                                                                     |
| 3. | Scale.                                                                                    |
| 4. | Symbol for the method of projection.                                                      |
| 5. | Drawing number.                                                                           |
| 6. | Initials with dates of persons who have designed, drawn, checked, standards and approved. |
| 7. | No. of sheet and total number of sheets of the drawing of the object.                     |

(8) **List of parts or the bill of materials:** When drawings of a number of constituent parts of an object are drawn in a single drawing sheet, a list of these parts should be placed above or beside the title block in a tabular form. It should provide the following minimum particulars for each part:

Part no., name or description, no. off i.e. quantity required, material and sometime stock size of raw material, remarks.

Additional information such as job and order number, instructions regarding finish, heat-treatment, tolerances (general) and references pertaining to jigs, fixtures, tools, gauges etc. may be, if necessary included in the title block or given separately in tabular form.

(9) **Revisions of drawing:** For locating a portion of the drawing for the purpose of revision etc., the sides of the three larger sizes of the drawing sheets viz. A0, A1 and A2 are divided into a number of equal zones.

Fig. 2-3 shows an A1 size sheet with zones marked on it. The zones along the length *l* are designated by numerals, while those along the width *w* are designated by letters. The location D8 is a rectangle formed by the intersection of the zones D and 8. The number of zones suggested by the B.I.S. for A0, A1 and A2 sizes of drawing sheets along the lengths *l* are 16, 12 and 8 respectively, while those along the widths *w* are 12, 8 and 6 respectively.

A revision panel is drawn either attached to the title block above it or in the top right-hand corner of the sheet. The revisions are recorded in it giving the revision number, date, zone etc. and also the initials of the approving authority.

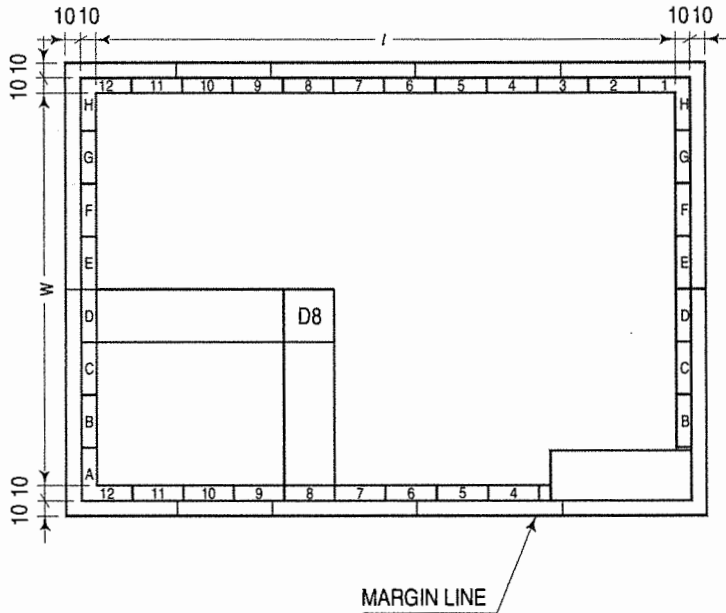
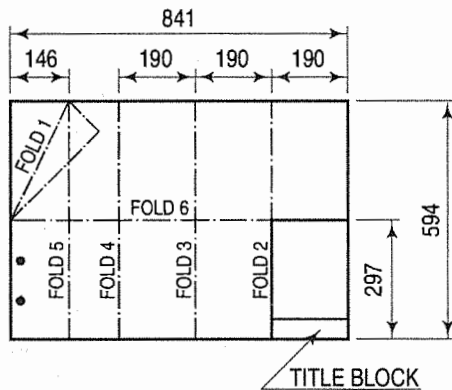


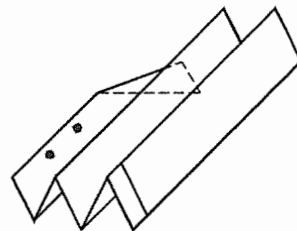
FIG. 2-3

**(10) Folding marks:** Folding marks are made in the drawing sheet as shown in fig. 2-1(a). They are helpful in folding of prints in proper and easy manner. Two methods of folding of prints are in general use. Method I is suitable for prints which are to be filed or bound. It allows prints to be unfolded or refolded without removing them from the files.

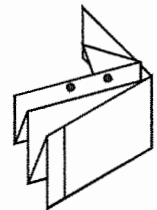
Fig. 2-4(i) shows the folding diagram for folding an A1 size sheet by method I. It is folded in two stages, viz. lengthwise [fig. 2-4(ii)] and crosswise [fig. 2-4(iii)].



(i)



(ii)



(iii)

FIG. 2-4

When prints are to be stored and preserved in cabinets they are folded by method II. The folding diagram for folding an A1 size sheet by method II is shown in fig. 2-5. The two stages of folding are similar to those for method I.

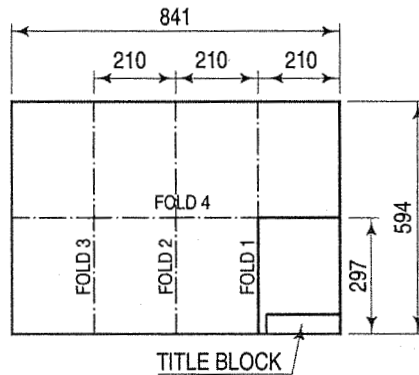


FIG. 2-5

Dimensions for folding of various sizes of drawing sheets by the two methods are given below:

**Method I****Method II**

| Sheet size | Horizontal dimensions from left (mm) | Vertical dimensions from bottom (mm) | No. of folds | Sheet size | Horizontal dimensions from left (mm) | Vertical dimensions from bottom (mm) | No. of folds |
|------------|--------------------------------------|--------------------------------------|--------------|------------|--------------------------------------|--------------------------------------|--------------|
| A0         | $130 + 109 + 190 \times 5$           | $297 \times 2 + 247$                 | 9            | A0         | $139 + 210 \times 5$                 | $297 + 297 + 247$                    | 7            |
| A1         | $146 + 125 + 190 \times 3$           | $297 + 297$                          | 6            | A1         | $211 + 210 \times 3$                 | $297 + 297$                          | 4            |
| A2         | $116 + 96 \times 3 + 190$            | 297                                  | 1            | A2         | $174 + 210 \times 2$                 | $297 + 123$                          | 3            |
| A3         | $125 + 105 + 190$                    | 297                                  | 1            | A3         | $210 + 210$                          | 297                                  | 1            |

The final size of the folded print in method I will be 297 mm  $\times$  190 mm, while that in method II will be 297 mm  $\times$  210 mm. In either case the title block is visible in the top part of the folded print.

**(11) Scales and scale drawing:** Sometime machine part is required to draw larger or smaller than their actual size. For example a crankshaft of an engine would be drawn to a reduced scale, while connecting rod's bolt is to be drawn to enlarged scale. The scale of drawing must be indicated in the title block as shown in fig. 2-2. When details are drawn to the different scale in the same drawing sheet corresponding scale should be mentioned under each such detail.

Table 2-3 shows the scales recommended in SP : 46 (2003).

TABLE 2-3

|     |                  |          |          |           |
|-----|------------------|----------|----------|-----------|
| (a) | Reducing scales  | 1 : 2    | 1 : 5    | 1 : 10    |
|     |                  | 1 : 20   | 1 : 50   | 1 : 100   |
|     |                  | 1 : 200  | 1 : 500  | 1 : 1000  |
|     |                  | 1 : 2000 | 1 : 5000 | 1 : 10000 |
| (b) | Enlarging scales | 50 : 1   | 20 : 1   | 10 : 1    |
|     |                  | 5 : 1    | 2 : 1    |           |
| (c) | Full size scales |          |          | 1 : 1     |

## 2-2. TYPES OF MACHINE DRAWINGS



Machine drawings are prepared for various purposes and they are further classified as under:

(1) **Production drawing:** A production drawing is legal document of company. It is used by the technicians on the shop-floor for manufacturing the parts. It must provide informations about part number, dimensions, tolerance, surface finish, material and stock size, manufacturing process, special finishing process if required, and no. off required for each assembly. It is further sub-classified as:

- (i) Part drawing or detailed drawing
- (ii) Assembly drawing.

(2) **Exploded assembly drawing:** It represents the details of machine in a pictorial form as it is assembled. It helps the mechanics for dismantling machine for repairing purpose.

(3) **Schematic assembly drawing:** This type of assembly drawing is used for explaining working principle of any machine.

(4) **Drawing for instruction manual:** This is assembly drawing without dimensions. Each part of machine is numbered so that it can be easily dismantled or assembled if required. This is also used for explaining working principle of each part.

(5) **Drawing for installation:** This is assembly drawing with overall dimensions. It is used for the preparation of foundation for installing machine.

(6) **Drawing for catalogue:** Special assembly drawings are prepared for catalogues, with overall and principal dimensions.

(7) **Tabular drawing:** This is part drawing. It is used when components of same shape but different dimensions are to be manufactured.

(8) **Patent drawing:** It is generally assembly drawing either in pictorial form or principal view of orthographic projection of machine. It is used for obtaining *patent* of the machine.

## 2-3. FREE-HAND SKETCHING



(1) **Sketching or freehand:** Sketching or freehand is a first step to the preparation of a scale-drawing, i.e., a drawing drawn with the aid of instruments. A designer records his ideas initially in the form of sketches which are later converted into drawings. Similarly, views of actual objects are in the first instance, sketched freehand. Scale-drawings are then prepared from these sketches. Ideas and objects can be described in words, but the description is made more expressive with the aid of sketches. Thus sketching is of great importance in engineering practice.

Sketching is always done freehand. It is in fact a freehand drawing made in correct proportions, but not to scale. A sketch should be so prepared as to give to others a clear idea, complete information and true impression of the object to be constructed. It should never be drawn too small. The size of a sketch should be such that all the features of the object, together with their dimensions, explanatory notes etc. are clearly incorporated in it. Proficiency in sketching can be achieved with constant practice only.

**(2) Sketching materials:**

- (i) A soft-grade pencil, preferably HB or H sharpened to a conical point.
- (ii) A soft rubber-eraser.
- (iii) A paper in form of a sketch-book or a pad.

Above the *three* things are absolutely essential for sketching.

Sometimes, cross-sectioned graph paper ruled with light lines is also used instead of a plain paper. These lines and squares help in drawing straight lines and also maintaining proportions.

As such papers may not always be readily available, it is advisable to learn sketching without their aid.

**(3) To sketch straight lines:** Horizontal lines are sketched with the motion of the wrist and the fore-arm. They are sketched from left to right [fig. 2-6(i)]. To sketch a horizontal line, mark the end points. Hold the pencil at about 30 mm distance from the lead point. Swing it from left to right and backwards, between the two points and without touching the surface of the paper, till the correct direction is achieved. Then begin to draw the line (with the wrist-motion) with short and light strokes. Shift the hand after each stroke. Keep your eyes on the point at which the line is to end. Finish finally with a dark and firm line. Take proper care to maintain straightness and correct direction of the line.

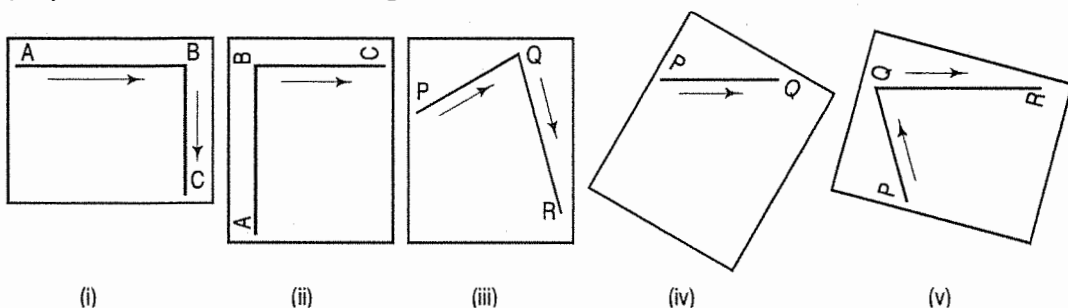


FIG. 2-6

Vertical lines are sketched downwards [fig. 2-6(i)] with the movement of fingers. They may also be sketched by converting them into horizontal lines by revolving the paper as shown in fig. 2-6(ii). Vertical or horizontal lines near the edges of a sketch-book may be drawn by sliding fingers along those edges, thus using them as guides. Inclined lines, when they are nearly horizontal, are sketched from left to right. When they are nearly vertical, they are sketched downwards [fig. 2-6(iii)]. These lines also may be sketched as horizontal lines by revolving the paper as shown in fig. 2-6(iv) and fig. 2-6(v).

**(4) To sketch circles and arcs:** Mark the centre and through it, draw horizontal and vertical centre lines [fig. 2-7(i)]. Add four radial lines between them. Mark points on these lines at radius-distance from the centre, judging by the eye or using a slip of paper as a trammel, on which the radius-distance has been approximately marked [fig. 2-7(i)]. Complete the circle with light strokes. The paper may be revolved after about each quarter-circle for easy wrist motion. Erase the additional radial lines completely. Dim all the lines before fairing the circle with a thin and black outline [fig. 2-7(iii)]. Keep the centre lines thin and light.

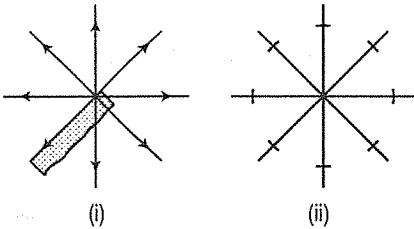


FIG. 2-7

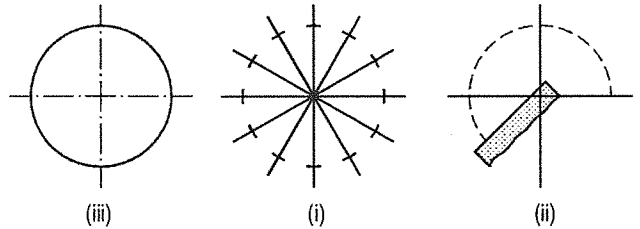


FIG. 2-8

Large circles may be sketched as described above by adding a few extra radial lines [fig. 2-8(i)]. An easier method is to mark a number of points by means of a trammel (at radius-distance from the centre) and to sketch the circle through these points [fig. 2-8(ii)].

Large circles can also be drawn by making a compass of fingers and a pencil. Keep the little finger as a pivot at the centre. Hold the pencil stationary so that its point is at radius-distance from the centre and touches the paper. Rotate the paper with the other hand. The pencil-point will mark the circle on the paper. Two pencils may also be used as a compass. One pencil is held as a pivot, while the other describes the circle as the paper is rotated.

A circle of small radius can be sketched within a square. Sketch the circumscribing square (length of the side equal to the diameter of the circle) and mark the diagonals [fig. 2-9(i)]. Mark the mid-points of the sides of the square and four points on the diagonals at radius-distance from the centre [fig. 2-9(ii)]. Sketch a neat circle through the eight points [fig. 2-9(iii)].

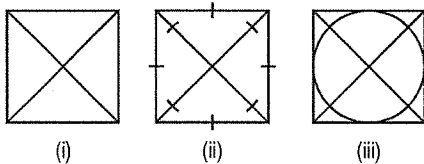


FIG. 2-9

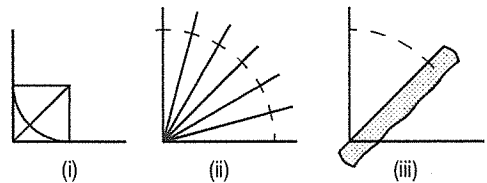


FIG. 2-10

Arcs of small radii are conveniently drawn by constructing squares [fig. 2-10(i)]. Large-radii arcs may be drawn by one of the methods described above for large circles. Radial-line and trammel methods are shown in fig. 2-10(ii) and fig. 2-10(iii) respectively.

(5) **Sketching procedure:** Sketches should never be prepared with the aid of a scale or a straight-edge. All lines must be absolutely free-hand and their measurements must be in proportion only. A sketch is considered to be good when its features are shown in correct proportions. Its outlines must be black and thin but rigidly firm. Dimension lines and centre lines should be comparatively light. Dimension figures must be inserted with good care, as if they are printed. Lettering also should be done in a similar manner.

(6) **Steps in sketching:** Following are the steps in sketching three orthographic views of a shaft-support, shown (in pictorial view) in fig. 2-11.

- (i) Determine the over-all dimensions of the views and sketch rectangles for the same in good proportions and correct projection, keeping sufficient space between them and from the border lines [fig. 2-12(i)].

- (ii) Insert centre lines for arcs and circles, and block-in the main shapes [fig. 2-12(ii)].
- (iii) Sketch arcs and circles, and build-up the details [fig. 2-12(iii)].
- (iv) Erase all construction lines. Dim all the lines and then make them bold and firm [fig. 2-12(iv)] starting with the curves. Keep the centre lines fainter. Insert all dimensions. Print the title.

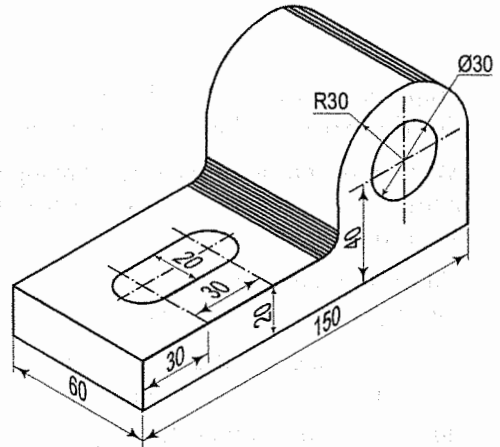
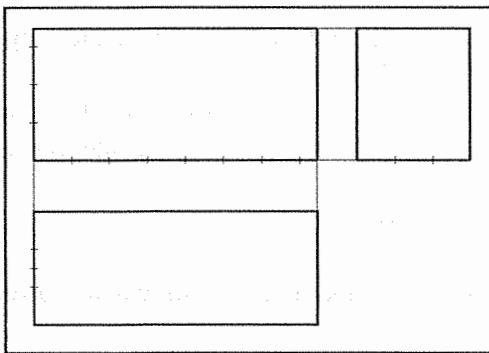
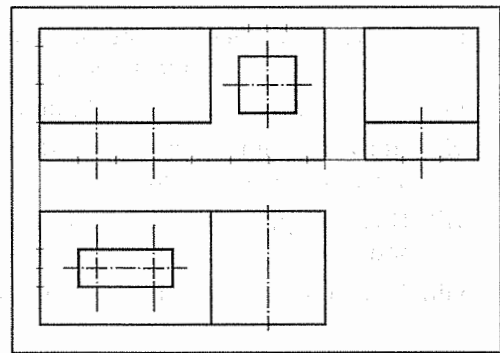


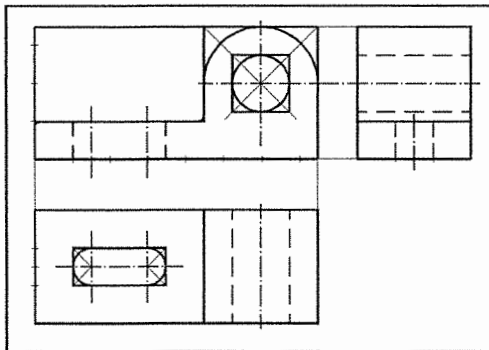
FIG. 2-11



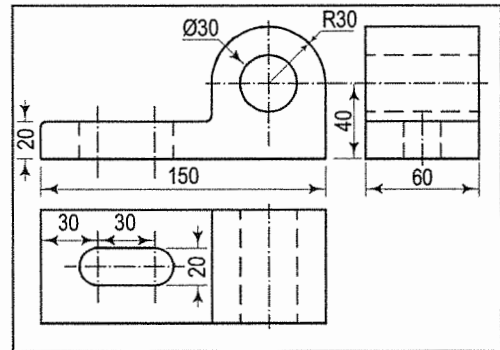
(i)



(ii)



(iii)



(iv)

FIG. 2-12

Please also refer topic 20-7 "Procedure for preparing a scale-drawing".

## EXERCISES 2



1. Prepare, giving all details, a neat sketch of a simple title block provided in a drawing sheet. What is its size, as recommended by the B.I.S.?
2. Make a sketch of an A1 size drawing sheet, showing in it, the border lines, margin lines, folding marks, zones and the title block. Give the untrimmed and trimmed sizes of this sheet.
3. Fill-up the blanks in the following by appropriate words or numerals selected from those given in the list as shown below.
  - (i) The size of the title block for all sizes of drawing sheets is \_\_\_\_\_ mm  $\times$  \_\_\_\_\_ mm.
  - (ii) The zones along the length of the sheet are designated by \_\_\_\_\_, while those along its width by \_\_\_\_\_.
  - (iii) For locating a portion of a drawing, the sheet is divided into a number of \_\_\_\_\_.
  - (iv) For A1 size sheet the number of zones suggested by B.I.S. along the length are \_\_\_\_\_ while those along the width are \_\_\_\_\_.
  - (v) The drawing sheet is so folded that the \_\_\_\_\_ is always on the top.
  - (vi) Horizontal lines are sketched from \_\_\_\_\_ to \_\_\_\_\_ while vertical lines are sketched from \_\_\_\_\_ to \_\_\_\_\_.
  - (vii) Three things absolutely essential for sketch-work are \_\_\_\_\_, \_\_\_\_\_ and \_\_\_\_\_.
  - (viii) A sketch is considered to be good when its features are shown in correct \_\_\_\_\_.

List of words and numerals for Ex. 3:

- |                |                |              |             |
|----------------|----------------|--------------|-------------|
| 1. Title block | 7. Proportions | 13. Eraser   | 19. Top     |
| 2. Pencil      | 8. Right       | 14. Left     | 20. Compass |
| 3. 210         | 9. 85          | 15. 16       | 21. 65      |
| 4. Numerals    | 10. Letters    | 16. Position | 22. Blocks  |
| 5. Paper       | 11. Bottom     | 17. Zones    | 23. 8       |
| 6. 12          | 12. 6          | 18. 185      |             |

Answer to Ex. 3:

- (i) 18 and 21, (ii) 4 and 10, (iii) 17, (iv) 6 and 23,
- (v) 1, (vi) 14, 8, 19 and 11, (vii) 2, 13 and 5, (viii) 7.
4. Study chapter 7 and 20 carefully and then solve the exercises given at the end of each, by means of sketches.
5. What are the different types of machine drawings?
6. Differentiate clearly production drawing and installation drawing.
7. Prepare freehand sketches of the views given in figures 2-13 to 2-26. Assume each unit to be of 10 mm length. Insert all dimensions.

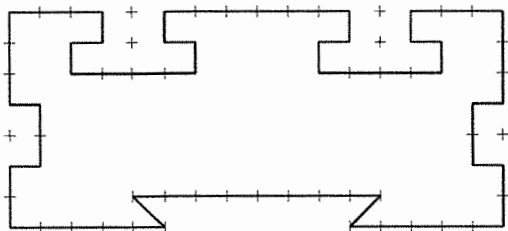


FIG. 2-13

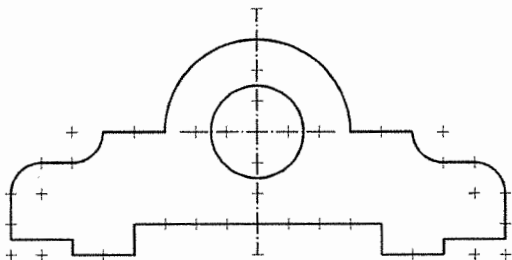


FIG. 2-14

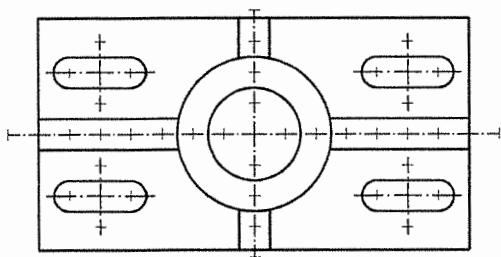


FIG. 2-15

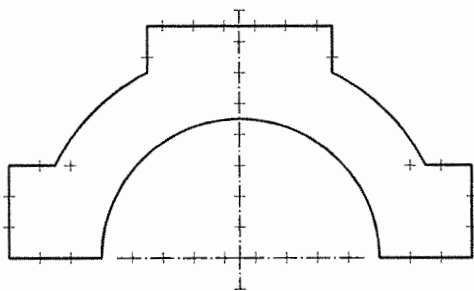


FIG. 2-16

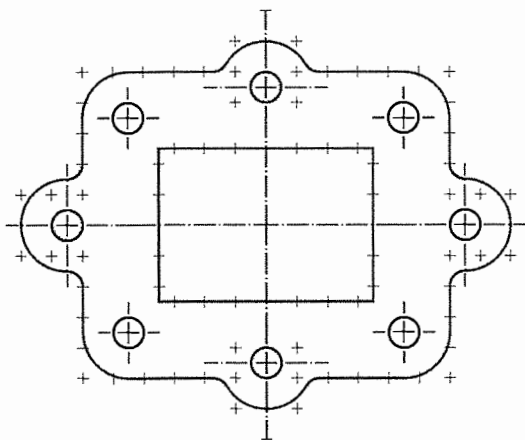


FIG. 2-17

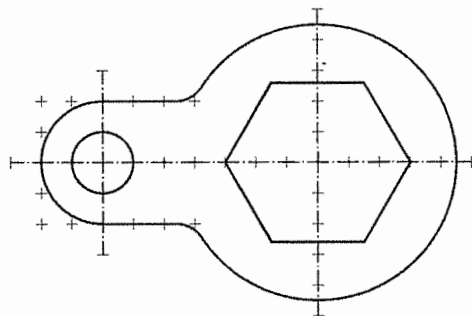


FIG. 2-18

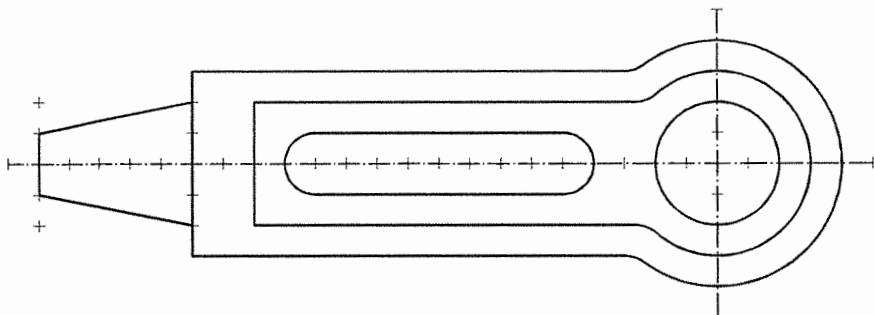


FIG. 2-19

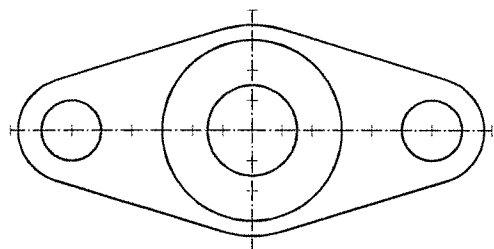


FIG. 2-20

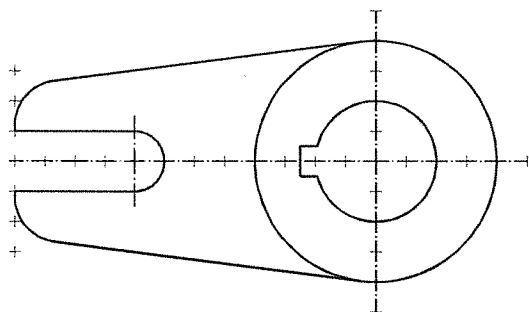


FIG. 2-21

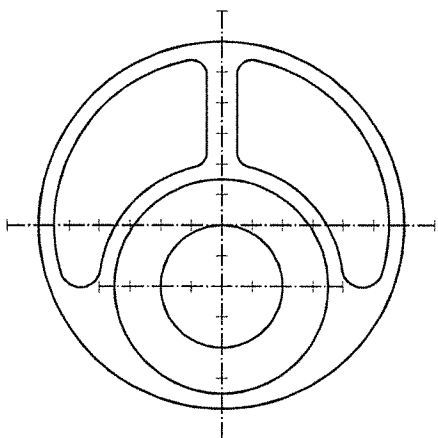


FIG. 2-22

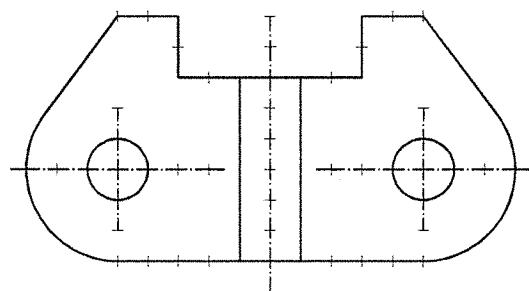


FIG. 2-23

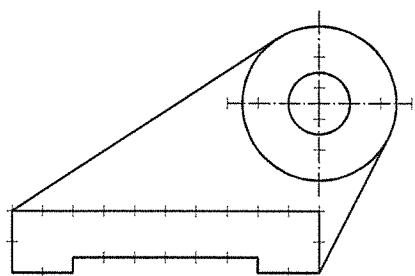


FIG. 2-24

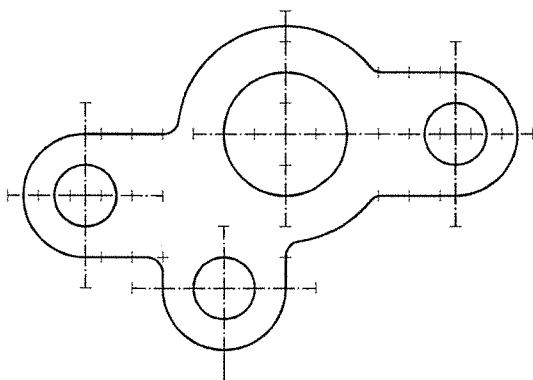


FIG. 2-25

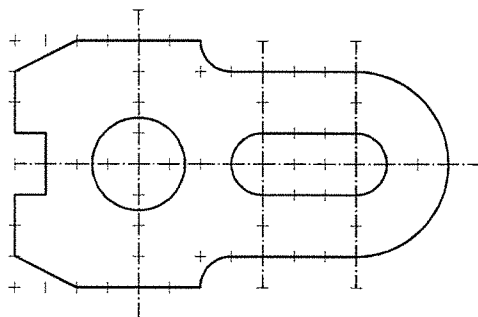


FIG. 2-26

# Chapter 3

## LINE, LETTERING AND DIMENSIONING



### 3-0. INTRODUCTION

This chapter deals with various types of lines, lettering and dimensioning which are used in engineering drawing.

### 3-1. LINES

Various types of lines used in general engineering drawing are shown in fig. 3-1 as described by SP 46:2003.

(1) **Line thickness:** The thicknesses of lines are varied according to the drawing and are finalized either *by ink* or *by pencil*.

(2) **Inked drawings:** The thicknesses of lines in various groups are shown in table 3-1. The line-group is designated according to the *thickness of the thickest line*. For any particular drawing a line-group is selected according to its size and type. All lines should be sharp and dense so that good prints can be reproduced.

TABLE 3-1

| Line-group<br>mm | Type of line |     |     |     |     |     |     |
|------------------|--------------|-----|-----|-----|-----|-----|-----|
|                  | A            | B   | C   | D   | E   | F   | G   |
| 1.2              | 1.2          | 0.6 | 0.4 | 0.4 | 1.2 | 0.4 | 0.4 |
| 0.8              | 0.8          | 0.4 | 0.3 | 0.3 | 0.8 | 0.3 | 0.3 |
| 0.5              | 0.5          | 0.3 | 0.2 | 0.2 | 0.5 | 0.2 | 0.2 |
| 0.3              | 0.3          | 0.2 | 0.1 | 0.1 | 0.3 | 0.1 | 0.1 |

(3) **Pencil drawings:** For drawings finalized with pencil, the lines can be divided into two line-groups as shown in table 3-2. It is important to note that in the *finished drawing*, all lines except construction lines should be *dense, clean* and *uniform*. Construction lines should be drawn very *thin* and *faint* and should be hardly visible in the finished drawing.

TABLE 3-2

| Line-group<br>mm | Thickness | Lines                                                                                                                                    |
|------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------|
| 0.2              | Medium    | Out lines, dotted lines, cutting plane-lines                                                                                             |
| 0.1              | Thin line | Centre lines, section lines, dimension lines, extension lines, construction lines, leader lines, short-break lines and long-break lines. |

TABLE 3-3

| Lines                                                                    | Pencil   |
|--------------------------------------------------------------------------|----------|
| Initial work and construction lines                                      | H        |
| Outlines, dotted lines, section-plane lines, dimension lines, arrowheads | 2H       |
| Centre lines, section lines                                              | 3H or 4H |

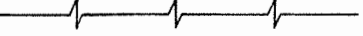
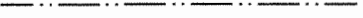
| Line                                                                                  | Description                                                                   | General applications                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A    | Continuous thick or Continuous wide                                           | Visible outlines, visible edges; crests of screw threads; limits of length of full depth thread, lines of cuts and section arrows; parting lines of moulds in views; main representations in diagrams, maps, flow charts; system lines (structural metal engg.)                 |
| B    | Continuous thin (narrow) (straight or curved)                                 | Imaginary lines of intersection; grid, dimension, extension, projection, short centre, leader, reference lines; hatching; outlines of revolved sections; root of screw threads; interpretation lines of tapered features; framing of details; indication of repetitive details; |
| C    | Continuous thin (narrow) freehand                                             | Limits of partial or interrupted views and sections, if the limit is not a chain thin line                                                                                                                                                                                      |
| D  | Continuous thin (narrow) with zigzags (straight)                              | Long-break line                                                                                                                                                                                                                                                                 |
| E  | Dashed thick (wide)                                                           | Line showing permissible of surface treatment                                                                                                                                                                                                                                   |
| F  | Dashed thin (narrow)                                                          | Hidden outlines; hidden edges                                                                                                                                                                                                                                                   |
| G  | Chain thin Long-dashed dotted (narrow)                                        | Centre line; lines of symmetry; trajectories; pitch circle of gears, pitch circle of holes,                                                                                                                                                                                     |
| H  | Chain thin (narrow) with thick (wide) at the ends and at changing of position | Cutting planes                                                                                                                                                                                                                                                                  |
| J  | Chain thick or Long-dashed dotted (wide)                                      | Indication of lines or surfaces to which a special requirement applies                                                                                                                                                                                                          |
| K  | Chain thin double-dashed or long-dashed double-dotted (narrow)                | Outlines of adjacent parts<br>Alternative and extreme positions of movable parts<br>Centroidal lines<br>Initial outlines prior to forming<br>Parts situated in front of the cutting plane                                                                                       |

FIG. 3-1

Drawing pencils are graded according to increase in relative hardness. They are marketed with the labelled as H, 2H, 3H, 4H, 5H, 6H etc. Students are advised to use pencils as recommended in table 3-3, for getting accurate, clean and neat drawings.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 3 for Type of lines, lettering and dimensioning.

### 3-1-1. TYPES OF LINES (FIG. 3-2 AND FIG. 3-3)



(1) **Outlines (A):** Lines drawn to represent visible edges and surface boundaries of objects are called outlines or principal lines. They are continuous thick or wide lines (fig. 3-2).

(2) **Margin lines (A):** They are continuous thick or wide lines along which the prints are trimmed [fig. 2-1(a)].

(3) **Dimension lines (B):** These lines are continuous thin lines. They are terminated at the outer ends by pointed arrowheads touching the outlines, extension lines or centre lines (fig. 3-2).

(4) **Extension or projection lines (B):** These lines also are continuous thin lines. They extend by about 3 mm beyond the dimension lines (fig. 3-2).

(5) **Construction lines (B):** These lines are drawn for constructing figures. They are shown in geometrical drawings only. They are continuous thin light lines.

(6) **Hatching or section lines (B):** These lines are drawn to make the section evident. They are continuous thin lines and are drawn generally at an angle of  $45^\circ$  to the main outline of the section. They are uniformly spaced about 1 mm to 2 mm apart (fig. 3-2).

(7) **Leader or pointer lines (B):** Leader line is drawn to connect a note with the feature to which it applies. It is a continuous thin line (fig. 3-2).

(8) **Border lines (B):** Perfectly rectangular working space is determined by drawing the border lines [fig. 2-1(a)]. They are continuous thin lines.

(9) **Short-break lines (C):** These lines are continuous, thin and wavy. They are drawn freehand and are used to show a short break, or irregular boundaries (fig. 3-3).

(10) **Long-break lines (D):** These lines are thin ruled lines with short zigzags within them. They are drawn to show long breaks (fig. 3-3).

(11) **Hidden or dotted lines (E or F):** Interior or hidden edges and surfaces are shown by hidden lines. They are also called dotted lines. They are of medium thickness and made up of short dashes of approximately equal lengths of about 2 mm spaced at equal distances of about 1 mm. When a hidden line meets or intersects another hidden line or an outline, their point of intersection or meeting should be clearly shown (fig. 3-2).

(12) **Centre lines (G):** Centre lines are drawn to indicate the axes of cylindrical, conical or spherical objects or details, and also to show the centres of circles and arcs. They are thin, long, chain lines composed of alternately long and dot spaced approximately 1 mm apart. The long dashes are about 9 to 12 mm. Centre lines should extend for a short distance beyond the outlines to which they refer. For the purpose of dimensioning or to correlate the views they may be extended as required. The point of intersection between two centre lines must always be indicated. Locus lines, extreme positions of movable parts and pitch circles are also shown by this type of line (fig. 3-2 and fig. 3-3).

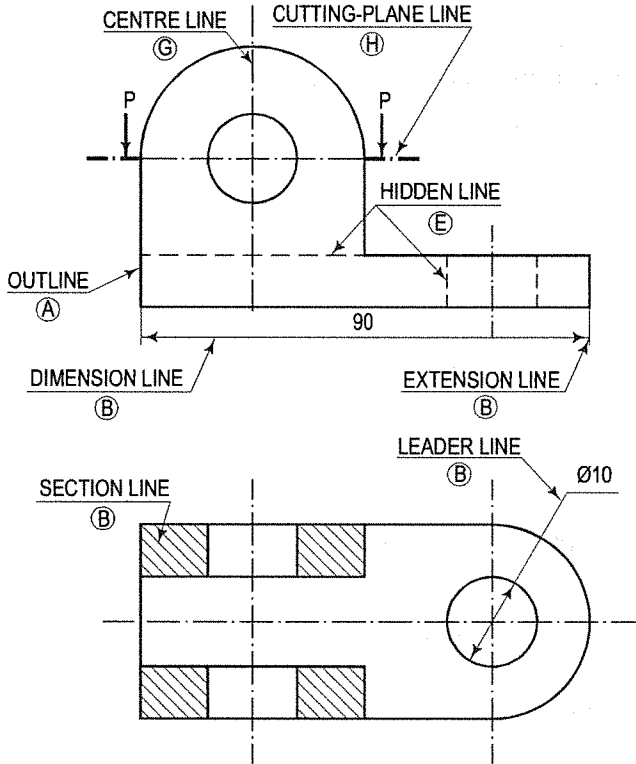


FIG. 3-2

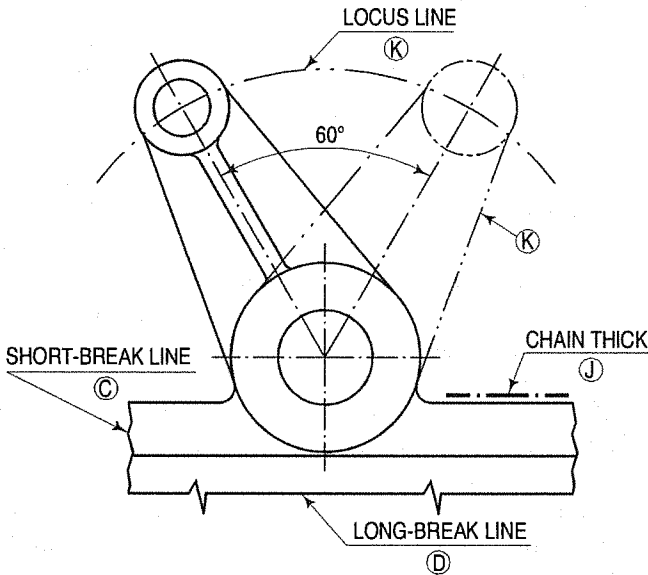


FIG. 3-3

(13) **Cutting-plane lines (H):** The location of a cutting plane is shown by this line. It is a long, thin, chain line, thick at ends only (fig. 3-2).

(14) **Chain thick (J):** These lines are used to indicate special treatment on the surface.

(15) **Chain thick double-dots (K):** This is chain thin double-dot line.

### 3-2. LETTERING (IS : 9609-2001)



Writing of titles, dimensions, notes and other important particulars on a drawing is called *lettering*. It is an important part of a drawing. However accurate and neat a drawing may be drawn. Its appearance is spoiled and sometimes, its usefulness is impaired by poor lettering. Lettering should, therefore, be done properly in clear, legible and uniform style. It should be in plain and simple style so that it could be done *freehand* and *speedily*.

**Note:** Use of drawing instruments in lettering takes considerable time and hence, it should be avoided. Efficiency in the art of lettering can be achieved by careful and continuous practice.

**(1) Single-stroke letters:** The Bureau of Indian Standards (IS : 9609-2001) recommends single-stroke lettering for use in engineering drawing. These are the simplest forms of letters and are usually employed in most of the engineering drawings.

*The word single-stroke should not be taken to mean that the letter should be made in one stroke without lifting the pencil.*

It actually means that the thickness of the line of the letter should be such as is obtained in one stroke of the pencil. The horizontal lines of letters should be drawn from left to right and vertical or inclined lines, from top to bottom.

Single-stroke letters are of two types:

- (i) vertical and
- (ii) inclined.

Inclined letters lean to the right, the slope being  $75^\circ$  with the horizontal. The size of a letter is described by its height. According to the height of letters, they are classified as:

- (i) Lettering 'A' (refer to table 3-4 and fig. 3-8)
- (ii) Lettering 'B' (refer to table 3-5 and fig. 3-8).

TABLE 3-4

LETTERING A  $\left(d = \frac{h}{10}\right)$

| Characteristic                |   | Ratio                          | Dimensions (mm) |      |      |     |     |    |     |  |
|-------------------------------|---|--------------------------------|-----------------|------|------|-----|-----|----|-----|--|
| Lettering height              |   |                                |                 |      |      |     |     |    |     |  |
| Height of capitals            | h | $\left(\frac{14}{14}\right) h$ | 2.5             | 3.5  | 5    | 7   | 10  | 14 | 20  |  |
| Height of lower-case letters  | c | $\left(\frac{10}{14}\right) h$ | —               | 2.5  | 3.5  | 5   | 7   | 10 | 14  |  |
| Spacing between characters    | a | $\left(\frac{2}{14}\right) h$  | 0.35            | 0.5  | 0.7  | 1   | 1.4 | 2  | 2.8 |  |
| Minimum spacing of base lines | b | $\left(\frac{20}{14}\right) h$ | 3.5             | 5    | 7    | 10  | 14  | 20 | 28  |  |
| Minimum spacing between words | e | $\left(\frac{6}{14}\right) h$  | 1.05            | 1.5  | 2.1  | 3   | 4.2 | 6  | 8.4 |  |
| Thickness of lines            | d | $\left(\frac{1}{14}\right) h$  | 0.18            | 0.25 | 0.35 | 0.5 | 0.7 | 1  | 1.4 |  |

TABLE 3-5  
LETTERING B  $\left(d = \frac{h}{10}\right)$

| Characteristic                | Ratio                          | Dimensions (mm) |      |      |     |     |    |     |    |
|-------------------------------|--------------------------------|-----------------|------|------|-----|-----|----|-----|----|
| Lettering height              | $\left(\frac{10}{10}\right) h$ | h               | 2.5  | 3.5  | 5   | 7   | 10 | 14  | 20 |
| Height of capitals            |                                |                 |      |      |     |     |    |     |    |
| Height of lower-case letters  |                                |                 |      |      |     |     |    |     |    |
| Spacing between characters    | $\left(\frac{2}{10}\right) h$  | a               | 0.5  | 0.7  | 1   | 1.4 | 2  | 2.8 | 4  |
| Minimum spacing of base lines | $\left(\frac{14}{10}\right) h$ | b               | 3.5  | 5    | 7   | 10  | 14 | 20  | 28 |
| Minimum spacing between words | $\left(\frac{6}{10}\right) h$  | e               | 1.5  | 2.1  | 3   | 4.2 | 6  | 8.4 | 12 |
| Thickness of lines            | $\left(\frac{1}{10}\right) h$  | d               | 0.25 | 0.35 | 0.5 | 0.7 | 1  | 1.4 | 2  |

In lettering 'A' type, the height of the capital letter is divided into 14 parts, while in lettering 'B' type it is divided into 10 parts. The height of the letters and numerals for engineering drawing can be selected from 2.5, 3.5, 5, 7, 10, 14 and 20 mm according to the size of drawing. The ratio of height to width varies but in case of most of the letters it is 6 : 5.

Lettering is generally done in capital letters. Different sizes of letters are used for different purposes.

The main titles are generally written in 6 mm to 8 mm size, sub-titles in 3 mm to 6 mm size, while notes, dimension figures etc. in 3 mm to 5 mm size. The drawing number in the title block is written in numerals of 10 mm to 12 mm size.

Fig. 3-4 shows single-stroke vertical capital letters and figures with approximate proportions.

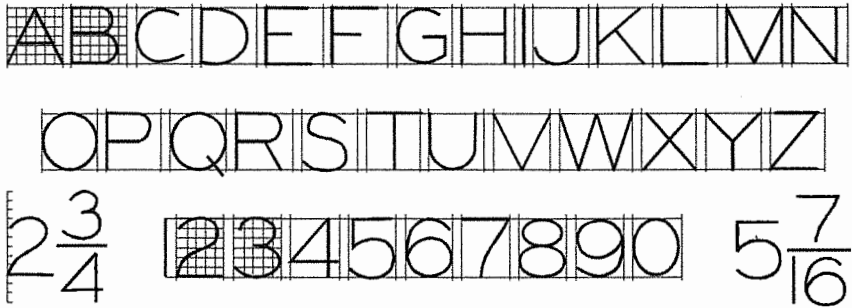


FIG. 3-4

Single-stroke inclined capital letters and figures are shown in fig. 3-5. The lower-case letters are usually used in architectural drawings.

Vertical and inclined lower-case alphabets are shown in fig. 3-6 and fig. 3-7 respectively. The width of the majority of letters is equal to the height.

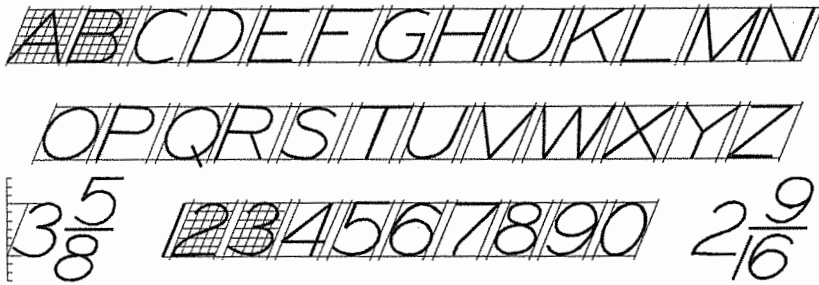


FIG. 3-5

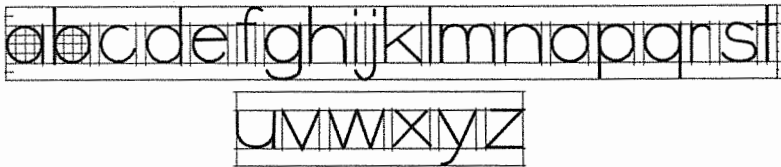


FIG. 3-6

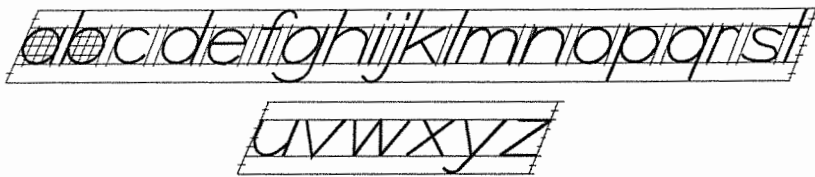


FIG. 3-7

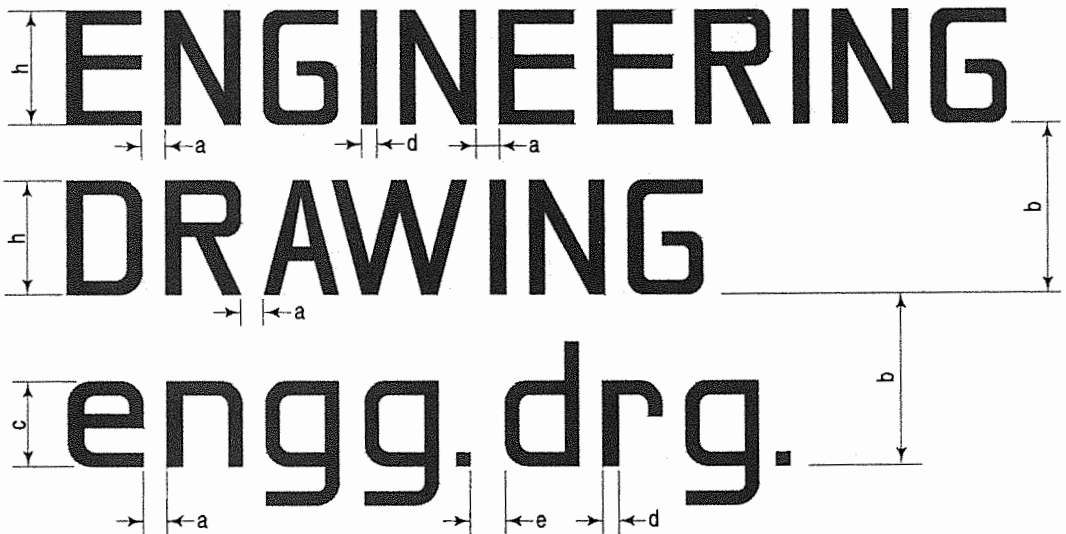


FIG. 3-8

All letters should be uniform in shape, slope, size, shade and spacing. The shape and slope of every letter should be uniform throughout a drawing. For maintaining uniformity in size, thin and light guide-lines may first be drawn, and lettering may then be done between them. The shade of every letter must be the same as that of the outlines of drawings, i.e. intensely black.

Therefore, H or HB grade of pencil is recommended for this purpose. The spacing between two letters should not necessarily be equal. The letters should be so spaced that they do not appear too close together or too much apart.

*Judging by the eye, the back ground areas between the letters should be kept approximately equal. The distance between the words must be uniform and at least equal to the height of the letters. Refer to fig. 3-8.*

Lettering should be so done as can be read from the front with the main title horizontal, i.e. when the drawing is viewed from the bottom edge.

All sub-titles should be placed below but not too close to the respective views. Lettering, except the dimension figures, should be underlined to make them more prominent.

**(2) Gothic letters:** Stems of single-stroke letters, if given more thickness, form what are known as *gothic* letters. These are mostly used for main titles of ink-drawings. The outlines of the letters are first drawn with the aid of instruments and then filled-in with ink.

The thickness of the stem may vary from  $\frac{1}{5}$  to  $\frac{1}{10}$  of the height of the letters.

Fig. 3-9 shows the alphabet and figures in gothic with thickness equal to  $\frac{1}{7}$  of the height.

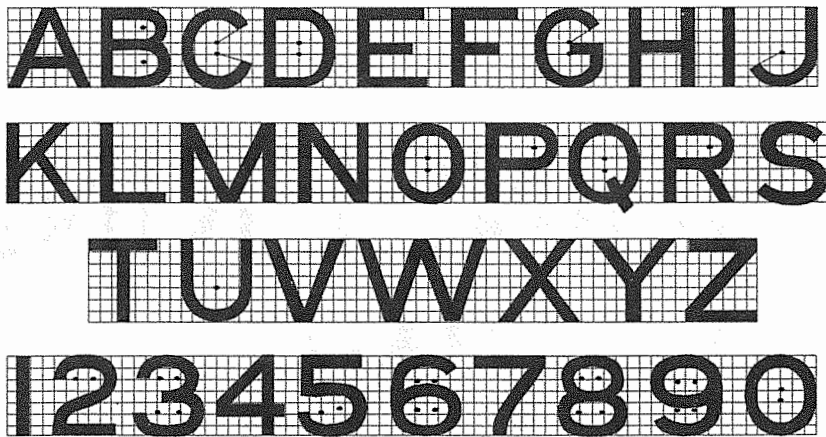


FIG. 3-9

### 3-3. DIMENSIONING

Every drawing, whether a scale drawing or a freehand drawing, besides showing the true shape of an object, must supply its exact length, breadth, height, sizes and positions of holes, grooves etc. and such other details relating to the manufacture of that object.

Providing this information on a drawing is called dimensioning. Lines, figures, numerals, symbols, notes etc. are used for this purpose.

**Types of dimensions** (fig. 3-10): Two types of dimensions needed on a drawing are:

- (i) size or functional dimensions and
- (ii) location or datum dimensions (shown by letters F and L respectively).

The former indicate sizes, viz. length, breadth, height, depth, diameter etc. The latter show locations or exact positions of various constructional details within the object. The letter F represents functional dimensions, while NF represents non-functional dimensions.

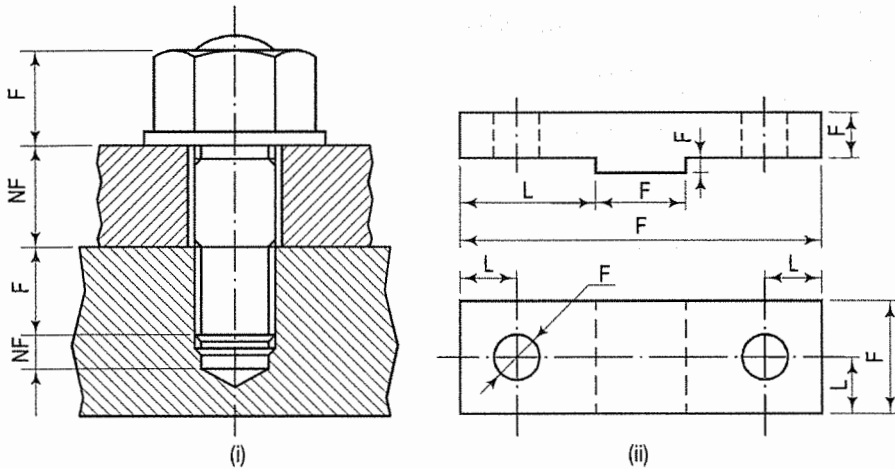


FIG. 3-10

### 3-4. DIMENSIONING TERMS AND NOTATIONS

(1) **Dimension line** (fig. 3-11): Dimension line is a thin continuous line. It is terminated by arrowheads touching the outlines, extension lines or centre lines.

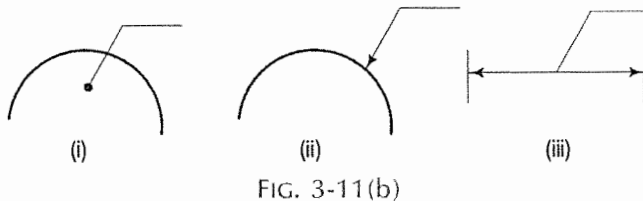
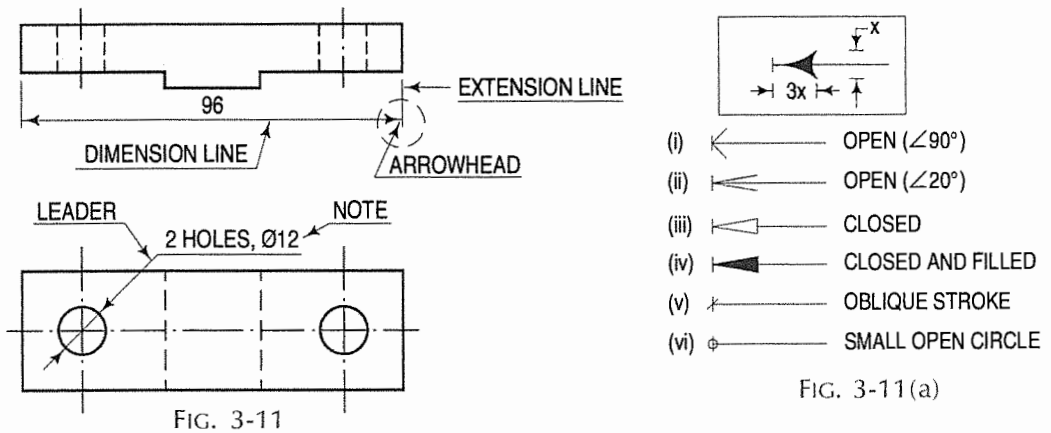
(2) **Extension line** (fig. 3-11): An extension line is also a thin continuous line drawn in extension of an outline. (Formerly, the B.I.S. had recommended that a gap of about 1 mm should be kept between the extension line and an outline or object boundary.) It extends by about 3 mm beyond the dimension line.

(3) **Arrowhead** (fig. 3-11): An arrowhead is placed at each end of a dimension line. Its pointed end touches an outline, an extension line or a centre line. The size of an arrowhead should be proportional to the thickness of the outlines. *The length of the arrowhead should be about three times its maximum width.* It is drawn freehand with two strokes made in the direction of its pointed end. The space between them is neatly filled up. Different types of arrow heads are shown in fig. 3-11(a). Generally closed and filled arrowhead is widely used in engineering drawing.

**Note** (fig. 3-11): A note gives information regarding specific operation relating to a feature. It is placed outside a view but adjacent to the feature concerned. It is so written that it may be read when the drawing is viewed from the bottom edge.

(4) **Leader** (fig. 3-11): A leader or a pointer is a thin continuous line connecting a note or a dimension figure with the feature to which it applies. One end of the leader terminates either in an arrowhead or a dot. The arrowhead touches the outline, *while the dot is placed within the outline of the object* [fig. 2-1(a) and fig. 3-11(b)].

The other end of the leader is terminated in a horizontal line at the bottom level of the first or the last letter of the note. *The leader is never drawn vertical or horizontal or curved. It is drawn at a convenient angle of not less than 30° to the line to which it touches.* When pointing to a circle or an arc it is drawn radially. Use of common leaders for more than one feature should never be made.

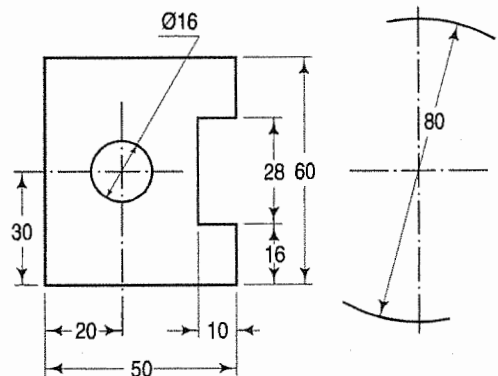
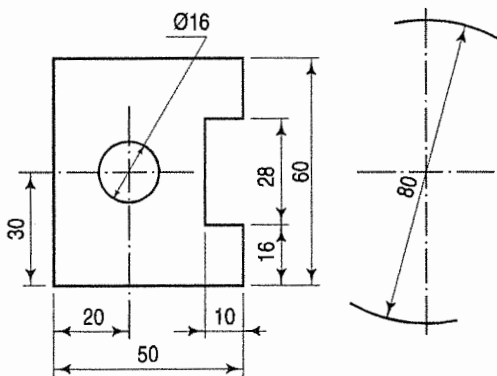


### 3-5. PLACING OF DIMENSIONS

The two systems of placing dimensions are:

- (1) Aligned system and
- (2) Unidirectional system.

(1) **Aligned system** (fig. 3-12): In the aligned system the dimension is placed perpendicular to the dimension line in such a way that it may be read from the bottom edge or the right-hand edge of the drawing sheet. The dimensions should be placed near the middle and above, but clear of the dimension lines.



(2) **Unidirectional system** (fig. 3-13): In unidirectional system all dimensions are so placed that they can be read from the bottom edge of the drawing sheet. The dimension lines are broken near the middle for inserting the dimensions. This system is mainly used on large drawings — as of aircrafts, automobiles etc. where it is inconvenient to read dimensions from the right-hand side.

### 3-6. UNIT OF DIMENSIONING

As far as possible all dimensions should be given in *millimetres*, omitting the abbreviation mm. Even when it is not convenient to give dimensions in millimetres and another unit is used, only the dimension figures are written. But a foot note such as '*all dimensions are in centimetres*' is inserted in a prominent place near the title block. The *height* of the *dimension figures* (as stated earlier) should be from 3 mm to 5 mm. The decimal point in a dimension should be quite distinct and written in line with the bottom line of the figure (fig. 3-27). A zero must always precede the decimal point when the dimension is less than unity.

### 3-7. GENERAL RULES FOR DIMENSIONING

- (1) Dimensioning should be done so completely that further calculation or assumption of any dimension, or direct measurement from the drawing is not necessary.
- (2) Every dimension must be given, but none should be given more than once.
- (3) A dimension should be placed on the view where its use is shown more clearly.
- (4) Dimensions should be placed outside the views, unless they are clearer and more easily read inside.
- (5) Mutual crossing of dimension lines and dimensioning between hidden lines should be avoided. Dimension lines should not cross any other line of the drawing.
- (6) An outline or a centre line should never be used as a dimension line. A centre line may be extended to serve as an extension line (fig. 3-13).
- (7) Aligned system of dimensioning is recommended.

### 3-8. PRACTICAL HINTS ON DIMENSIONING

- (1) Dimension lines should be drawn at least 8 mm away from the outlines and from each other.
- (2) Dimensions in a series may be placed in any one of the following two ways:
  - (i) *Continuous or chain dimensioning* (fig. 3-14): Dimensions are arranged in a straight line. An overall dimension is placed outside the smaller dimensions. One of the smaller dimensions (the least important) is generally omitted.
  - (ii) *Progressive or parallel dimensioning* (fig. 3-15): All dimensions are shown from a common base line. Cumulative error is avoided by this method. This method is preferable.
- (3) Smaller dimensions should be placed nearer the view and the larger further away so that extension lines do not cross dimension lines. Extension lines may cross each other (fig. 3-16) or the outlines (fig. 3-13).

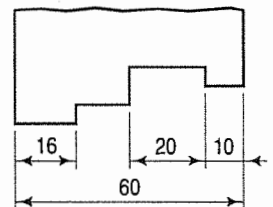


FIG. 3-14

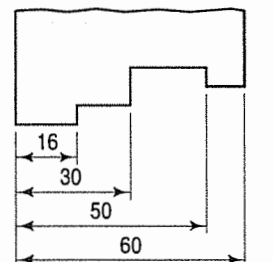


FIG. 3-15

- (4) When a number of parallel dimension lines are to be shown near each other, the dimensions should be staggered (fig. 3-16).
- (5) Dimensions should be shown where the shape is easily identified.
- (6) Arrowheads should ordinarily be drawn within the limits of the dimensioned feature. But when the space is too narrow, they may be placed outside (fig. 3-17). A dot may also be used to replace an arrowhead. Due to lack of space, the dimension figure may be written above the extended portion of the dimension line, but *preferably on the right-hand side* (fig. 3-17).
- (7) Dimensions of cylindrical parts should as far as possible be placed in the views in which they are seen as rectangles (fig. 3-18). The dimension indicating a diameter should always be preceded by the symbol  $\varnothing$ . *Dimension of a cylinder should not be given as a radius.*

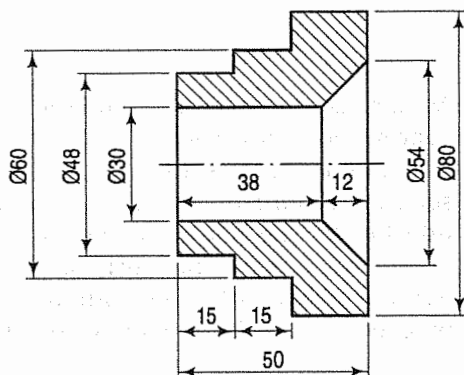


FIG. 3-16

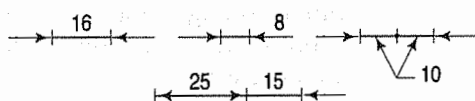


FIG. 3-17

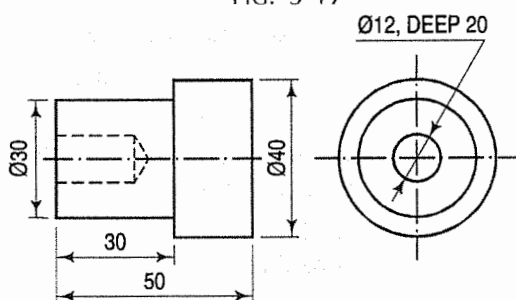


FIG. 3-18

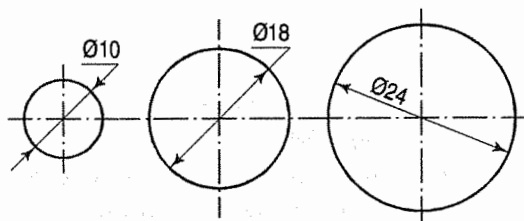


FIG. 3-19

Fig. 3-19 shows various methods of dimensioning different sizes of circles. Dimensions should be shown in one view only, the same dimension must not be repeated in other view.

Holes should be dimensioned in the view in which they appear as circles (fig. 3-20). They should be located by their centre lines.

As far as possible, all dimensions for one particular operation, such as diameter and depth of a drilled hole (fig. 3-18), or size and depth of a threaded hole (fig. 3-29) should be given in one view only.

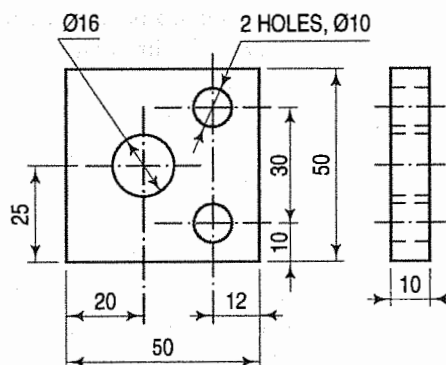


FIG. 3-20

In case of a large-size bore or a pitch circle, the dimension may be shown by a diagonal diameter (fig. 3-21).

*But (in aligned system) a dimension should not be placed within 30° zone of the vertical centre line as shown by the shaded space in fig. 3-21.*

Holes on pitch circles when equally spaced should be dimensioned as shown in fig. 3-21. When holes are not equally or uniformly spaced on the pitch circle, they should be located by angles with one of the two main centre lines (fig. 3-22).

*Arcs of circles should be dimensioned by their respective radii. Dimension line for the radius should pass through the centre of the arc. The dimension figure must be preceded by the letter R.* Fig. 3-23 shows different methods of showing the radii of arcs.

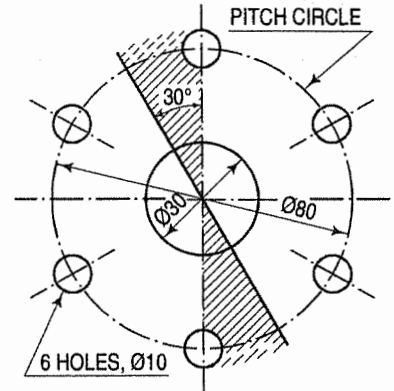


FIG. 3-21

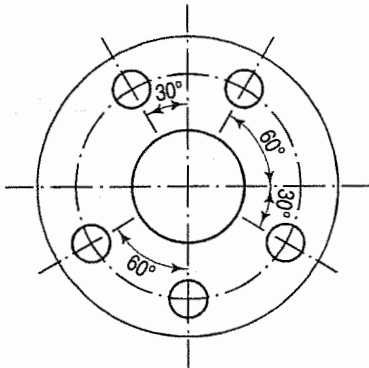


FIG. 3-22

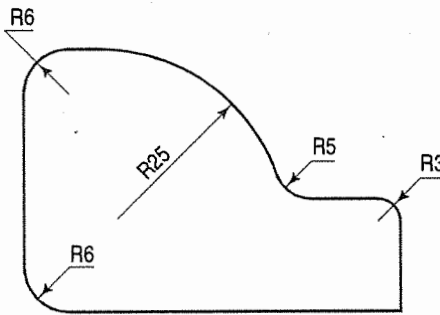


FIG. 3-23

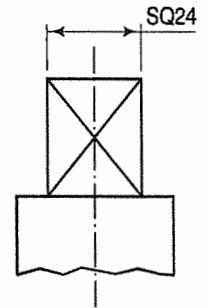


FIG. 3-24

- (8) Letters SQ should precede the dimension for a rod of square cross-section (fig. 3-24). The word SPHERE should be placed before the dimension (radius  $R$  or diameter  $\phi$ ) of a spherical part (fig. 3-25).
- (9) Angular dimensions may be given by any one of the methods shown in fig. 3-26.
- (10) Fig. 3-27 shows a method of dimensioning a countersunk hole. The maximum diameter is also sometimes given.

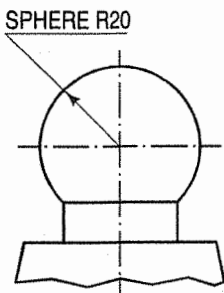


FIG. 3-25

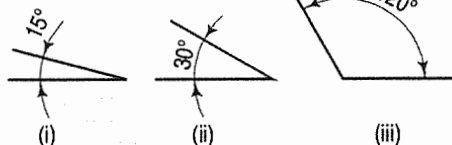


FIG. 3-26

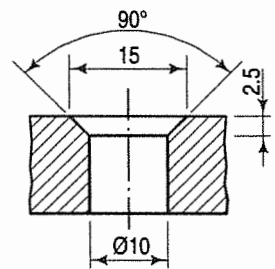


FIG. 3-27

(11) Methods of dimensioning a chamfer is shown in fig. 3-28.

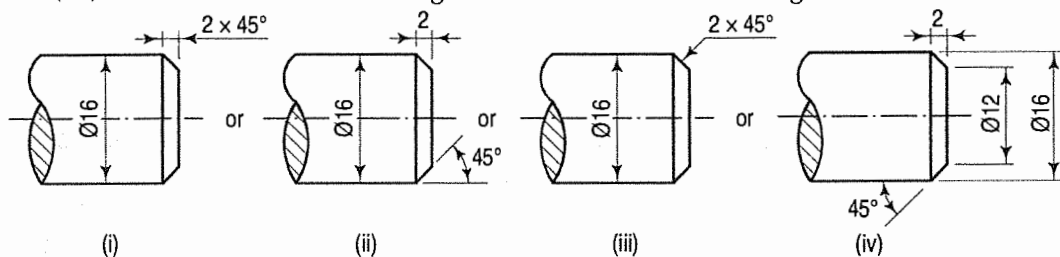


FIG. 3-28

(12) Designation and size, along with the useful length must be given while dimensioning an external screw thread (fig. 3-29). In case of internal screw thread, in addition to the size and type, the depth of the drilled hole before tapping must also be given (fig. 3-30).

(13) Left-hand thread and multiple-start thread should be dimensioned as shown in fig. 3-31 and fig. 3-32 respectively.

(14) A slope or taper is defined as unit alteration in a specified length. The specified length is measured along the base line in case of flat pieces and along the axis in case of shafts.

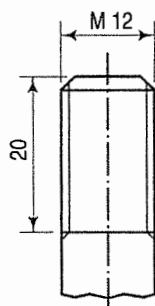


FIG. 3-29

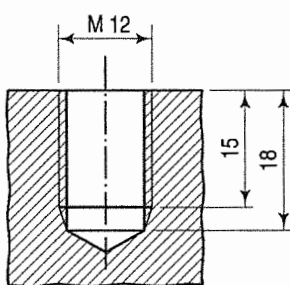


FIG. 3-30

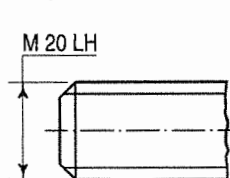


FIG. 3-31

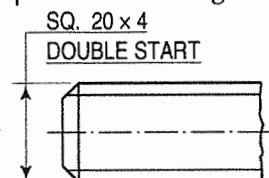


FIG. 3-32

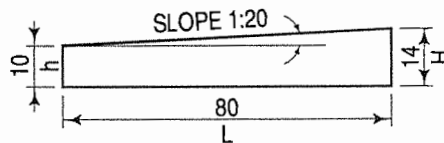


FIG. 3-33

(15) Fig. 3-33 shows the method of indicating slope on a flat piece. It is written parallel to the sloping line.

$$\text{Flat taper} = \frac{H - h}{L} = \frac{1}{20}$$

(16) The taper on a shaft is indicated along the centre line and is accompanied by one or both the diameters (fig. 3-34).

$$\text{Taper on diameter} = \frac{D - d}{L} = \frac{1}{10}$$

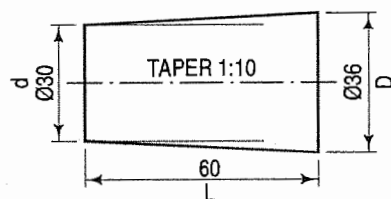


FIG. 3-34

(17) Fig. 3-35 shows method of dimensioning for frustum. It is drawn obliquely but parallel to each other.

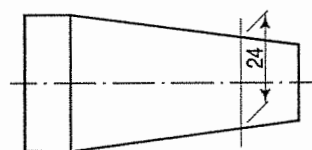


FIG. 3-35

(18) Fig. 3-36 shows methods of dimensioning chords, arcs and angles.

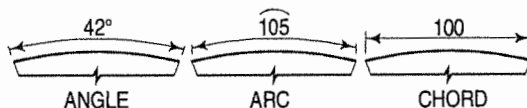


FIG. 3-36

(19) Study carefully fig. 3-37 for correct dimensioning.

Incorrect dimensioning is shown by cross (X) in the figure.

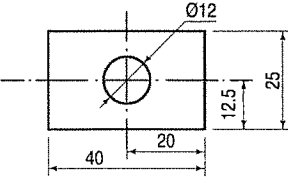
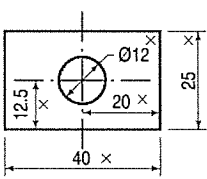
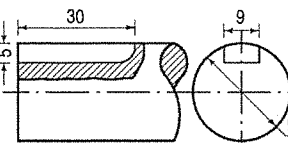
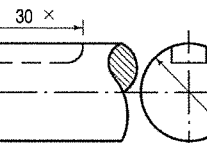
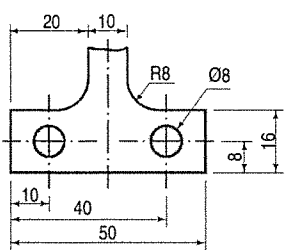
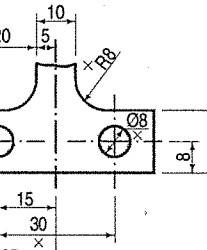
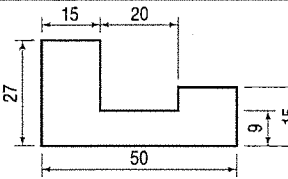
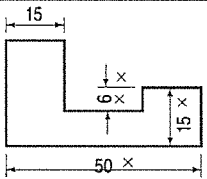
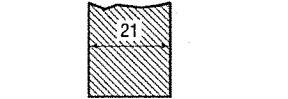
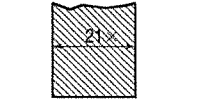
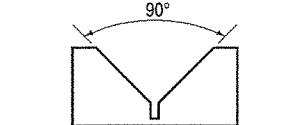
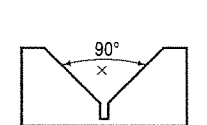
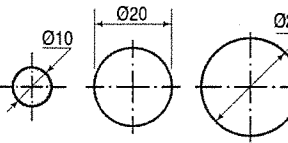
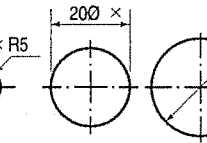
|       | CORRECT                                                                                                                                                                             | INCORRECT                                                                           | REASONS FOR INCORRECT                                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   |  <p>Dimensions should be placed outside view</p>                                                   |    | <ol style="list-style-type: none"> <li>1. Arrow head not proportionate.</li> <li>2. Hole dimension shown in figure. Leader line not ends horizontally.</li> <li>3. Dimension '40' is too close.</li> <li>4. Placing dimensions methods mix. Dimension '40' is according to aligned method.</li> </ol>                                             |
| (ii)  |  <p>Dim. should be marked from visible outlines</p>                                                |    | <ol style="list-style-type: none"> <li>1. A key-way is shown with dotted line where the dimensions are placed.</li> <li>2. Leader line for the shaft diameter is drawn horizontal touching the boundary line.</li> </ol>                                                                                                                          |
| (iii) |  <p>Dimensions should be given from the outlines (finished surface) or a centre line of a hole</p> |    | <ol style="list-style-type: none"> <li>1. Dimensions are given from the mid-line of the object.</li> <li>2. Dimensions of holes are shown inside the figure.</li> <li>3. Dimensions are shown in vertical line.</li> <li>4. Smaller dimensions (25 mm) precedes the larger dimensions (30 mm).</li> <li>5. Fillet radius is not shown.</li> </ol> |
| (iv)  |                                                                                                  |  | <ol style="list-style-type: none"> <li>1. Dimension lines are used as extension.</li> <li>2. Dimensions are placed inside the view.</li> <li>3. Dimension 27 and 50 not written according to aligned system.</li> </ol>                                                                                                                           |
| (v)   |                                                                                                  |  | Section overlap the dimension 21.                                                                                                                                                                                                                                                                                                                 |
| (vi)  |                                                                                                  |  | The outlines of the object are used as the extension lines.                                                                                                                                                                                                                                                                                       |
| (vii) |                                                                                                  |  | <ol style="list-style-type: none"> <li>1. Smaller circle is designated with radius.</li> <li>2. Convention 'Ø' for diameter is placed after dimension.</li> <li>3. Leader has arrow and it is drawn horizontal.</li> </ol>                                                                                                                        |

FIG. 3-37

## EXERCISES 3



1. Write freehand, in single-stroke (i) vertical capital letters and (ii) inclined capital letters of 3 mm height, the following paragraph from page 39.  
'All letters should \_\_\_\_\_ i.e. intensely black'.
2. Write freehand, in single-stroke vertical lower-case letters of 3 mm height, the following two paragraphs from page 37.  
'Writing of titles, \_\_\_\_\_ freehand and speedily'.
3. Write freehand, the paragraphs stated in Ex. 2 above, in single-stroke inclined lower-case letters of 3 mm height.
4. Print in gothic letters of 12 mm height, the titles of Chapters 1, 2 and 3.
5. Fill-in the blanks with appropriate words selected from the list given below:
  - (a) Writing of titles, notes etc. on a drawing is called \_\_\_\_\_.
  - (b) Efficiency in the art of lettering is achieved by continuous \_\_\_\_\_.
  - (c) Lettering should be in plain and simple style so that it could be done \_\_\_\_\_ and \_\_\_\_\_.
  - (d) For maintaining uniformity in size, thin and light \_\_\_\_\_ may first be drawn.
  - (e) All letters should be uniform in \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_ and \_\_\_\_\_.
  - (f) The inclined letters slope to the \_\_\_\_\_ at an angle of \_\_\_\_\_ degrees.
  - (g) The size of the letter is described by its \_\_\_\_\_.
  - (h) Main title of inked drawing is generally written in \_\_\_\_\_ letters.
  - (i) The two types of single-stroke letters are \_\_\_\_\_ and \_\_\_\_\_.
  - (j) Lettering is usually done in \_\_\_\_\_ letters.
  - (k) Lower-case letters are usually used in \_\_\_\_\_ drawings.
  - (l) Lettering should be so done as can be read from the \_\_\_\_\_ with the main title \_\_\_\_\_.

### List of words for Ex. 5:

- |               |                  |                 |                   |
|---------------|------------------|-----------------|-------------------|
| 1. Inclined   | 6. Vertical      | 11. Right       | 16. Practice      |
| 2. Shade      | 7. Freehand      | 12. Front       | 17. Lettering     |
| 3. Horizontal | 8. Shape         | 13. Guide lines | 18. Spacing       |
| 4. Speedily   | 9. Gothic        | 14. Slope       | 19. Architectural |
| 5. Height     | 10. Seventy five | 15. Capital     | 20. Size.         |
6. Complete each statement given in A, by selecting an appropriate one from those given in B:
    - A (a) Outlines or principal lines are drawn as \_\_\_\_\_
    - (b) Lines for hidden edges are drawn as \_\_\_\_\_
    - (c) Thin and long chain line is made up of \_\_\_\_\_
    - (d) Centre lines, locus lines and pitch circles are drawn as \_\_\_\_\_
    - (e) Dashed line of medium thickness is made up of \_\_\_\_\_
    - (f) Dimension lines, hatching and extension lines are drawn as \_\_\_\_\_
    - (g) The position of cutting plane is shown by \_\_\_\_\_
    - (h) Irregular boundaries and short breaks are shown by \_\_\_\_\_
    - (i) Long breaks are shown by \_\_\_\_\_

- B (1) long and thin chain lines  
 (2) thick and long chain line, thick at ends only  
 (3) thick continuous lines  
 (4) short dashes of approximately equal lengths  
 (5) thin continuous lines  
 (6) continuous, thin and wavy lines, drawn freehand  
 (7) thin ruled lines with short zigzags within them  
 (8) alternately long and short dashes  
 (9) dashed lines of medium thickness.
7. Fill-up the blanks in the following sentences by appropriate words selected from the list of words given below:
- Two types of dimensions needed on a drawing are \_\_\_\_\_ and \_\_\_\_\_ dimensions.
  - The two systems of placing dimensions are \_\_\_\_\_ and \_\_\_\_\_.
  - In \_\_\_\_\_ system, the dimension is placed \_\_\_\_\_ to and near the \_\_\_\_\_ but clear of dimension line.
  - In \_\_\_\_\_ system, all dimensions are so placed that they are readable from the \_\_\_\_\_ edge of the drawing sheet. Dimension lines are \_\_\_\_\_ near the middle for inserting dimensions.
  - As far as possible dimensions should be given in one unit, preferably in \_\_\_\_\_.
  - Dimension lines should be drawn at least 8 mm away from the \_\_\_\_\_ and from \_\_\_\_\_.
  - An \_\_\_\_\_ or \_\_\_\_\_ should never be used as a dimension line.
  - Mutual \_\_\_\_\_ of dimension lines and dimensioning between \_\_\_\_\_ should be avoided.
  - All dimensions are shown from a common base line in \_\_\_\_\_ dimensioning.
  - In \_\_\_\_\_ dimensioning, dimensions are arranged in a straight line, and an \_\_\_\_\_ dimension is placed outside the small dimensions.
  - When a number of parallel dimensions are to be shown near each other, the dimensions should be \_\_\_\_\_.
  - The extension line should extend about 3 mm beyond the \_\_\_\_\_.
  - The line connecting a view to a note is called a \_\_\_\_\_.
  - The dimension indicating a diameter should be \_\_\_\_\_ by the symbol \_\_\_\_\_.
  - The symbol \_\_\_\_\_ should be \_\_\_\_\_ by the dimension indicating a radius.
  - Dimensions of cylindrical parts should as far as possible be shown in the views in which they are seen as \_\_\_\_\_.
  - Dimension of a cylinder should never be given as a \_\_\_\_\_.
  - The taper on a shaft is indicated along the \_\_\_\_\_.
  - The specified length for taper is measured along the \_\_\_\_\_ in case of a flat piece and along the \_\_\_\_\_ in case of a shaft.
  - The section lines are continuous \_\_\_\_\_ lines and are drawn at an \_\_\_\_\_ of \_\_\_\_\_ to the main outline of the section.

List of words for Ex. 7:

- |                |                    |                   |                     |
|----------------|--------------------|-------------------|---------------------|
| 1. Thin        | 9. Crossing        | 17. Millimetres   | 25. Leader          |
| 2. Axis        | 10. Dimension line | 18. Overall       | 26. Radius          |
| 3. Aligned     | 11. Dashed line    | 19. Outlines      | 27. Angle           |
| 4. Broken      | 12. Each other     | 20. Outline       | 28. R               |
| 5. Base line   | 13. Followed       | 21. Perpendicular | 29. 45°             |
| 6. Bottom      | 14. Location       | 22. Progressive   | 30. Size            |
| 7. Centre line | 15. Middle         | 23. Rectangles    | 31. Staggered       |
| 8. Continuous  | 16. Ø              | 24. Preceded      | 32. Unidirectional. |

Answers to Ex. 5, 6 and 7:

- |                      |                         |               |                   |
|----------------------|-------------------------|---------------|-------------------|
| Ex. 5. (a) 17        | (d) 13                  | (g) 5         | (j) 15            |
| (b) 16               | (e) 20, 8, 14, 2 and 18 | (h) 9         | (k) 19            |
| (c) 7 and 4          | (f) 11 and 10           | (i) 6 and 1   | (l) 12 and 3.     |
| Ex. 6. (a) 3         | (d) 1                   | (g) 2         |                   |
| (b) 9                | (e) 4                   | (h) 6         |                   |
| (c) 8                | (f) 5                   | (i) 7.        |                   |
| Ex. 7. (a) 30 and 14 | (f) 19 and 12           | (k) 31        | (p) 23            |
| (b) 3 and 32         | (g) 20 and 7            | (l) 10        | (q) 26            |
| (c) 3, 21 and 15     | (h) 9 and 11            | (m) 25        | (r) 7             |
| (d) 32, 6 and 4      | (i) 22                  | (n) 24 and 16 | (s) 5 and 2       |
| (e) 17               | (j) 8 and 18            | (o) 28 and 13 | (t) 1, 27 and 29. |

8. What are the two systems of placing dimensions on a drawing? Illustrate your answer with sketches.
9. Show by sketches the difference between (i) continuous or chain dimensioning and (ii) progressive or parallel dimensioning. What are the advantages of one above the other?
10. The orthographic views of the objects are drawn according to the First-angle projection method to 1 : 2 scale size and shown below in fig. 3-38(i), fig. 3-38(ii) and fig. 3-38(iii). Insert dimensions.

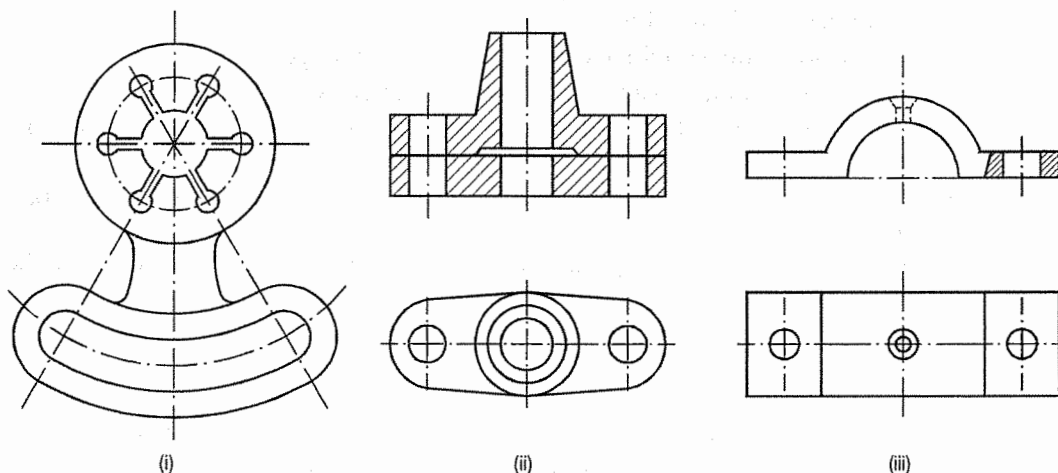


FIG. 3-38



## 4-1. INTRODUCTION

Drawings of small objects can be prepared of the same size as the objects they represent. A 150 mm long pencil may be shown by a drawing of 150 mm length. Drawings drawn of the same size as the objects, are called full-size drawings. The ordinary full-size scales are used for such drawings.

A scale is defined as the ratio of the linear dimensions of element of the object as represented in a drawing to the actual dimensions of the same element of the object itself.

## 4-2. SCALES

The scales generally used for general engineering drawings are shown in table 4-1 [SP : 46].

TABLE 4-1

|       |                  |          |          |           |
|-------|------------------|----------|----------|-----------|
| (i)   | Reducing scales  | 1 : 2    | 1 : 5    | 1 : 10    |
|       |                  | 1 : 20   | 1 : 50   | 1 : 100   |
|       |                  | 1 : 200  | 1 : 500  | 1 : 1000  |
|       |                  | 1 : 2000 | 1 : 5000 | 1 : 10000 |
| (ii)  | Enlarging scales | 50 : 1   | 20 : 1   | 10 : 1    |
|       |                  | 5 : 1    | 2 : 1    |           |
| (iii) | Full size scales |          |          | 1 : 1     |

All these scales are usually 300 mm long and sub-divided throughout their lengths. The scale is indicated on the drawing at a suitable place near the title. The complete designation of a scale consists of word scale followed by the ratio, i.e. scale 1 : 1 or scale, full size.

It may not be always possible to prepare full-size drawings. They are, therefore, drawn proportionately smaller or larger. When drawings are drawn smaller than the actual size of the objects (as in case of buildings, bridges, large machines etc.) the scale used is said to be a *reducing scale* (1 : 5). Drawings of small machine parts, mathematical instruments, watches etc. are made larger than their real size. These are said to be drawn on an *enlarging scale* (5 : 1).

The scales can be expressed in the following *three* ways:

(1) **Engineer's scale:** In this case, the relation between the dimension on the drawing and the actual dimension of the object is mentioned numerically in the style as 10 mm = 5 m etc.

(2) **Graphical scale:** The scale is drawn on the drawing itself. As the drawing becomes old, the engineer's scale may shrink and may not give accurate results.

However, such is not the case with graphical scale because if the drawing shrinks, the scale will also shrink. Hence, the graphical scale is commonly used in survey maps.

(3) **Representative fraction:** The ratio of the length of the object represented on drawing to the actual length of the object represented is called the Representative Fraction (i.e. R.F.).

$$\text{R.F.} = \frac{\text{Length of the drawing}}{\text{Actual length of object}}$$

When a 1 cm long line in a drawing represents 1 metre length of the object, the R.F. is equal to  $\frac{1 \text{ cm}}{1 \text{ m}} = \frac{1 \text{ cm}}{1 \times 100 \text{ cm}} = \frac{1}{100}$  and the scale of the drawing will be 1 : 100 or  $\frac{1}{100}$  full size. *The R.F. of a drawing is greater than unity when it is drawn on an enlarging scale.* For example, when a 2 mm long edge of an object is shown in a drawing by a line 1 cm long, the R.F. is  $\frac{1 \text{ cm}}{2 \text{ mm}} = \frac{10 \text{ mm}}{2 \text{ mm}} = 5$ . Such a drawing is said to be drawn on scale 5 : 1 or *five times full-size*.

#### 4-3. SCALES ON DRAWINGS

When an unusual scale is used, it is constructed on the drawing sheet. To construct a scale the following information is essential:

- (1) The R.F. of the scale.
- (2) The units which it must represent, for example, millimetres and centimetres, or feet and inches etc.
- (3) The maximum length which it must show.

The length of the scale is determined by the formula:

Length of the scale = R.F.  $\times$  maximum length required to be measured.

It may not be always possible to draw as long a scale as to measure the longest length in the drawing. The scale is therefore drawn 15 cm to 30 cm long, longer lengths being measured by marking them off in parts.

#### 4-4. TYPES OF SCALES

The scales used in practice are classified as under:

- |                        |                      |
|------------------------|----------------------|
| (1) Plain scales       | (4) Vernier scales   |
| (2) Diagonal scales    | (5) Scale of chords. |
| (3) Comparative scales |                      |

(1) **Plain scales:** A plain scale consists of a line divided into suitable number of equal parts or units, the first of which is sub-divided into smaller parts. Plain scales represent either two units or a unit and its sub-division.

In every scale,

- (i) The zero should be placed at the end of the first main division, i.e. between the unit and its sub-divisions.

- (ii) From the zero mark, the units should be numbered to the right and its sub-divisions to the left.
- (iii) The names of the units and the sub-divisions should be stated clearly below or at the respective ends.
- (iv) The name of the scale (e.g. scale, 1 : 10) or its R.F. should be mentioned below the scale.

**Problem 4-1.** (fig. 4-1): Construct a scale of 1 : 4 to show centimetres and long enough to measure upto 5 decimetres.

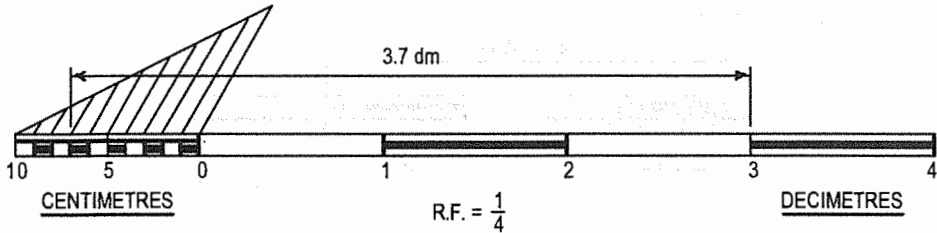


FIG. 4-1

- (i) Determine R.F. of the scale. Here it is  $\frac{1}{4}$ .
- (ii) Determine length of the scale.  
Length of the scale = R.F.  $\times$  maximum length =  $\frac{1}{4} \times 5 \text{ dm} = 12.5 \text{ cm}$ .
- (iii) Draw a line 12.5 cm long and divide it into 5 equal divisions, each representing 1 dm.
- (iv) Mark 0 at the end of the first division and 1, 2, 3 and 4 at the end of each subsequent division to its right.
- (v) Divide the first division into 10 equal sub-divisions, each representing 1 cm.
- (vi) Mark cms to the left of 0 as shown in the figure.

To distinguish the divisions clearly, show the scale as a rectangle of small width (about 3 mm) instead of only a line. Draw the division-lines showing decimetres throughout the width of the scale. Draw the lines for the sub-divisions slightly shorter as shown. Draw thick and dark horizontal lines in the middle of all alternate divisions and sub-divisions. This helps in taking measurements. Below the scale, print DECIMETRES on the right-hand side, CENTIMETRES on the left-hand side, and the R.F. in the middle.

To set-off any distance, say 3.7 dm, place one leg of the divider on 3 dm mark and the other on 7 cm mark. The distance between the ends of the two legs will represent 3.7 dm.

**Problem 4-2.** (fig. 4-2): Draw a scale of 1 : 60 to show metres and decimetres and long enough to measure upto 6 metres.

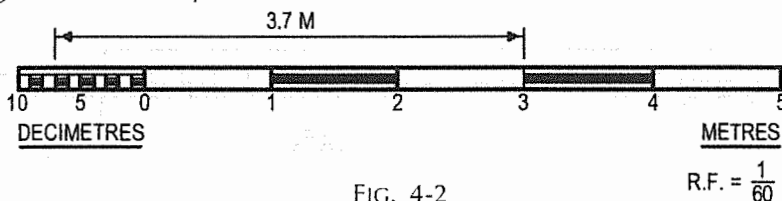


FIG. 4-2

- (i) Determine R.F. of the scale, here  $R.F. = \frac{1}{60}$ .
- (ii) Determine length of the scale.

$$\text{Length of the scale} = \frac{1}{60} \times 6 \text{ m} = \frac{1}{10} \text{ metre} = 10 \text{ cm.}$$

- (iii) Draw a line 10 cm long and divide it into 6 equal parts.
- (iv) Divide the first part into 10 equal divisions and complete the scale as shown. The length 3.7 metres is shown on the scale.

**Problem 4-3.** (fig. 4-3): Construct a scale of 1.5 inches = 1 foot to show inches and long enough to measure upto 4 feet.

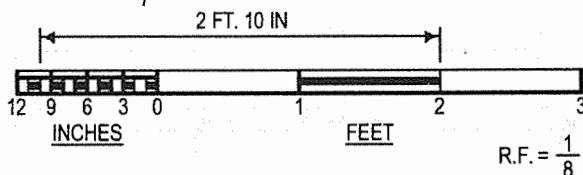


FIG. 4-3

- (i) Determine R.F. of the scale.  $R.F. = \frac{1.5 \text{ inches}}{1 \times 12 \text{ inches}} = \frac{1}{8}$ .
- (ii) Draw a line,  $1.5 \times 4 = 6$  inches long.
- (iii) Divide it into four equal parts, each part representing one foot.
- (iv) Divide the first division into 12 equal parts, each representing 1". Complete the scale as explained in problem 4-1. The distance 2'-10" is shown measured in the figure.

**Problem 4-4.** (fig. 4-4): Construct a scale of  $R.F. = \frac{1}{60}$  to read yards and feet, and long enough to measure upto 5 yards.

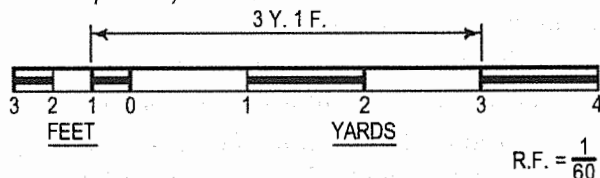


FIG. 4-4

- (i) Length of the scale =  $R.F. \times \text{max. length} = \frac{1}{60} \times 5 \text{ yd} = \frac{1}{12} \text{ yd} = 3 \text{ inches.}$
- (ii) Draw a line 3 inches long and divide it into 5 equal parts.
- (iii) Divide the first part into 3 equal divisions.
- (iv) Mark the scale as shown in the figure.

**Problem 4-5.** (fig. 4-5): Construct a scale of  $R.F. = \frac{1}{84480}$  to show miles and furlongs and long enough to measure upto 6 miles.

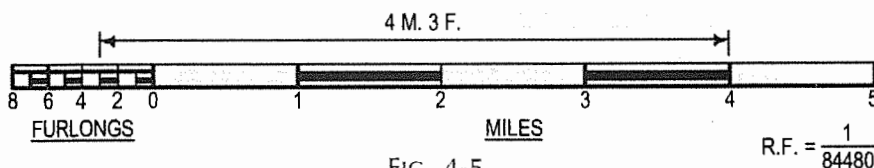


FIG. 4-5

(i) Length of the scale =  $\frac{1}{84480} \times 6 = \frac{1}{14080}$  miles =  $4\frac{1}{2}$ ''.

(ii) Draw a line  $4\frac{1}{2}$ " long and divide it into 6 equal parts. Divide the first part into 8 equal divisions and complete the scale as shown.

The distance 4 miles and 3 furlongs is shown measured in the figure.

(2) **Diagonal scales:** A diagonal scale is used when very minute distances such as 0.1 mm etc. are to be accurately measured or when measurements are required in three units; for example, dm, cm and mm, or yard, foot and inch.

Small divisions of short lines are obtained by the principle of diagonal division, as explained below.

**Principle of diagonal scale:** To obtain divisions of a given short line  $AB$  in multiples of  $\frac{1}{10}$  its length, e.g.  $0.1 AB$ ,  $0.2 AB$ ,  $0.3 AB$  etc. (fig. 4-6).

- (i) At one end, say  $B$ , draw a line perpendicular to  $AB$  and along it, step-off ten equal divisions of any length, starting from  $B$  and ending at  $C$ .
- (ii) Number the division-points, 9, 8, 7,.....1 as shown.
- (iii) Join  $A$  with  $C$ .
- (iv) Through the points 1, 2 etc. draw lines parallel to  $AB$  and cutting  $AC$  at  $1'$ ,  $2'$  etc. It is evident that triangles  $1'C$ ,  $2'C$  ...  $ABC$  are similar.

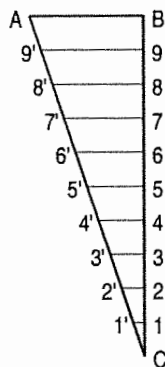


FIG. 4-6

Since  $C5 = 0.5BC$ , the line  $5'5 = 0.5AB$ .

Similarly,  $1'1 = 0.1AB$ ,  $2'2 = 0.2AB$  etc.

Thus, each horizontal line below  $AB$  becomes progressively shorter in length by  $\frac{1}{10} AB$  giving lengths in multiples of  $0.1AB$ .

**Problem 4-6.** (fig. 4-7): Construct a diagonal scale of 3 : 200 i.e.  $1 : 66 \frac{2}{3}$  showing metres, decimetres and centimetres and to measure upto 6 metres.

$$\text{Length of the scale} = \frac{3}{200} \times 6 \text{ m} = 9 \text{ cm.}$$

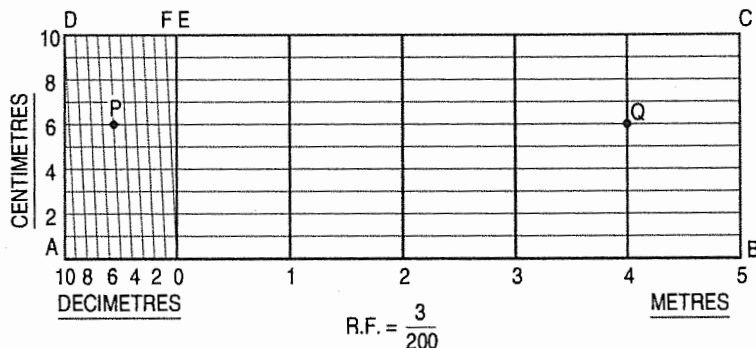


FIG. 4-7

- (i) Draw a line  $AB$  9 cm long and divide it into 6 equal parts. Each part will show a metre.
- (ii) Divide the first part  $A0$  into 10 equal divisions, each showing a decimetre or 0.1 m.
- (iii) At  $A$  erect a perpendicular and step-off along it, 10 equal divisions of any length, ending at  $D$ . Complete the rectangle  $ABCD$ .
- (iv) Erect perpendiculars at metre-divisions 0, 1, 2, 3 and 4.
- (v) Draw horizontal lines through the division-points on  $AD$ .
- (vi) Join  $D$  with the end of the first division along  $A0$ , viz. the point 9.
- (vii) Through the remaining points i.e. 8, 7, 6 etc. draw lines parallel to  $D9$ .

In  $\triangle OFE$ ,  $FE$  represents 1 dm or 0.1 m. Each horizontal line below  $FE$  progressively diminishes in length by  $0.1FE$ . Thus, the next line below  $FE$  is equal to  $0.9FE$  and represents 0.9 A 1 dm = 0.9 dm or 0.09 m or 9 cm.

Any length between 1 cm or 0.01 m and 6 m can be measured from this scale. To show a distance of 4.56 metres, i.e. 4 m, 5 dm and 6 cm, place one leg of the divider at  $Q$  where the vertical through 4 m meets the horizontal through 6 cm and the other leg at  $P$  where the diagonal through 5 dm meets the same horizontal.

**Problem 4-7.** (fig. 4-8): Construct a diagonal scale of R.F. =  $\frac{1}{4000}$  to show metres and long enough to measure upto 500 metres.

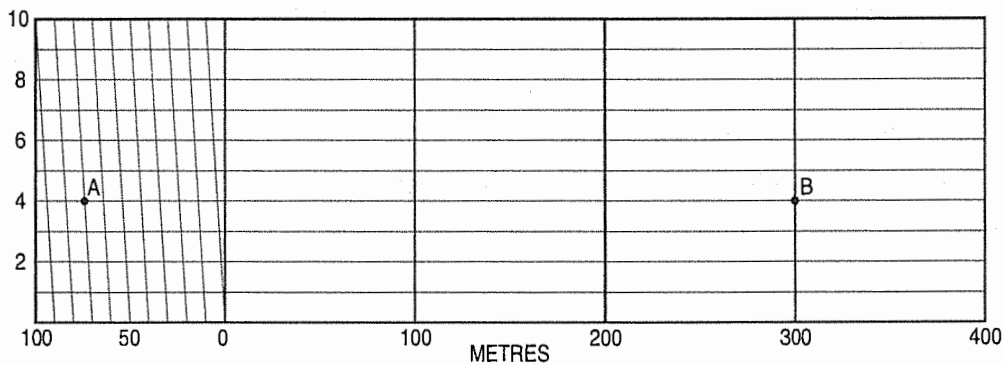


FIG. 4-8

$$R.F. = \frac{1}{4000}$$

$$\text{Length of the scale} = \frac{1}{4000} \times 500 \text{ m} = \frac{1}{8} \text{ metre} = 12.5 \text{ cm.}$$

- (i) Draw a line 12.5 cm long and divide it into 5 equal parts. Each part will show 100 metres.
  - (ii) Divide the first part into ten equal divisions. Each division will show 10 metres.
  - (iii) At the left-hand end, erect a perpendicular and on it, step-off 10 equal divisions of any length.
  - (iv) Draw the rectangle and complete the scale as explained in problem 4-6.
- The distance between points  $A$  and  $B$  shows 374 metres.

**Problem 4-8.** (fig. 4-9): Draw a diagonal scale of 1 : 2.5, showing centimetres and millimetres and long enough to measure upto 20 centimetres.

Length of the scale =  $\frac{1}{2.5} \times 20 \text{ cm} = 8 \text{ cm}$ .

- (i) Draw a line 8 cm long and divide it into 4 equal parts. Each part will represent a length of 5 cm.
- (ii) Divide the first part into 5 equal divisions. Each division will show 1 cm.
- (iii) At the left-hand end of the line, draw a vertical line and on it step-off 10 equal divisions of any length.

Complete the scale as explained in problem 4-6. The distance between points C and D shows 13.4 cm.

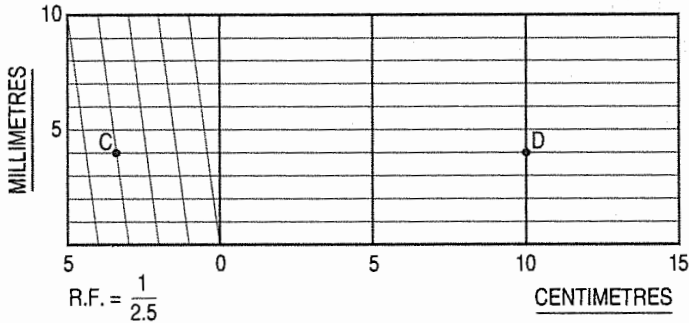


FIG. 4-9

**Problem 4-9.** (fig. 4-10): Construct a diagonal scale of R.F. =  $\frac{1}{32}$  showing yards, feet and inches and to measure upto 4 yards.

Length of the scale =  $\frac{1}{32} \times 4 \text{ yd} = 18 \text{ yd} = 4 \frac{1}{2}''$ .

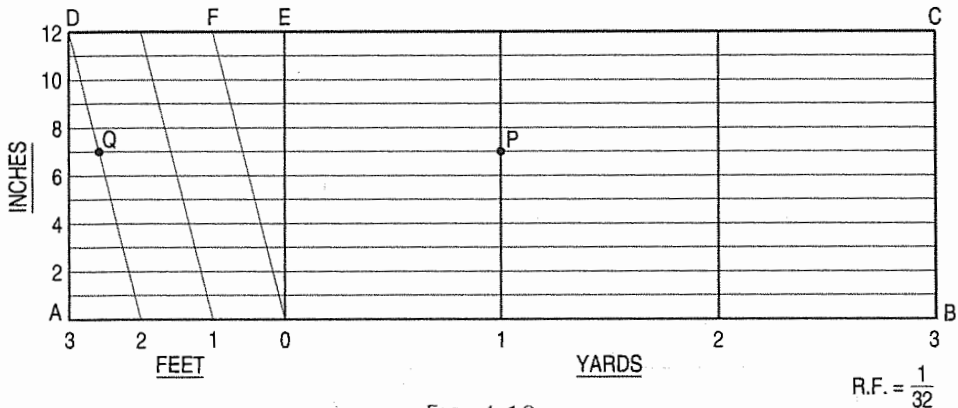


FIG. 4-10

- (i) Draw a line AB  $4 \frac{1}{2}''$  long.
- (ii) Divide it into 4 equal parts to show yards. Divide the first part A0 into 3 equal divisions showing feet.
- (iii) At A, erect a perpendicular and step-off along it, 12 equal divisions of any length, ending at D. Complete the scale as explained in problem 4-6.

To show a distance of 1 yard, 2 feet and 7 inches, place one leg of the divider at P, where the horizontal through 7" meets the vertical from 1 yard and the other leg at Q where the diagonal through 2' meets the same horizontal.

**Problem 4-10.** (fig. 4-11): Draw a scale of full-size, showing  $\frac{1}{100}$  inch and to measure upto 5 inches.

- Draw a line AB 5" long and divide it into five equal parts. Each part will show one inch.
- Sub-divide the first part into 10 equal divisions. Each division will measure  $\frac{1}{10}$  inch.
- At A, draw a perpendicular to AB and on it, step-off ten equal divisions of any length, ending at D.
- Draw the rectangle ABCD and complete the scale as explained in problem 4-6. The line QP shows 2.68 inches.

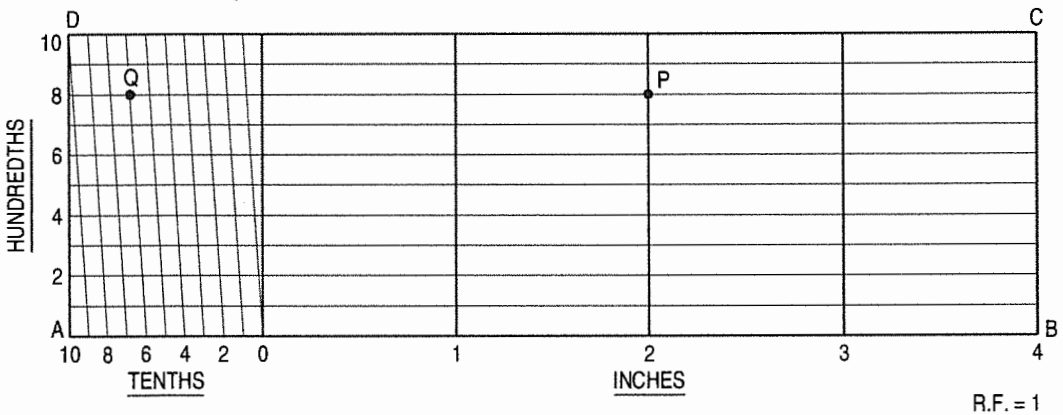


FIG. 4-11

**Problem 4-11.** (fig. 4-12): The area of a field is 50,000 sq m. The length and the breadth of the field, on the map is 10 cm and 8 cm respectively. Construct a diagonal scale which can read upto one metre. Mark the length of 235 metre on the scale. What is the R.F. of the scale?

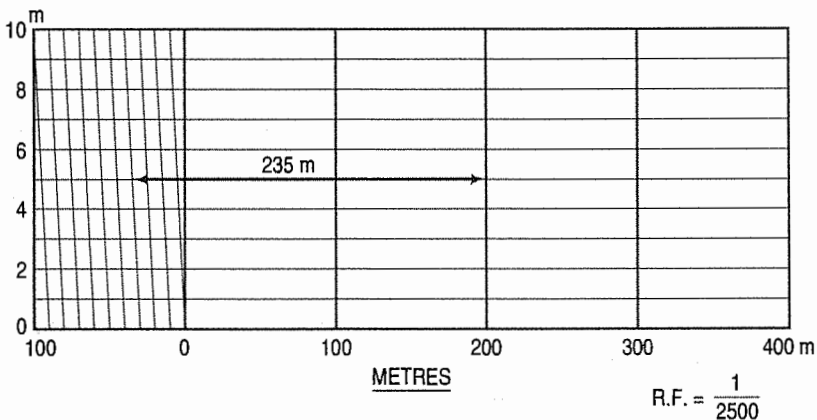


FIG. 4-12

The area of the field = 50,000 sq m.

The area of the field on the map = 10 cm  $\times$  8 cm = 80 cm<sup>2</sup>.

$$\therefore 1 \text{ sq cm} = \frac{50000}{80} = 625 \text{ sq m.}$$

$$\therefore 1 \text{ cm} = 25 \text{ m.}$$

$$\text{Now representative fraction} = \frac{1 \text{ cm}}{25 \text{ m}} = \frac{1}{2500}.$$

$$\text{Length of the scale} = \frac{1}{2500} \times \frac{500 \times 100}{1} = \frac{50000}{2500} = 20 \text{ cm.}$$

Take 20 cm length and divide it into 5 equal parts. Complete the scale as shown in fig. 4-12.

(3) **Comparative scales:** Scales having same representative fraction but graduated to read different units are called *comparative scales*. A drawing drawn with a scale reading inch units can be read in metric units by means of a metric comparative scale, constructed with the same representative fraction. Comparative scales may be plain scales or diagonal scales and may be constructed separately or one above the other.

**Problem 4-12.** [fig. 4-13(i) and fig. 4-13(ii)]: A drawing is drawn in inch units to a scale  $\frac{3}{8}$  full size. Draw the scale showing  $\frac{1}{8}$  inch divisions and to measure upto 15 inches. Construct a comparative scale showing centimetres and millimetres, and to read upto 40 centimetres.

(i) *Inch scale:*

Length of the scale

$$= \frac{3}{8} \times 15 = \frac{45}{8}$$

$$= 5 \frac{5}{8} \text{ inches.}$$

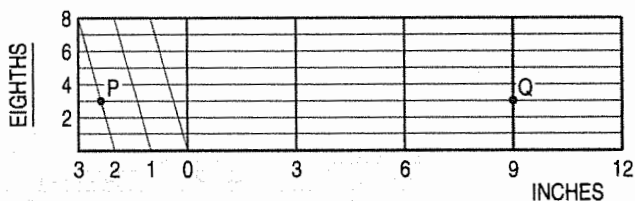


FIG. 4-13(i)

Construct the diagonal scale as shown in fig. 4-13(i).

(ii) *Comparative scale:*

Length of the scale

$$= \frac{3}{8} \times 40$$

$$= 15 \text{ cm.}$$

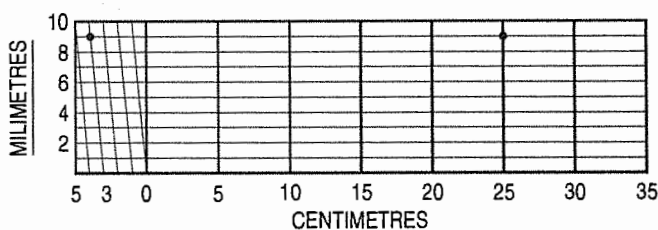


FIG. 4-13(ii)

Construct the diagonal scale as shown in fig. 4-13(ii).

The line PQ on the inch scale shows a length equal to  $11 \frac{3}{8}$ ". Its equivalent, when measured on the comparative scale is 28.9 cm.

**Problem 4-13.** (fig. 4-14): Draw comparative scales of R.F. =  $\frac{1}{485000}$  to read upto 80 kilometres and 80 versts. 1 verst = 1.067 km.

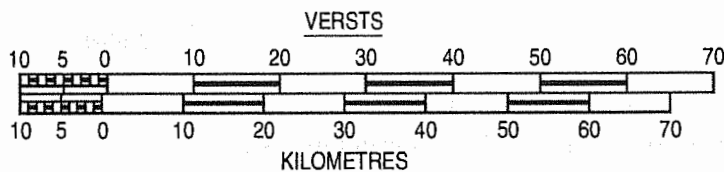


FIG. 4-14

$$\text{Length of kilometre scale} = \frac{1}{485000} \times 80 \times 1000 \times 100 = 16.5 \text{ cm.}$$

$$\text{Length of verst scale} = \frac{1}{485000} \times 80 \times 1.067 \times 1000 \times 100 = 17.6 \text{ cm.}$$

Draw the two scales one above the other as shown in the figure.

**Problem 4-14.** (fig. 4-15): On a road map, a scale of miles is shown. On measuring from this scale, a distance of 25 miles is shown by a line 10 cm long. Construct this scale to read miles and to measure upto 40 miles. Construct a comparative scale, attached to this scale, to read kilometres upto 60 kilometres. 1 mile = 1.609 km.

(i) Scale of miles:

$$\text{Length of the scale} = \frac{10 \times 40}{25} = 16 \text{ cm.}$$

Draw a line 16 cm long and construct a plain scale to show miles.

(ii) Scale of kilometres:

$$\text{R.F.} = \frac{10}{25 \times 1.609 \times 1000 \times 100} = \frac{1}{402250}$$

$$\text{Length of the scale} = \frac{1}{402250} \times 60 \times 1000 \times 100 = 14.9 \text{ cm.}$$

Construct the plain scale 14.9 cm long, above the scale of miles and attached to it, to read kilometres.

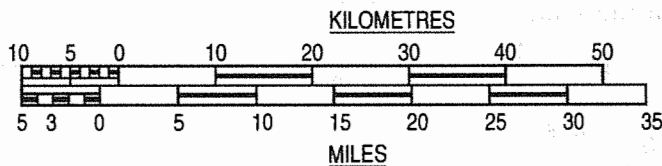


FIG. 4-15

**Problem 4-15.** (fig. 4-16): The distance between Bombay and Poona is 180 km. A passenger train covers this distance in 6 hours. Construct a plain scale to measure time upto a single minute. The R.F. of the scale is  $\frac{1}{200000}$ . Find the distance covered by the train in 36 minutes.

$$\text{Speed of the train} = \frac{180}{6} = 30 \text{ km/hour.}$$

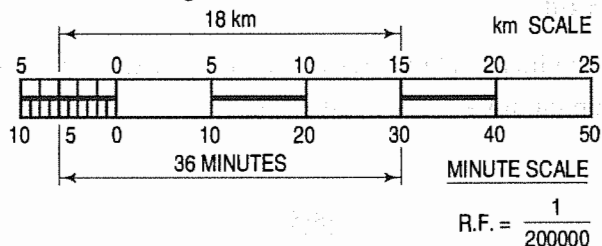


FIG. 4-16

(i) Distance scale (kilometres scale):

$$\text{Length of the scale} = \text{R.F.} \times \text{maximum distance}$$

$$= \frac{1}{200000} \times 30 \times 1000 \times 100 = 15 \text{ cm.}$$

## (ii) Time scale (minute scale):

Speed of the train = 30 km/hour.

i.e. 30 km is covered in 60 minutes.

As length of the scale of 15 cm represents 30 km, 60 minutes which is the time required to cover 30 km, can be represented on the same length of the scale.

- (iii) Draw a line 15 cm long and divide it into 6 equal parts. Each part represents 5 km for the distance scale and 10 minutes for the time scale.
- (iv) Divide the first part of the distance scale and the time scale into 5 and 10 equal parts respectively. Complete the scales as shown. The distance covered in 36 minutes is shown on the scale.

**Problem 4-16.** (fig. 4-17): On a Russian map, a scale of versts is shown. On measuring it with a metric scale, 150 versts are found to measure 15 cm. Construct comparative scales for the two units to measure upto 200 versts and 200 km respectively. 1 verst = 1.067 km.

## (i) Scale of verst:

$$\text{Length of the scale} = \frac{15 \times 200}{150} = 20 \text{ cm.}$$

Draw a line 20 cm long and construct a plain scale to show versts.

## (ii) Scale of kilometres:

$$\text{R.F.} = \frac{15}{150 \times 1.609 \times 1000 \times 10} = \frac{1}{160900}$$

$$\text{Length of the scale} = \frac{1}{160900} \times 200 \times 1000 \times 10 = 12.4 \text{ cm.}$$

Construct the plain scale 12.4 cm long, above the scale of versts and attached to it, to read kilometres (fig. 4-17).

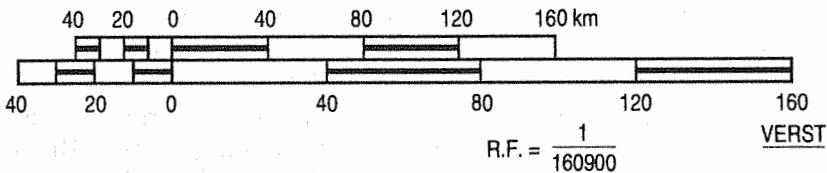


FIG. 4-17

(4) **Vernier scales:** Vernier scales, like diagonal scales, are used to read to a very small unit with great accuracy. A vernier scale consists of two parts — a primary scale and a vernier. The primary scale is a plain scale fully divided into minor divisions.

As it would be difficult to sub-divide the minor divisions in the ordinary way, it is done with the help of the vernier. The graduations on the vernier are derived from those on the primary scale.

(a) **Principle of vernier:** Fig. 4-18 shows a part of a plain scale in which the length A0 represents 10 cm. If we divide A0 into ten equal parts, each part will represent 1 cm. It would not be easy to divide each of these parts into ten equal divisions to get measurements in millimetres.

Now, if we take a length  $BO$  equal to  $10 + 1 = 11$  such equal parts, thus representing 11 cm, and divide it into ten equal divisions, each of these divisions will represent  $\frac{11}{10} = 1.1$  cm or 11 mm.

The difference between one part of  $AO$  and one division of  $BO$  will be equal  $1.1 - 1.0 = 0.1$  cm or 1 mm.

Similarly, the difference between two parts of each will be 0.2 cm or 2 mm.

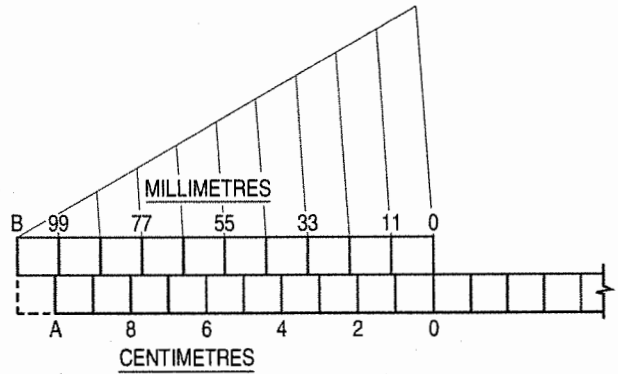


FIG. 4-18

The *upper scale*  $BO$  is the vernier. The combination of the plain scale and the vernier is the vernier scale.

In general, if a line representing  $n$  units is divided into  $n$  equal parts, each part will show  $\frac{n}{n} = 1$  unit. But, if a line equal to  $n + 1$  of these units is taken and then divided into  $n$  equal parts, each of these parts will be equal to  $\frac{n + 1}{n} = 1 + \frac{1}{n}$  units.

The difference between one such part and one former part will be equal to  $\frac{n + 1}{n} - \frac{n}{n} = \frac{1}{n}$  unit.

Similarly, the difference between two parts from each will be  $\frac{2}{n}$  unit.

(b) **Least count of a vernier:** It is the difference of 1 primary scale division and 1 vernier scale division. It is denoted by  $\angle C$ .

$\angle C = 1$  primary scale division  $- 1$  vernier scale division.

The vernier scales are classified as under:

- Forward vernier:** In this case, the length of one division of the vernier scale is smaller than the length of one division of the primary scale. The vernier divisions are marked in the same direction as that of the main scale.
- Backward vernier:** The length of each division of vernier scale is greater than the length of each division of the primary scale. The numbering is done in the opposite direction as that of the primary scale.

**Problem 4-17.** (fig. 4-19): Draw a vernier scale of R.F. =  $\frac{1}{25}$  to read centimetres upto 4 metres and on it, show lengths representing 2.39 m and 0.91 m.

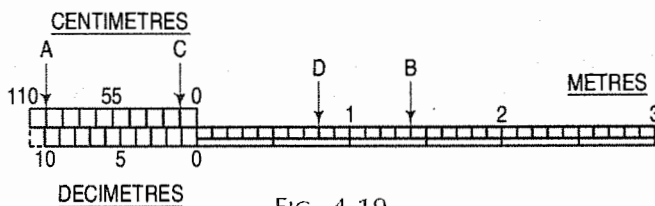


FIG. 4-19

Length of the scale =  $\frac{1}{25} \times 4 \times 100 = 16$  cm.

- (i) Draw a line 16 cm long and divide it into 4 equal parts to show metres. Divide each of these parts into 10 equal parts to show decimetres.
- (ii) To construct a vernier, take 11 parts of dm length and divide it into 10 equal parts. Each of these parts will show a length of 1.1 dm or 11 cm.

To measure a length representing 2.39 m, place one leg of the divider at A on 99 cm mark and the other leg at B on 1.4 m mark. The length AB will show 2.39 metres ( $0.99 + 1.4 = 2.39$ ).

Similarly, the length, CD shows 0.91 metre ( $0.8 + 0.11 = 0.91$ ).

The necessity of dividing the plain scale into minor divisions throughout its length is quite evident from the above measurements.

**Problem 4-18.** (fig. 4-20): Construct a full-size vernier scale of inches and show on it lengths 3.67", 1.54" and 0.48".

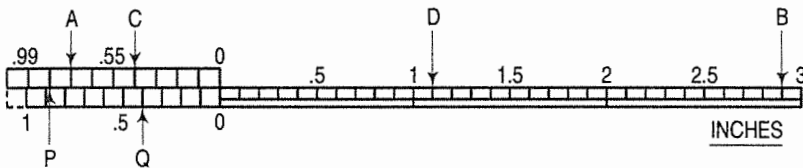


FIG. 4-20

- (i) Draw a plain full-size scale 4" long and divide it fully to show 0.1" lengths.
- (ii) Construct a vernier of length equal to  $10 + 1 = 11$  parts and divide it into 10 equal parts. Each of these parts will  $\frac{11 \times 0.1}{10} = 0.11"$ .

The line AB shows a length of 3.67" ( $0.77" + 2.9" = 3.67"$ ). Similarly, lines CD and PQ show lengths of 1.54" ( $0.44" + 1.1" = 1.54"$ ) and 0.48" ( $0.88" - 0.4" = 0.48"$ ) respectively.

**Problem 4-19.** (fig. 4-21): Construct a vernier scale of R.F. =  $\frac{1}{80}$  to read inches and to measure upto 15 yards.

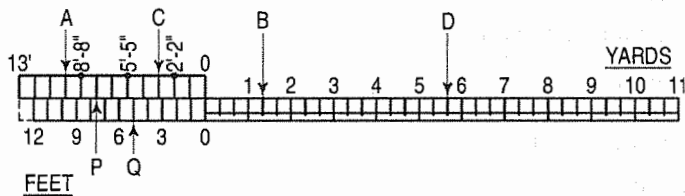


FIG. 4-21

$$\text{Length of the scale} = \frac{1}{80} \times 15 \text{ yd} = \frac{3}{16} \text{ yd} = 6 \frac{3}{4} \text{ yd}.$$

- (i) Draw the plain scale  $6 \frac{3}{4}$  long and divide it fully to show yards and feet.
- (ii) To construct the vernier, take a length of  $12 + 1 = 13$  feet-divisions and divide it into 12 equal parts. Each part will represent  $\frac{13}{12}$  ft or 1'-1".

Lines AB, CD and PQ show respectively lengths representing 4 yd 1 ft 9 in ( $9' - 9'' + 4'$ ), 6 yd 2 ft 3 in ( $3' - 3'' + 17'$ ) and 0 yd 2 ft 7 in ( $7' - 7'' - 5''$ ).

(c) **Circular vernier:** The circular vernier are used in surveying instruments to measure angle to the required accuracy. In the case of mechanical engineering, it is used in measuring instruments such as micrometer.

**Problem 4-20.** A theodolite has main scale plate of 240 mm diameter graduated to  $0^\circ$  to  $360^\circ$  with the vernier scale to read degree and minute. The main scale can read to accuracy of  $0.5^\circ$  and the vernier is to read one minute. Draw the scale.

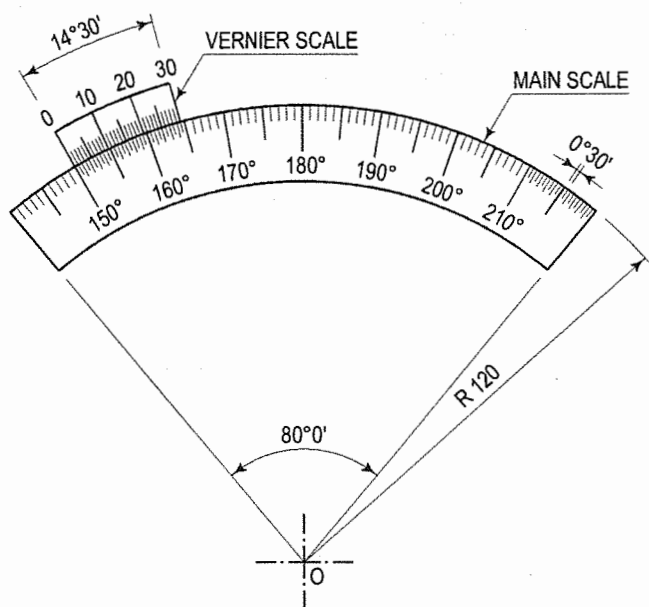


FIG. 4-22

- (i) Draw an arc of radius 120 mm. (Diameter 240 mm)
- (ii) Mark with the protractor graduation as shown on the main scale, each division is  $\frac{1^\circ}{2}$ .
- (iii) Now one division of the main scale will read to accuracy of  $\frac{1^\circ}{2}$ .
- (iv) The vernier can read upto 1 minute  
i.e. 1 minute =  $\frac{1^\circ}{60} = \left(\frac{1}{30}\right) \left(\frac{1}{2}\right)$   
One vernier scale can read =  $\frac{1}{30}$  of main scale division.
- (v) But least count of the vernier  
 $\angle C = (1 \text{ m.s.d} - 1 \text{ v.s.d})$  where m.s.d. = main scale division  
 $\therefore \angle C = 1 \text{ m.s.d.} - \frac{1}{30} \text{ m.s.d.}$  v.s.d. = vernier scale division  
 $1 \text{ v.s.d.} = \frac{29}{30} \text{ m.s.d.}$   
 $\therefore 30 \text{ division of v.s.d.} = 29 \text{ division of m.s.d.}$
- (vi) Draw clockwise arc from  $150^\circ$  above the main scale of the convenient radius and arc length equal 29 divisions of m.s.d (i.e.  $14^\circ - 30'$ ).
- (vii) Divide this arc (vernier) into 30 division by protractor or divider. The last division will coincide in this case at  $164^\circ - 30'$ .

(5) **Scale of chords:** The scale of chords is used to set out or measure angles when a protractor is not available. It is based on the lengths of chords of different angles measured on the same arc and is constructed as shown below.

- (i) Draw a line  $AB$  of any length (fig. 4-23).
- (ii) At  $B$ , erect a perpendicular.
- (iii) With  $B$  as centre, describe an arc  $AC$  cutting the perpendicular at a point  $C$ . Then, the arc  $AC$  (or the chord  $AC$ ) subtends an angle of  $90^\circ$  at the centre  $B$ .
- (iv) Divide  $AC$  into nine equal parts. This may be done
  - (a) by dividing the arc  $AC$  into three equal parts by drawing arcs with centres  $A$  and  $C$  and radius  $AB$ , and then
  - (b) by dividing each of these parts into three equal parts by trial and error method. Each of the nine equal parts subtends an angle of  $10^\circ$  at the centre  $B$ .
- (v) Transfer each division-point from the arc to the straight line  $AB$ -produced, by taking  $A$  as centre and radii equal to chords  $A-10$ ,  $A-20$  etc.
- (vi) Complete the scale by drawing a rectangle below  $AD$ . The divisions obtained are unequal, decreasing gradually from  $A$  to  $D$ . It is quite evident that the distance from  $A$  to a division-point on the scale is equal to the length of the chord of the angle subtended by it at the centre  $B$ . It may be noted that the chord  $A-60$  is equal to the radius  $AB$ .

The scale may be fully divided, i.e. each division divided into ten equal parts to show degrees. In the figure, degrees are shown in multiples of 5.

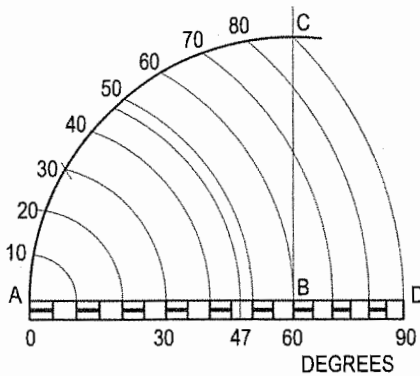


FIG. 4-23

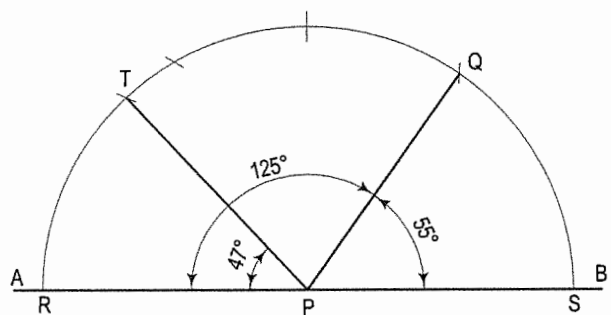


FIG. 4-24

**Problem 4-21.** (fig. 4-24): Construct angles of  $47^\circ$  and  $125^\circ$  by means of the scale of chords.

- (i) Draw any line  $AB$ .
- (ii) With any point  $P$  on it as centre and radius equal to 0-60 (from the scale of chords), draw an arc cutting  $AP$  at a point  $R$ .
- (iii) With  $R$  as centre and radius equal to 0-47 (chord of  $47^\circ$ ) cut the arc at a point  $T$ .
- (iv) Draw a line joining  $P$  with  $T$ . Then  $\angle RPT = 47^\circ$ .

As the scale of chords gives angles upto only  $90^\circ$ , angle of  $125^\circ$  may be set-off in two parts viz.  $60^\circ + 65^\circ$  or  $90^\circ + 35^\circ$  as shown in the figure.

It may also be obtained by setting-off a chord  $SQ$  of  $55^\circ$  ( $180^\circ - 125^\circ$ ) from the right side.

**Problem 4-22.** (fig. 4-25): Measure the given angle  $PQR$  by means of the scale of chords.

- (i) With  $Q$  as centre and radius equal to 0-60, draw an arc cutting  $PQ$  at  $A$  and  $RQ$  at  $B$ .
- (ii) Take the chord length  $AB$  and apply it to the scale of chords which shows the angle to be of  $37^\circ$ .

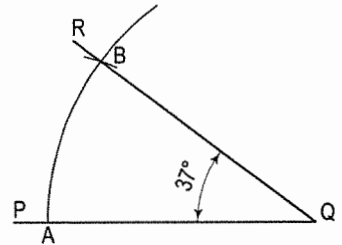


FIG. 4-25

## EXERCISES 4

1. Fill-up the blanks in the following sentences, using appropriate words, selected from those given in the brackets:
  - (a) The ratio of the length of the drawing of the object to the actual length of the object is called \_\_\_\_\_ (resulting fraction, representative figure, representative fraction).
  - (b) When the drawing is drawn of the same size as that of the object, the scale used is \_\_\_\_\_ (diagonal scale, full-size scale, vernier scale).
  - (c) For drawings of small instruments, watches etc. \_\_\_\_\_ scale is always used (reducing, full-size, enlarging).
  - (d) Drawings of buildings are drawn using \_\_\_\_\_ (full-size scale, reducing scale, scale of chords).
  - (e) The R.F. in case of (b), (c) and (d) above would be, \_\_\_\_\_, \_\_\_\_\_, and \_\_\_\_\_ respectively (equal to 1, less than 1, greater than 1).
  - (f) When measurements are required in three units \_\_\_\_\_ scale is used (diagonal, plain, comparative).
  - (g) The scale of chords is used to set out or measure \_\_\_\_\_ (chords, lines, angles).

Answers to Ex. (1):

- a-3, b-2, c-3, d-2, e-1, 3 and 2, f-1, g-3.
2. Construct a scale of 1:5 to show decimetres and centimetres and to read upto 1 metre. Show the length of 7.6 dm on it.
  3. Construct a scale of 1.5 cm = 1 dm to read upto 1 metre and show on it a length of 0.6 metre.
  4. Draw a diagonal scale of R.F. =  $\frac{3}{100}$ , showing metres, decimetres and centimetres, and to measure upto 5 metres. Show the length of 3.69 metres on it.
  5. Draw a scale of 1:50 showing metres and decimetres, and to measure upto 8 metres.

6. A 3.2 cm long line represents a length of 4 metres. Extend this line to measure lengths upto 25 metres and show on it units of metre and 5 metres. Show the length of 17 metres on this line.
7. Construct a diagonal scale of R.F. =  $\frac{1}{6250}$  to read upto 1 kilometre and to read metres on it. Show a length of 653 metres on it.
8. On a map, the distance between two points is 14 cm. The real distance between them is 20 km. Draw a diagonal scale of this map to read kilometres and hectametres, and to measure upto 25 km. Show a distance of 17.6 km on this scale.
9. An area of 144 sq cm on a map represents an area of 36 sq km on the field. Find the R.F. of the scale for this map and draw a diagonal scale to show kilometres, hectametres and decametres and to measure upto 10 kilometres. Indicate on the scale a distance of 7 kilometres, 5 hectametres and 6 decametres.
10. Construct the following scales and show below each, its R.F. and the units which its divisions represent:
  - (a) Scale of  $1\frac{1}{4}" = 1$  foot, to measure upto 5 feet and showing feet and inches.
  - (b) Scale of  $\frac{3}{4}" = 1$  yard, to measure upto 10 yards and showing yards and feet.
  - (c) Scale of  $1\frac{3}{8}" = 1$  mile, to measure upto 4 miles and showing miles and furlongs.
11. Construct a scale of  $1" = 1$  foot to read upto 6 feet and show on it,  $4' - 7''$  length.
12. The R.F. of a scale showing miles, furlongs and chains is  $\frac{1}{50688}$ . Draw a scale to read upto 5 miles and show on it, the length representing 3 m 5 f 3 ch.
13. Draw a 4" long diagonal scale of  $1" = 1'$  and show on it, the length of 2.14" and 3.79".
14. The distance between two points on a map is  $5\frac{3}{4}"$ . The points are actually 20 miles apart. Construct a diagonal scale of the map, showing miles and furlongs and to read upto 25 miles.
15. Construct comparative diagonal scales of metres and yards having R.F. =  $\frac{1}{2700}$  to show upto 400 metres. 1 metre = 1.0936 yards.
16. Define simple and comparative scales. What is the R.F. of a scale which measures 2.5 inches to a mile? Draw a comparative scale of kilometres to read upto 10 km. 1 mile = 1.609 km.
17. On a map showing a scale of kilometres, 60 km are found to equal 4.5 inches. What is the R.F.? Construct a comparative scale of English miles. 1 km = 1093.6 yards.
18. On a Russian map, a scale of versts is shown. On measuring it with a metric scale, 120 versts are found to measure 10 cm. Construct comparative scales for the two units to measure upto 150 versts and 150 km respectively. 1 verst = 1.067 km.

19. Prepare a scale of knots comparative to a scale of 1 cm = 5 km. Assume suitable lengths. 1 knot = 1.85 km.
20. Draw a full-size vernier scale to read  $\frac{1}{8}$ " and  $\frac{1}{64}$ " lengths and mark on it lengths of  $5\frac{7}{32}$ ",  $2\frac{51}{64}$ " and  $\frac{29}{64}$ ".
21. Construct a scale of R.F. =  $\frac{1}{2.5}$  to show decimetres and centimetres and by a vernier to read millimetres, to measure upto 4 decimetres.
22. Construct a vernier scale to show yards, the R.F. being  $\frac{1}{3300}$ . Show the distance representing 2 furlongs 99 yards.
23. Construct a scale of chords showing  $5^\circ$  divisions and with its aid set-off angles of  $25^\circ$ ,  $40^\circ$ ,  $55^\circ$  and  $130^\circ$ .
24. Draw a triangle having sides 8 cm, 9 cm and 10 cm long respectively and measure its angles with the aid of a scale of chords.
25. The distance between Vadodara and Surat is 130 km. A train covers this distance in 2.5 hours. Construct a plain scale to measure time upto a single minute. The R.F. of the scale is  $\frac{1}{260000}$ . Find the distance covered by the train in 45 minutes.
26. On a building plan, a line 20 cm long represents a distance of 10 m. Devise a diagonal scale for the plan to read upto 12 m, showing metres, decimetres and centimetres. Show on your scale the lengths 6.48 m and 11.14 m.
27. A room of  $1728 \text{ m}^3$  volume is shown by a cube of  $216 \text{ cm}^3$  volume. Find R.F. and construct a plain scale to measure upto 42 m. Mark a distance of 22 m on the scale.
28. An old plan was drawn to a scale of 1 cm = 24 m. It has shrunk so that actual length of 100 m at site now works out to 96 m as per scale on plan. Find out the shrinkage factor and the corrected R.F. of the plan. (Hint: Shrinkage factor = present length on scale/original length on scale.)
29. The actual length of 500 m is represented by a line of 15 cm on a drawing. Construct a vernier scale to read upto 600 m. Mark on the scale a length of 549 m.
30. Draw a vernier scale of R.F. = 5 to read  $\frac{1}{5}$  cm and  $\frac{1}{25}$  cm and to measure upto 5 cm. Mark on the scale distances of 2.12 cm.
31. A car is running at a speed of 50 km/hour. Construct a diagonal scale to show 1 km by 3 cm and to measure upto 6 km. Mark also on the scale the distance covered by the car in 5 minutes 28 seconds.

## Chapter

# 5



# GEOMETRICAL CONSTRUCTION

## 5-0. INTRODUCTION



In this chapter, we shall deal with problems on geometrical construction which are mostly based on plane geometry and which are very essential in the preparation of engineering drawings. They are described as under:

- |                                           |                                                             |
|-------------------------------------------|-------------------------------------------------------------|
| (1) Bisecting a line                      | (11) To construct squares                                   |
| (2) To draw perpendiculars                | (12) To construct regular polygons                          |
| (3) To draw parallel lines                | (13) Special methods of drawing regular polygons            |
| (4) To divide a line                      | (14) Regular polygons inscribed in circles                  |
| (5) To divide a circle                    | (15) To draw regular figures using T-square and set-squares |
| (6) To bisect an angle                    | (16) To draw tangents                                       |
| (7) To trisect an angle                   | (17) Lengths of arcs                                        |
| (8) To find the centre of an arc          | (18) Circles and lines in contact                           |
| (9) To construct an ogee or reverse curve | (19) Inscribed circles.                                     |
| (10) To construct equilateral triangles   |                                                             |

## 5-1. BISECTING A LINE



**Problem 5-1.** To bisect a given straight line (fig. 5-1).

- (i) Let  $AB$  be the given line. With centre  $A$  and radius greater than half  $AB$ , draw arcs on both sides of  $AB$ .

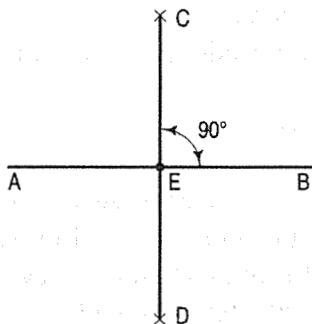


FIG. 5-1

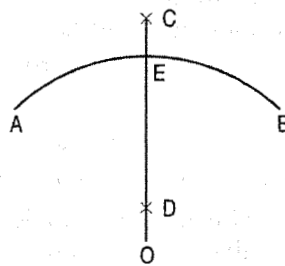


FIG. 5-2

(ii) With centre  $B$  and the same radius, draw arcs intersecting the previous arcs at  $C$  and  $D$ .

(iii) Draw a line joining  $C$  and  $D$  and cutting  $AB$  at  $E$ .

Then  $AE = EB = \frac{1}{2} AB$ .

Further,  $CD$  bisects  $AB$  at right angles.

**Problem 5-2.** To bisect a given arc (fig. 5-2).

Let  $AB$  be the arc drawn with centre  $O$ . Adopt the same method as shown in problem 5-1. The bisector  $CD$ , if produced, will pass through the centre  $O$ .

## 5-2. TO DRAW PERPENDICULARS

**Problem 5-3.** To draw a perpendicular to a given line from a point within it.

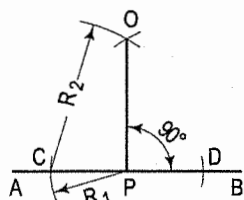


FIG. 5-3

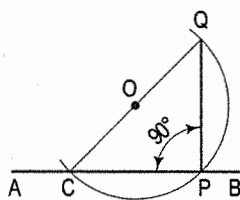


FIG. 5-4

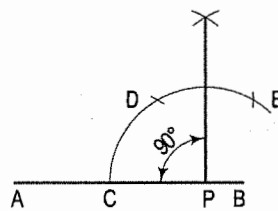


FIG. 5-5

**Method I** (fig. 5-3):

(a) When the point is near the middle of the line.

Let  $AB$  be the given line and  $P$  the point in it.

(i) With  $P$  as centre and any convenient radius  $R_1$ , draw an arc cutting  $AB$  at  $C$  and  $D$ .

(ii) With any radius  $R_2$  greater than  $R_1$  and centres  $C$  and  $D$ , draw arcs intersecting each other at  $O$ .

(iii) Draw a line joining  $P$  and  $O$ .

Then  $PO$  is the required perpendicular.

(b) When the point is near an end of the line.

Let  $AB$  be the given line and  $P$  the point in it.

**Method II** (fig. 5-4):

(i) With any point  $O$  as centre and radius equal to  $OP$ , draw an arc greater than the semi-circle, cutting  $AB$  at  $C$ .

(ii) Draw a line joining  $C$  and  $O$ , and produce it to cut the arc at  $Q$ . Draw a line joining  $P$  and  $Q$ .

Then  $PQ$  is the required perpendicular.

**Method III** (fig. 5-5):

(i) With  $P$  as centre and any convenient radius, draw an arc cutting  $AB$  at  $C$ .

(ii) With the same radius cut (from the arc) two equal divisions  $CD$  and  $DE$ .

(iii) Again with the same radius and centres  $D$  and  $E$ , draw arcs intersecting each other at  $Q$ . Draw a line joining  $P$  and  $Q$ .

Then  $PQ$  is the required perpendicular.

**Problem 5-4.** To draw a perpendicular to a given line from a point outside it.

(a) When the point is nearer the centre than the end of the line (fig. 5-6).

Let  $AB$  be the given line and  $P$  the point.

- (i) With centre  $P$  and any convenient radius, draw an arc cutting  $AB$  at  $C$  and  $D$ .
- (ii) With any radius greater than half  $CD$  and centres  $C$  and  $D$ , draw the arcs intersecting each other at  $E$ .
- (iii) Draw a line joining  $P$  and  $E$  and cutting  $AB$  at  $Q$ .

Then  $PQ$  is the required perpendicular.

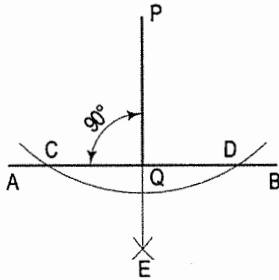


FIG. 5-6

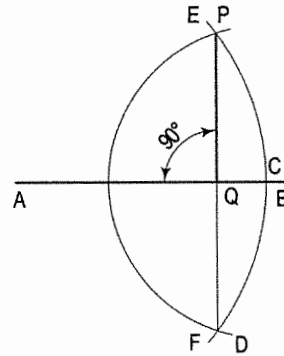


FIG. 5-7

(b) When the point is nearer the end than the centre of the line (fig. 5-7).

Let  $AB$  be the given line and  $P$  the point.

- (i) With centre  $A$  and radius equal to  $AP$ , draw an arc  $EF$  cutting  $AB$  or  $AB$ -produced, at  $C$ .
- (ii) With centre  $C$  and radius equal to  $CP$ , draw an arc cutting  $EF$  at  $D$ .
- (iii) Draw a line joining  $P$  and  $D$  and intersecting  $AB$  at  $Q$ .

Then  $PQ$  is the required perpendicular.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 4 for the following problem.

**Problem 5-5.** To draw perpendicular to a given line from a point outside it (fig. 5-8).

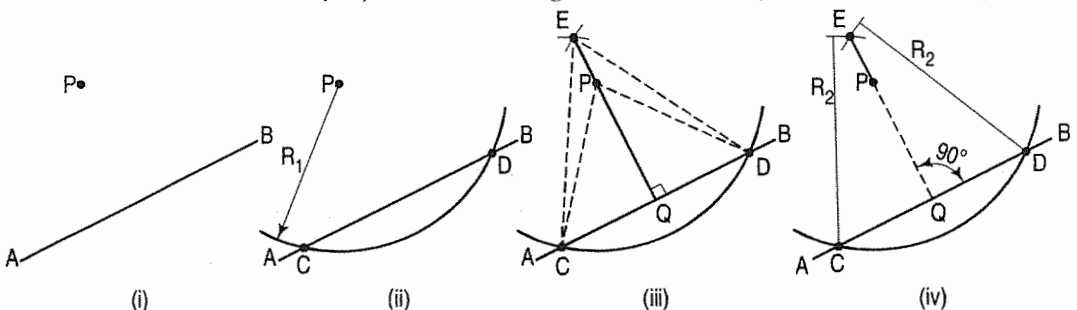


FIG. 5-8

When the point is nearer the centre than the end of the line.

Let  $AB$  be the given line and  $P$  the point.

- (i) Using compass and with  $P$  as centre draw an arc of any radius  $R_1$  cutting the line  $AB$  at points  $C$  and  $D$ .
- (ii) With points  $C$  and  $D$  as centres, and with a larger radius  $R_2 (> R_1)$ , draw arcs to cut on the side of the line  $AB$  in which perpendicular is to draw. The arcs intersect in point  $E$ .
- (iii) Now join points  $E$  and  $P$ . (If required) Line  $EP$  may be extended to meet the line  $AB$  at point  $Q$ . Line  $EPQ$  will be perpendicular to line  $AB$ .
- (iv) Verify the an angle  $AQP$  or  $BQP$  using a protractor. The angle  $AQP$  or  $BQP$  is the required perpendicular.

### 5-3. TO DRAW PARALLEL LINES



**Problem 5-6.** To draw a line through a given point, parallel to a given straight line (fig. 5-9).

Let  $AB$  be the given line and  $P$  the point.

- (i) With centre  $P$  and any convenient radius, draw an arc  $CD$  cutting  $AB$  at  $E$ .
- (ii) With centre  $E$  and the same radius, draw an arc cutting  $AB$  at  $F$ .
- (iii) With centre  $E$  and radius equal to  $FP$ , draw an arc to cut  $CD$  at  $Q$ .
- (iv) Draw a straight line through  $P$  and  $Q$ . Then this is the required line.

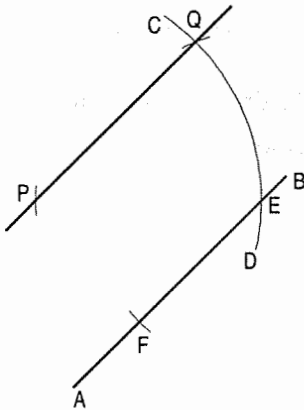


FIG. 5-9

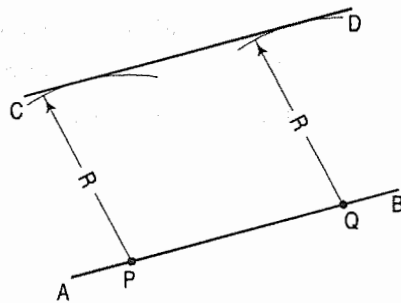


FIG. 5-10

**Problem 5-7.** To draw a line parallel to and at a given distance from a given straight line (fig. 5-10).

Let  $AB$  be the given line and  $R$  the given distance.

- (i) Mark points  $P$  and  $Q$  on  $AB$ , as far apart as convenient.
- (ii) With  $P$  and  $Q$  as centres and radius equal to  $R$ , draw arcs on the same side of  $AB$ .
- (iii) Draw the line  $CD$ , just touching the two arcs.  $CD$  is the required line.

## 5-4. TO DIVIDE A LINE



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 5 for the following problem.

**Problem 5-8.** To divide a given straight line into any number of equal parts (fig. 5-11).

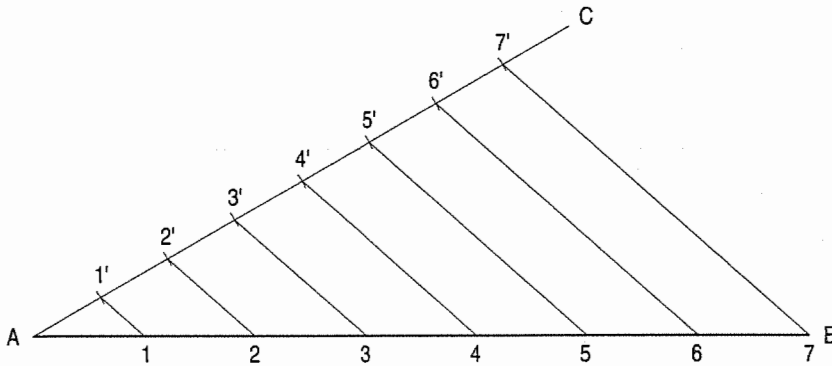


FIG. 5-11

Let  $AB$  be the given line to be divided into say, seven equal parts.

- (i) Draw the line  $AB$  of given length.
- (ii) Draw another line  $AC$  making an angle of less than  $30^\circ$  with  $AB$ .
- (iii) With the help of dividers mark 7 equal parts of any suitable length on line  $AC$  and mark them by points  $1'$ ,  $2'$ ,  $3'$ ,  $4'$ ,  $5'$ ,  $6'$  and  $7'$  as shown.
- (iv) Join the last point  $7'$  with point  $B$  of the line  $AB$ .
- (v) Now, from each of the other marked points  $6'$ ,  $5'$ ,  $4'$ ,  $3'$ ,  $2'$  and  $1'$ , draw lines parallel to  $7'B$  cutting the line  $AB$  at  $6$ ,  $5$ ,  $4$ ,  $3$ ,  $2$  and  $1$  respectively.
- (vi) Now the line  $AB$  has been divided into 7 equal parts. You can verify this by measuring the lengths.

**Problem 5-9.** To divide a given straight line into unequal parts (fig. 5-12).

Let  $AB$  be the given line to be divided into unequal parts say  $\frac{1}{6}$ ,  $\frac{1}{5}$ ,  $\frac{1}{4}$ ,  $\frac{1}{3}$  and  $\frac{1}{2}$ .

- (i) Draw a line  $AB$  of given length, say, 120 mm.
- (ii) Erect perpendicular  $AD$  and  $BC$  at the ends  $A$  and  $B$ . Complete rectangular  $ABCD$ .
- (iii) Join diagonals  $AC$  and  $BD$  intersecting at  $E$ .
- (iv) Draw perpendicular from  $E$  on  $AB$  as shown.
- (v) Then  $AF = \frac{1}{2} AB$ .
- (vi) Join  $D$  and  $F$ . The line  $FD$  intersects the diagonal  $AC$  at  $G$ . Drop perpendicular from  $G$  to  $AB$ . Then  $AH = \frac{1}{3} AB$ .
- (vii) Similarly make construction in figure, for obtaining  $\frac{1}{4} AB$ ,  $\frac{1}{5} AB$  and  $\frac{1}{6} AB$  as shown.

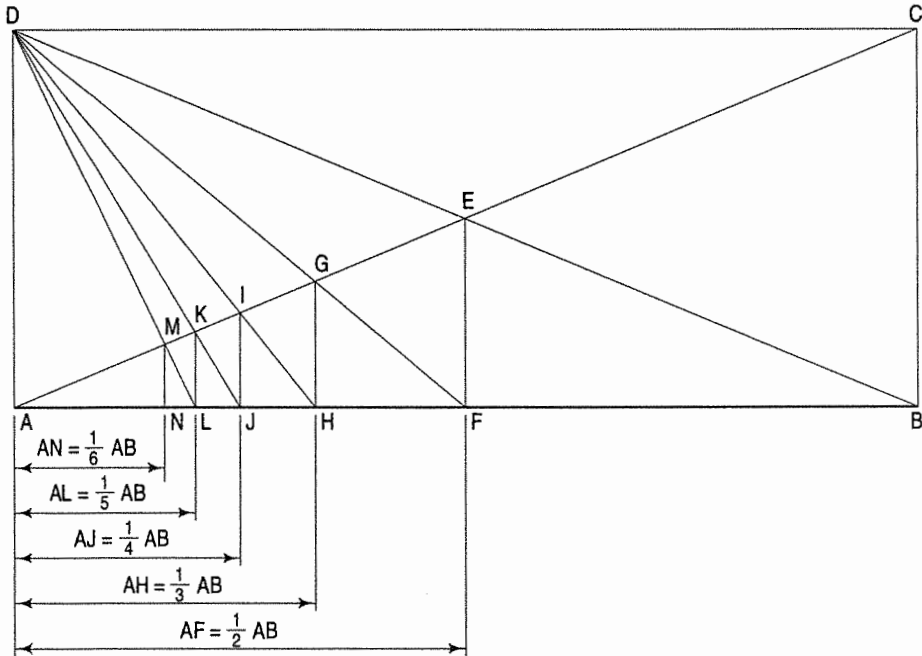


FIG. 5-12

### 5-5. TO DIVIDE A CIRCLE



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 6 for the following problem.

**Problem 5-10.** To divide a circle of a given radius into  $N$  equal parts (fig. 5-13).

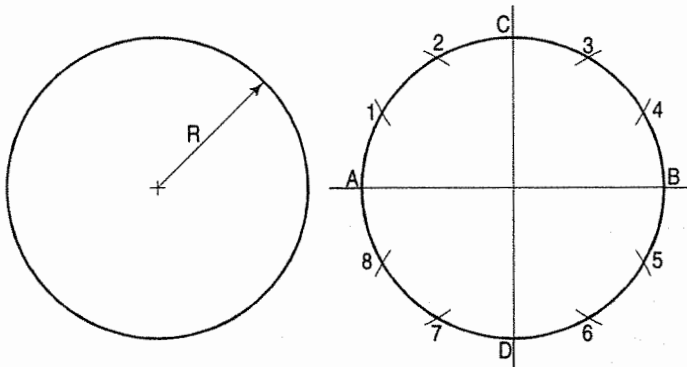


FIG. 5-13

Given a circle of radius  $R$ , to divide it into 12 equal parts along its circumference, the geometric construction procedure is as follows.

- (i) Draw two diagonals  $AB$  and  $CD$  at right angles to each other cutting the circle at  $A$ ,  $B$  and  $C$ ,  $D$ .

- (ii) With  $A$  as centre, and radius equal to the given radius of the circle, draw arcs cutting the circle at 2 and 7.
- (iii) Similarly, with  $B$  as centre, and with the given circle radius, draw arcs to cut the circle at 3 and 6.
- (iv) Similarly, from points  $C$  and  $D$  as centre, and with the given radius of circle, draw arcs to cut the circle at 1, 4 and 5, 8 respectively.
- (v) Thus dividing it at the cuts are the required twelve equal parts of the circle.

## 5-6. TO BISECT AN ANGLE

**Problem 5-11.** To bisect a given angle (fig. 5-14).

Let  $ABC$  be the given angle.

- (i) With  $B$  as centre and any radius, draw an arc cutting  $AB$  at  $D$  and  $BC$  at  $E$ .
- (ii) With centres  $D$  and  $E$  and the same or any convenient radius, draw arcs intersecting each other at  $F$ .
- (iii) Draw a line joining  $B$  and  $F$ .  $BF$  bisects the angle  $ABC$ , i.e.  $\angle ABF = \angle FBC$ .

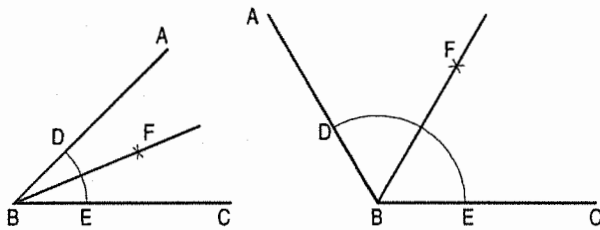


FIG. 5-14

**Problem 5-12.** To draw a line inclined to a given line at an angle equal to a given angle (fig. 5-15).

Let  $PQ$  be the given line and  $AOB$  the given angle.

- (i) With  $O$  as centre and any radius, draw an arc cutting  $OA$  at  $C$  and  $OB$  at  $D$ .
- (ii) With the same radius and centre  $P$ , draw an arc  $EF$  cutting  $PQ$  at  $F$ .
- (iii) With  $F$  as centre and radius equal to  $CD$ , draw an arc cutting the arc  $EF$  at  $G$ .
- (iv) From  $P$ , draw a line passing through  $G$ . This is the required line.

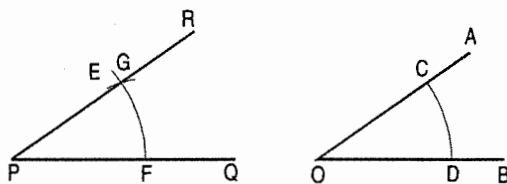


FIG. 5-15

## 5-7. TO TRISECT AN ANGLE



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 7 for the following problem.

**Problem 5-13.** To trisect a given right angle (fig. 5-16).

Let  $ABC$  be the given right angle.

- (i) With centre  $B$  and any radius, draw an arc cutting  $AB$  at  $D$  and  $BC$  at  $E$ .
- (ii) With the same radius and centres  $D$  and  $E$ , draw arcs cutting the arc  $DE$  at points  $Q$  and  $P$ .

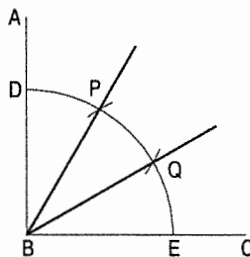


FIG. 5-16

(iii) Draw lines joining  $B$  with  $P$  and  $Q$ .  $BP$  and  $BQ$  trisect the right angle  $ABC$ .

Thus,  $\angle ABP = \angle PBQ = \angle QBC = \frac{1}{3} \angle ABC$ .

## 5-8. TO FIND THE CENTRE OF AN ARC

**Problem 5-14.** To find the centre of a given arc (fig. 5-17).

Let  $AB$  be the given arc.

- In  $AB$ , draw two chords  $CD$  and  $EF$  of any lengths.
- Draw perpendicular bisectors of  $CD$  and  $EF$  intersecting each other at  $O$ . Then  $O$  is the required centre.

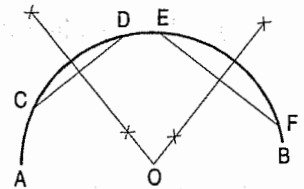


FIG. 5-17

**Problem 5-15.** To draw an arc of a given radius, touching a given straight line and passing through a given point (fig. 5-18).

Let  $AB$  be the given line,  $P$  the point and  $R$  the radius.

- Draw a line  $CD$  parallel to and at a distance equal to  $R$  from  $AB$  (Problem 5-7).
- With  $P$  as centre and radius equal to  $R$ , draw an arc cutting  $CD$  at  $O$ .
- With  $O$  as centre, draw the required arc.

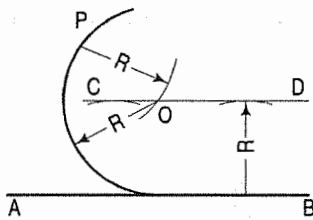


FIG. 5-18

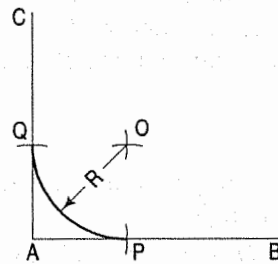


FIG. 5-19

**Problem 5-16.** To draw an arc of a given radius touching two given straight lines at right angles to each other (fig. 5-19).

Let  $AB$  and  $AC$  be the given lines and  $R$  the given radius.

- With centre  $A$  and radius equal to  $R$ , draw arcs cutting  $AB$  at  $P$  and  $AC$  at  $Q$ .
- With  $P$  and  $Q$  as centres and the same radius, draw arcs intersecting each other at  $O$ .
- With  $O$  as centre and radius equal to  $R$ , draw the required arc.

**Problem 5-17.** To draw an arc of a given radius touching two given straight lines which make any angle between them (fig. 5-20 and fig. 5-21).

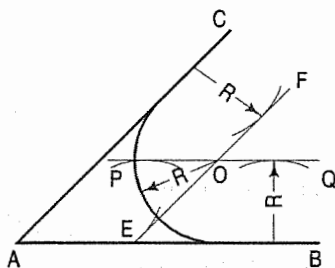


FIG. 5-20

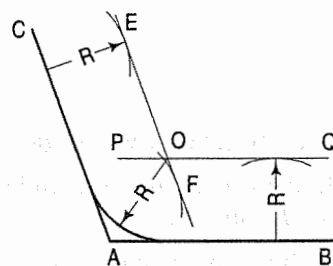


FIG. 5-21

Let  $AB$  and  $AC$  be the given lines and  $R$  the given radius.

- (i) Draw a line  $PQ$  parallel to and at a distance equal to  $R$  from  $AB$ .
- (ii) Similarly, draw a line  $EF$  parallel to and at a distance equal to  $R$  from  $AC$ , intersecting  $PQ$  at  $O$ .
- (iii) With  $O$  as centre and radius equal to  $R$ , draw the required arc.

**Problem 5-18.** To draw an arc of a given radius touching a given arc and a given straight line.

**Case I:** (fig. 5-22): Let  $AB$  be the given line,  $CD$  the given arc drawn with centre  $O$  and radius equal to  $R_1$ , and  $R_2$  the given radius.

- (i) With  $O$  as centre and radius equal to  $(R_1 - R_2)$ , draw an arc  $EF$ .
- (ii) Draw a line parallel to and at a distance equal to  $R_2$  from  $AB$  and intersecting  $EF$  at a point  $P$ .
- (iii) With  $P$  as centre and radius equal to  $R_2$ , draw the required arc.

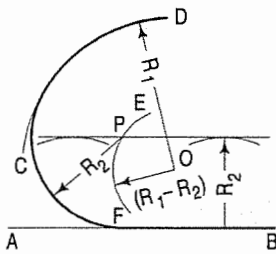


FIG. 5-22

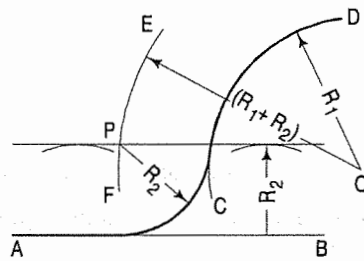


FIG. 5-23

**Case II:** (fig. 5-23): Let  $AB$  be the given line,  $CD$  the given arc drawn with centre  $O$  and radius equal to  $R_1$ , and  $R_2$  the given radius.

- (i) With  $O$  as centre and radius equal to  $(R_1 + R_2)$ , draw an arc  $EF$ .
- (ii) Draw a line parallel to and at a distance equal to  $R_2$  from  $AB$  and intersecting  $EF$  at a point  $P$ .
- (iii) With  $P$  as centre and radius equal to  $R_2$ , draw the required arc.

**Problem 5-19.** To draw an arc of a given radius touching two given arcs.

Let  $AB$  be the given arc drawn with centre  $O$  and radius equal to  $R_1$ ;  $CD$  the arc drawn with centre  $P$  and radius equal to  $R_2$ , and  $R_3$  the given radius.

**Case I** (fig. 5-24):

- (i) With  $O$  as centre and radius equal to  $(R_1 + R_3)$ , draw an arc  $EF$ .
- (ii) With  $P$  as centre and radius equal to  $(R_2 + R_3)$ , draw an arc intersecting  $EF$  at a point  $Q$ .
- (iii) With  $Q$  as centre and radius equal to  $R_3$ , draw the required arc.

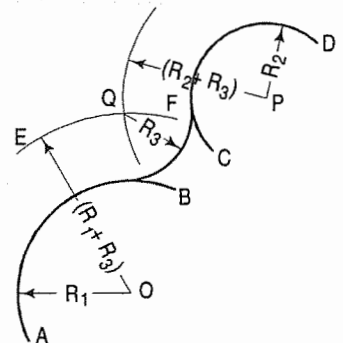


FIG. 5-24

**Case II** (fig. 5-25):

- (i) With  $O$  as centre and radius equal to  $(R_1 - R_3)$ , draw an arc  $EF$ .
- (ii) With  $P$  as centre and radius equal to  $(R_2 + R_3)$ , draw an arc intersecting  $EF$  at a point  $Q$ .
- (iii) With  $Q$  as centre and radius equal to  $R_3$ , draw the required arc.

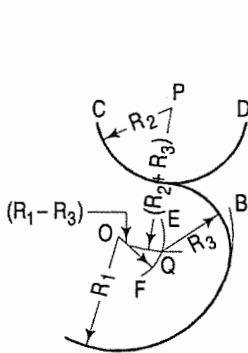


FIG. 5-25

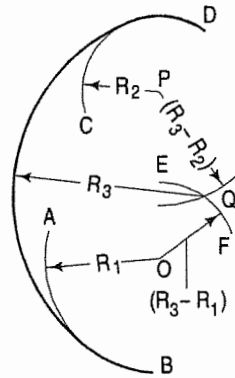


FIG. 5-26

**Case III** (fig. 5-26):

- (i) With  $O$  as centre and radius equal to  $(R_3 - R_1)$ , draw an arc  $EF$ .
- (ii) With  $P$  as centre and radius equal to  $(R_3 - R_2)$ , draw an arc intersecting  $EF$  at a point  $Q$ .
- (iii) With  $Q$  as centre and radius equal to  $R_3$ , draw the required arc.

**Problem 5-20.** To draw an arc passing through three given points not in a straight line (fig. 5-27).

Let  $A$ ,  $B$  and  $C$  be the given points.

- (i) Draw lines joining  $B$  with  $A$  and  $C$ .
- (ii) Draw perpendicular bisectors of  $AB$  and  $BC$  intersecting each other at a point  $O$ .
- (iii) With  $O$  as centre and radius equal to  $OA$  or  $OB$  or  $OC$ , draw the required arc.

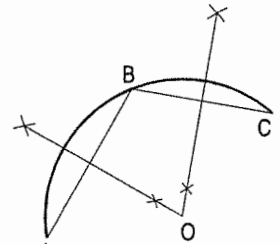


FIG. 5-27

**Problem 5-21.** To draw a continuous curve of circular arcs passing through any number of given points not in a straight line (fig. 5-28).

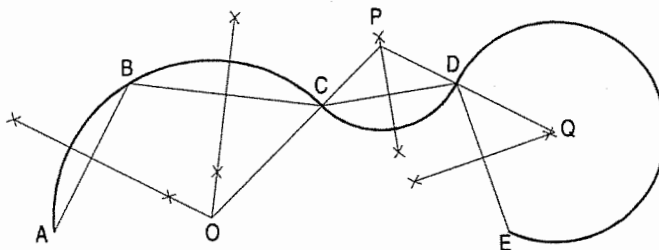


FIG. 5-28

Let  $A, B, C, D$  and  $E$  be the given points.

- (i) Draw lines joining  $A$  with  $B$ ,  $B$  with  $C$ ,  $C$  with  $D$  etc.
- (ii) Draw perpendicular bisectors of  $AB$  and  $BC$  intersecting at  $O$ .
- (iii) With  $O$  as centre and radius equal to  $OA$ , draw an arc  $ABC$ .
- (iv) Draw a line joining  $O$  and  $C$ .
- (v) Draw the perpendicular bisector of  $CD$  intersecting  $OC$  or  $OC$  produced, at  $P$ .
- (vi) With  $P$  as centre and radius equal to  $PC$ , draw an arc  $CD$ .
- (vii) Repeat the same construction. Note that the centre of the arc is at the intersection of the perpendicular bisector and the line, or the line-produced, joining the previous centre with the last point of the previous arc.

### 5-9. TO CONSTRUCT AN OGEE OR REVERSE CURVE

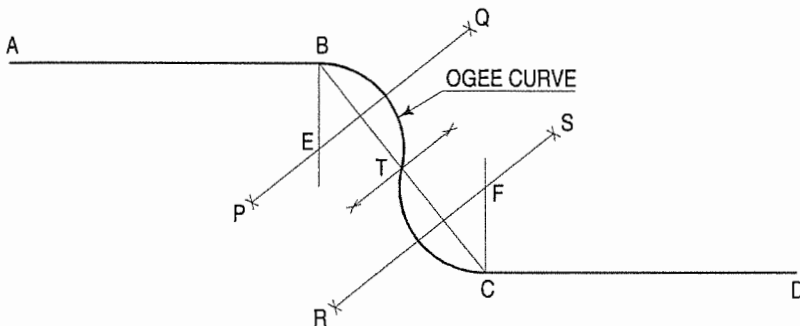


An ogee curve or a reverse curve is a combination of two same curves in which the second curve has a reverse shape to that of the first curve. In other words, any curve or line or mould consist of a continuous double curve with the upper part convex and lower part concave, to some extent having shape of "S".



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 8 for the following problem.

**Problem 5-22.** To draw an ogee shaped arc tangent between two parallel lines (fig. 5-29).



Ogee shaped arc tangent to two parallel lines

FIG. 5-29

- (i) Let  $AB$  and  $CD$  be the two given parallel lines which are to be connected by an ogee curve.
- (ii) Join points  $B$  and  $C$ . Bisect the line  $BC$  and obtain its centre point  $T$ .
- (iii) Bisect line segment  $BT$  and draw the bisector line  $PQ$ . Similarly, bisect segment  $CT$  and draw the bisector line  $RS$ .
- (iv) From point  $B$ , draw a perpendicular line to cut the bisector line  $PQ$  at  $E$ .

- (v) Similarly, from point  $C$ , draw a perpendicular line to cut the bisector line  $RS$  at  $F$ .
- (vi) Points  $E$  and  $F$  are the centre points of the ogee curve. With  $E$  as centre draw an arc  $BT$ . With  $F$  as centre draw another arc  $CT$ . Arc  $BTC$  is the required ogee curve.

## 5-10. TO CONSTRUCT EQUILATERAL TRIANGLES



**Problem 5-23.** To construct an equilateral triangle, given the length of the side (fig. 5-30 and fig. 5-31).

(a) With T-square and set-square only (fig. 5-30).

- (i) With the T-square, draw a line  $AB$  of given length.
- (ii) With  $30^\circ$ – $60^\circ$  set-square and T-square, draw a line through  $A$  making  $60^\circ$  angle with  $AB$ .
- (iii) Similarly, through  $B$ , draw a line making the same angle with  $AB$  and intersecting the first line at  $C$ .

Then  $ABC$  is the required triangle.

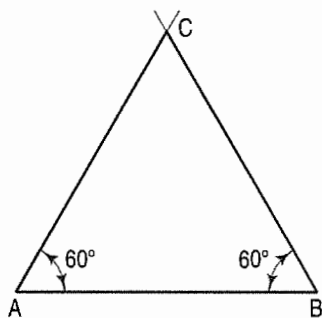


FIG. 5-30

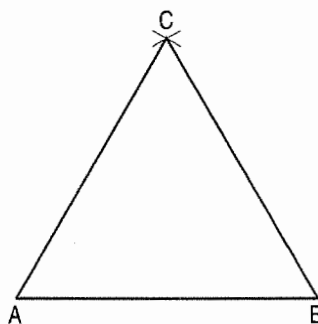


FIG. 5-31

(b) With the aid of a compass (fig. 5-31).

- (i) With centres  $A$  and  $B$  and radius equal to  $AB$ , draw arcs intersecting each other at  $C$ .
- (ii) Draw lines joining  $C$  with  $A$  and  $B$ .

Then  $ABC$  is the required triangle.

**Problem 5-24.** To construct an equilateral triangle of a given altitude (fig. 5-32 and fig. 5-33).

(a) With T-square and set-square only (fig. 5-32).

- (i) With the T-square, draw a line  $AB$  of any length.
- (ii) From a point  $P$  in  $AB$ , draw with a set-square, the vertical  $PQ$  equal to the given altitude.

(iii) With T-square and  $30^{\circ}$ - $60^{\circ}$  set-square, draw lines through  $Q$  on both sides of and making  $30^{\circ}$  angles with  $PQ$  and cutting  $AB$  in  $R$  and  $T$ .  
Then  $QRT$  is the required triangle.

(b) *With the aid of a compass* (fig. 5-33).

- (i) Draw a line  $AB$  of any length.
- (ii) At any point  $P$  in  $AB$ , draw the perpendicular  $PQ$  equal to the given altitude (Problem 5-3).
- (iii) With centre  $Q$  and any radius, draw an arc intersecting  $PQ$  at  $C$ .
- (iv) With centre  $C$  and the same radius, draw arcs cutting the first arc at  $E$  and  $F$ .
- (v) Draw bisectors of  $CE$  and  $CF$  to intersect  $AB$  at  $R$  and  $T$  respectively.  
Then  $QRT$  is the required triangle.

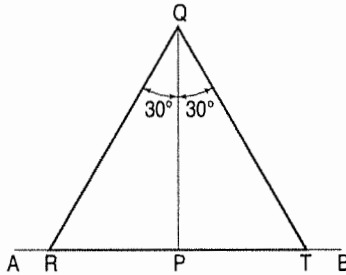


FIG. 5-32

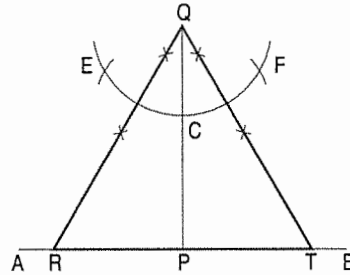


FIG. 5-33

## 5-11. TO CONSTRUCT SQUARES



**Problem 5-25.** *To construct a square, length of a side given* (fig. 5-34 and fig. 5-35).

(a) *With T-square and set-square only* (fig. 5-34).

- (i) With the T-square, draw a line  $AB$  equal to the given length.
- (ii) At  $A$  and  $B$ , draw verticals  $AE$  and  $BF$ .
- (iii) From point  $A$  draw a line inclined at  $45^{\circ}$  to  $AB$ , cutting  $BF$  at  $C$ .
- (iv) From point  $B$  draw a line inclined at  $45^{\circ}$  to  $AB$ , cutting  $AE$  at  $D$ .
- (v) Draw a line joining  $C$  with  $D$ .

Then  $ABCD$  is the required square.

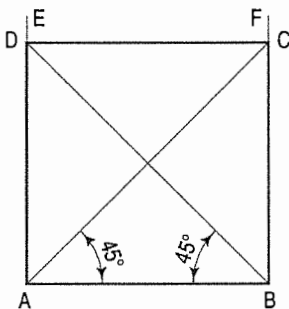


FIG. 5-34

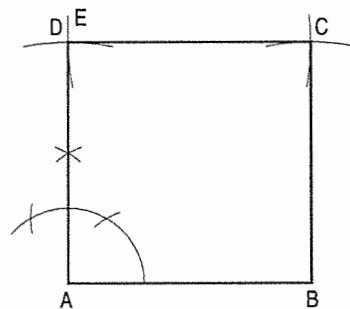


FIG. 5-35

(b) *With the aid of a compass* (fig. 5-35).

- (i) Draw a line  $AB$  equal to the given length.
  - (ii) At  $A$ , draw a line  $AE$  perpendicular to  $AB$ . (Refer problem 5-3, Method III, fig. 5-5).
  - (iii) With centre  $A$  and radius  $AB$ , draw an arc cutting  $AE$  at  $D$ .
  - (iv) With centres  $B$  and  $D$  and the same radius, draw arcs intersecting at  $C$ .
  - (v) Draw lines joining  $C$  with  $B$  and  $D$ .
- Then  $ABCD$  is the required square.

## 5-12. TO CONSTRUCT REGULAR POLYGONS



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 9 for the following problem.

**Problem 5-26.** *To construct a regular polygon, given the length of its side.*

Let the number of sides of the polygon be seven (i.e. heptagon).

**Method I:** (fig. 5-36 and fig. 5-37):

- (i) Draw a line  $AB$  equal to the given length.
- (ii) With centre  $A$  and radius  $AB$ , draw a semi-circle  $BP$ .
- (iii) With a divider, divide the semi-circle into seven equal parts (same as the number of sides). Number the division-points as 1, 2, etc. starting from  $P$ .
- (iv) Draw a line joining  $A$  with the second division-point 2.

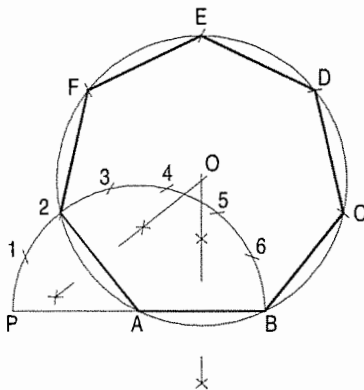


FIG. 5-36

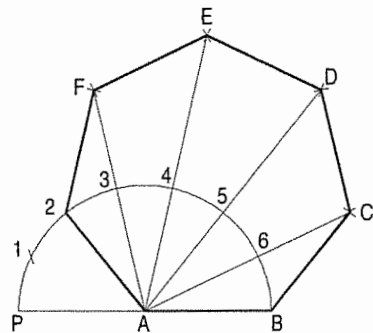


FIG. 5-37

(a) *Inscribe circle method* (fig. 5-36).

- (i) Draw perpendicular bisectors of  $AB$  intersecting each other at  $O$ .
- (ii) With centre  $O$  and radius  $OA$ , describe a circle.
- (iii) With radius  $AB$  and starting from  $B$ , cut the circle at points,  $C, D, \dots$ .
- (iv) Draw lines  $BC, CD$  etc. thus completing the required heptagon.

(b) *Arc method* (fig. 5-37).

- (i) With centre  $B$  and radius  $AB$ , draw an arc cutting the line  $A6$ -produced at  $C$ .
- (ii) With centre  $C$  and the same radius, draw an arc cutting the line  $A5$ -produced at  $D$ .
- (iii) Find points  $E$  and  $F$  in the same manner.
- (iv) Draw lines  $BC$ ,  $CD$  etc. and complete the heptagon.

**Method II: General method for drawing any polygon** (fig. 5-38):

- (i) Draw a line  $AB$  equal to the given length.
  - (ii) At  $B$ , draw a line  $BP$  perpendicular and equal to  $AB$ .
  - (iii) Draw a line joining  $A$  with  $P$ .
  - (iv) With centre  $B$  and radius  $AB$ , draw the quadrant  $AP$ .
  - (v) Draw the perpendicular bisector of  $AB$  to intersect the straight line  $AP$  in 4 and the arc  $AP$  in 6.
- (a) A square of a side equal to  $AB$  can be inscribed in the circle drawn with centre 4 and radius  $A4$ .
  - (b) A regular hexagon of a side equal to  $AB$  can be inscribed in the circle drawn with centre 6 and radius  $A6$ .
  - (c) The mid-point 5 of the line 4-6 is the centre of the circle of the radius  $A5$  in which a regular pentagon of a side equal to  $AB$  can be inscribed.
  - (d) To locate centre 7 for the regular heptagon of side  $AB$ , step-off a division 6-7 equal to the division 5-6.
    - (i) With centre 7 and radius equal to  $A7$ , draw a circle.
    - (ii) Starting from  $B$ , cut it in seven equal divisions with radius equal to  $AB$ .
    - (iii) Draw lines  $BC$ ,  $CD$  etc. and complete the heptagon.

Regular polygons of any number of sides can be drawn by this method.

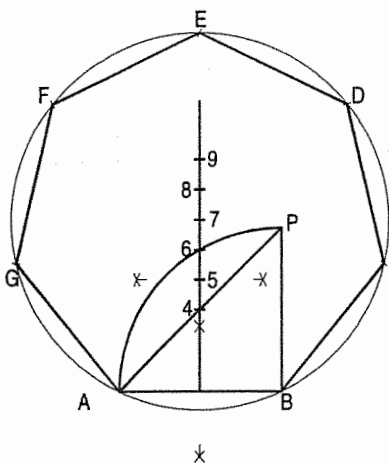


FIG. 5-38

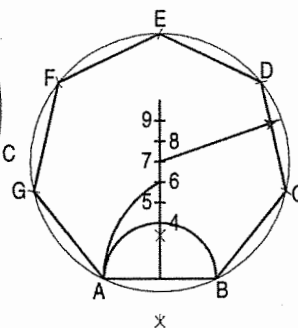


FIG. 5-39

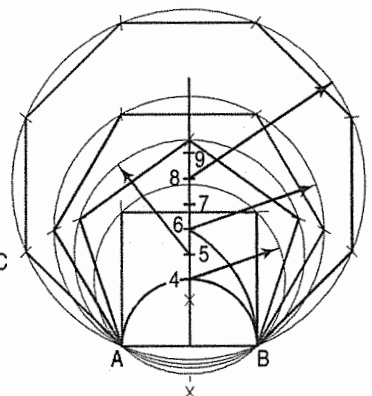


FIG. 5-40



- (iv) With centre  $C$  and radius  $CD$ , draw an arc to intersect the line  $AB$ -produced at  $E$ .
- (v) Then  $AE$  is the length of the diagonal of the pentagon.
- (vi) Therefore, with centre  $A$  and radius  $AB$ , draw an arc intersecting the arc drawn with centre  $B$  and radius  $AE$  at  $R$ .
- (vii) Again with centre  $A$  and radius  $AE$ , draw an arc intersecting the arc drawn with centre  $B$  and radius  $AB$  at  $P$ .
- (viii) With centres  $A$  and  $B$  and radius  $AE$ , draw arcs intersecting each other at  $Q$ .
- (ix) Draw lines  $BP$ ,  $PQ$ ,  $QR$  and  $RA$ , thus completing the pentagon.

**Problem 5-28.** To construct a hexagon, length of a side given (fig. 5-43 and fig. 5-44).

- (a) With *T-square* and  $30^\circ$ - $60^\circ$  set-square only (fig. 5-43).
  - (i) Draw a line  $AB$  equal to the given length.
  - (ii) From  $A$ , draw lines  $A1$  and  $A2$  making  $60^\circ$  and  $120^\circ$  angles respectively with  $AB$ .
  - (iii) From  $B$ , draw lines  $B3$  and  $B4$  making  $60^\circ$  and  $120^\circ$  angles respectively with  $AB$ .
  - (iv) From  $O$  the point of intersection of  $A1$  and  $B3$ , draw a line parallel to  $AB$  and intersecting  $A2$  at  $F$  and  $B4$  at  $C$ .
  - (v) From  $F$ , draw a line parallel to  $BC$  and intersecting  $B3$  at  $E$ .
  - (vi) From  $C$ , draw a line parallel to  $AF$  and intersecting  $A1$  at  $D$ .
  - (vii) Draw a line joining  $E$  and  $D$ .

Then  $ABCDEF$  is the required hexagon.

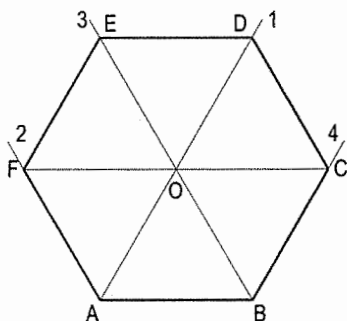


FIG. 5-43

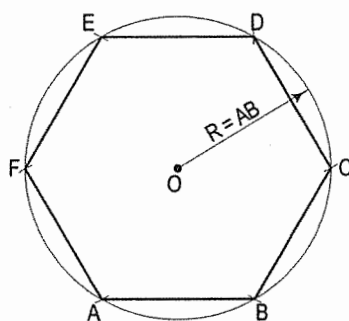


FIG. 5-44

- (b) With the aid of a compass (fig. 5-44).
  - (i) With Point  $O$  as centre, draw a circle of radius equal to the given side length of the required polygon.
  - (ii) Draw a horizontal line passing through the centre of the circle and cutting the circle at opposite ends, say at points  $F$  and  $C$ . Mark the centre of circle as  $O$ .

- (iii) Starting with either  $F$  or  $C$  as centre and side as length, go on marking the points on the circumference,  $A, B, D$  and  $E$ .
- (iv) Join points  $A-B-C-D-E-F$ . You will get the required Hexagon (6 sided polygon).



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 11 for the following problem.

**Problem 5-29.** To inscribe a regular octagon in a given square (fig. 5-45).

- (i) Draw the given square  $ABCD$ .
- (ii) Draw diagonals  $AC$  and  $BD$  intersecting each other at  $O$ .
- (iii) With centre  $A$  and radius  $AO$ , draw an arc cutting  $AB$  at 2 and  $AD$  at 7.
- (iv) Similarly, with centres  $B, C$  and  $D$  and the same radius, draw arcs and obtain points 1, 3, 4 etc. as shown.

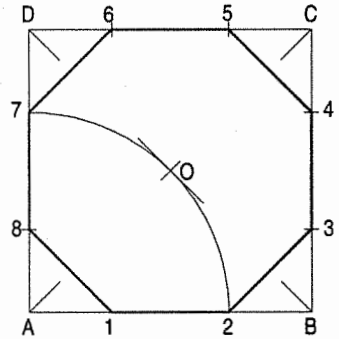


FIG. 5-45

Draw lines 2-3, 4-5, 6-7 and 8-1, thus completing the octagon.

## 5-14. REGULAR POLYGONS INSCRIBED IN CIRCLES

**Problem 5-30.** To inscribe a regular polygon of any number of sides, say 5, in a given circle (fig. 5-46).

- (i) With centre  $O$ , draw the given circle.
- (ii) Draw a diameter  $AB$  and divide it into five equal parts (same number of parts as the number of sides) and number them as shown.
- (iii) With centres  $A$  and  $B$  and radius  $AB$ , draw arcs intersecting each other at  $P$ .
- (iv) Draw a line  $PO$  and produce it to meet the circle at  $C$ . Then  $AC$  is the length of the side of the pentagon.
- (v) Starting from  $C$ , step-off on the circle, divisions  $CD, DE$  etc., equal to  $AC$ .
- (vi) Draw lines  $CD, DE$  etc., thus completing the pentagon.

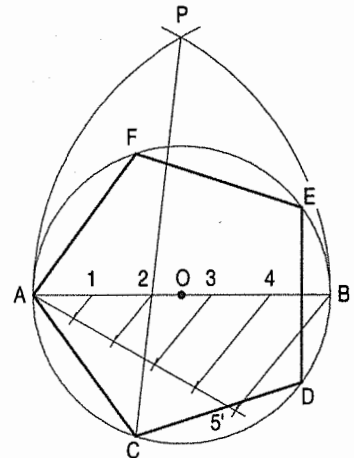


FIG. 5-46

**Problem 5-31.** To inscribe a square in a given circle (fig. 5-47).

- (i) With centre  $O$ , draw the given circle.
- (ii) Draw diameters  $AB$  and  $CD$  perpendicular to each other.
- (iii) Draw lines  $AC, CB, BD$  and  $DA$ , thus completing the square.

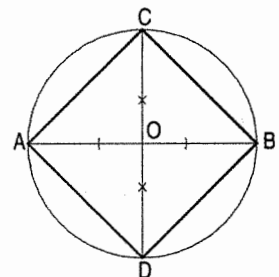


FIG. 5-47

**Problem 5-32.** To inscribe a regular pentagon in a given circle (fig. 5-48).

- (i) With centre  $O$ , draw the given circle.
- (ii) Draw diameters  $AB$  and  $CD$  perpendicular to each other.
- (iii) Bisect  $AO$  in a point  $P$ . With centre  $P$  and radius  $PC$ , draw an arc cutting  $OB$  in  $Q$ .
- (iv) With centre  $C$  and radius  $CQ$ , draw an arc cutting the circle in  $E$  and  $F$ .
- (v) With centres  $E$  and  $F$  and the same radius, draw arcs cutting the circle in  $G$  and  $H$  respectively.
- (vi) Draw lines  $CE$ ,  $EG$ ,  $GH$ ,  $HF$  and  $FC$ , thus completing the required pentagon.

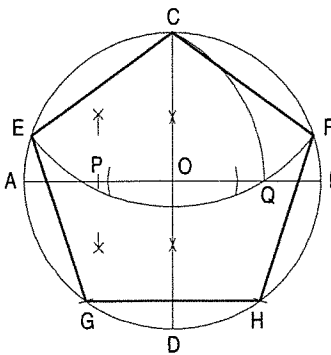
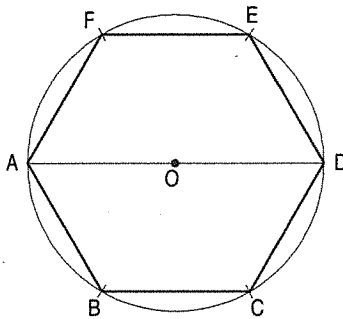
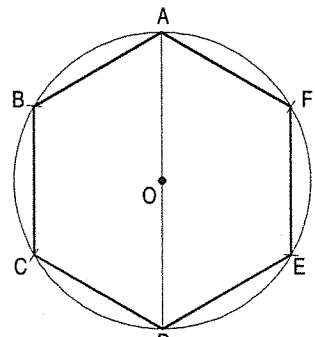


FIG. 5-48



(i)



(ii)

FIG. 5-49

**Problem 5-33.** To inscribe a regular hexagon in a given circle (fig. 5-49).

Apply the same method as shown in Problem 5-28(b).

*Note:* (a) When two sides of the hexagon are required to be horizontal the starting point for stepping-off equal divisions should be on an end of the horizontal diameter.

(b) If they are to be vertical, the starting point should be on an end of the vertical diameter.

In either case, to avoid inaccuracy, the points should be joined with the aid of T-square and 30°-60° set-square.

**Problem 5-34.** To inscribe a regular heptagon in a given circle (fig. 5-50).

- (i) With centre  $O$ , draw the given circle.
- (ii) Draw a diameter  $AB$ . With centre  $A$  and radius  $AO$ , draw an arc cutting the circle at  $E$  and  $F$ .
- (iii) Draw a line  $EF$ , cutting  $AO$  in  $G$ .

Then  $EG$  or  $FG$  is the length of the side of the heptagon.

Therefore, from any point on the circle, say  $A$ , step-off divisions equal to  $EG$ , around the circle. Join the division-points and obtain the heptagon.

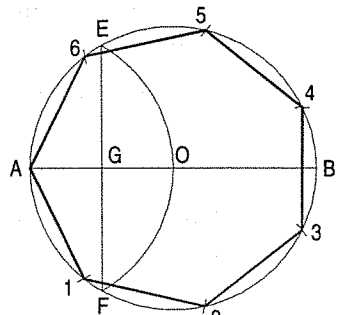


FIG. 5-50

**Problem 5-35.** To inscribe a regular octagon in a given circle (fig. 5-51).

- (i) With centre  $O$ , draw the given circle.
- (ii) Draw diameters  $AB$  and  $CD$  at right angles to each other.
- (iii) Draw diameters  $EF$  and  $GH$  bisecting angles  $AOC$  and  $COB$ .
- (iv) Draw lines  $AE$ ,  $EC$  etc. and complete the octagon.

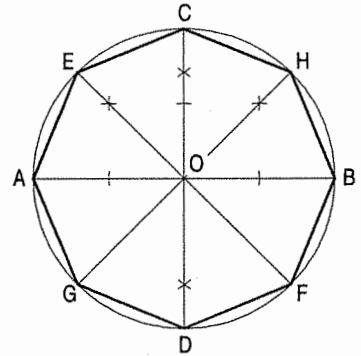


FIG. 5-51

### 5-15. TO DRAW REGULAR FIGURES USING T-SQUARE AND SET-SQUARES

**Problem 5-36.** To describe an equilateral triangle about a given circle (fig. 5-52).

- (i) With centre  $O$ , draw the given circle.
- (ii) Draw a vertical radius  $OA$ .
- (iii) Draw radii  $OB$  and  $OC$  with a  $30^\circ$ – $60^\circ$  set-square, such that  $\angle AOB = \angle AOC = 120^\circ$ .
- (iv) At  $A$ ,  $B$  and  $C$ , draw tangents to the circle, i.e. a horizontal line  $EF$  through  $A$ , and lines  $FG$  and  $GE$  through  $B$  and  $C$  respectively with a  $30^\circ$ – $60^\circ$  set-square.

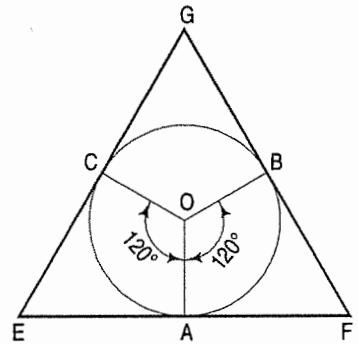


FIG. 5-52

Then  $EFG$  is the required triangle.

**Problem 5-37.** To draw a square about a given circle (fig. 5-53).

- (i) With centre  $O$ , describe the given circle.
- (ii) Draw diameters  $AB$  and  $CD$  at right angles to each other as shown.
- (iii) At  $A$  and  $B$ , draw vertical lines, and at  $C$  and  $D$ , draw horizontal lines intersecting at  $E$ ,  $F$ ,  $G$  and  $H$ .

$EFGH$  is the required square.

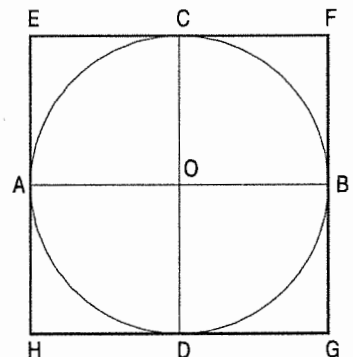


FIG. 5-53

**Problem 5-38.** To describe a regular hexagon about a given circle (fig. 5-54).

- (i) With centre  $O$ , draw the given circle.
- (ii) Draw horizontal diameter  $AB$ , and diameters  $CD$  and  $EF$  making  $60^\circ$  angle with  $AB$ .
- (iii) Draw tangents at all the six ends, i.e. verticals at  $A$  and  $B$ , and lines with a  $30^\circ$ – $60^\circ$  set-square at the remaining points intersecting at 1, 2,.....6.

A hexagon with two sides horizontal can be drawn by drawing a vertical diameter  $AB$  and the other lines as shown in fig. 5-55.

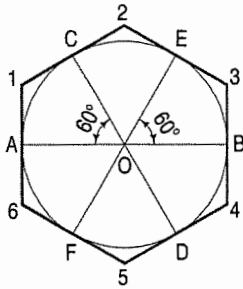


FIG. 5-54

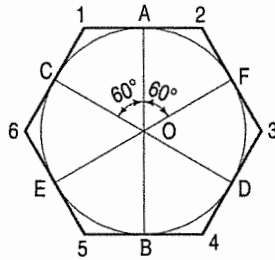


FIG. 5-55

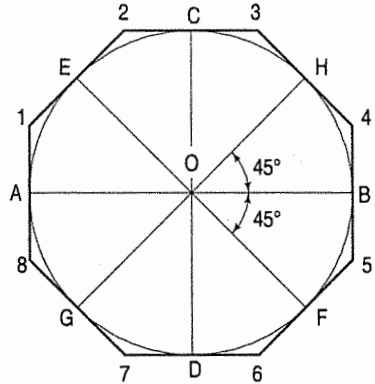


FIG. 5-56

**Problem 5-39.** To describe a regular octagon about a given circle (fig. 5-56).

- (i) With centre  $O$ , describe the given circle.
- (ii) Draw a horizontal diameter  $AB$ , a vertical diameter  $CD$  and diameters  $EF$  and  $GH$  at  $45^\circ$  to the first two.
- (iii) Draw tangents at the eight points  $A, B, \dots, H$  intersecting one another at  $1, 2, \dots, 8$ . Then  $1, 2, \dots, 8$  is the required octagon.

## 5-16. TO DRAW TANGENTS

**Problem 5-40.** (fig. 5-57): To draw a tangent to a given circle at any point on it.

- (i) With centre  $O$ , draw the given circle and mark a point  $P$  on it.
- (ii) Draw a line joining  $O$  and  $P$ .
- (iii) Produce  $OP$  to  $Q$  so that  $PQ = OP$ .
- (iv) With centres  $O$  and  $Q$  and with any convenient radius, draw arcs intersecting each other at  $R$ .
- (v) Draw a line through  $P$  and  $R$ . Then this line is the required tangent.

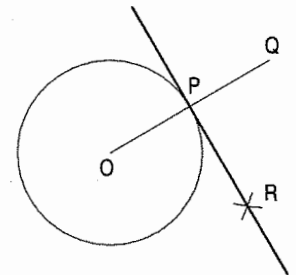


FIG. 5-57

**Problem 5-41.** (fig. 5-58): To draw a tangent to a given circle from any point outside it.

- (i) With centre  $O$ , draw the given circle.
- (ii) Mark a point  $P$  outside it.
- (iii) Draw a line joining  $O$  and  $P$ .
- (iv) With  $OP$  as diameter, draw a semi-circle cutting the given circle at  $R$  and  $R_1$ .
- (v) Draw a line through  $P$  and  $R$ . Then this line is the required tangent. The line through  $P$  and  $R_1$  is the other tangent which can be drawn from the same point.

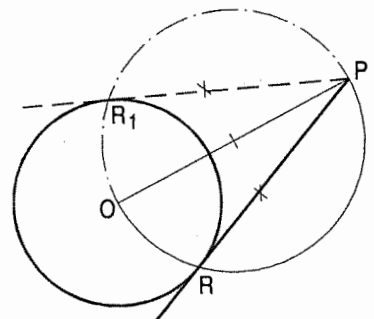


FIG. 5-58

**Problem 5-42.** (fig. 5-59): To draw a tangent to a given arc of inaccessible centre at any point on it.

Let  $AB$  be the given arc and  $P$  the point on it.

- (i) With centre  $P$  and any radius, draw arcs cutting the arc  $AB$  at  $C$  and  $D$ . Draw  $EF$ , the bisector of the arc  $CD$ . It will pass through  $P$ .
- (ii) Through  $P$ , draw a line  $RS$  perpendicular to  $EF$ .  $RS$  is the required tangent.

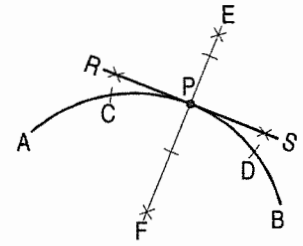


FIG. 5-59

**Problem 5-43.** (fig. 5-60): To draw a tangent to a given circle and parallel to a given line.

The circle with centre  $O$  and the line  $AB$  are given.

- (i) From  $O$ , draw a line perpendicular to  $AB$  and cutting the circle at a point  $P$  or  $Q$ .
- (ii) Through  $P$  or  $Q$ , draw the required tangent  $CD$  or  $C_1 D_1$  (problem 5-40).

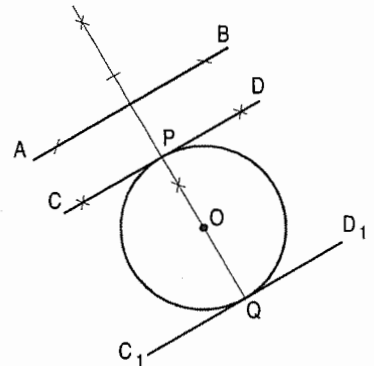


FIG. 5-60

**Problem 5-44.** To draw a common tangent to two given circles of equal radii (fig. 5-61).

Draw the given circles with centres  $O$  and  $P$ .

(a) *External tangents* (fig. 5-61):

- (i) Draw a line joining  $O$  and  $P$ .
- (ii) At  $O$  and  $P$ , erect perpendiculars to  $OP$  on the same side of it and intersecting the circles at  $A$  and  $B$ .
- (iii) Draw a line through  $A$  and  $B$ . This line is the required tangent.  $A_1 B_1$  is the other tangent.

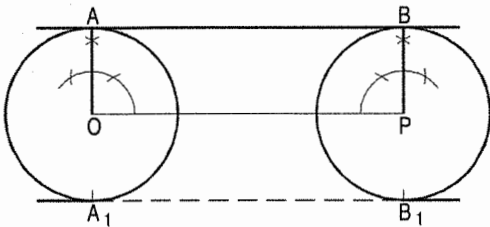


FIG. 5-61

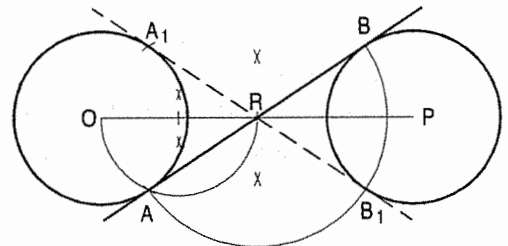


FIG. 5-62

(b) *Internal tangents* (fig. 5-62):

- (i) Draw a line joining  $O$  and  $P$ .
- (ii) Bisect  $OP$  in  $R$ . Draw a semi-circle with  $OR$  as diameter to cut the circle at  $A$ .
- (iii) With centre  $R$  and radius  $RA$ , draw an arc to intersect the other circle on the other side of  $OP$  at  $B$ .
- (iv) Draw a line through  $A$  and  $B$ . This line is the required tangent.

The other tangent through  $A_1$  and  $B_1$  can also be similarly drawn.

**Problem 5-45.** To draw a common tangent to two given circles of unequal radii (fig. 5-63 and fig. 5-64).

Draw the given circles with centres  $O$  and  $P$ , and radii  $R_1$  and  $R_2$  respectively, of which  $R_1$  is greater than  $R_2$ .

(a) *External tangents* (fig. 5-63):

- (i) Draw a line joining centres  $O$  and  $P$ .
- (ii) With centre  $O$  and radius equal to  $(R_1 - R_2)$ , draw a circle.
- (iii) From  $P$ , draw a tangent  $PT$  to this circle (Problem 5-41).
- (iv) Draw a line  $OT$  and produce it to cut the outer circle at  $A$ .
- (v) Through  $P$ , draw a line  $PB$  parallel to  $OA$ , on the same side of  $OP$  and cutting the circle at  $B$ .
- (vi) Draw a line through  $A$  and  $B$ . Then this line is the required tangent.

The other similar tangent will pass through  $A_1$  and  $B_1$ .

(b) *Internal tangents* (fig. 5-64):

- (i) Draw a line joining the centres  $O$  and  $P$ .
- (ii) With centre  $O$  and radius equal to  $(R_1 + R_2)$ , draw a circle.
- (iii) From  $P$ , draw a line  $PT$  tangent to this circle.
- (iv) Draw a line  $OT$  cutting the circle at  $A$ .
- (v) Through  $P$ , draw a line  $PB$  parallel to  $OA$ , on the other side of  $OP$  and cutting the circle at  $B$ .
- (vi) Draw a line through  $A$  and  $B$ . Then this line is the required tangent. The second tangent will pass through  $A_1$  and  $B_1$ .

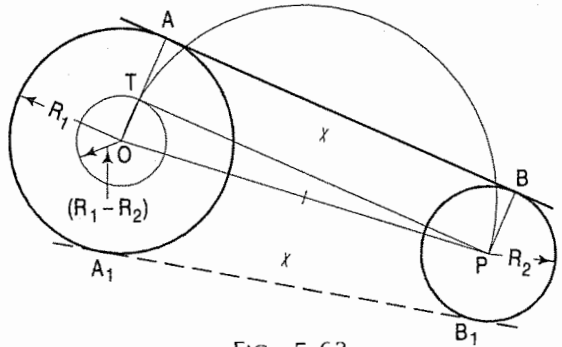


FIG. 5-63

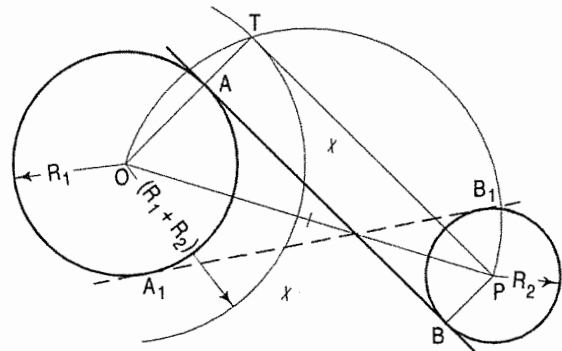


FIG. 5-64

## 5-17. LENGTHS OF ARCS

**Problem 5-46.** To determine the length of a given arc (fig. 5-65).

Let  $AB$  be the given arc drawn with centre  $O$ .

- (i) At  $A$ , draw a tangent to the arc.
- (ii) Draw the chord  $AB$  and produce it beyond  $A$  to a point  $C$  such that  $AC = \frac{1}{2} AB$ .
- (iii) With centre  $C$  and radius equal to  $CB$ , describe an arc cutting the tangent at  $D$ .
- (iv) Then the length  $AD$  is approximately equal to the length of the arc  $AB$ .

*This method is satisfactory for arcs which subtend at the centre, angles smaller than  $60^\circ$ .*

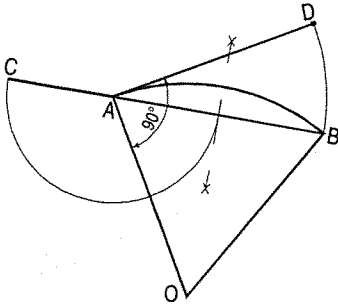


FIG. 5-65

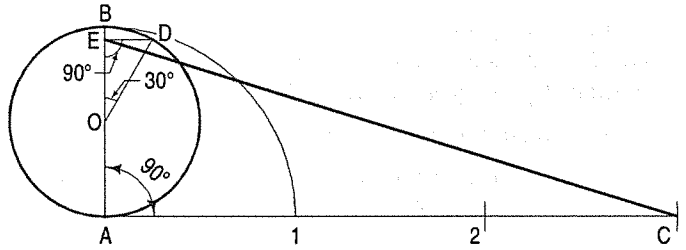


FIG. 5-66

**Problem 5-47.** To determine the length of the circumference of a given circle (fig. 5-66).

Let the circle with centre O be given.

- Draw a diameter AB.
- At A, draw a tangent AC equal to 3 times AB.
- Draw a radius OD making an angle of  $30^\circ$  with OB.
- From D, draw a line DE perpendicular to OB.
- Draw a line joining E and C. Then EC is approximately equal in length to the circumference of the circle.

## 5-18. CIRCLES AND LINES IN CONTACT

**Problem 5-48.** To draw a circle passing through a given point and tangent to a given line at a given point on it (fig. 5-67).

A point P and a line AB with a point Q in it are given. At Q, draw a line perpendicular to AB.

- Draw a line joining P and Q.
- Draw a perpendicular bisector of PQ to intersect the perpendicular from Q at O.
- With centre O and radius OP or OQ, draw the required circle.

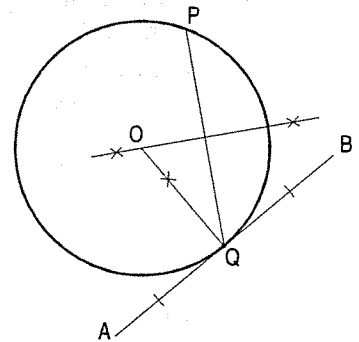


FIG. 5-67

**Problem 5-49.** To draw a circle passing through a given point and touching a given circle at a given point on it (fig. 5-68).

A point P, a circle with centre A and a point Q on the circle are given.

- Draw a line joining P and Q.
- Draw a perpendicular bisector of PQ, to intersect the line through A and Q at O.
- With centre O and radius OP, draw the required circle.

The required circle includes the given circle when the point is in a position such as P'.

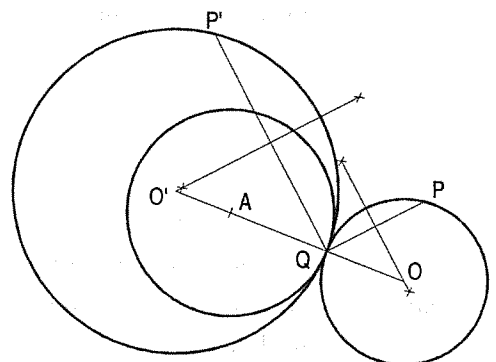


FIG. 5-68

**Problem 5-50.** To draw a circle to touch a given line and a given circle at a given point on it (fig. 5-69).

A line  $AB$ , a circle with centre  $C$  and a point  $P$  on the circle are given.

From  $P$ , draw a tangent to the circle intersecting  $AB$  in  $D$ .

- (a) Draw a bisector of  $\angle PDB$ , to intersect the line through  $C$  and  $P$  at  $O$ . With centre  $O$  and radius  $OP$ , draw the required circle.

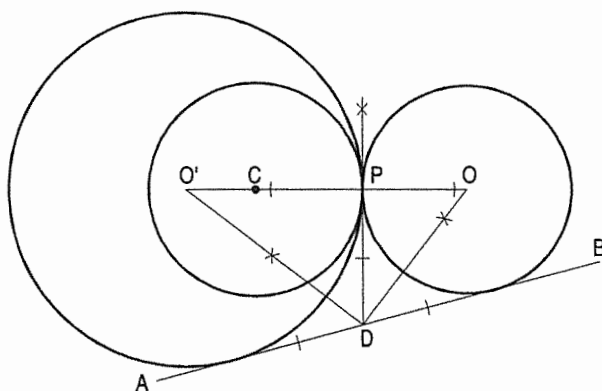


FIG. 5-69

- (b) Draw a bisector of  $\angle PDA$  to meet the line through  $C$  and  $P$  at  $O'$ . Then  $O'$  is the centre of another circle which will include the given circle within it.

**Problem 5-51.** To draw a circle to touch a given circle and a given line at a given point on it (fig. 5-70).

A circle with centre  $C$  and a line  $AB$  with a point  $P$  in it are given.

Through  $C$ , draw a line perpendicular to  $AB$  and cutting the circle in  $E$  or  $F$ .

- (a) Draw a line joining  $P$  and  $F$  and intersecting the circle at  $G$ .

At  $P$ , draw a perpendicular to  $AB$  intersecting the line through  $C$  and  $G$  at  $O$ . With centre  $O$  and radius  $OP$ , draw the required circle.

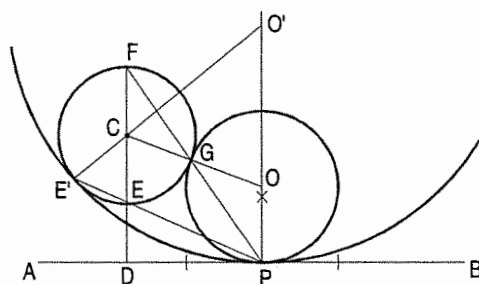


FIG. 5-70

- (b) Draw a line through  $P$  and  $E$  and obtain centre  $O'$  for another circle in the same manner. It will include the given circle within it.

**Problem 5-52.** To draw a circle touching two given circles, one of them at a given point on it (fig. 5-71).

Circles with centres  $A$  and  $B$ , and a point  $P$  on the circle  $A$  are given (fig. 5-71).

- (i) Draw a line joining  $A$  and  $P$ .
- (ii) Through  $B$ , draw a line parallel to  $AP$  and intersecting the circle in  $C$ .
- (iii) Draw a line  $PC$  and produce it (if necessary) to cut the circle (with centre  $B$ ) in  $D$ .
- (iv) Draw a line through  $D$  and  $B$  to intersect  $AP$  or  $AP$ -produced, at  $O$ .
- (v) With centre  $O$  and radius  $OP$ , draw the required circle.

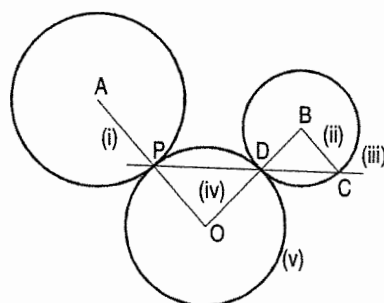


FIG. 5-71

**Problem 5-53.** To draw a circle touching two given circles, one of them at a given point on it [fig. 5-72).

- (i) Circles with centres  $A$  and  $B$ , and a point  $P$  on the circle  $A$  are given.
- (ii) Draw a line joining centre  $A$  and the point  $P$ .
- (iii) Through  $B$ , draw a line parallel to  $AP$  (if extended) and intersecting circle  $B$  in  $C''$  and  $D'$ .
- (iv) Join  $PC''$  and extend to intersect circle  $B$  at  $D''$ .
- (v) Draw a line through  $D''$  and  $B$  to intersect the line  $AP$  at  $O''$ .
- (vi) Join  $PD'$  which intersect circle  $B$  at  $C'$ . Join  $C'B$  and extend to intersect  $AP$  at  $O'$ . Draw a circle with  $O'$  as centre and  $O'P$  as radius. It is the circle (1).
- (vii) Draw another circle with centre as  $O''$  and the radius  $O''P$ . It is the circle (2).

Circle-(1) which includes one of the given circles, and circle-(2) which includes both of them (fig. 5-72).

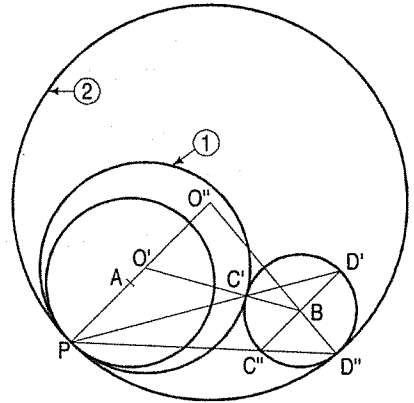


FIG. 5-72

## 5-19. INSCRIBED CIRCLES

**Problem 5-54.** To inscribe a circle in given triangle (fig. 5-73).

Let  $ABC$  be the triangle.

- (i) Bisect any two angles by lines intersecting each other at  $O$ .
- (ii) Draw a perpendicular from  $O$  to any one side of the triangle, meeting it at  $P$ .
- (iii) With centre  $O$  and radius  $OP$ , describe the required circle.

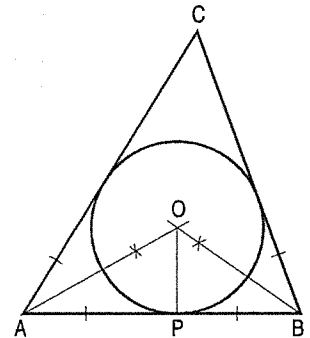


FIG. 5-73

**Problem 5-55.** To draw a circle touching three lines inclined to each other but not forming a triangle (fig. 5-74).

Let  $AB$ ,  $BC$  and  $AD$  be the given lines.

- (i) Draw bisectors of the two angles intersecting each other at  $O$ .
- (ii) From  $O$ , draw a perpendicular to any one line intersecting it at  $P$ .
- (iii) With centre  $O$  and radius  $OP$ , draw the required circle.

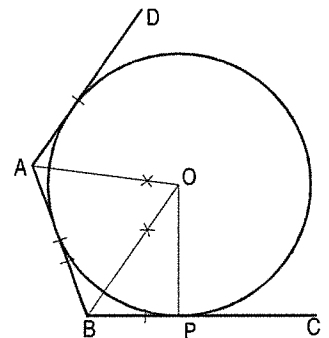


FIG. 5-74

**Problem 5-56.** To inscribe a circle in a regular polygon of any number of sides, say a pentagon (fig. 5-75).

Let  $ABCDE$  be the pentagon.

- (i) Bisect any two angles by lines intersecting each other at  $O$ .
- (ii) From  $O$ , draw a perpendicular to any one side of the pentagon cutting it at  $P$ .
- (iii) With centre  $O$  and radius  $OP$ , draw the required circle.

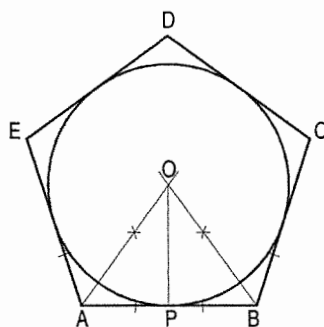


FIG. 5-75

**Problem 5-57.** To draw in a regular polygon, the same number of equal circles as the sides of the polygon, each circle touching one side of the polygon and two of the other circles (fig. 5-76).

- (i) Let  $ABCD$  be the given square.
- (ii) Draw bisectors of all the angles of the square. They will meet at  $O$ , thus dividing the square into four equal triangles.

In each triangle inscribe a circle (Problem 5-54). Each circle will touch a side of the square and two other circles as required.

Fig. 5-77 shows five equal circles inscribed in a regular pentagon in the same manner.

**Problem 5-58.** To draw in a regular polygon, the same number of equal circles as the sides of the polygon, each circle touching two adjacent sides of the polygon and two of the other circles (fig. 5-78).

Let  $ABCDEF$  be the given hexagon.

- (i) Draw the perpendicular bisectors of all sides of the hexagon. They will meet at  $O$  and will divide the hexagon into six equal quadrilaterals.
- (ii) Inscribe a circle in each quadrilateral as shown in case of  $A_1O_2$  and as explained below.
- (iii) Bisect any two adjacent angles with bisectors intersecting each other at  $P$ .
- (iv) From  $P$ , draw a perpendicular to any one side of the quadrilateral, meeting it at  $Q$ . With centre  $P$  and radius  $PQ$ , draw one of the required circles.
- (v) Draw other circles in the same manner.

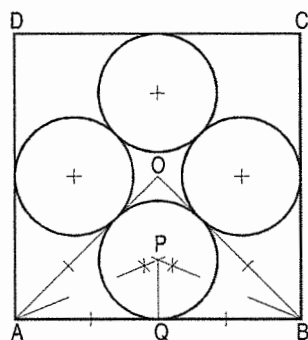


FIG. 5-76

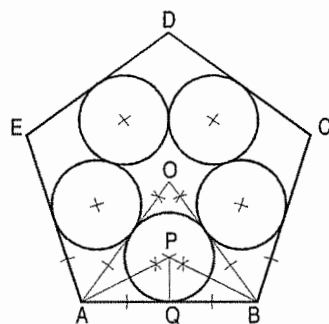


FIG. 5-77

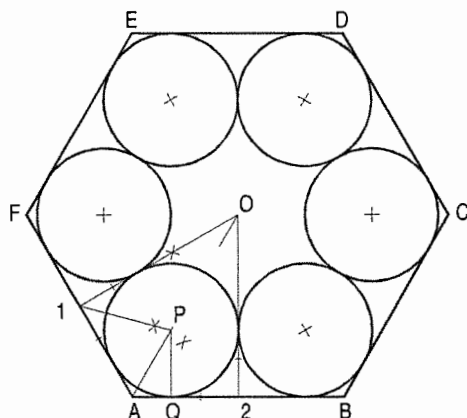


FIG. 5-78

**Problem 5-59.** To draw in a given regular hexagon, three equal circles, each touching one side and two other circles (fig. 5-79).

- (i) Draw the given hexagon.
- (ii) Draw perpendicular bisectors of its two alternate sides, to intersect each other at  $O$  and to meet the middle side produced on both sides at 1 and 2.
- (iii) Inscribe a circle in triangle  $O 1 2$ . Similarly, draw the other two required circles.

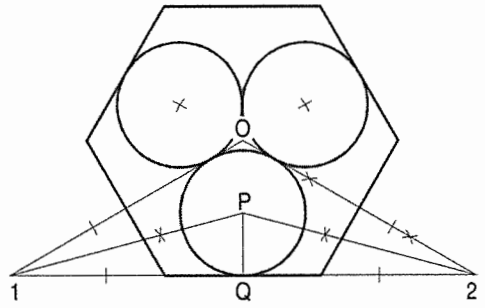


FIG. 5-79

**Problem 5-60.** To draw in a given circle, any number of equal circles, say four, each touching the given circle and two of the other circles (fig. 5-80).

- (i) Divide the given circle into four equal parts by diameters  $AB$  and  $CD$ .
- (ii) Draw a tangent to the circle at  $D$ . Draw lines bisecting  $\angle AOD$  and  $\angle BOD$  and meeting the tangent at 1 and 2.
- (iii) Inscribe a circle in the triangle  $O 1 2$ .

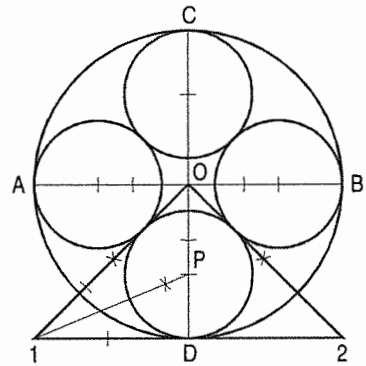


FIG. 5-80

Draw the other circles in the same manner. The centres for the remaining circles may also be determined by drawing a circle with centre  $O$  and radius  $OP$  to cut the diameters at the required points.

**Problem 5-61.** To draw outside a given regular polygon, the same number of equal circles as the sides of the polygon, each circle touching one side and two of the other circles (fig. 5-81).

- (i) Let  $ABCDE$  be the given pentagon.
- (ii) Draw bisectors of two adjacent angles, say  $\angle A$  and  $\angle B$ , and produce them outside the pentagon.
- (iii) Draw a circle touching the extended bisectors and the side  $AB$  (Problem 5-55). Obtain the other four required circles in the same manner.

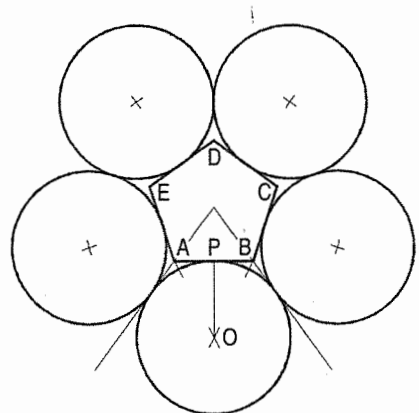


FIG. 5-81

**Problem 5-62.** To draw outside a given circle any number of equal circles, say six, each touching the given circle and two other circles (fig. 5-82).

- (i) Draw the given circle and describe a regular hexagon about it.
- (ii) Draw the required six equal circles outside the hexagon as shown in the previous problem.

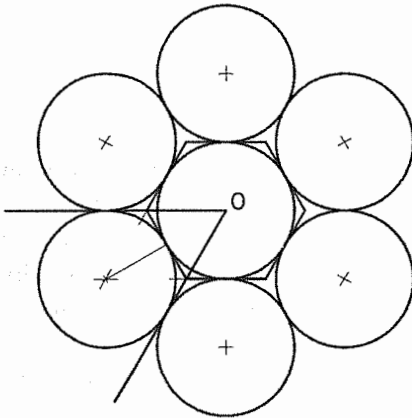


FIG. 5-82

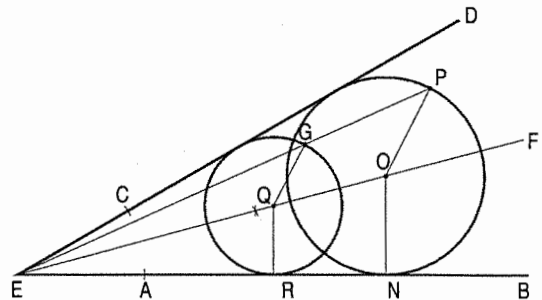


FIG. 5-83

**Problem 5-63.** To draw a circle touching two converging lines and passing through a given point between them (fig. 5-83).

- (i) Lines  $AB$  and  $CD$ , and the point  $P$  are given.
- (ii) Produce lines  $AB$  and  $CD$  to intersect at a point  $E$ . Draw the bisector  $EF$  of  $\angle AEC$ .
- (iii) Mark any point  $Q$  on  $EF$  and from it draw a perpendicular  $QR$  on  $AB$ .
- (iv) With  $Q$  as centre and  $QR$  as radius draw a circle which will touch the line  $CD$  also.
- (v) Draw a line joining  $P$  with  $E$ , cutting the circle at a point  $G$ .
- (vi) Draw the line  $QG$ .
- (vii) From  $P$ , draw a line parallel to  $QG$  intersecting  $EF$  at a point  $O$ .
- (viii) From  $O$ , draw a perpendicular  $ON$  to either  $AB$  or  $CD$ .
- (ix) With  $O$  as centre and  $ON$  as radius, draw the required circle.

**Problem 5-64.** To draw two circles touching each other and two converging lines, the smaller circle being of given radius (fig. 5-84).

- (i) Lines  $AB$  and  $CD$  and radius  $R$  of the smaller circle are given.
- (ii) Produce lines  $AB$  and  $CD$  to intersect at a point  $E$ . Draw the bisector  $EF$  of  $\angle AEC$ .
- (iii) Draw a line parallel to and at distance  $R$  from  $AB$  to intersect  $EF$  in a point  $Q$ .
- (iv) Draw the perpendicular  $QP$  on the line  $AB$ .
- (v) With  $Q$  as centre and  $QP$  as radius, draw the smaller circle.
- (vi) Mark points  $T$  and  $N$  at which the circle cuts  $EF$ .
- (vii) Draw the line joining  $T$  with  $P$ . Draw a line  $NS$  parallel to  $TP$  intersecting  $AB$  in the point  $S$ .

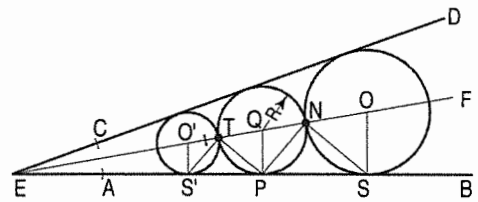


FIG. 5-84

From  $S$ , draw the perpendicular to  $AB$  cutting  $EF$  in the point  $O$ . With  $O$  as centre and  $OS$  as radius, draw the required circle.

$O'$  is the centre of smaller circle, obtained in the same manner, touching the two given lines and the given circle.

## EXERCISES 5



1. Draw a line 125 mm long and quadrisect it.
2. Draw a line  $AB$  80 mm long and divide it into five parts, one of them 20 mm long and the remaining each 15 mm long, by the method of bisection.
3. With centre  $O$  and radius equal to 50 mm, draw two arcs of any lengths on opposite sides of  $O$ . Bisect the two arcs and produce the bisectors till they meet.
4. Draw a line  $AB$  75 mm long. At  $B$ , erect a perpendicular  $BC$  100 mm long. Draw a line joining  $A$  and  $C$ , and measure its length. Construct a square on each line as a side.
5. Draw a line  $PQ$  100 mm long. At any point  $O$  in it near its centre, erect a perpendicular  $OA$  65 mm long. Through  $A$ , draw a line parallel to  $PQ$ .
6. Mark any point  $O$ . Draw a line  $AB$ , such that its shortest distance from  $O$  is 50 mm.
8. Draw a line  $AB$  75 mm long. Mark a point  $C$ , 65 mm from  $A$  and 90 mm from  $B$ . Join  $C$  with  $A$  and  $B$ . Through the points  $A$ ,  $B$  and  $C$ , draw lines (i) perpendicular and (ii) parallel to their opposite lines.
7. Construct a rectangle of sides 65 mm and 40 mm long.
9. Construct a square of 75 mm side. Draw the diagonals intersecting at  $O$ . From  $O$ , draw lines perpendicular to the sides of the square.
10. Draw a circle of 50 mm radius. Divide it (i) into 8 equal parts by continued bisection and (ii) into 12 equal parts by bisection of a line and trisection of a right angle methods.
11. Draw two lines  $AB$  and  $AC$  making an angle of  $75^\circ$ . Draw a circle of 25 mm radius touching them.
12. Construct a right angle  $PQR$ . Describe a circle of 20 mm radius touching the sides  $PQ$  and  $QR$ .
13. Draw a line  $AB$  of any length. Mark a point  $O$  at a distance of 25 mm from  $AB$ . With  $O$  as centre, draw a circle of 40 mm diameter. Describe another circle (i) of 20 mm radius, touching the circle and  $AB$ ; (ii) of 35 mm radius, touching  $AB$  and the circle, and including the circle within it.
14. Draw two circles of 20 mm and 30 mm radii respectively with centres 65 mm apart. (i) Describe a third circle of 50 mm radius touching the two circles and (a) outside them; (b) including 20 mm circle; (c) including 30 mm circle. (ii) Describe a circle of 75 mm radius, touching both circles and including both of them within it.
15. Mark points  $A$  and  $B$ , 50 mm apart. Mark a third point 75 mm from both  $A$  and  $B$ . Describe a circle passing through the three points.
16. Draw the machine handle shown in fig. 5-85. All dimensions are in millimetres.

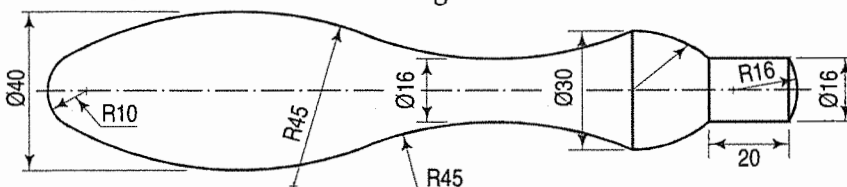


FIG. 5-85

17. The distance between the centres of two circles of 65 mm and 90 mm diameters is 120 mm. Draw an internal and an external common tangent to the two circles.
18. Draw a circle with centre  $O$  and radius equal to 30 mm. From a point  $P$ , 75 mm from  $O$ , draw a line joining  $P$  and  $O$ , and produce it to cut the circle at  $Q$ . From  $P$  and  $Q$  draw tangents to the circle.
19. Two shafts carry pulleys of 900 mm and 1350 mm diameters respectively. The distance between their centres is 2700 mm. Draw the arrangement showing the two pulleys connected by (i) direct belt (ii) crossed belt. Take 1 mm = 20 mm.
20. An arc  $AB$  drawn with 50 mm radius subtends an angle of  $45^\circ$  at the centre. Determine approximately the length of  $AB$ .
21. Determine the length of the circumference of a 75 mm diameter circle.
22. A point  $P$  is 25 mm from a line  $AB$ .  $Q$  is a point in  $AB$  and is 50 mm from  $P$ . Draw a circle passing through  $P$  and touching  $AB$  at  $Q$ .
23. Construct an equilateral triangle  $ABC$  of 40 mm side. Construct a square, a regular pentagon and a regular hexagon on its sides  $AB$ ,  $BC$  and  $CA$  respectively.
24. The centre  $O$  of a circle of 30 mm diameter is 25 mm from a line  $AB$ . Draw a circle (i) to touch the given circle and the line  $AB$  at a point  $P$ , 50 mm from  $O$ ; (ii) to touch  $AB$  and the given circle at a point  $Q$ , 20 mm from  $AB$ .
25. Two circles of 40 mm and 50 mm diameters have their centres 60 mm apart. Draw a circle to touch both circles and (i) to include the bigger circle, the point of contact on it being 75 mm from the centre of the other circle; (ii) to include both the circles, the point of contact being the same as in (i).
26. Construct a regular pentagon of 30 mm side by three different methods.
27. On a line  $AB$  40 mm long, construct a regular heptagon by two different methods.
28. Construct a regular octagon of 40 mm side. Inscribe another octagon with its corners on the mid-points of the sides of the first octagon.
29. Construct the following regular polygons in circles of 100 mm diameter, using a different method in each case: (i) Pentagon (ii) Heptagon.
30. Draw the following regular figures, the distance between their opposite sides being 75 mm: (i) Square; (ii) Hexagon; (iii) Octagon.
31. Construct a regular octagon in a square of 75 mm side.
32. Describe a regular pentagon about a circle of 100 mm diameter.
33. Construct a triangle having sides 25 mm, 30 mm and 40 mm long. Draw three circles, each touching one of the sides and the other two sides produced.
34. Inscribe a circle in a triangle having sides 50 mm, 65 mm and 75 mm long.
35. Construct a regular heptagon of 25 mm side and inscribe a circle in it.
36. Construct a regular hexagon of 40 mm side and draw in it, six equal circles, each touching one side of the hexagon and two other circles.
37. Construct a square of 50 mm side and draw in it, four equal circles, each touching two adjacent sides and two other circles.
38. In a regular octagon of 40 mm side, draw four equal circles, each touching one side of the octagon and two other circles.
39. Draw a circle of 125 mm diameter and draw in it, five equal circles, each touching the given circle and two other circles.

40. Construct a square of 25 mm side. Draw outside it four equal circles, each touching a side of the square and two other circles.
41. Outside a circle of 25 mm diameter, draw five equal circles, each touching the given circle and two other circles.
42. Two lines converge to a point making an angle of  $30^\circ$  between them. Draw three circles to touch both these lines, the middle circle being of 25 mm radius and touching the other two circles.
43. Two lines converge to a point making an angle of  $30^\circ$  between them. A point  $P$  is between these lines 15 mm from one line and 25 mm from the other. Draw a circle to touch both the lines and pass through  $P$ .
44. Draw a series of four circles, each touching the preceding circle and two converging lines which make an angle of  $25^\circ$  between them. Take the radius of the smallest circle as 10 mm.
45. A vertical straight line  $AB$  is at a distance of 90 mm from the centre of a circle of 75 mm diameter. A straight line  $PQ$  passes through the centre of the circle and makes an angle of  $60^\circ$  with the vertical. Draw circles having their centres on  $PQ$  and to touch the straight line  $AB$  and the circle. Measure the radius of each circle.
46. Draw a semi-circle of 125 mm diameter and inscribe in it the largest equilateral triangle having a corner at the centre. The semi-circle is the development of a cone and the triangle that of a line on its surface. Draw the projections of the cone resting on its base on the ground showing the line in both views.
47. Construct a lever as shown in fig. 5-86.

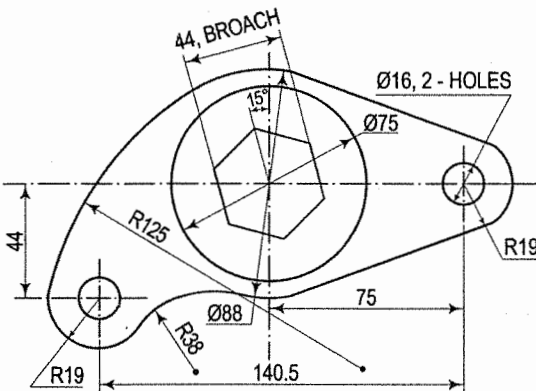


FIG. 5-86

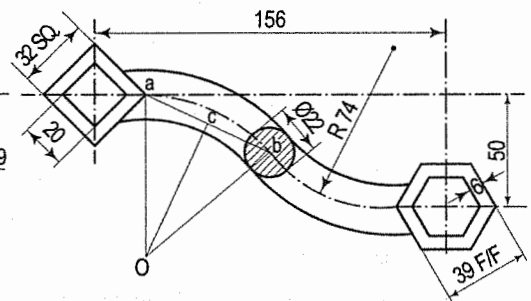


FIG. 5-87

48. Construct a special spanner as shown in fig. 5-87.  
[Hint: The method of drawing curve is shown on left-hand side. The same is to be followed on the right-hand side. This curve is known as ogee (reverse) curve.]
49. Two shafts, 1200 mm apart are connected by flat belt. The flat belt pulleys of 300 mm diameter and 600 mm diameter are fixed on the shafts. Draw the arrangement and determine approximately length of the belt.
50. Draw plan-view of a hexagonal nut of 20 mm using standard dimensions.
51. (i) Draw a number '8' of height 105 mm and 15 mm thick.  
(ii) Draw an alphabet 'S' of height 105 mm and 15 mm thick.

## Chapter

# 6



# CURVES USED IN ENGINEERING PRACTICE

## 6-0. INTRODUCTION

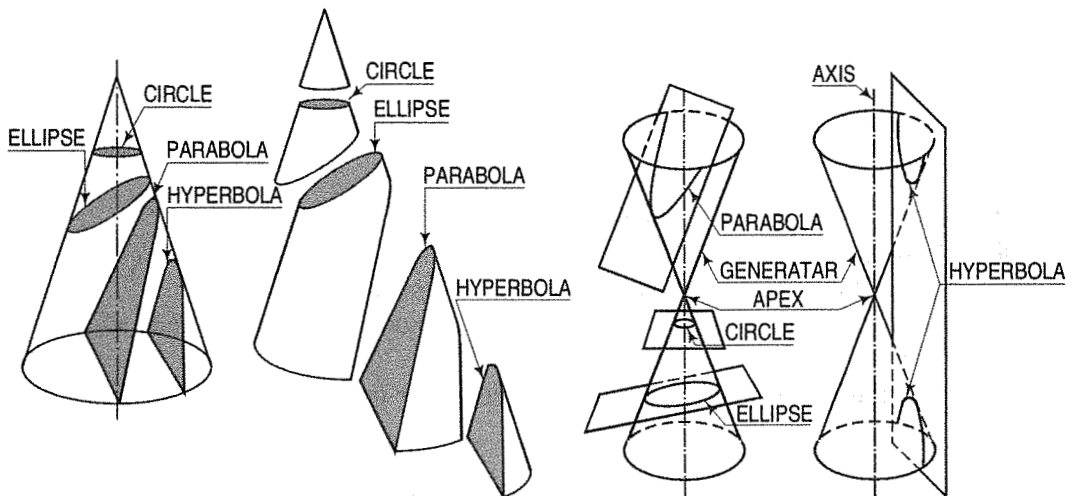
The profile of number of objects consists of various types of curves. This chapter deals with various types of curves which are commonly used in engineering practice as shown below:

1. Conic sections
2. Cycloidal curves
3. Involute
4. Evolutes
5. Spirals
6. Helix.

We shall now discuss the above in details with reference to their construction and applications.

## 6-1. CONIC SECTIONS

The sections obtained by the intersection of a right circular cone by a plane in different positions relative to the axis of the cone are called *conics*. Refer to fig. 6-1.



Conic sections

FIG. 6-1

- (i) When the section plane is inclined to the axis and cuts all the generators on one side of the apex, the section is an *ellipse* [fig. 6-1].

- (ii) When the section plane is inclined to the axis and is parallel to one of the generators, the section is a *parabola* [fig. 6-1].
- (iii) A *hyperbola* is a plane curve having two separate parts or branches, formed when two cones that point towards one another are intersected by a plane that is parallel to the axes of the cones.

The conic may be defined as the locus of a point moving in a plane in such a way that the ratio of its distances from a fixed point and a fixed straight line is always constant. The fixed point is called the *focus* and the fixed line, the *directrix*.

The ratio  $\frac{\text{distance of the point from the focus}}{\text{distance of the point from the directrix}}$  is called *eccentricity* and is denoted by  $e$ . It is always less than 1 for ellipse, equal to 1 for parabola and greater than 1 for hyperbola i.e.

- (i) ellipse :  $e < 1$
- (ii) parabola :  $e = 1$
- (iii) hyperbola :  $e > 1$ .

The line passing through the focus and perpendicular to the directrix is called the *axis*. The point at which the conic cuts its axis is called the *vertex*.

### 6-1-1. ELLIPSE

Use of elliptical curves is made in arches, bridges, dams, monuments, man-holes, glands and stuffing-boxes etc. Mathematically an ellipse can be described by equation  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ . Here 'a' and 'b' are half the length of major and minor axes of the ellipse and x and y co-ordinates.

#### (1) General method of construction of an ellipse:



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 12 for the following problem.

**Problem 6-1.** (fig. 6-2): To construct an ellipse when the distance of the focus from the directrix is equal to 50 mm and eccentricity is  $\frac{2}{3}$ .

- (i) Draw any vertical line AB as directrix.
- (ii) At any point C on it, draw the axis perpendicular to the AB (directrix).
- (iii) Mark a focus F on the axis such that CF = 50 mm.
- (iv) Divide CF into 5 equal divisions (sum of numerator and denominator of the eccentricity.).
- (v) Mark the vertex V on the third division-point from C.

$$\text{Thus, eccentricity, } e = \frac{VF}{VC} = \frac{2}{3}.$$

- (vi) A scale may now be constructed on the axis (as explained below), which will directly give the distances in the required ratio.
- (vii) At V, draw a perpendicular VE equal to VF. Draw a line joining C and E.

$$\text{Thus, in triangle CVE, } \frac{VE}{VC} = \frac{VF}{VC} = \frac{2}{3}.$$

- (viii) Mark any point 1 on the axis and through it, draw a perpendicular to meet  $CE$ -produced at  $1'$ .
- (ix) With centre  $F$  and radius equal to  $1-1'$ , draw arcs to intersect the perpendicular through 1 at points  $P_1$  and  $P'_1$ .

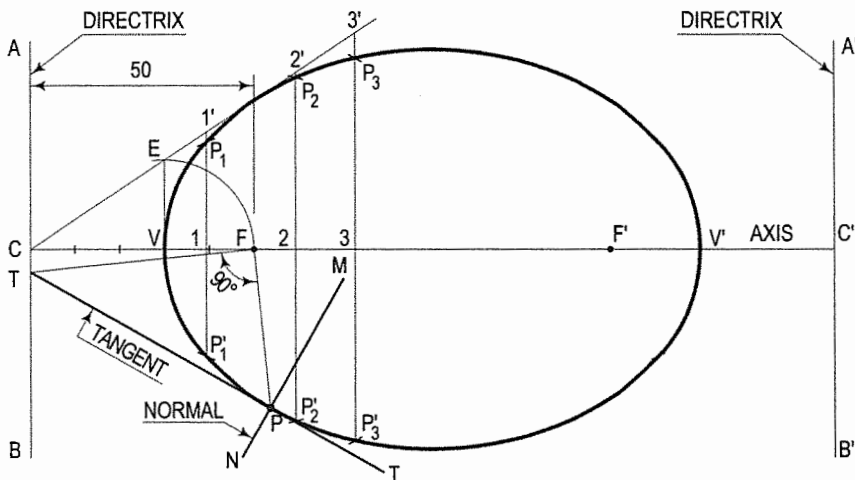
These are the points on the ellipse, because the distance of  $P_1$  from  $AB$  is equal to  $C1$ ,

$$P_1 F = 1-1'$$

and 
$$\frac{1-1'}{C1} = \frac{VF}{VC} = \frac{2}{3}.$$

Similarly, mark points 2, 3 etc. on the axis and obtain points  $P_2$  and  $P'_2$ ,  $P_3$  and  $P'_3$  etc.

- (x) Draw the ellipse through these points. It is a closed curve having two foci and two directrices.



Directrix and focus

FIG. 6-2

## (2) Construction of ellipse by other methods:

Ellipse is also defined as a curve traced out by a point, moving in the same plane as and in such a way that the sum of its distances from two fixed points is always the same.

- Each of the two fixed points is called the *focus*.
- The line passing through the two foci and terminated by the curve, is called the *major axis*.
- The line bisecting the major axis at right angles and terminated by the curve, is called the *minor axis*.

**Conjugate axes:** Those axes are called *conjugate axes* when they are parallel to the tangents drawn at their extremities.

In fig. 6-3,  $AB$  is the major axis,  $CD$  the minor axis and  $F_1$  and  $F_2$  are the foci. The foci are equidistant from the centre  $O$ .

The points  $A, P, C$  etc. are on the curve and hence, according to the definition,

$$(AF_1 + AF_2) = (PF_1 + PF_2) = (CF_1 + CF_2) \text{ etc.}$$

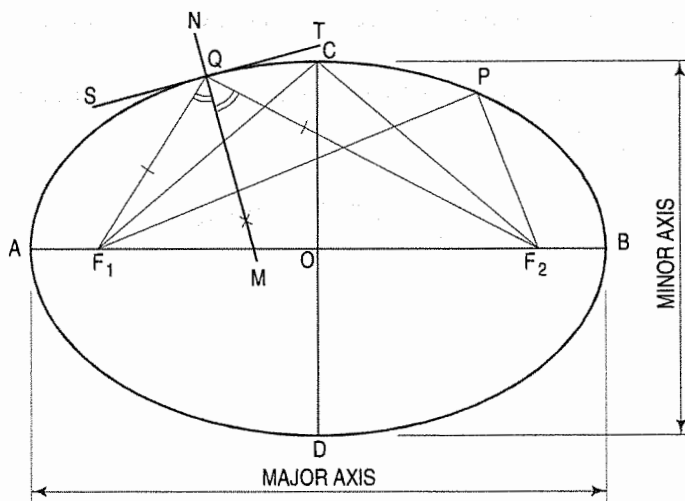
But  $(AF_1 + AF_2) = AB. \therefore (PF_1 + PF_2) = AB$ , the major axis.

Therefore, the sum of the distances of any point on the curve from the two foci is equal to the major axis.

Again,  $(CF_1 + CF_2) = AB$ .

But  $CF_1 = CF_2 \therefore CF_1 = CF_2 = \frac{1}{2} AB$ .

Hence, the distance of the ends of the minor axis from the foci is equal to half the major axis.



Conjugate axes

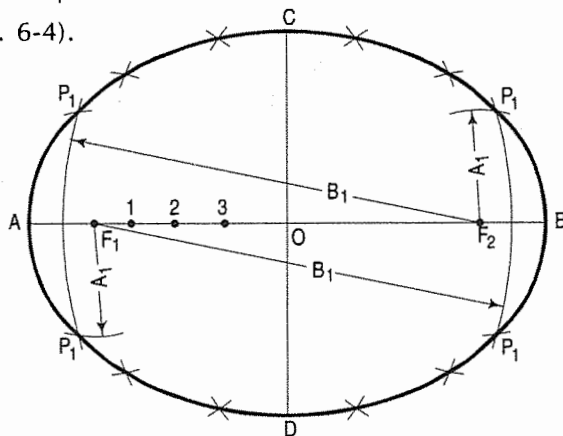
FIG. 6-3

**Problem 6-2.** To construct an ellipse, given the major and minor axes.

The ellipse is drawn by, first determining a number of points through which it is known to pass and then, drawing a smooth curve through them, either freehand or with a french curve. Larger the number of points, more accurate the curve will be.

**Method I:** Arcs of circles method (fig. 6-4).

- (i) Draw a line  $AB$  equal to the major axis and a line  $CD$  equal to the minor axis, bisecting each other at right angles at  $O$ .
- (ii) With centre  $C$  and radius equal to half  $AB$  (i.e.  $AO$ ) draw arcs cutting  $AB$  at  $F_1$  and  $F_2$ , the foci of the ellipse.
- (iii) Mark a number of points 1, 2, 3 etc. on  $AB$ .
- (iv) With centres  $F_1$  and  $F_2$  and radius equal to  $A1$ , draw arcs on both sides of  $AB$ .



Arc of circle method

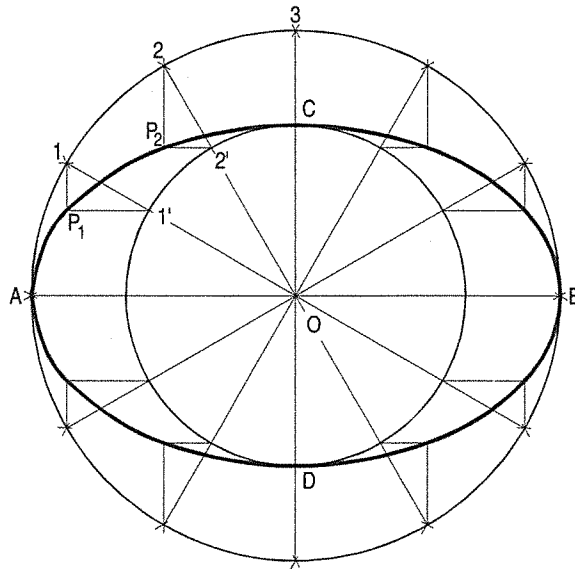
FIG. 6-4

- (v) With same centres and radius equal to  $B1$ , draw arcs intersecting the previous arcs at four points marked  $P_1$ .
- (vi) Similarly, with radii  $A2$  and  $B2$ ,  $A3$  and  $B3$  etc. obtain more points.
- (vii) Draw a smooth curve through these points. This curve is the required ellipse.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 13 for the following method II.

**Method II:** Concentric circles method (fig. 6-5).



Concentric circle method

FIG. 6-5

- (i) Draw the major axis  $AB$  and the minor axis  $CD$  intersecting each other at  $O$ .
- (ii) With centre  $O$  and diameters  $AB$  and  $CD$  respectively, draw two circles.
- (iii) Divide the major-axis-circle into a number of equal divisions, say 12 and mark points 1, 2 etc. as shown.
- (iv) Draw lines joining these points with the centre  $O$  and cutting the minor-axis-circle at points  $1'$ ,  $2'$  etc.
- (v) Through point 1 on the major-axis-circle, draw a line parallel to  $CD$ , the minor axis.
- (vi) Through point  $1'$  on the minor-axis-circle, draw a line parallel to  $AB$ , the major axis. The point  $P_1$ , where these two lines intersect is on the required ellipse.
- (vii) Repeat the construction through all the points. Draw the ellipse through  $A$ ,  $P_1$ ,  $P_2$ ... etc.

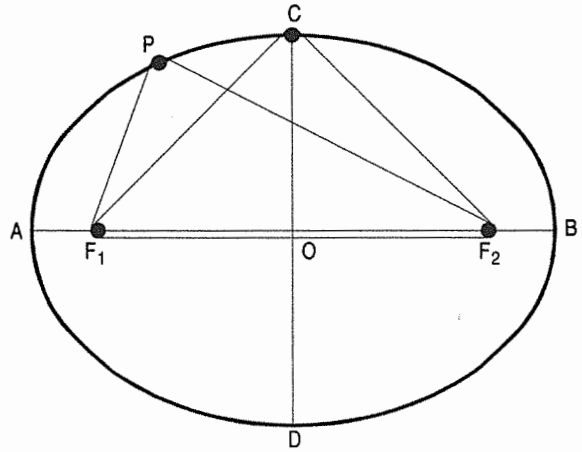
**Method III:** Loop of the thread method (fig. 6-6).

This is practical application of the first method.

- (i) Draw the two axes  $AB$  and  $CD$  intersecting at  $O$ . Locate the foci  $F_1$  and  $F_2$ .
- (ii) Insert a pin at each focus-point and tie a piece of thread in the form of a loop around the pins, in such a way that the pencil point when placed in the loop (keeping the thread tight), is just on the end  $C$  of the minor axis.

- (iii) Move the pencil around the foci, maintaining an even tension in the thread throughout and obtain the ellipse.

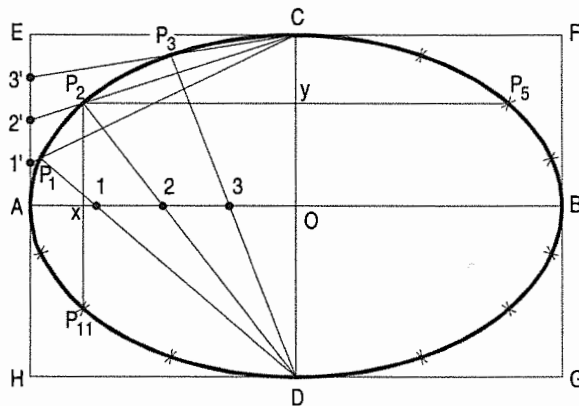
It is evident that  $PF_1 + PF_2 = CF_1 + CF_2$  etc.



Loop of the thread method

FIG. 6-6

*Method IV: Oblong method (fig. 6-7).*



Oblong method

FIG. 6-7

- (i) Draw the two axes  $AB$  and  $CD$  intersecting each other at  $O$ .
- (ii) Construct the oblong  $EFGH$  having its sides equal to the two axes.
- (iii) Divide the semi-major-axis  $AO$  into a number of equal parts, say 4, and  $AE$  into the same number of equal parts, numbering them from  $A$  as shown.
- (iv) Draw lines joining  $1'$ ,  $2'$  and  $3'$  with  $C$ .
- (v) From  $D$ , draw lines through  $1$ ,  $2$  and  $3$  intersecting  $C_1$ ,  $C_2$  and  $C_3$  at points  $P_1$ ,  $P_2$  and  $P_3$  respectively.
- (vi) Draw the curve through  $A$ ,  $P_1$ , ...,  $C$ . It will be one quarter of the ellipse.
- (vii) Complete the curve by the same construction in each of the three remaining quadrants.

As the curve is symmetrical about the two axes, points in the remaining quadrants may be located by drawing perpendiculars and horizontals from  $P_1$ ,  $P_2$  etc. and making each of them of equal length on both the sides of the two axes.

For example,  $P_2x = xP_{11}$  and  $P_2y = yP_5$ .

An ellipse can be inscribed within a parallelogram by using the above method as shown in fig. 6-8.

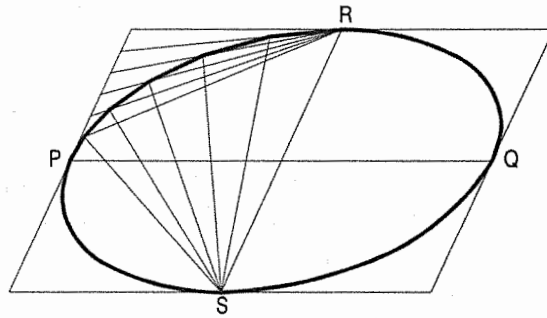
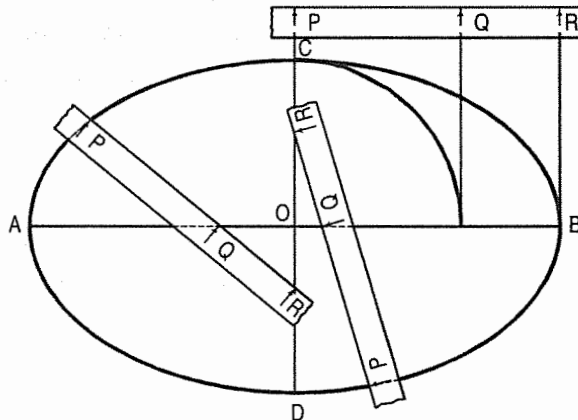


FIG. 6-8

Lines  $PQ$  and  $RS$ , joining the mid-points of the opposite sides of the parallelogram are called conjugate axes.

**Method V: Trammel method** (fig. 6-9).



Trammel method

FIG. 6-9

- (i) Draw the two axes  $AB$  and  $CD$  intersecting each other at  $O$ . Along the edge of a strip of paper which may be used as a trammel, mark  $PQ$  equal to half the minor axis and  $PR$  equal to half the major axis.
- (ii) Place the trammel so that  $R$  is on the minor axis  $CD$  and  $Q$  on the major axis  $AB$ . Then  $P$  will be on the required ellipse. By moving the trammel to new positions, always keeping  $R$  on  $CD$  and  $Q$  on  $AB$ , obtain other points. Draw the ellipse through these points.

**Problem 6-3.** (fig. 6-10):  $ABC$  is a triangle such that  $AB = 100$  mm,  $AC = 80$  mm and  $BC = 60$  mm. Draw an ellipse passing through points  $A$ ,  $B$  and  $C$ .

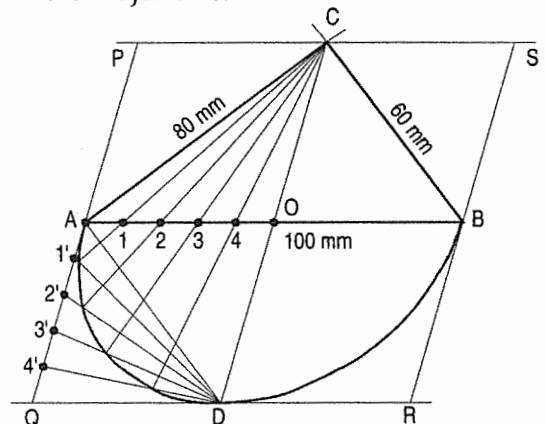


Fig. 6-10

- (i) Draw horizontal line  $AB = 100$  mm. Take  $A$  as centre draw an arc of 80 mm. Similarly  $B$  as centre and the radius equal to 60 mm, draw the arc such that it intersects previously drawn arc at the point  $C$ . Join  $ABC$  to complete triangle.
- (ii) Mark the mid point of  $AB$  such that  $OA = OB = 50$  mm. Join  $OC$  and extend  $CO$  such that  $CO = OD$ .
- (iii) Draw parallel lines from  $C$  and  $D$  to the line  $AB$ . Similarly draw parallel lines from  $A$  and  $B$  to the line  $CD$  and complete the rhombus ( $PQRS$ ).
- (iv) Divide  $AO$  into convenient number of equal parts  $A1 = 12 = 23 = 34 = 4O$  and  $AQ$  to same number of equal parts  $A1' = 1'2' = 2'3' = 3'4' = 4'O$ . Join  $A, 1', 2', 3', 4'$  with  $D$ . Join  $C1$  and extend it to intersect line  $D1'$ . Similarly join  $C2, C3, C4$  and extend it to intersect  $D2', D3', D4'$  respective. Draw smooth curve passing through all intersection.
- (v) Complete the ellipse by above method for the remaining part.

(3) **Normal and tangent to an ellipse:** The normal to an ellipse at any point on it bisects the angle made by lines joining that point with the foci.

The tangent to the ellipse at any point is perpendicular to the normal at that point.

**Problem 6-4.** (fig. 6-3): To draw a normal and a tangent to the ellipse at a point  $Q$  on it.

Join  $Q$  with the foci  $F_1$  and  $F_2$ .

- (i) Draw a line  $NM$  bisecting  $\angle F_1 Q F_2$ .  $NM$  is the normal to the ellipse.
- (ii) Draw a line  $ST$  through  $Q$  and perpendicular to  $NM$ .  $ST$  is the tangent to the ellipse at the point  $Q$ .

**Problem 6-5.** (fig. 6-11): To draw a curve parallel to an ellipse and at distance  $R$  from it.

This may be drawn by two methods:

- (a) A large number of arcs of radius equal to the required distance  $R$ , with centres on the ellipse, may be described. The curve drawn touching these arcs will be parallel to the ellipse.
- (b) It may also be obtained by drawing a number of normals to the ellipse, making them equal to the required distance  $R$  and then drawing a smooth curve through their ends.

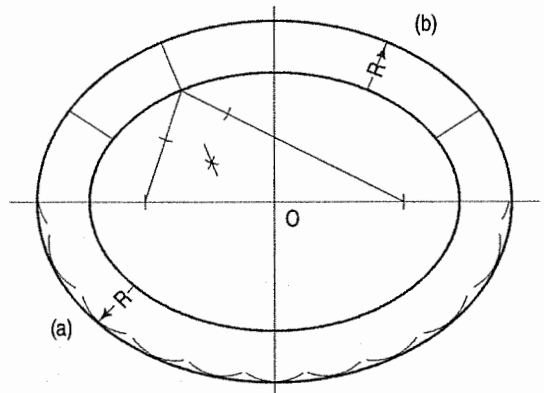


FIG. 6-11

**Problem 6-6.** (fig. 6-12): To find the major axis and minor axis of an ellipse whose conjugate axes and angle between them are given.

Conjugate axes  $PQ$  and  $RS$ , and the angle  $\alpha$  between them are given.

- (i) Draw the two axes intersecting each other at  $O$ .
- (ii) Complete the parallelogram and inscribe the ellipse in it as described in problem 6-2, method (iv).

- (iii) With  $O$  as centre and  $OR$  as radius, draw the semi-circle cutting the ellipse at a point  $E$ .
- (iv) Draw the line  $RE$ .
- (v) Through  $O$  draw a line parallel to  $RE$  and cutting the ellipse at points  $C$  and  $D$ .  $CD$  is the minor axis.
- (vi) Through  $O$ , draw a line perpendicular to  $CD$  and cutting the ellipse at points  $A$  and  $B$ .  $AB$  is the major axis.

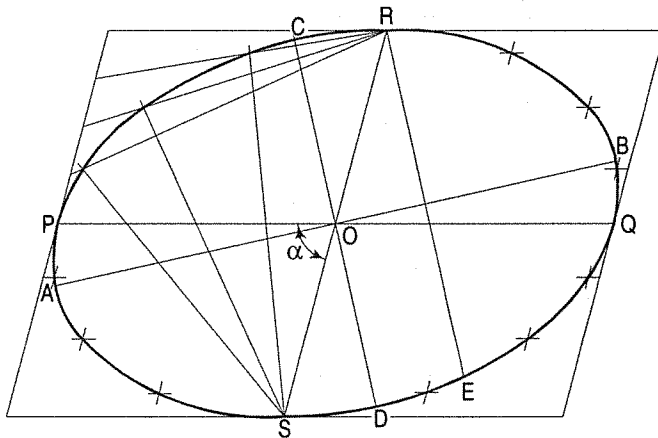


FIG. 6-12

**Problem 6-7.** (fig. 6-13): To find the centre, major axis and minor axis of a given ellipse.

- (i) Draw any two chords 1-2 and 3-4 parallel to each other.
- (ii) Find their mid-points  $P$  and  $Q$ , and draw a line passing through them, cutting the ellipse at points  $R$  and  $S$ . Bisect the line  $RS$  in the point  $O$  which is the centre of the ellipse.

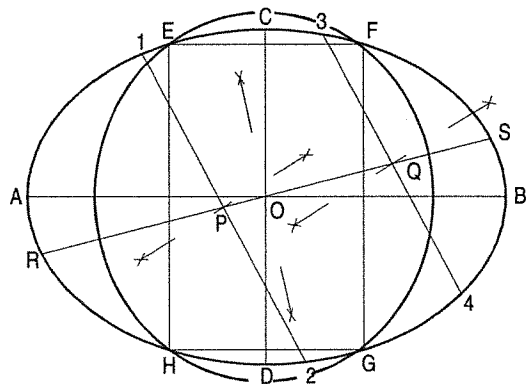


FIG. 6-13

With  $O$  as centre and any convenient radius, draw a circle cutting the ellipse in points  $E, F, G$  and  $H$ . Complete the rectangle  $EFGH$ . Through  $O$ , draw a line parallel to  $EF$  cutting the ellipse in points  $A$  and  $B$ . Again through  $O$ , draw a line parallel to  $FG$  cutting the ellipse at points  $C$  and  $D$ .  $AB$  and  $CD$  are respectively the major axis and the minor axis.

## 6-1-2. PARABOLA



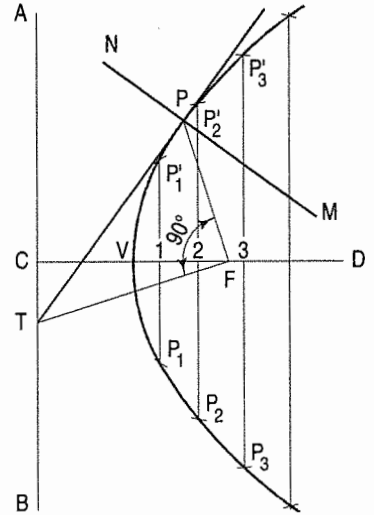
This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 14 for the following problem.

Use of parabolic curves is made in arches, bridges, sound reflectors, light reflectors etc. Mathematically a parabola can be described by an equation  $y^2 = 4ax$  or  $x^2 = 4ay$ .

**(1) General method of construction of a parabola:**

**Problem 6-8.** (fig. 6-14): To construct a parabola, when the distance of the focus from the directrix is 50 mm.

- (i) Draw the directrix  $AB$  and the axis  $CD$ .
- (ii) Mark focus  $F$  on  $CD$ , 50 mm from  $C$ .
- (iii) Bisect  $CF$  in  $V$  the vertex (because eccentricity = 1).
- (iv) Mark a number of points 1, 2, 3 etc. on the axis and through them, draw perpendiculars to it.
- (v) With centre  $F$  and radius equal to  $CF$ , draw arcs cutting the perpendicular through 1 at  $P_1$  and  $P'_1$ .
- (vi) Similarly, locate points  $P_2$  and  $P'_2$ ,  $P_3$  and  $P'_3$  etc. on both the sides of the axis.
- (vii) Draw a smooth curve through these points. This curve is the required parabola. It is an open curve.



Directrix and focus  
FIG. 6-14

**Problem 6-9.** (fig. 6-15): To find the axis of a given parabola.

- (i) Draw any two chords  $AB$  and  $CD$  across the parabola, parallel to each other and any distance apart.
- (ii) Bisect  $AB$  and  $CD$  in points  $E$  and  $F$  respectively and draw a line  $GH$  passing through them. The line  $GH$  will be parallel to the axis.
- (iii) Draw a chord  $PQ$ , perpendicular to  $GH$ .
- (iv) Bisect  $PQ$  in the point  $O$  and through it draw a line  $XY$  parallel to  $GH$ . Then  $XY$  is the required axis of the parabola.

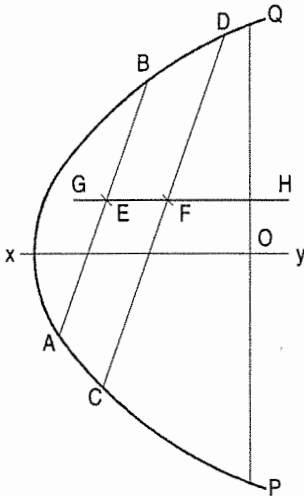


FIG. 6-15

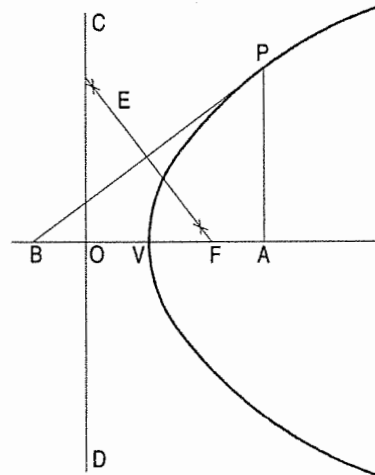


FIG. 6-16

**Problem 6-10.** (fig. 6-16): To find the focus and the directrix of a parabola whose axis is given.

- (i) Mark any point  $P$  on the parabola and draw a perpendicular  $PA$  to the axis. Mark a point  $B$  on the axis such that  $BV = VA$ .
- (ii) Draw a line joining  $B$  with  $P$ .
- (iii) Draw a perpendicular bisector  $EF$  of  $BP$ , intersecting the axis at a point  $F$ . Then  $F$  is the focus of the parabola.
- (iv) Mark a point  $O$  on the axis such that  $OV = VF$ . Through  $O$ , draw a line  $CD$  perpendicular to the axis. Then  $CD$  is the directrix of the parabola.

**(2) Construction of parabola by other methods:**

**Method I: Rectangle method** (fig. 6-17).

**Problem 6-11.** To construct a parabola given the base and the axis.

- (i) Draw the base  $AB$ .
- (ii) At its mid-point  $E$ , draw the axis  $EF$  at right angles to  $AB$ .
- (iii) Construct a rectangle  $ABCD$ , making side  $BC$  equal to  $EF$ .
- (iv) Divide  $AE$  and  $AD$  into the same number of equal parts and name them as shown (starting from  $A$ ).
- (v) Draw lines joining  $F$  with points 1, 2 and 3. Through 1', 2' and 3', draw perpendiculars to  $AB$  intersecting  $F1$ ,  $F2$  and  $F3$  at points  $P_1$ ,  $P_2$  and  $P_3$  respectively.
- (vi) Draw a curve through  $A$ ,  $P_1$ ,  $P_2$  etc. It will be a half parabola.

Repeat the same construction in the other half of the rectangle to complete the parabola. Or, locate the points by drawing lines through the points  $P_1$ ,  $P_2$  etc. parallel to the base and making each of them of equal length on both the sides of  $EF$ , e.g.  $P_1O = OP'_1$ .  $AB$  and  $EF$  are called the base and the axis respectively of the parabola.

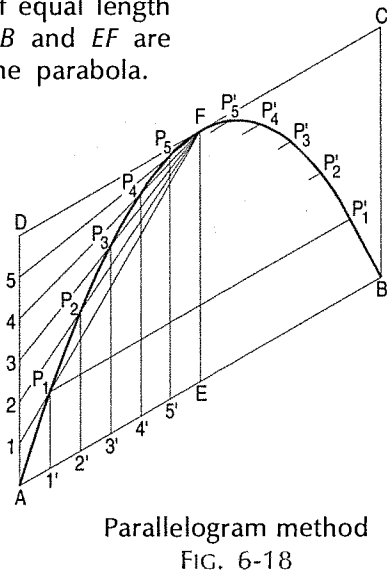
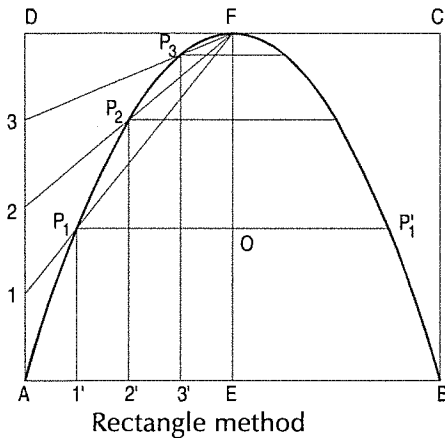
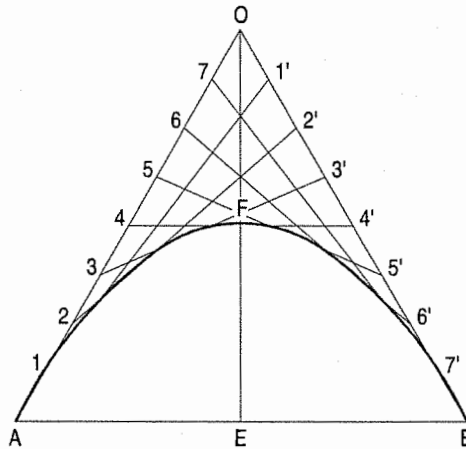


Fig. 6-18 shows a parabola drawn in a *parallelogram* by this method.

**Method II: Tangent method** (fig. 6-19).

- (i) Draw the base  $AB$  and the axis  $EF$ . (These are taken different from those in method I.)
- (ii) Produce  $EF$  to  $O$  so that  $EF = FO$ .

- (iii) Join  $O$  with  $A$  and  $B$ . Divide lines  $OA$  and  $OB$  into the same number of equal parts, say 8.
- (iv) Mark the division-points as shown in the figure.
- (v) Draw lines joining 1 with 1', 2 with 2' etc. Draw a curve starting from  $A$  and tangent to lines 1-1', 2-2' etc. This curve is the required parabola.



Tangent method

FIG. 6-19

### 6-1-3. HYPERBOLA

Use of hyperbolic curves is made in cooling towers, water channels etc.

**Rectangular hyperbola:** It is a curve traced out by a point moving in such a way that the product of its distances from two fixed lines at right angles to each other is a constant. The fixed lines are called *asymptotes*.

This curve graphically represents the Boyle's Law, viz.  $P \times V = a$ ,  $P$  = pressure,  $V$  = volume and  $a$  is constant. It is also useful in design of water channels.

#### General method of construction of a hyperbola:

Mathematically, we can describe a hyperbola by

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1. \text{ (Fig. 6-20 and fig. 6-21.)}$$

**Problem 6-12.** (fig. 6-20): Construct a hyperbola, when the distance of the focus from the directrix is 65 mm and eccentricity is  $\frac{3}{2}$ .

- (i) Draw the directrix  $AB$  and the axis  $CD$ .
- (ii) Mark the focus  $F$  on  $CD$  and 65 mm from  $C$ .
- (iii) Divide  $CF$  into 5 equal divisions and mark  $V$  the vertex, on the second division from  $C$ .

$$\text{Thus, eccentricity} = \frac{VF}{VC} = \frac{3}{2}.$$

To construct the scale for the ratio  $\frac{3}{2}$  draw a line  $VE$  perpendicular to  $CD$  such that  $VE = VF$ . Join  $C$  with  $E$ .

Thus, in triangle  $CVE$ ,  $\frac{VE}{VC} = \frac{VF}{VC} = \frac{3}{2}$ .

- (iv) Mark any point 1 on the axis and through it, draw a perpendicular to meet  $CE$ -produced at  $1'$ .
- (v) With centre  $F$  and radius equal to  $1-1'$ , draw arcs intersecting the perpendicular through 1 at  $P_1$  and  $P'_1$ .
- (vi) Similarly, mark a number of points 2, 3 etc. and obtain points  $P_2$  and  $P'_2$ ,  $P_3$  and  $P'_3$  etc.
- (vii) Draw the hyperbola through these points.

**Problem 6-13.** (fig. 6-21): To draw a hyperbola when its foci and vertices are given, and to locate its asymptotes.

- (i) Draw a horizontal line as axis and on it, mark the given foci  $F$  and  $F_1$ , and vertices  $V$  and  $V_1$ .
- (ii) Mark any number of points 1, 2, 3 etc. to the right of  $F_1$ .
- (iii) With  $F$  and  $F_1$  as centres and radius, say  $V_12$ , draw four arcs.
- (iv) With the same centres and radius  $V_12$ , draw four more arcs intersecting the first four arcs at points  $P_2$ . Then these points lie on the hyperbola.
- (v) Repeat the process with the same centres and radii  $V_1$  and  $V_11$ ,  $V_3$  and  $V_13$  etc. Draw the required hyperbola through the points thus obtained.
- (vi) With  $FF_1$  as diameter, draw a circle.
- (vii) Through the vertices  $V$  and  $V_1$ , draw lines perpendicular to the axis, cutting the circle at four points  $A$ . From  $O$ , the centre of the circle, draw lines passing through points  $A$ . These lines are the required asymptotes.

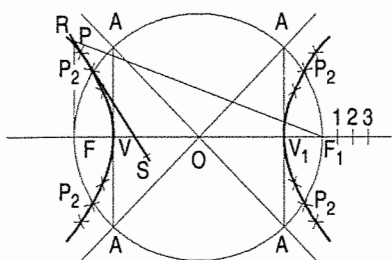


FIG. 6-21

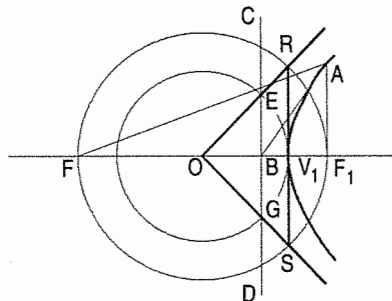


FIG. 6-22

**Problem 6-14.** (fig. 6-22): To locate the directrix and asymptotes of a hyperbola when its axis and foci are given.

From the focus  $F_1$ , draw a perpendicular to the axis intersecting the hyperbola at a point  $A$ .

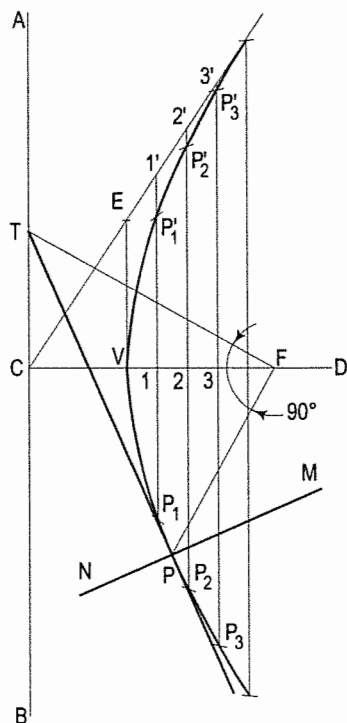


FIG. 6-20

- (i) Draw a line joining  $A$  with the other focus  $F$ . Draw the bisector of  $\angle FAF_1$ , cutting the axis at a point  $B$ .
- (ii) Through  $B$ , draw a line  $CD$  perpendicular to the axis.  $CD$  is the required directrix.
- (iii) With  $O$  the mid-point of  $FF_1$  as centre and  $OV_1$  as radius, draw a circle cutting  $CD$  at points  $E$  and  $G$ .
- (iv) Lines drawn from  $O$  and passing through  $E$  and  $G$  are the required asymptotes.
- (v) The asymptotes will also pass through the points of intersection ( $R$  and  $S$ ) between the circle of radius  $OF_1$  and the vertical through  $V_1$ .

**Problem 6-15.** (fig. 6-23): To draw a rectangular hyperbola, given the position of a point  $P$  on it.

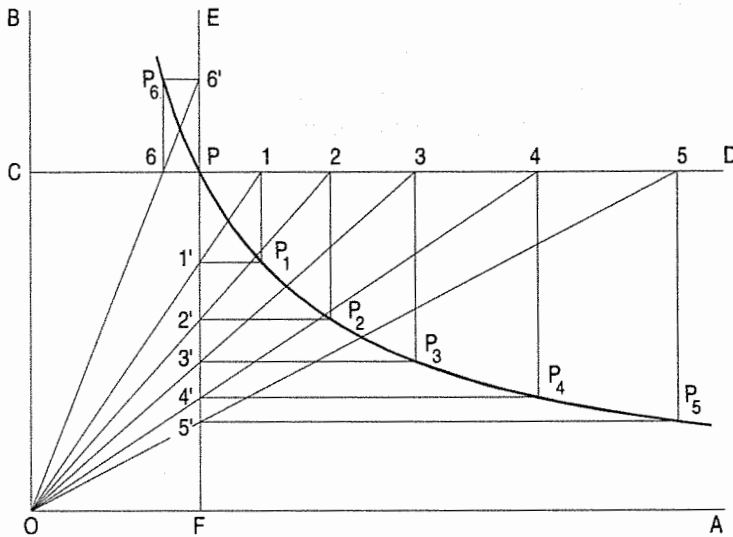


FIG. 6-23

- (i) Draw lines  $OA$  and  $OB$  at right angles to each other.
- (ii) Mark the position of the point  $P$ .
- (iii) Through  $P$ , draw lines  $CD$  and  $EF$  parallel to  $OA$  and  $OB$  respectively.
- (iv) Along  $PD$ , mark a number of points 1, 2, 3 etc. not necessarily equidistant.
- (v) Draw lines  $O1$ ,  $O2$  etc. cutting  $PF$  at points  $1'$ ,  $2'$  etc.
- (vi) Through point 1, draw a line parallel to  $OB$  and through  $1'$ , draw a line parallel to  $OA$ , intersecting each other at a point  $P_1$ .
- (vii) Obtain other points in the same manner.

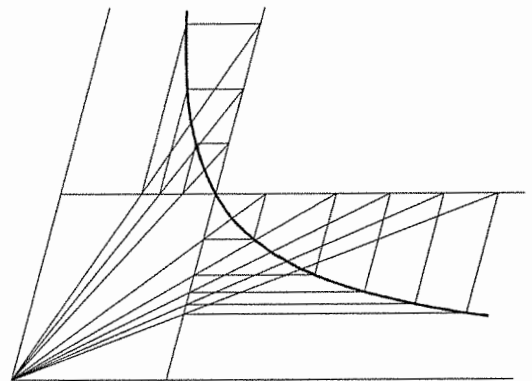


FIG. 6-24

For locating the point, say  $P_6$ , to the left of  $P$ , the line  $O6$  should be extended to meet  $PE$  at  $6'$ . Draw the hyperbola through the points  $P_6$ ,  $P$ ,  $P_1$  etc.

A hyperbola, through a given point situated between two lines making any angle between them, can similarly be drawn, as shown in fig. 6-24.

### 6-1-4. TANGENTS AND NORMALS TO CONICS



#### (1) General method:

The common rule for drawing tangents and normals:

When a tangent at any point on the curve is produced to meet the directrix, the line joining the focus with this meeting point will be at right angles to the line joining the focus with the point of contact.

The normal to the curve at any point is perpendicular to the tangent at that point.

**Problem 6-16.** To draw a tangent at any point  $P$  on the conics (fig. 6-2, fig. 6-14 and fig. 6-20).

- (i) Join  $P$  with  $F$ .
- (ii) From  $F$ , draw a line perpendicular to  $PF$  to meet  $AB$  at  $T$ .
- (iii) Draw a line through  $T$  and  $P$ . This line is the tangent to the curve.
- (iv) Through  $P$ , draw a line  $NM$  perpendicular to  $TP$ .  $NM$  is the normal to the curve.

#### (2) Other methods of drawing tangents to conics:

##### Method I:

**Problem 6-17.** (fig. 6-25): To draw a tangent to an ellipse at any point  $P$  on it.

- (i) With  $O$ , the centre of the ellipse as centre, and one half the major axis as radius, draw a circle.
- (ii) From  $P$ , draw a line parallel to the minor axis, cutting the circle at a point  $Q$ .

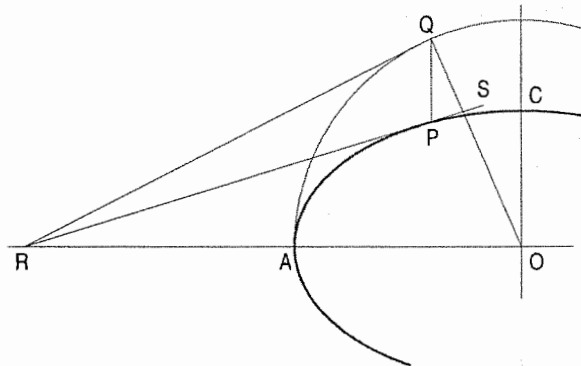


FIG. 6-25

- (iii) Draw a tangent to the circle at the point  $Q$  cutting the extended major axis at a point  $R$ .

From  $R$ , draw a line  $RS$  passing through  $P$ .  $RS$  is the required tangent to the ellipse.

##### Method II: When an axis of parabola is given.

**Problem 6-18.** To draw a tangent to a parabola at any point  $P$  on it.

- (i) From  $P$ , draw a perpendicular  $PA$  to the axis, intersecting it at a point  $A$  (fig. 6-26).

Mark a point  $B$  on the axis such that  $BV = VA$ . Draw a line from  $B$  passing through  $P$ .

Then this line is the required tangent.

- (ii) Through  $P$ , draw a line  $PQ$  parallel to the axis (fig. 6-27).

Draw two lines  $AB$  and  $CD$  parallel to, equidistant from and on opposite sides of  $PQ$ , and cutting the parabola at points  $A$  and  $C$ . Draw a line joining  $A$  with  $C$ .

Through  $P$ , draw a line  $RS$  parallel to  $AC$ .  $RS$  is the required tangent.

**Method III:** When the focus and the directrix are given (fig. 6-28).

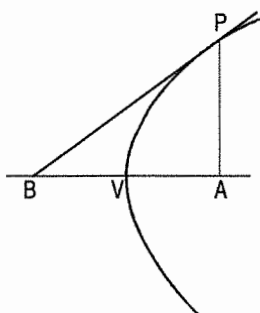


FIG. 6-26

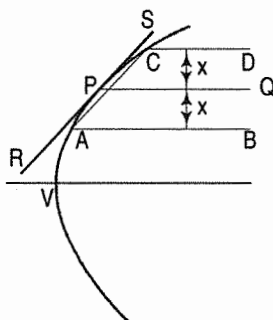


FIG. 6-27

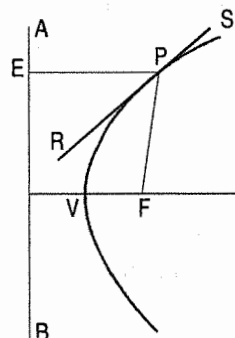


FIG. 6-28

- (i) From  $P$  draw a line  $PE$  perpendicular to the directrix  $AB$ , meeting it at a point  $E$ .
- (ii) Draw a line joining  $P$  with the focus  $F$ .
- (iii) Draw the bisector  $RS$  of angle  $EPF$ .

Then  $RS$  is the required tangent.

**Problem 6-19.** (fig. 6-21): To draw a tangent to a hyperbola at a point  $P$  on it when the axis and the foci are given.

Draw lines joining  $P$  with foci  $F$  and  $F_1$ . Draw the bisector  $RS$  of  $\angle FPF_1$ .  $RS$  is the required tangent to the hyperbola.

**Note:** In fig. 6-22, the line  $AB$  is the tangent to the hyperbola at the point  $A$ .

## 6-2. CYCLOIDAL CURVES

These curves are generated by a fixed point on the circumference of a circle, which rolls without slipping along a fixed straight line or a circle. The rolling circle is called *generating circle* and the fixed straight line or circle is termed *directing line* or *directing circle*. Cycloidal curves are used in tooth profile of gears of a dial gauge.

### 6-2-1. CYCLOID

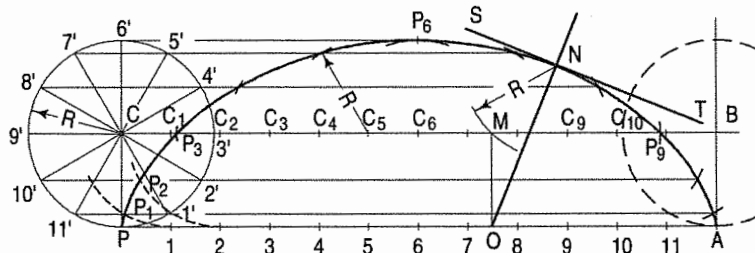
Cycloid is a curve generated by a point on the circumference of a circle which rolls along a straight line. It can be described by an equation,

$$y = a(1 - \cos\theta) \text{ or } x = a(\theta - \sin\theta).$$

**Problem 6-20.** (fig. 6-29): To construct a cycloid, given the diameter of the generating circle.

- (i) With centre  $C$  and given radius  $R$ , draw a circle. Let  $P$  be the generating point.

- (ii) Draw a line  $PA$  tangential to and equal to the circumference of the circle.
- (iii) Divide the circle and the line  $PA$  into the same number of equal parts, say 12, and mark the division-points as shown.
- (iv) Through  $C$ , draw a line  $CB$  parallel and equal to  $PA$ .
- (v) Draw perpendiculars at points 1, 2 etc. cutting  $CB$  at points  $C_1, C_2$  etc.



Cycloid  
FIG. 6-29

Assume that the circle starts rolling to the right. When point  $1'$  coincides with 1, centre  $C$  will move to  $C_1$ . In this position of the circle, the generating point  $P$  will have moved to position  $P_1$  on the circle, at a distance equal to  $P_1$  from point 1. It is evident that  $P_1$  lies on the horizontal line through  $1'$  and at a distance  $R$  from  $C_1$ .

Similarly,  $P_2$  will lie on the horizontal line through  $2'$  and at the distance  $R$  from  $C_2$ .

#### Construction:

- (vi) Through the points  $1', 2'$  etc. draw lines parallel to  $PA$ .
- (vii) With centres  $C_1, C_2$  etc. and radius equal to  $R$ , radius of generating circle, draw arcs cutting the lines through  $1', 2'$  etc. at points  $P_1, P_2$  etc. respectively. Draw a smooth curve through points  $P, P_1, P_2, \dots, A$ . This curve is the required cycloid.

**Normal and tangent to a cycloid curve:** The rule for drawing a normal to all cycloidal curves:

The normal at any point on a cycloidal curve will pass through the corresponding point of contact between the generating circle and the directing line or circle.

The tangent at any point is perpendicular to the normal at that point.

**Problem 6-21.** (fig. 6-29): To draw a normal and a tangent to a cycloid at a given point  $N$  on it.

- (i) With centre  $N$  and radius equal to  $R$ , draw an arc cutting  $CB$  at  $M$ .
- (ii) Through  $M$ , draw a line  $MO$  perpendicular to the directing line  $PA$  and cutting it at  $O$ .  
 $O$  is the point of contact and  $M$  is the position of the centre of the generating circle, when the generating point  $P$  is at  $N$ .
- (iii) Draw a line through  $N$  and  $O$ . This line is the required normal.
- (iv) Through  $N$ , draw a line  $ST$  at right angles to  $NO$ .  $ST$  is the tangent to the cycloid.

**Problem 6-22.** (fig. 6-30): A thin circular disc of 50 mm diameter is allowed to roll without slipping from upper edge of sloping plank which is inclined at  $15^\circ$  with the horizontal plane. Draw the curve traced by the point on the circumference of the disc.

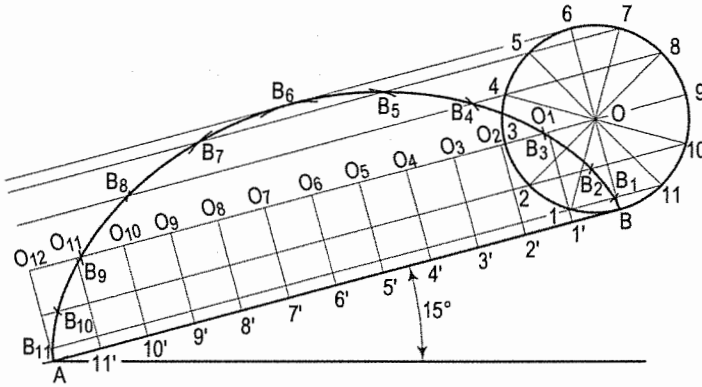


FIG. 6-30

- (i) Draw a line  $AB$  of length 157 mm (i.e.  $\pi D$ ) at angle of  $15^\circ$  with the horizontal.
- (ii) Draw a circle of 50 mm diameter at upper edge  $B$  as shown and divide the circumference of the circle in twelve equal parts. Name 1, 2, 3, 4 to 11. Draw parallel lines from 1, 2, 3, 4 to 11 to the  $AB$ . Take  $B$  as generating point on the circumference.
- (iii) Draw a parallel line from the centre of circle  $O$  to line  $AB$  and equal to length of  $AB$ . Divide this line in 12 parts. Name  $O_1$  to  $O_{12}$  as shown.
- (iv) As the disc rolls from position  $B$ , the centre of disc simultaneously moves from  $O$  to  $O_1$ . Draw an arc with radius as 25 mm and centre as  $O_1$  to intersect the line drawn from the point 1 parallel to  $AB$ . Mark intersection point as  $B_1$ . This point is on the cycloid.
- (v) Similarly draw arc from the points  $O_2$  to  $O_{12}$  such that it intersects each line drawn from 2 to 11 respectively. Join the points by smooth curve. This curve is the cycloid.

### 6-2-2. TROCHOID

Trochoid is a curve generated by a point fixed to a circle, within or outside its circumference, as the circle rolls along a straight line.

When the point is within the circle, the curve is called an *inferior trochoid* and when outside the circle, it is termed a *superior trochoid*.

**Problem 6-23.** (fig. 6-31, fig. 6-32 and fig. 6-33): To draw a trochoid, given the rolling circle and the generating point.

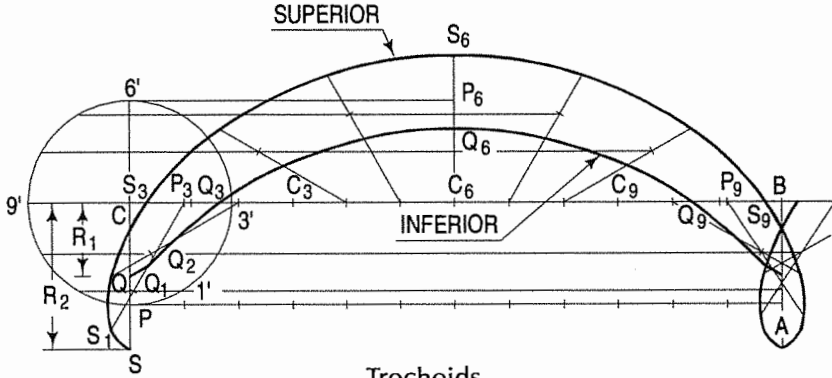
#### (a) *Inferior trochoid:*

Let  $Q$  be the point within the circle and at a distance  $R_1$  from the centre  $C$ .

- (i) Draw the circle and mark a point  $Q$  on the line  $CP$  and at a distance  $R_1$  from  $C$ .
- (ii) Draw a tangent  $PA$  equal to the circumference of the circle and a line  $CB$  equal and parallel to  $PA$ .
- (iii) Divide the circle and the line  $CB$  into 12 equal parts.

**Method I:** (fig. 6-31): Determine the positions  $P_1, P_2$  etc. for the cycloid as shown in problem 6-20. Draw lines  $C_1P_1, C_2P_2$  etc. With centres  $C_1, C_2$  etc. and radius equal to  $R_1$ , draw arcs cutting  $C_1P_1, C_2P_2$  etc. at points  $Q_1, Q_2$  etc. respectively.

Draw a curve through these points. This curve is the inferior trochoid.

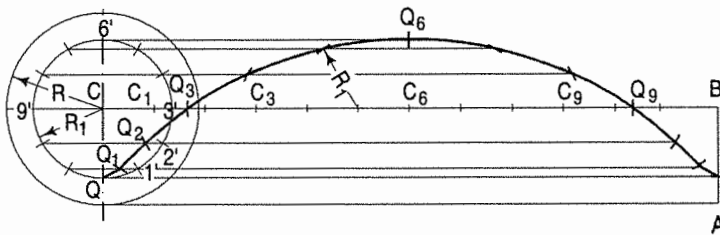


Trochoids

FIG. 6-31

**Method II:** (fig. 6-32): With centre  $C$  and radius equal to  $R_1$ , draw a circle and divide it into 12 equal parts.

Through the division-points, draw lines parallel to  $PA$ . With centres  $C_1, C_2$  etc. and radius equal to  $R_1$ , draw arcs to cut the lines through  $1', 2'$  etc. at points  $Q_1, Q_2$  etc. Draw the trochoid through these points.

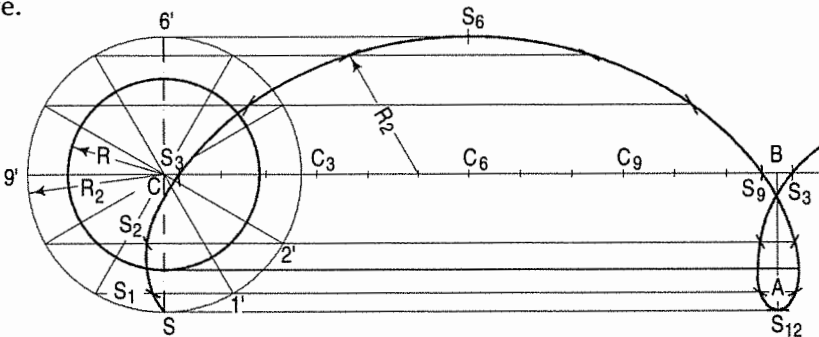


Inferior trochoid

FIG. 6-32

**(b) Superior trochoid:**

Let  $S$  be the point outside circumference of the circle and at a distance  $R_2$  from the centre.



Superior trochoid

FIG. 6-33

**Method I:** (fig. 6-31): Adopt the same method as method I used for inferior trochoid. Point  $S$  will lie on the line  $CP$ -produced, at distance  $R_2$  from  $C$ . Points  $S_1, S_2$  etc. are obtained by cutting the lines  $C_1P_1$ -produced,  $C_2P_2$ -produced etc. with arcs drawn with centres  $C_1, C_2$  etc. and radius equal to  $R_2$ .  $S, S_1, S_2$  etc. are the points on the superior trochoid.

**Method II:** (fig. 6-33): Same as method II for inferior trochoid. Note that the radius of the circle is equal to  $R_2$ . A loop is formed when the circle rolls for more than one revolution.

### 6-2-3. EPICYCLOID AND HYPOCYCLOID

The curve generated by a point on the circumference of a circle, which rolls without slipping along another circle *outside* it, is called an *epicycloid*. The *epicycloid* can be represented mathematically by

$$x = (a + b) \cos \theta - a \cos \left( \frac{a + b}{a} \theta \right),$$

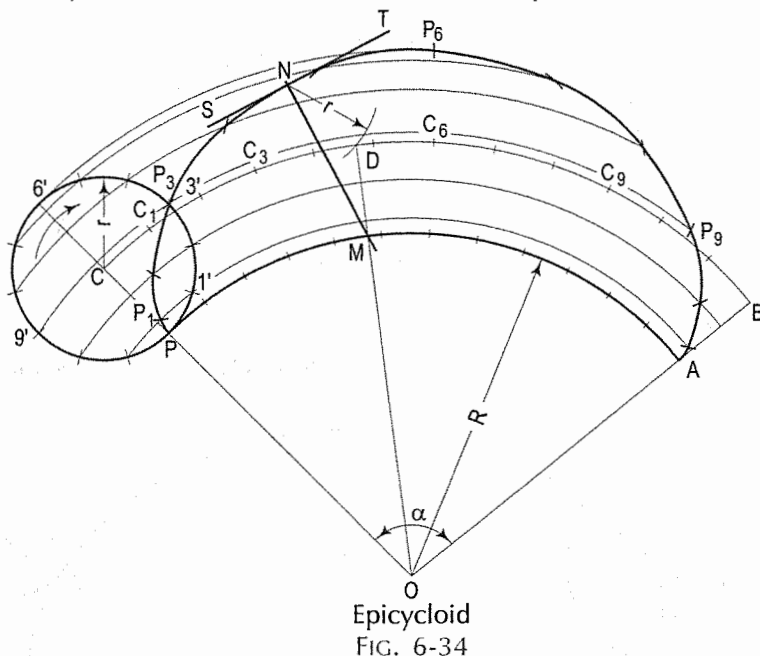
$$y = (a + b) \sin \theta - a \sin \left( \frac{a + b}{a} \theta \right)$$

where  $a$  is the radius of rolling circle.

When the circle rolls along another circle *inside* it, the curve is called a *hypocycloid*. It can be represented by mathematically  $x = a \cos^3 \theta$ ,  $y = a \sin^3 \theta$ .

**Problem 6-24.** To draw an *epicycloid* and a *hypocycloid*, given the generating and directing circles of radii  $r$  and  $R$  respectively.

**Epicycloid** (fig. 6-34): With centre  $O$  and radius  $R$ , draw the directing circle (only a part of it may be drawn). Draw a radius  $OP$  and produce it to  $C$ , so that  $CP = r$ .



With  $C$  as centre, draw the generating circle. Let  $P$  be the generating point.

In one revolution of the generating circle, the point  $P$  will move to a point  $A$ , so that the arc  $PA$  is equal to the circumference of the generating circle.

The position of  $A$  may be located by calculating the angle subtended by the arc  $PA$  at centre  $O$ , by the formula,

$$\frac{\angle POA}{360^\circ} = \frac{\text{arc } PC}{\text{circumference of directing circle}} = \frac{2\pi r}{2\pi R} = \frac{r}{R}$$

$$\therefore \angle POA = 360^\circ \times \frac{r}{R}$$

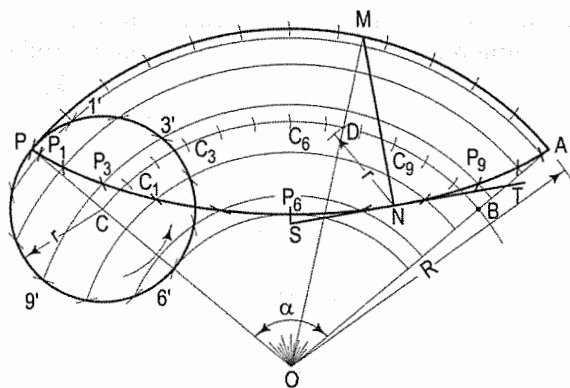
(i) Set-off this angle and obtain the position of  $A$ .

(ii) With centre  $O$  and radius equal to  $OC$ , draw an arc intersecting  $OA$ -produced at  $B$ . This arc  $CB$  is the locus of the centre  $C$ .

(iii) Divide  $CB$  and the generating circle into twelve equal parts.

(iv) With centre  $O$ , describe arcs through points  $1', 2', 3'$  etc.

(v) With centres  $C_1, C_2$  etc. and radius equal to  $r$ , draw arcs cutting the arcs through  $1', 2'$  etc. at points  $P_1, P_2$  etc.

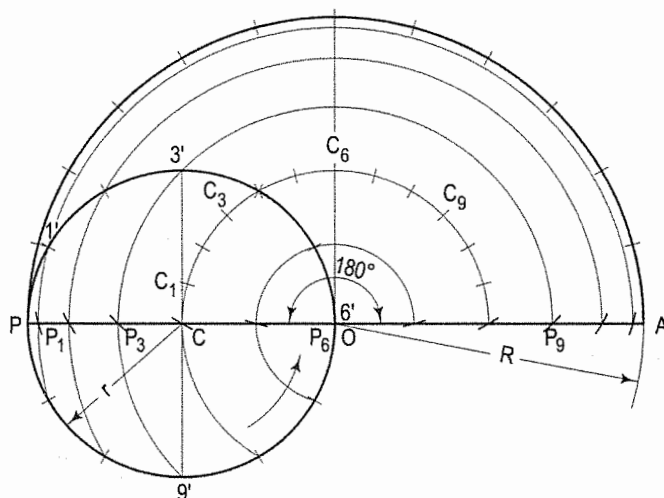


Hypocycloid

FIG. 6-35

Draw the required epicycloid through the points  $P, P_1, P_2, \dots, A$ .

**Hypocycloid** (fig. 6-35): The method for drawing the hypocycloid is same as for epicycloid. Note that the centre  $C$  of the generating circle is inside the directing circle.



Hypocycloid

FIG. 6-36

When the diameter of the rolling circle is half the diameter of the directing circle, the hypocycloid is a straight line and is a diameter of the directing circle (fig. 6-36).

**Normal and tangent to an epicycloid and a hypocycloid:**

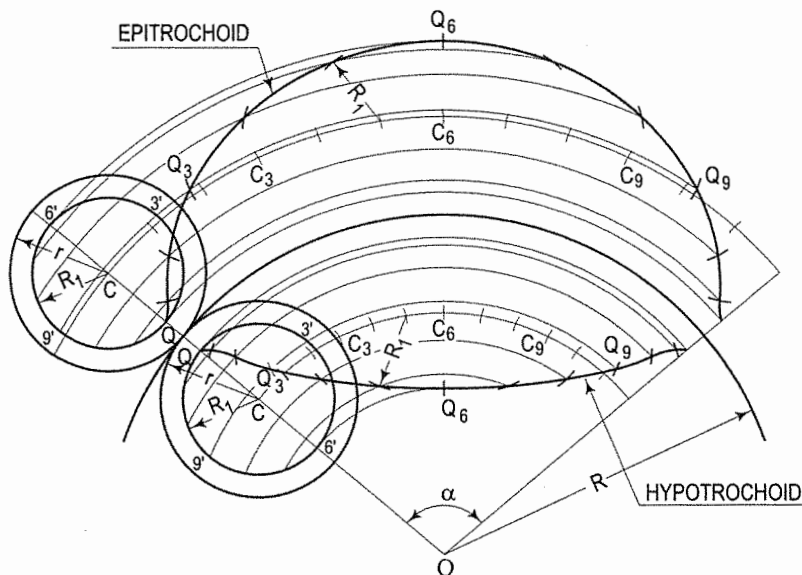
**Problem 6-25.** (fig. 6-34 and fig. 6-35): To draw a normal and a tangent to an epicycloid and a hypocycloid at a point  $N$  in each of them.



**6-2-5. HYPOTROCHOID** (FIG. 6-39)

When the circle rolls inside another circle, the curve is called a hypotrochoid.

The curve is termed inferior or superior, according to the position of the point being inside or outside the rolling circle.



Inferior epitrochoid and hypotrochoid

FIG. 6-39

**Problem 6-26.** To draw an epitrochoid and a hypotrochoid, given the rolling and directing circles and the generating points.

These curves are drawn by applying the methods used for trochoids. Note that arcs are drawn instead of horizontal lines.

**Epitrochoids:**

Method I: Superior and inferior

– see fig. 6-37

Method II: Superior – see fig. 6-38

Inferior – see fig. 6-39

**Hypotrochoids:**

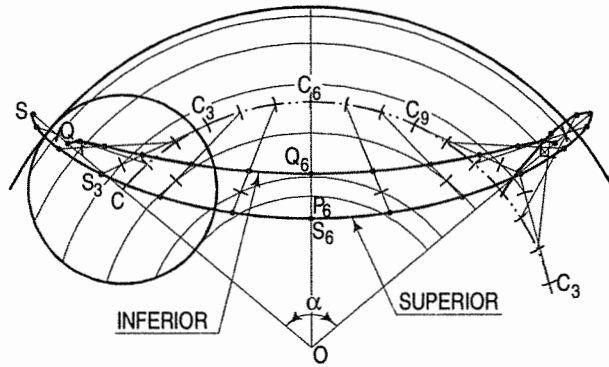
Method I: Superior and inferior

– see fig. 6-40

Method II: Superior – see fig. 6-41

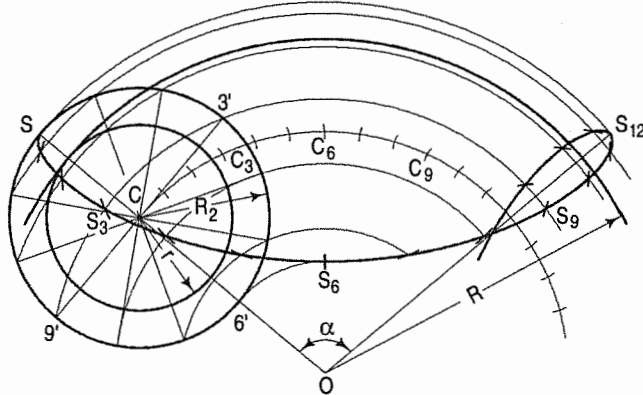
Inferior – see fig. 6-39

**Note:** When the diameter of the rolling circle is half the diameter of the directing circle, the hypotrochoid will be an ellipse.



Hypotrochoids

FIG. 6-40



Superior hypotrochoids

FIG. 6-41

### 6-3. INVOLUTE

The involute is a curve traced out by an end of a piece of thread unwound from a circle or a polygon, the thread being kept tight. It may also be defined as a curve traced out by a point in a straight line which rolls without slipping along a circle or a polygon. Involute of a circle is used as teeth profile of gear wheel. Mathematically it can be described by  $x = r \cos \theta + r \theta \sin \theta$ ,  $y = r \sin \theta - r \theta \cos \theta$ , where "r" is the radius of a circle.

**Problem 6-27.** (fig. 6-42): To draw an involute of a given circle.

With centre C, draw the given circle. Let P be the starting point, i.e. the end of the thread.

Suppose the thread to be partly unwound, say upto a point 1. P will move to a position  $P_1$  such that  $1P_1$  is tangent to the circle and is equal to the arc  $1P$ .  $P_1$  will be a point on the involute.

**Construction:**

- (i) Draw a line PQ, tangent to the circle and equal to the circumference of the circle.
- (ii) Divide PQ and the circle into 12 equal parts.

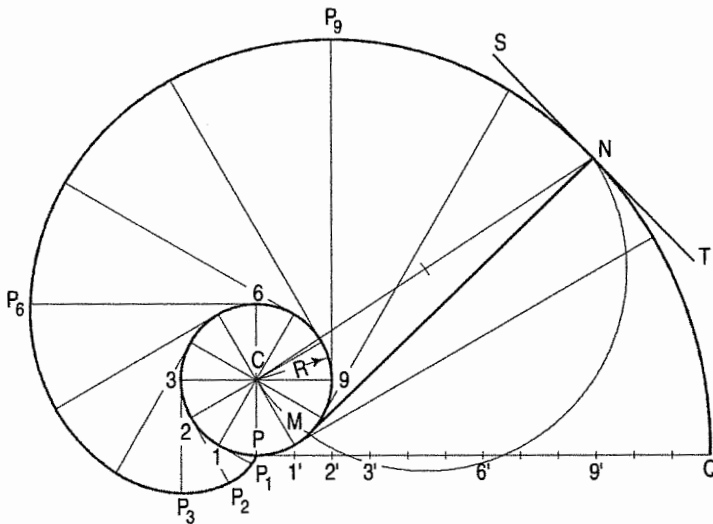
- (iii) Draw tangents at points 1, 2, 3 etc. and mark on them points  $P_1, P_2, P_3$  etc. such that  $1P_1 = P_1', 2P_2 = P_2', 3P_3 = P_3'$  etc.

Draw the involute through the points  $P, P_1, P_2, \dots$  etc.

#### Normal and tangent to an involute:

The normal to an involute of a circle is tangent to that circle.

**Problem 6-28.** (fig. 6-42): To draw a normal and a tangent to the involute of a circle at a point  $N$  on it.



Involute

FIG. 6-42

- (i) Draw a line joining  $C$  with  $N$ .
- (ii) With  $CN$  as diameter describe a semi-circle cutting the circle at  $M$ .
- (iii) Draw a line through  $N$  and  $M$ . This line is the normal. Draw a line  $ST$ , perpendicular to  $NM$  and passing through  $N$ .  $ST$  is the tangent to the involute.

**Problem 6-29.** (fig. 6-43): To draw an involute of a given square.

Let  $ABCD$  be the given square.

- (i) With centre  $A$  and radius  $AD$ , draw an arc to cut the line  $BA$ -produced at a point  $P_1$ .
- (ii) With centre  $B$  and radius  $BP_1$  (i.e.  $BA + AD$ ) draw an arc to cut the line  $CB$ -produced at a point  $P_2$ . Similarly, with centres  $C$  and  $D$  and radii  $CP_2$  (i.e.  $CB + BA + AD$ ) and  $DP_3$  (i.e.  $DC + CB + BA + AD = \text{perimeter}$ ) respectively, draw arcs to cut  $DC$ -produced at a point  $P_3$  and  $AD$ -produced at a point  $P_4$ .

The curve thus obtained is the involute of the square.

**Problem 6-30.** (fig. 6-44): Trace the paths of the ends of a straight line  $AP$ , 100 mm long, when it rolls, without slipping, on a semi-circle having its diameter  $AB$ , 75 mm long. (Assume the line  $AP$  to be tangent to the semi-circle in the starting position.)



- (ii) From  $A$ , draw a line passing through 1.  $A$  as centre and  $AP$  as radius, draw the arc intersecting extended line  $A1'$  at  $P_0$ . Extend the side 1-2, 1 as centre and  $1'P_0$  as radius, draw the arc to intersect extended line 1-2 at  $P_1$ .
- (iii) Divide the circumference of the semicircle into six equal parts and label it as 2, 3, 4, 5, 6, 7 and 8.
- (iv) Draw a tangent to semicircle from 2 such that  $2'-P_1 = 2'-P_2$ . Mark  $8'$  on this tangent such that  $2'-8' = \pi R$ . Divide  $2'-8'$  into six equal parts.
- (v) Similarly draw tangents at 3, 4, 5, 6, 7 and 8 in anti-clockwise direction such that  $3-P_3 = 3'-9'$ ,  $4-P_4 = 4'-9'$ ,  $5-P_5 = 5'-9'$ ,  $6-P_6 = 6'-9'$ ,  $7-P_7 = 7'-9'$ ,  $8-P_8 = 8'-9'$  and  $8-P_9 = 8'-9'$  respectively.
- (vi) Join  $P_1, P_2, \dots, P_9$  with smooth curve.

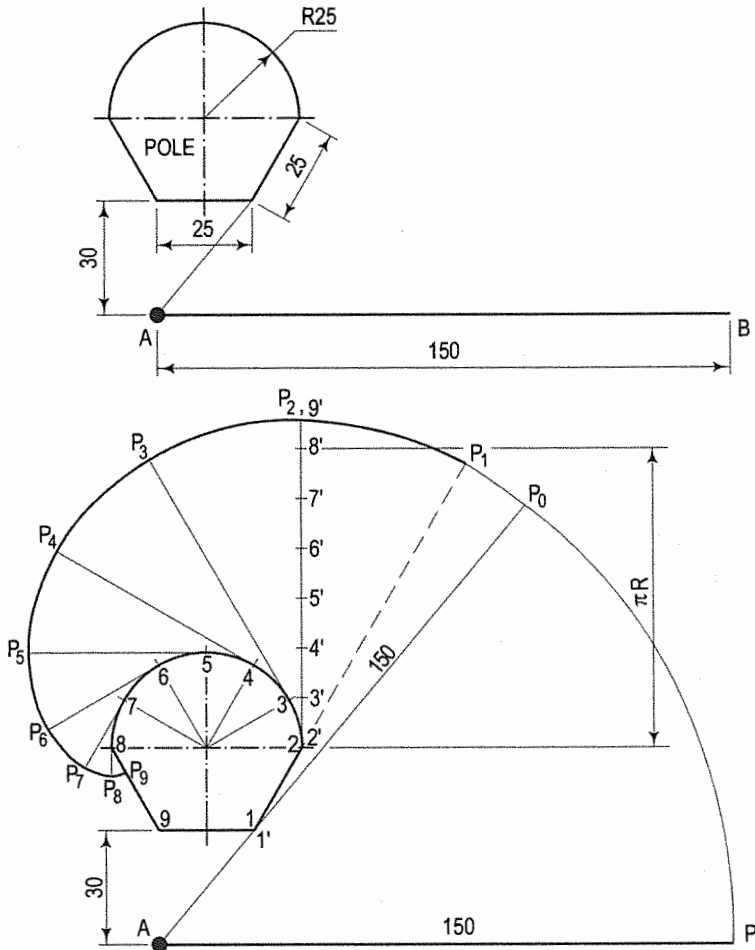


Fig. 6-46

**Problem 6-32.** A thin semi-circular plate with  $C$  as centre and radius equal to 32 mm is fixed.  $OP$  is the inelastic rope of 140 mm horizontal length. End  $O$  of the rope is fixed. The end  $O$  is 20 mm above and 20 mm on the left of  $C$ . The rope is wound in anti-clockwise direction around the circumference of the plate. Draw the locus of free end  $P$  of the rope.

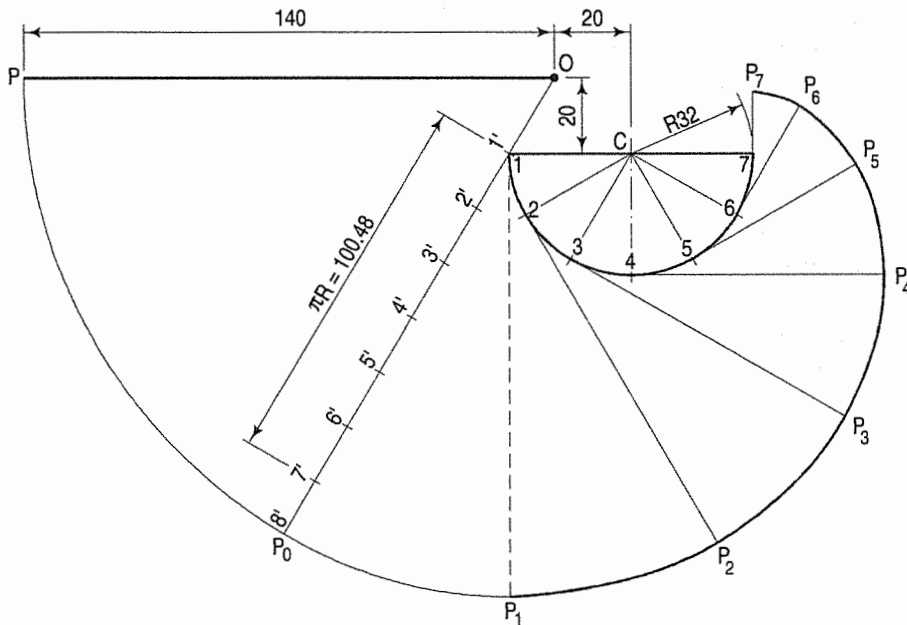


Fig. 6-47

- (i) Taking C as centre, draw the semicircle of radius 32 mm as shown in fig. 6-47. Draw the horizontal line 20 mm above C. Mark O, 20 mm on left of C. Draw OP parallel to the diameter of the plate equal to 140 mm.
- (ii) Divide the semi-circle into six equal parts and label it 1, 2, 3, 4, 5, 6 and 7. Join all points with C. Now rotate the line OP about O, till it touches the semicircular plate at point 1. Mark the point 1 as point 1' on the rope.
- (iii) Mark 7' on line 1'P<sub>0</sub> from 1' such that 1'-7' =  $\pi R = 100.48$  mm. Divide 1'-7' into six equal parts and name it as 1', 2', 3', 4', 5', 6' and 7'.
- (iv) Draw tangents on semicircular plate at 1, 2, 3, 4, 5, 6 and 7 in anticlockwise direction such that 1-P<sub>1</sub> = 1'8', 2-P<sub>2</sub> = 2'8', 3-P<sub>3</sub> = 3'8', 4-P<sub>4</sub> = 4'8', 5-P<sub>5</sub> = 5'8', 6-P<sub>6</sub> = 6'8', and 7-P<sub>7</sub> = 7'8' respectively.
- (v) Join points P<sub>1</sub>, P<sub>2</sub>, ..., P<sub>7</sub> by smooth curve. It is involute.

**Problem 6-33.** (fig. 6-48): A thin triangular equilateral plate of 20 mm side is pinned at its centroid O. An inelastic string circumscribes complete perimeter of the plate. One end of the string is attached to one of the apex of the plate. Draw the curve traced out by other end of the string keeping it tight, when the string is unwound.

- (i) Draw an equilateral triangle of 20 mm sides. Determine its centroid by drawing perpendicular at mid point of each sides.
- (ii) Consider starting point P. Taking PR = 20 mm as radius and R as centre, draw the arc to intersect the extended line QR at point 1, of 40 mm as radius.
- (iii) At the point Q, extend the line PQ equal to 2 QR = 40 mm. Q as centre draw the arc to cut at the point 2.
- (iv) Extend line RP equal to 3 PQ = 60 mm P as centre, 60 mm as radius draw the arc to intersect the extended line RP at 3.
- (v) Thus the curve obtained is involute.



- (iii) Consider starting point  $E$ . Extend line  $BA$  through  $A$ .  $A$  as centre,  $AE$  as radius draw an arc starting from  $E$  and intersecting the extended line  $BA$  at 1.
- (iv) Similarly at  $B$ ,  $C$ , and  $D$  extend lines. At  $B$  as centre and  $B1 = (AE + AB)$  as radius, draw the arc cutting extended line  $CB$  at 2. At  $C$  and  $C3 = 3 A1$  as radius draw arc cutting extended line  $CB$  at 3. Similarly draw arcs for extended line  $ED$  and  $AE$  cutting at 4, 5 respectively.
- (v) Thus the curve obtained is involute.

## 6-4. EVOLUTES



$APB$  is a given curve (fig. 6-50).  $O$  is the centre of a circle drawn through three points  $C$ ,  $P$  and  $D$  on this curve. If the points  $C$  and  $D$  are moved to converge towards  $P$ , until they are indefinitely close together, then in the limit, the circle becomes the *circle of curvature* of the curve  $APB$  at  $P$ . The centre  $O$  of the circle of curvature lies on the normal to the curve at  $P$ . This centre is called the *centre of curvature* at  $P$ . The locus of the centre of curvature of a curve is called the *evolute* of the curve. A curve has only one evolute.

**Problem 6-35** (fig. 6-51): To determine the centre of curvature at a given point on a conic.

Let  $P$  be the given point on the conic and  $F$ , the focus.

- (i) Join  $P$  with  $F$ .
- (ii) At  $P$ , draw a normal  $PN$ , cutting the axis at  $N$ .
- (iii) Draw a line  $NR$  perpendicular to  $PN$  and cutting  $PF$  or  $PF$ -produced at  $R$ .
- (iv) Draw a line  $RO$  perpendicular to  $PR$  and cutting  $PN$ -produced at  $O$ . Then  $O$  is the centre of curvature of the conic at the point  $P$ .

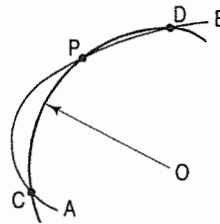


FIG. 6-50

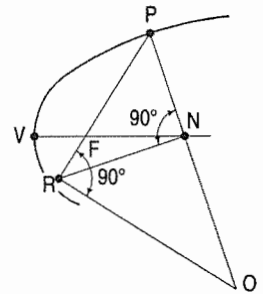


FIG. 6-51

The above construction does not hold good when the given point coincides with the vertex. As the point  $P$  approaches the vertex, the points  $R$ ,  $N$  and  $O$  move nearer to one another, so that when  $P$  is at the vertex, the three points coincide on the axis.

In a parabola,  $PF$  will be equal to  $FR$ . Hence, when  $P$  is at  $V$  the vertex, the centre of curvature  $O$  is on the axis so that  $OF = VF$ .

In an ellipse or a hyperbola, the ratio of the distances of  $P$  from the foci is equal to the ratio of focal distances of  $N$ , i.e.  $\frac{PF}{PF_1} = \frac{NF}{NF_1}$ .

Hence, when  $P$  coincides with  $V$  the vertex, the ratio becomes  $\frac{VF}{VF_1} = \frac{OF}{OF_1}$ .

Similarly, when  $P$  coincides with the other vertex  $V_1$ , the ratio becomes  $\frac{V_1F_1}{V_1F} = \frac{O_1F_1}{O_1F}$ .

**Problem 6-36.** To determine the centre of curvature  $O$ , when the point  $P$  is at the vertex  $V$  of a conic.

- (a) **Parabola** (fig. 6-52): Mark the centre of curvature  $O$  on the axis such that  $OF = VF$ .
- (b) **Ellipse** (fig. 6-53):

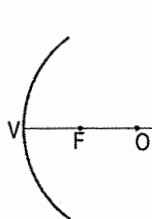


FIG. 6-52

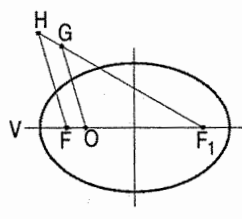


FIG. 6-53

**Method I:**

- (i) Draw a line  $F_1G$  inclined to the axis and equal to  $VF_1$ .
- (ii) Produce  $F_1G$  to  $H$  so that  $GH = VF$ . Join  $H$  with  $F$ .
- (iii) Draw a line  $GO$  parallel to  $HF$  and intersecting the axis at  $O$ . Then  $O$  is the required centre of curvature.

**Method II:** (fig. 6-54):

- (i) Draw a rectangle  $AOCE$  in which  
 $AO = \frac{1}{2}$  major axis and  
 $CO = \frac{1}{2}$  minor axis.

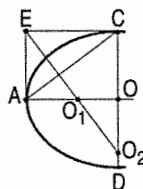


FIG. 6-54

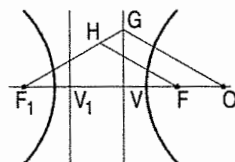


FIG. 6-55

- (ii) Join  $A$  with  $C$ .
- (iii) Through  $E$ , draw a line perpendicular to  $AC$  and cutting the major axis at  $O_1$  and the minor axis  $O_2$ . Then  $O_1$  and  $O_2$  are the centres of curvature when the point  $P$  is at  $A$  and  $C$  respectively.

(c) **Hyperbola** (fig. 6-55):

- (i) Draw a line  $F_1G$  inclined to the axis and equal to  $FV_1$ .
- (ii) On  $F_1G$ , mark a point  $H$  such that  $HG = VF$ . Join  $H$  with  $F$ .
- (iii) Draw a line  $GO$  parallel to  $HF$  and cutting the axis at  $O$ . Then  $O$  is the centre of curvature at the vertex  $V$ .

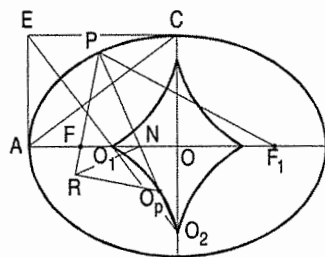


FIG. 6-56

**Problem 6-37.** (fig. 6-56 and fig. 6-57): To draw the evolute of an ellipse.

The ellipse with major axis  $AB$  and minor axis  $CD$  is given.

- (i) Mark a number of points on the ellipse.
- (ii) Determine the centres of curvatures at these points (as shown at the point  $P$ ) and draw a smooth curve through them. This is the evolute of the ellipse. The evolute may sometimes go outside the ellipse as shown in fig. 6-57. The centres of curvature at points  $A$  and  $C$  are shown by method II of problem 6-36.

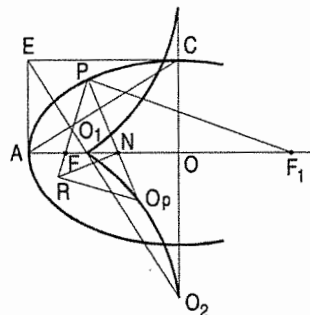


FIG. 6-57

**Problem 6-38.** (fig. 6-58): To draw the evolute of a parabola.

- (i) Mark a number of points on the parabola and determine the centres of curvature at these points (as shown at the point  $P$ ).
- (ii) Draw the evolute through these centres. Note that  $PF = FR$ .

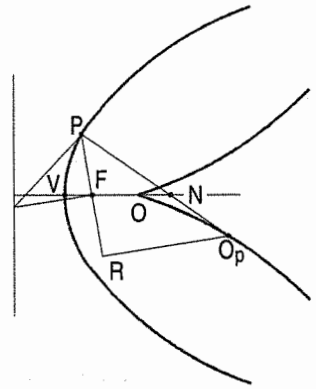


FIG. 6-58

**Problem 6-39.** (fig. 6-59): To draw the evolute of a hyperbola.

- (i) Mark a number of points on the hyperbola and determine the centres of curvature at these points (as shown at the point  $P$ ).
- (ii) Draw the evolute through these centres. To obtain the centre of curvature at the vertex, the position of the other focus  $F_1$  must be found.

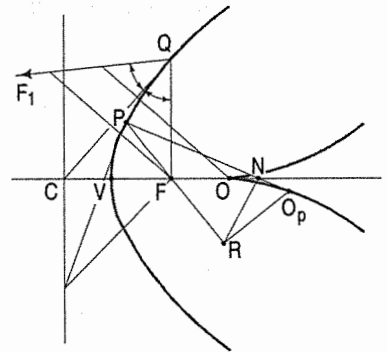


FIG. 6-59

It is determined by making use of the following rule:

The tangent at any point on the curve bisects the angle made by lines joining that point with the two foci, i.e.  $\angle F_1QC = \angle FQC$ .

**Problem 6-40.** (fig. 6-60): To draw the evolute of a cycloid.

- (i) Mark a point  $P$  on the cycloid and draw the normal  $PN$  to it (Problem 6-21).
- (ii) Produce  $PN$  to  $O_p$  so that  $NO_p = PN$ .  $O_p$  is the centre of curvature at the point  $P$ .
- (iii) Similarly, mark a number of points on the cycloid and determine centres of curvature at these points. The curve drawn through these centres is the evolute of the cycloid. It is an equal cycloid.

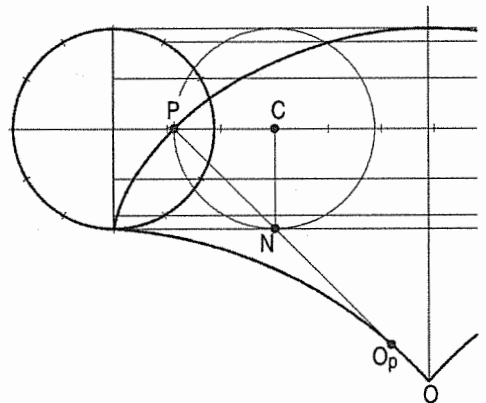


FIG. 6-60

**Problem 6-41.** (fig. 6-61): To draw the evolute of an epicycloid.

- (i) Mark a point  $P$  on the epicycloid and draw the normal  $PN$  to it (Problem 6-24 and problem 6-25).
- (ii) Draw the diameter  $PQ$  of the rolling circle. Join  $Q$  with  $O$ , the centre of the directing circle.
- (iii) Produce  $PN$  to cut  $QO$  at  $O_p$ , which is the centre of curvature at the point  $P$ .
- (iv) Mark a number of points on the epicycloid and similarly, obtain centres of curvature at these points. The curve drawn through these centres is the evolute of the epicycloid.

Through  $O_p$ , draw a line perpendicular to  $PO_p$  and intersecting the line joining  $C$  (the centre of the rolling circle) with  $O$  at a point  $R$ . The evolute is the epicycloid of the circle of diameter  $NR$ , rolling along the circle of radius  $OR$ .

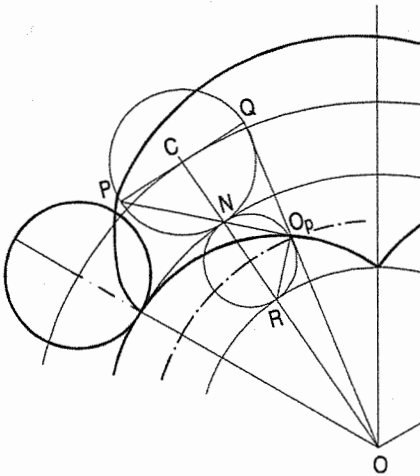


FIG. 6-61

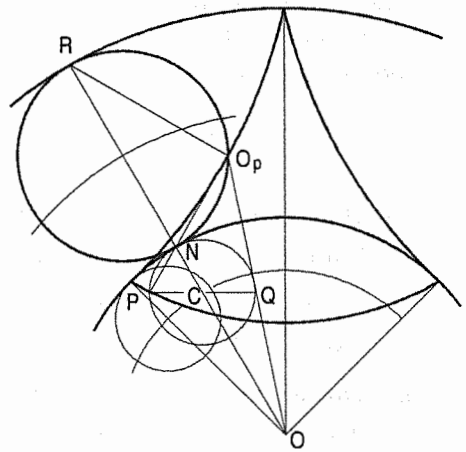


FIG. 6-62

**Problem 6-42.** (fig. 6-62): To draw the evolute of a hypocycloid.

- (i) Mark a point  $P$  on the hypocycloid and draw the normal  $PN$  to it (Problem 6-24 and problem 6-25).
- (ii) Draw the diameter  $PQ$  of the rolling circle. Join  $Q$  with  $O$ , the centre of the directing circle.
- (iii) Produce  $PN$  to cut  $OQ$ -produced at  $O_p$ , which is the centre of curvature at the point  $P$ .
- (iv) Mark a number of points on the hypocycloid and similarly, obtain centres of curvature at these points. The curve drawn through these centres is the evolute of the hypocycloid.

Through  $O_p$ , draw a line perpendicular to  $PO_p$  and intersecting  $OC$ -produced at a point  $R$ . The evolute is the hypocycloid of the circle of diameter  $NR$  rolling along the circle of radius  $OR$ .

**Problem 6-43.** (fig. 6-42): To draw the evolute of an involute of a circle.

In the involute of a circle, the normal  $NM$  at any point  $N$  is tangent to the circle at the point of contact  $M$ .  $M$  is the centre of curvature at the point  $N$ . Hence, the evolute of the involute is the circle itself.

## 6-5. SPIRALS

If a line rotates in a plane about one of its ends and if at the same time, a point moves along the line continuously in one direction, the curve traced out by the moving point is called a *spiral*. The point about which the line rotates is called a *pole*.

The line joining any point on the curve with the pole is called the *radius vector* and the angle between this line and the line in its initial position is called the *vectorial angle*.

Each complete revolution of the curve is termed the *convolution*. A spiral may make any number of convolutions before reaching the pole.

### 6-5-1. ARCHEMEDIAN SPIRAL

It is a curve traced out by a point moving in such a way that its movement towards or away from the pole is uniform with the increase of the vectorial angle from the starting line.

The use of this curve is made in teeth profiles of helical gears, profiles of cams etc.

**Problem 6-44.** (fig. 6-63): To construct an Archemedian spiral of  $1\frac{1}{2}$  convolutions, given the greatest and the shortest radii.

Let  $O$  be the pole,  $OP$  the greatest radius and  $OQ$  the shortest radius.

(i) With centre  $O$  and radius equal to  $OP$ , draw a circle.  $OP$  revolves around  $O$  for  $1\frac{1}{2}$  revolutions. During this period,  $P$  moves towards  $O$ , the distance equal to  $(OP - OQ)$  i.e.  $QP$ .

(ii) Divide the angular movements of  $OP$ , viz  $1\frac{1}{2}$  revolutions i.e.  $540^\circ$ , and the line  $QP$  into the same number of equal parts, say 18 (one revolution divided into 12 equal parts).

When the line  $OP$  moves through one division, i.e.  $30^\circ$ , the point  $P$  will move towards  $O$  by a distance equal to one division of  $QP$  to a point  $P_1$ .

(iii) To obtain points systematically draw arcs with centre  $O$  and radii  $O1, O2, O3$  etc. intersecting lines  $O1', O2', O3'$  etc. at points  $P_1, P_2, P_3$  etc. respectively.

In one revolution,  $P$  will reach the 12th division along  $QP$  and in the next half revolution it will be at the point  $P_Q$  on the line  $18'-O$ .

(iv) Draw a curve through points  $P, P_1, P_2, P_Q$ . This curve is the required Archemedian spiral.

**Normal and tangent to an Archemedian spiral:** The normal to an Archemedian spiral at any point is the hypotenuse of the right-angled triangle having the other two sides equal in length to the radius vector at that point and the constant of the curve respectively.

The constant of the curve is equal to the difference between the lengths of any two radii divided by the circular measure of the angle between them.

$OP$  and  $OP_3$  (fig. 6-63) are two radii making  $90^\circ$  angle between them. In circular measure in radian,  $90^\circ = \frac{\pi}{2} = 1.57$ . Therefore, the constant of the curve,  $C = \frac{OP - OP_3}{1.57}$ .

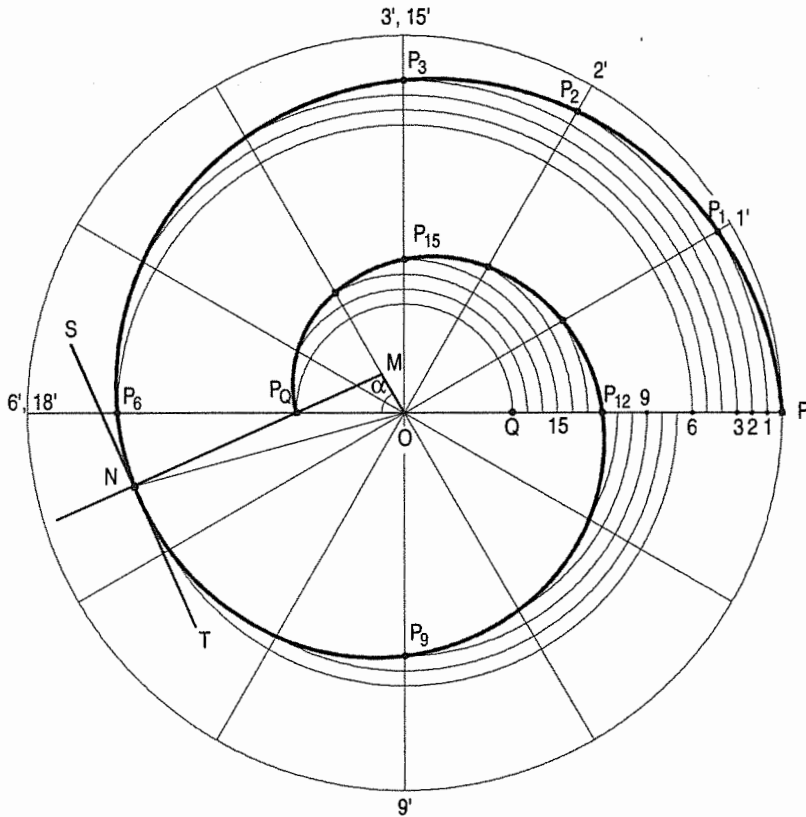
**Problem 6-45.** (fig. 6-63): To draw a normal to the Archemedian spiral at a point  $N$  on it.

(i) Draw the radius vector  $NO$ .

(ii) Draw a line  $OM$  equal in length to the constant of the curve  $C$  and perpendicular to  $NO$ .

(iii) Draw the line  $NM$  which is the normal to the spiral.

(iv) Through  $N$ , draw a line  $ST$  perpendicular to  $NM$ .  $ST$  is the tangent to the spiral.



Archimedean spiral

FIG. 6-63

**Problem 6-46.** (fig. 6-64): A link 225 mm long, swings on a pivot  $O$  from its vertical position of rest to the right through an angle of  $75^\circ$  and returns to its initial position at uniform velocity. During that period, a point  $P$  moving at uniform speed along the centre line of the link from a point at a distance of 25 mm from  $O$ , reaches the end of the link. Draw the locus of the point  $P$ .

- (i) Draw a vertical line  $OA$ , 225 mm long.
- (ii) With centre  $O$  and radius equal to  $OA$ , draw an arc.
- (iii) Draw a line  $OB$  making  $\angle AOB$  equal to  $75^\circ$  and cutting the arc at  $B$ .
- (iv) Mark a point  $P$  along  $OA$  and at a distance of 25 mm from  $O$ .
- (v) Divide the angular movement of the link and the line  $PA$  into the same number of equal parts, say 8.

The end  $A$  of the link moves to  $B$  and returns to its original position. Hence, the arc  $AB$  must be divided into four equal parts.

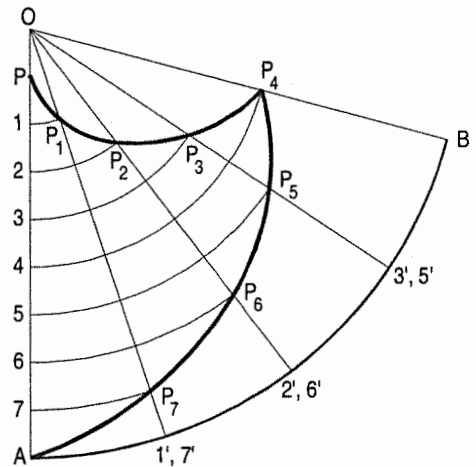


FIG. 6-64

- (vi) With centre  $O$  and radii  $O1, O2, O3$  etc., draw arcs intersecting lines  $O1', O2', O3'$  etc. at points  $P_1, P_2, P_3$  etc. respectively. Draw a curve through  $P, P_1, P_2, P_3, \dots, P_n$ . This curve is the locus of the point  $P$ .

**Problem 6-47.** (fig. 6-65): A straight link  $PQ$  of 60 mm length revolves one complete revolution with uniform motion in anti-clockwise direction about hinged  $P$ . During this period an insect moves along the link from  $P$  to  $Q$  and  $Q$  to  $P$  with uniform linear motion. Draw the path of the insect and name the curve.

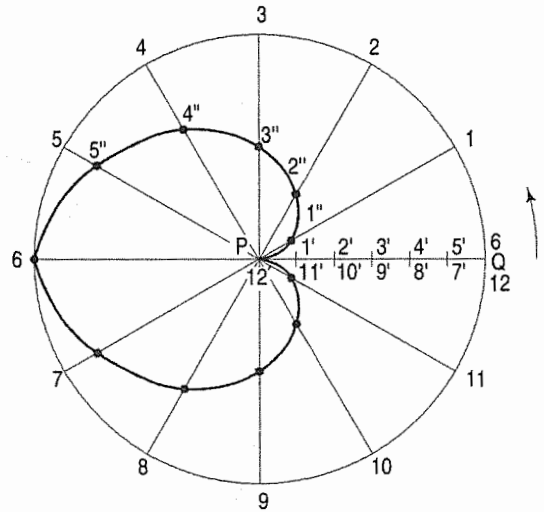


FIG. 6-65

- (i) Draw circle of 60 mm radius at  $P$  as centre and divide into twelve equal parts. Name in anti-clockwise direction 1, 2, 3, 4, 5..... 12.
- (ii) Divide line  $PQ$  into six equal part and mark as  $1', 2', 3', 4', 5'$  and  $6'$  and from  $Q$  as  $7'$  to  $12'$ .
- (iii) Mark  $1'', 2'', 3'', 4'', 5''$  and  $6''$  along radius  $P1, P2, P3, P4, P5$  and  $P6$  respectively increasing each division of the line  $PQ$  as shown in fig 6-65.
- (iv) Bottom portion is the mirror image of top portion. So drop the respective point on bottom radii.
- (v) Join them with the smooth curve. The curve is archmedian spiral.

### 6-5-2. LOGARITHMIC OR EQUIANGULAR SPIRAL



In a *logarithmic spiral*, the ratio of the lengths of consecutive radius vectors enclosing equal angles is always constant. In other words the values of vectorial angles are in *arithmetical progression* and the corresponding values of *radius vectors* are in *geometrical progression*.

The logarithmic spiral is also known as *equiangular spiral* because of its property that the angle which the tangent at any point on the curve makes with the radius vector at that point is constant.

**Problem 6-48.** (fig. 6-66): To construct a logarithmic spiral of one convolution, given the length of the shortest radius and the ratio of the lengths of radius vectors enclosing an angle  $\theta$ .

Assume the shortest radius be 1 cm long,  $\theta$  equal to  $30^\circ$  and the ratio  $\frac{10}{9}$ .

The lengths of radius vectors are determined from the scale which is constructed as shown below:

- (i) Draw lines  $AB$  and  $AC$  [fig. 6-66(ii)] making an angle of  $30^\circ$  between them.
- (ii) On  $AB$ , mark  $AD$ , 1 cm long. On  $AC$ , mark a point  $E$  such that  $AE = \frac{10}{9} \times 1 \text{ cm} = \frac{10}{9} \text{ cm}$ .

(iii) Draw a line joining  $D$  with  $E$ .

(iv) With centre  $A$  and radius  $AE$ , draw an arc cutting  $AB$  at a point 1. Through 1, draw a line  $1-1'$  parallel to  $DE$  and cutting  $AC$  at  $1'$ . Again, with centre  $A$  and radius  $A1'$  draw an arc cutting  $AB$  at 2. Through 2, draw a line  $2-2'$  parallel to  $DE$  and cutting  $AC$  at  $2'$ . Repeat the construction and obtain points 3, 4.....12.  $A1, A2$  etc. are the lengths of consecutive radius vectors.

(v) Draw a horizontal line  $OQ$  and on it, mark  $OP$ , 1 cm long [fig. 6-66(i)]. Through  $O$ , draw radial lines making  $30^\circ$  angles between two consecutive lines. These are the radius vectors.

(vi) Mark points  $P_1, P_2, \dots, P_{12}$  on consecutive radius vectors such that  $OP_1 = A1, OP_2 = A2, \dots, OP_{12} = A12$ .

(vii) Draw a smooth curve through  $P_1, P_2, \dots, P_{12}$ . This curve is the required logarithmic spiral.

The equation to the logarithmic spiral is  $r = a^\theta$  where  $r$  is the radius vector,  $\theta$  is the vectorial angle and  $a$  is a constant.

Hence,  $\log r = \theta \log a$ .

If  $\theta = 0, \log r = 0 \therefore r = 1$ .

When  $\theta = 30^\circ = \frac{\pi}{6}$  radians,  $r = \frac{10}{9} \times 1 = \frac{10}{9} \therefore \log \frac{10}{9} = \frac{\pi}{6} \log a$

i.e.  $\log a = \frac{6}{\pi} \log \frac{10}{9} \therefore a = 1.22$ .

**Problem 6-49.** [fig. 6-66(i)]: To draw a normal and a tangent at a given point  $N$  on the logarithmic spiral in problem 6-48.

The value of the tangent to the constant angle is obtained by the formula given below:

$$\tan \alpha = \frac{\log e}{\log a}, \quad \text{where } e = 2.718$$

$$\text{i.e. } \tan \alpha = \frac{\log 2.718}{\frac{6}{\pi} \log \frac{10}{9}} \therefore \alpha = 78^\circ - 38'$$

(i) Draw a line joining  $N$  with  $O$ .

(ii) Draw a line  $NT$  making an angle of  $78^\circ - 38'$  with  $NO$ .  $NT$  is the required tangent.  $NR$  drawn perpendicular to  $NT$  is the normal to the spiral.

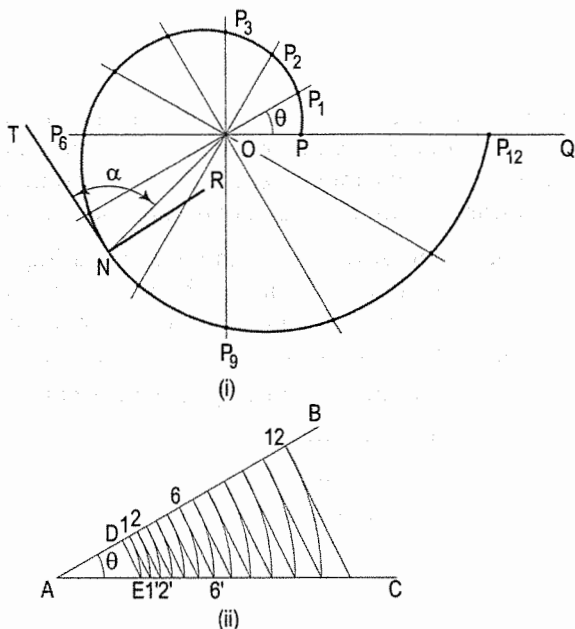


FIG. 6-66

## 6-6. HELIX

*Helix* is defined as a curve, generated by a point, moving around the surface of a right circular cylinder or a right circular cone in such a way that, its axial advance, i.e. its movement in the direction of the axis of the cylinder or the cone is uniform with its movement around the surface of the cylinder or the cone.

The axial advance of the point during one complete revolution is called the *pitch* of the helix. If the pitch is say 20 mm and the point starts upwards from the base of the cylinder, in one-fourth of a revolution, the point will move up a distance of 5 mm from the base. We shall now learn in the articles to follow:

1. A method of drawing a helical curve
2. Helical springs
3. Screw threads
4. Helix upon a cone.

### 6-6-1. A METHOD OF DRAWING A HELICAL CURVE

**Problem 6-50.** (fig. 6-67): To draw a helix of one convolution, given the pitch and the diameter of the cylinder. Also to develop the surface of the cylinder showing the helix in it.

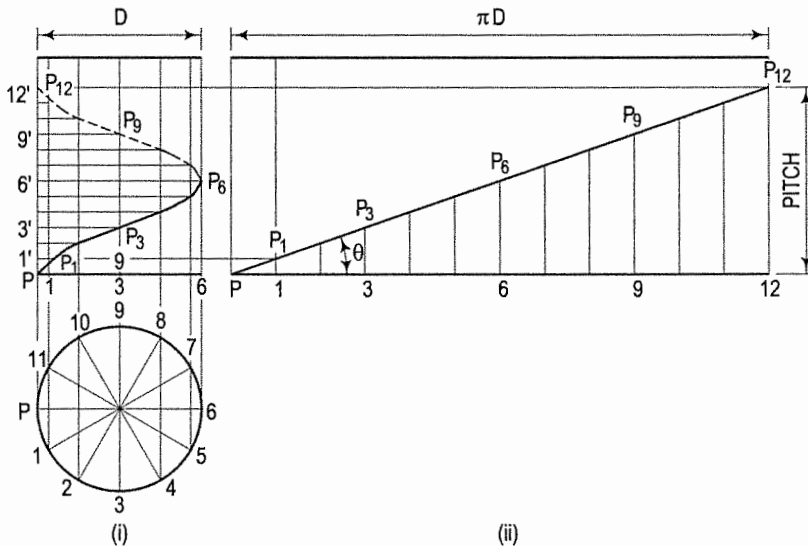


FIG. 6-67

- (i) Draw the projections of the cylinder [fig. 6-67(i)]. Divide the circle (in the top view) into any number of equal parts, say 12.
- (ii) Mark a length  $P = 12'$  equal to the given pitch, along a vertical side of the rectangle in the front view and divide it into the same number of equal parts, viz. 12.
- (iii) Assume the point  $P$  to move upwards and in anti-clockwise direction. When it has moved through  $30^\circ$  around the circle, it will have moved up by one division. To locate this position, draw a vertical line through the point 1 and a horizontal line through the point 1', both intersecting each other at a point  $P_1$ , which will be on the helix.

- (iv) Obtain other points in the same manner and draw the helix through them. The portion of the curve from  $P_6$  to  $P_{12}$  will be on the back side of the cylinder and hence it will not be visible.

Fig. 6-67(ii) shows the development of the surface of the cylinder. The helix is seen as a straight line and is the hypotenuse of a right-angled triangle having base equal to the circumference of the circle and the vertical side equal to the pitch of the helix. The angle  $\theta$  which it makes with the base, is called the *helix angle*. The helix angle can be expressed as  $\tan\theta = \frac{\text{pitch}}{\text{circumference of the circle}}$ .

## 6-6-2. HELICAL SPRINGS

In a spring having a wire of square cross-section, the outer two corners of the section may be assumed to be moving around the axis, on the surface of a cylinder having a diameter equal to the outside diameter of the spring. The inner two corners of the section will move on the surface of a cylinder having a diameter, equal to the inside diameter of the spring. The pitch in case of each corner will be the same.

- (1) Helical spring of a wire of square cross-section.
- (2) Helical spring of a wire of circular cross-section.

### (1) Helical spring of a wire of square cross-section.

**Problem 6-51.** (fig. 6-68): Draw the projections of two complete turns of a spring of a wire of square section of 20 mm side. Outside diameter of the spring = 110 mm; pitch = 60 mm.

Instead of full circles, semi-circles of 110 mm and 70 mm diameters for the outside and the inside diameters of the spring may be drawn in the top view.

- (i) Divide each semi-circle into  $\frac{12}{2}$ , i.e. 6 equal parts.
- (ii) Project up the division-points as lines in the front view. Let PQSR be the square section of the wire.
- (iii) In one revolution, the point P will move to  $P'$  so that  $PP' = 60$  mm. Similarly,  $RR'$ ,  $QQ'$  and  $SS'$  will each be equal to 60 mm.
- (iv) Divide these distances into 12 equal parts and plot the helices traced out by P and R on the outer surface and by Q and S on the inner surface, as shown. The outer helices will be parallel to each other. Note carefully the visible and the hidden parts of the curves.

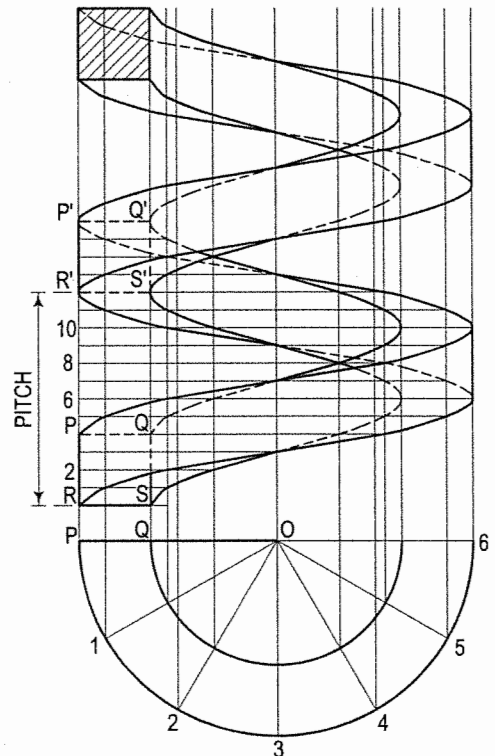


FIG. 6-68

**(2) Helical spring of a wire of circular cross-section.**

When the wire is of circular cross-section, a helical curve for the centre of the cross-section is first traced out. A number of circles of diameter equal to that of the cross-section are then drawn with a number of points on this curve as centres. Curves, tangent to these circles, will give the front view of the spring.

**Problem 6-52.** (fig. 6-69): Draw two complete turns of a helical spring of circular cross-section of 20 mm diameter. Spring index and pitch are 6 and 60 mm respectively.

$$\text{Spring index} = \frac{\text{Mean diameter of a coil}}{\text{Diameter of wire}}$$

$$\begin{aligned} \text{(a) Mean diameter of a coil} &= \text{Spring index} \times \text{Diameter of wire} \\ &= 6 \times 20 \\ &= 120 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{(b) Outside diameter of a coil} &= \text{Mean diameter} + \text{Diameter of wire} \\ &= 120 \text{ mm} + 20 \text{ mm} \\ &= 140 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{(c) Inside diameter of a coil} &= \text{Mean diameter} - \text{Diameter of wire} \\ &= 120 \text{ mm} - 20 \text{ mm} \\ &= 100 \text{ mm.} \end{aligned}$$

- (i) Draw the three concentric circles of diameters 140 mm, 120 mm and 100 mm representing the outside, mean and the inside circles of coil of the spring respectively in the top view.
- (ii) Divide the circles into 12 divisions. Mark the divisions on the circle of the mean diameter as 1, 2, 3 etc.
- (iii) Draw the projectors passing through these divisions in the front view. Mark the pitch length along one of the projectors which is parallel to the axis of the spring.
- (iv) Divide the pitch length into 12 divisions and name these division points as 1', 2', 3' etc. Draw the horizontal lines intersecting the previously drawn projectors from 1, 2, 3 etc.
- (v) With the intersection point as centre and radius equal to 10 mm, draw circles. Draw the curve touching the top and bottom of circle of the wire as shown in fig. 6-69. Repeat the process for one more pitch length.

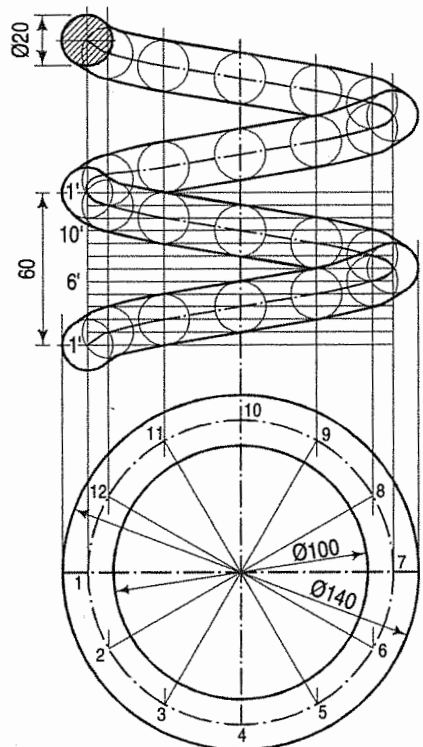


FIG. 6-69

### 6-6-3. SCREW THREADS

These are also constructed on the principle of the helix.

In a screw thread, the *pitch* is defined as the distance from a point on a thread to a corresponding point on the adjacent thread, measured parallel to the axis. The axial advance of a point on a thread, per revolution, is called the *lead* of the screw.

In the single-threaded screws, which are most commonly used in practice, the pitch is equal to lead. Therefore, the pitch of the screw is equal to the pitch of the helix. Unless stated otherwise, screws are *always assumed* to be single-threaded.

**Problem 6-53.** (fig. 6-70): Project two complete turns of a square thread having outside diameter 120 mm and pitch 45 mm.

In a square thread, the thickness of the thread = the depth of the thread =  $0.5 \times \text{pitch}$ . Hence, the section of the thread will be a square of 22.5 mm side and the diameter at the bottom of the thread will be 75 mm.

Project the threads in the same manner as the spring in problem 6-51. The screw differs from the spring in having a solid cylinder inside, which completely hides the back portions of the curves.

In double-threaded screws, two threads of the same size run parallel to each other. The axial advance per revolution, viz. the lead, is made twice the lead of the single-threaded screw, the pitch of the thread being kept the same in both cases.

Hence, in double-threaded screws,

Pitch of the helix = Lead = Twice the pitch of the screw.

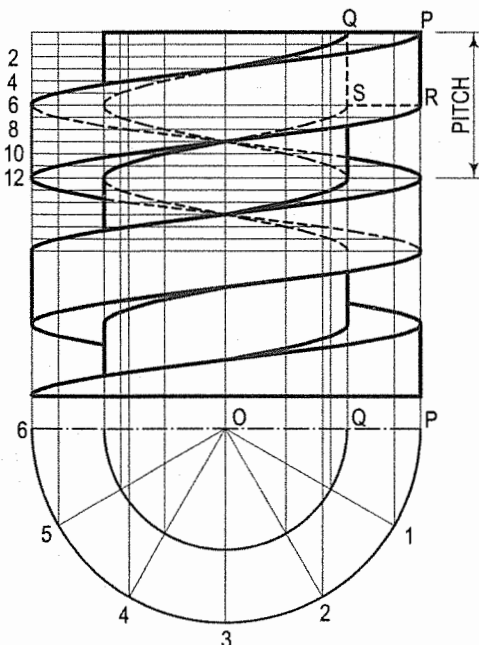


FIG. 6-70

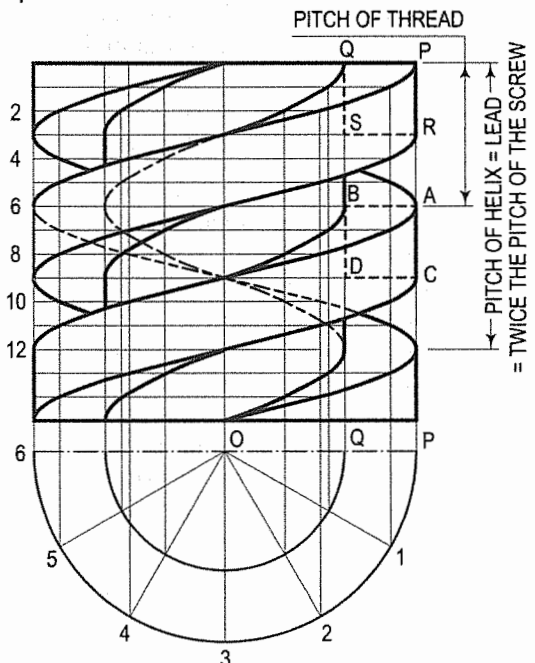


FIG. 6-71

Fig. 6-71 shows a double-threaded screw, having the same cross-section and the same pitch of the screw as in problem 6-53.

### 6-6-4. HELIX UPON A CONE



This curve is traced out by a point which, while moving around the axis and on the surface of the cone, approaches the apex. The movement around the axis is uniform with its movement towards the apex, measured parallel to the axis. The pitch of the helix is measured parallel to the axis of the cone.

As the whole surface of the cone is visible in its top view, the helix will be fully seen in it.

**Problem 6-54.** (fig. 6-72): Draw a helix of one convolution upon a cone, diameter of base 75 mm, height 100 mm and pitch 75 mm. Also develop the surface of the cone, showing the helix on it.

- (i) Draw the projections of the cone as shown. Divide the circle into twelve equal parts and join points 1, 2 etc. with  $o$ . Project these points to the base-line in the front view and join them with  $o'$ .
- (ii) Mark a point  $A$  on the axis at a distance of 75 mm from the base. Draw a horizontal line through  $A$  to cut the generators  $o'P$  at  $A'$ . Divide  $PA'$  into twelve equal parts.
- (iii) Let  $P$  be the starting point. When it has moved around through  $30^\circ$ , it will have moved up through one division to a point  $p'_1$  on the generator  $o'1'$ , obtained by drawing a horizontal line through  $1''$ .  $p'_1$  will be a point on the helix in the front view. Its projection to  $p_1$  on  $o1$ , will be the position of the point in the top view.
- (iv) Obtain all the points in this manner and draw smooth curves through them in both the views.

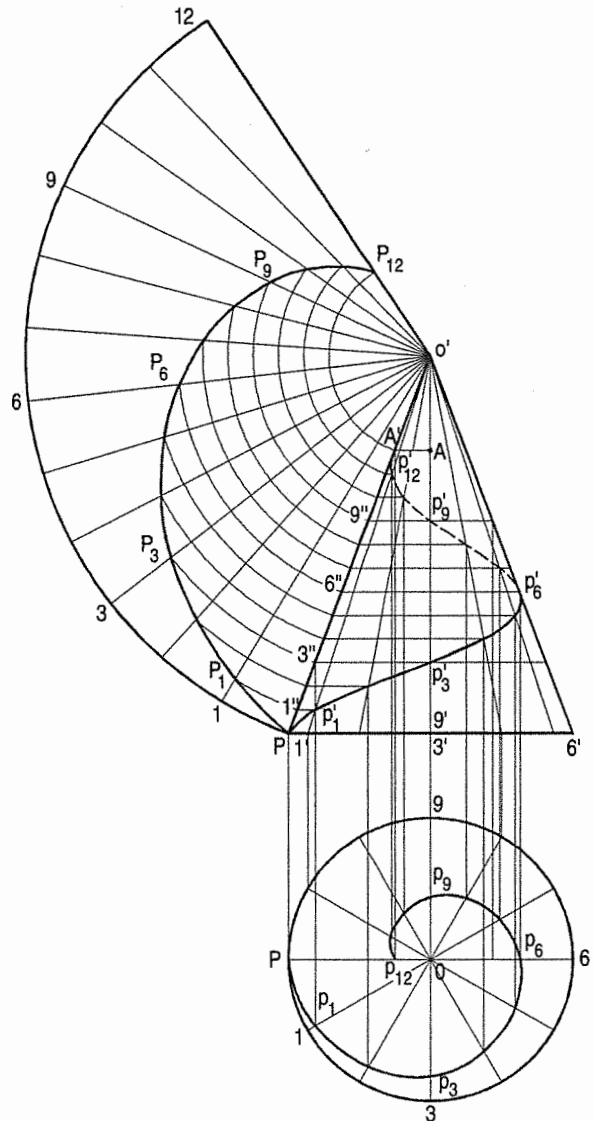


FIG. 6-72

- (v) Draw the development of the surface of the cone and locate points  $p_1$ ,  $p_2$  etc.

**Problem 6-55.** (fig. 6-73): A conical spring of a bicycle's seat has following specifications. Draw the front view and the top view of the spring. Only one-turn is sufficient.

- (i) Outer diameter of the coil at the bottom ..... 72 mm
- (ii) Outer diameter of the coil at the top ..... 42 mm
- (iii) Wire diameter ..... 10 mm
- (iv) Pitch ..... 60 mm.
- (a) Draw a circle of mean diameter of 72 mm in the top view.
- (b) Divide the circle into 12 divisions. Mark the divisions 1, 2, 3 etc. as shown in fig. 6-73.
- (c) Draw the front view of the cone with the base length 72 mm.
- (d) Draw pitch length 60 mm along the line parallel to the axis of the cone. Divide this pitch length into 12 divisions and number them as 1', 2', 3' etc.
- (e) Draw the vertical projectors from these points 1, 2, 3 etc. to intersect the horizontal lines from the points 1', 2', 3' etc. With the intersection point as centre and radius equal to 5 mm, draw the circles and the smooth curve touching to the top and bottom of the circles as shown in fig. 6-73.

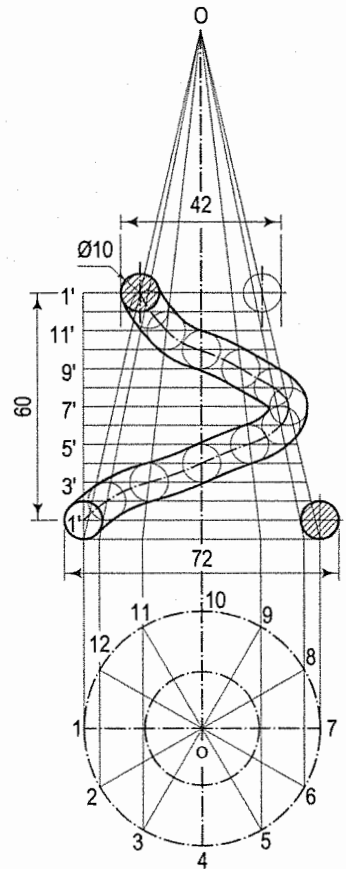


FIG. 6-73

## 6-7. CAM

A cam is a machine-part which, while rotating at uniform velocity, imparts reciprocating linear motion or oscillating motion to another machine-part called a follower. The motion imparted may be either uniform or variable, depending upon the shape of the cam profile.

The shape of the cam to transmit uniform linear motion is determined by the application of the principle of Archimedean spiral, as shown in problem 6-44.

The cams are widely used in automates, printing machines, an I.C. engines etc. The shape of cam depends upon the motion required for the followers. The terminology of the cam and follower is shown in fig. 6-74.

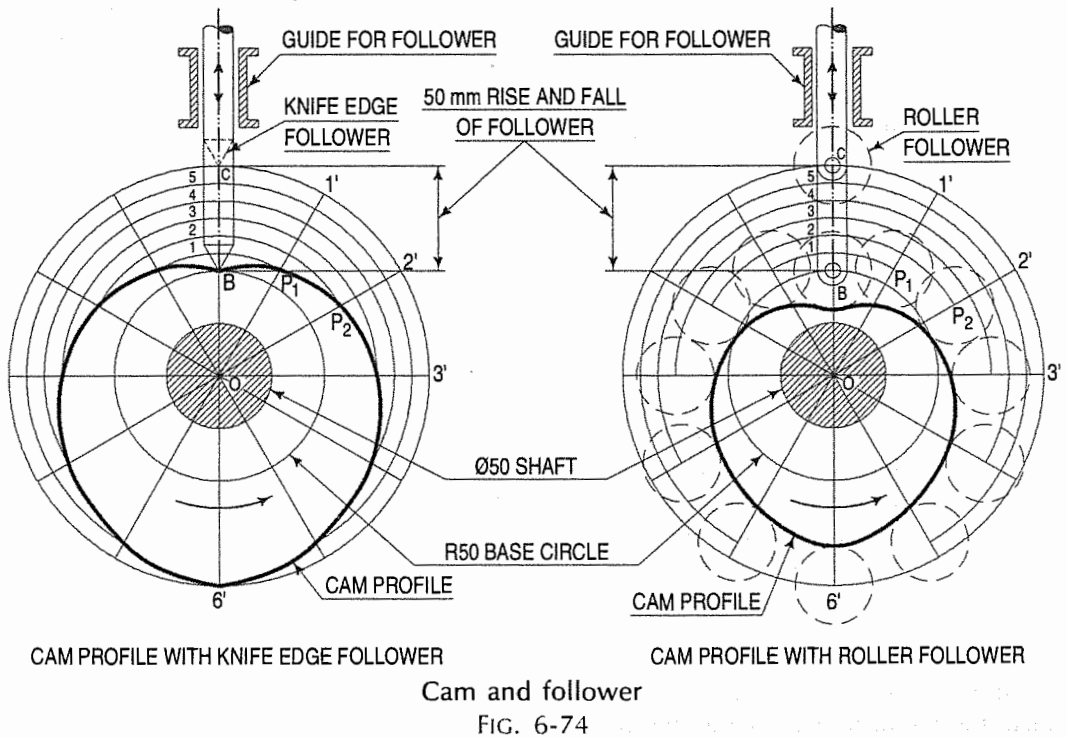
**Problem 6-56.** (fig. 6-74): Draw the shape of a cam to give a uniform rise and fall of 50 mm to a point, during each revolution of the cam. Diameter of the shaft = 50 mm; least radius of the cam = 50 mm.

- (i) Draw a circle (for the shaft) with centre O. With the same centre and radius OB equal to 50 mm (the least radius of the cam), draw another circle.
- (ii) Produce OB to C making BC equal to 50 mm (the rise of the point). The point is to be uniformly raised during  $\frac{1}{2}$  revolution of the shaft.
- (iii) Therefore, divide BC and angle  $180^\circ$  into the same number of equal parts, say 6.

Obtain points  $P_1$ ,  $P_2$  etc. as in the Archimedean spiral and draw the curve through them.

The point is to fall the same distance during the same period. Hence, the curve will be exactly of the same shape as for the rise.

The followers are generally provided with rollers to give smooth working. In such cases, the profile of the cam is designed initially, to transmit motion to a point (the centre of the roller) and then, a parallel curve is drawn inside it at a distance equal to the radius of the roller. This is done by first drawing a number of arcs with points on the curve as centres and radius equal to radius of the roller, and then drawing a smooth curve touching these arcs, as shown in fig. 6-74.



Cam and follower

FIG. 6-74

## EXERCISES 6

1. Draw a straight line  $AB$  of any length. Mark a point  $F$ , 65 mm from  $AB$ . Trace the paths of a point  $P$  moving in such a way, that the ratio of its distance from the point  $F$ , to its distance from  $AB$  is (i) 1; (ii) 3:2; (iii) 2:3. Plot at least 8 points. Name each curve. Draw a normal and a tangent to each curve at a point on it, 50 mm from  $F$ .
2. A fixed point is 75 mm from a fixed straight line. Draw the locus of a point  $P$  moving such a way that its distance from the fixed straight line is (i) twice its distance from the fixed point; (ii) equal to its distance from the fixed point. Name the curves.
3. The vertex of a hyperbola is 65 mm from its focus. Draw the curve if the eccentricity is  $\frac{3}{2}$ . Draw a normal and a tangent at a point on the curve, 75 mm from the directrix.

4. The major axis of an ellipse is 150 mm long and the minor axis is 100 mm long. Find the foci and draw the ellipse by 'arcs of circles' method. Draw a tangent to the ellipse at a point on it 25 mm above the major axis.
5. The foci of an ellipse are 90 mm apart and the minor axis is 65 mm long. Determine the length of the major axis and draw half the ellipse by concentric-circles method and the other half by oblong method. Draw a curve parallel to the ellipse and 25 mm away from it.
6. The major axis of an ellipse is 100 mm long and the foci are at a distance of 15 mm from its ends. Find the minor axis. Prepare a trammel and draw the ellipse using the same.
7. Two fixed points *A* and *B* are 100 mm apart. Trace the complete path of a point *P* moving (in the same plane as that of *A* and *B*) in such a way that, the sum of its distances from *A* and *B* is always the same and equal to 125 mm. Name the curve. Draw another curve parallel to and 25 mm away from this curve.
8. Inscribe an ellipse in a parallelogram having sides 150 mm and 100 mm long and an included angle of  $120^\circ$ .
9. Two points *A* and *B* are 100 mm apart. A point *C* is 75 mm from *A* and 60 mm from *B*. Draw an ellipse passing through *A*, *B* and *C*.
10. Draw a rectangle having its sides 125 mm and 75 mm long. Inscribe two parabolas in it with their axis bisecting each other.
11. A ball thrown up in the air reaches a maximum height of 45 metres and travels a horizontal distance of 75 metres. Trace the path of the ball, assuming it to be parabolic.
12. A point *P* is 30 mm and 50 mm respectively from two straight lines which are at right angles to each other. Draw a rectangular hyperbola from *P* within 10 mm distance from each line.
13. Two straight lines *OA* and *OB* make an angle of  $75^\circ$  between them. *P* is a point 40 mm from *OA* and 50 mm from *OB*. Draw a hyperbola through *P*, with *OA* and *OB* as asymptotes, marking at least ten points.
14. Two points *A* and *B* are 50 mm apart. Draw the curve traced out by a point *P* moving in such a way that the difference between its distances from *A* and *B* is always constant and equal to 20 mm.
15. A circle of 50 mm diameter rolls along a straight line without slipping. Draw the curve traced out by a point *P* on the circumference, for one complete revolution of the circle. Name the curve. Draw a tangent to the curve at a point on it 40 mm from the line.
16. Two points *Q* and *S* lie on a straight line through the centre *C* of a circle of 50 mm diameter, rolling along a fixed straight line. Draw and name the curves traced out by the points *Q* and *S* during one revolution of the circle.  $CQ = 20$  mm;  $CS = 35$  mm.
17. A circle of 50 mm diameter rolls on the circumference of another circle of 175 mm diameter and outside it. Trace the locus of a point on the circumference of the rolling circle for one complete revolution. Name the curve. Draw a tangent and a normal to the curve at a point 125 mm from the centre of the directing circle.
18. Construct a hypocycloid, rolling circle 50 mm diameter and directing circle 175 mm diameter. Draw a tangent to it at a point 50 mm from the centre of the directing circle.

19. A circle of 115 mm diameter rolls on another circle of 75 mm diameter with internal contact. Draw the locus of a point on the circumference of the rolling circle for its one complete revolution.
20. A circle of 50 mm diameter rolls on another circle of 175 mm diameter. Draw and name the curves traced out by two points  $Q$  and  $S$  lying on a straight line through the centre  $C$  of the rolling circle and respectively 20 mm and 35 mm from it, when it rolls (i) outside and (ii) inside the other circle.
21. Show by means of a drawing that when the diameter of the directing circle is twice that of the generating circle, the hypocycloid is a straight line. Take the diameter of the generating circle equal to 50 mm.
22. A circle of 50 mm diameter rolls on a horizontal line for a half revolution and then on a vertical line for another half revolution. Draw the curve traced out by a point  $P$  on the circumference of the circle.
23. Draw an involute of a circle of 40 mm diameter. Also, draw a normal and a tangent to it at a point 100 mm from the centre of the circle.
24. An inelastic string 145 mm long, has its one end attached to the circumference of a circular disc of 40 mm diameter. Draw the curve traced out by the other end of the string, when it is completely wound around the disc, keeping the string always tight.
25. Draw a circle with diameter  $AB$  equal to 65 mm. Draw a line  $AC$  150 mm long and tangent to the circle. Trace the path of  $A$ , when the line  $AC$  rolls on the circle, without slipping.
26.  $AB$  is a rope 1.6 metres long, tied to a peg at  $B$  (fig. 6-75). Keeping it always tight, the rope is wound round the pole  $O$ . Draw the curve traced out by the end  $A$ . Scale  $\frac{1}{10}$  full size.
27. Draw an Archimedean spiral of two convolutions, the greatest and the least radii being 115 mm and 15 mm respectively. Draw a tangent and a normal to the spiral at a point, 65 mm from the pole.
28. A point  $P$  moves towards another point  $O$ , 75 mm from it and reaches it while moving around it once, its movement towards  $O$  being uniform with its movement around it. Draw the curve traced out by the point  $P$ .
29. A link  $OA$ , 100 mm long rotates about  $O$  in anti-clockwise direction. A point  $P$  on the link, 15 mm away from  $O$ , moves and reaches the end  $A$ , while the link has rotated through  $\frac{2}{5}$  of a revolution. Assuming the movements of the link and the point to be uniform, trace the path of the point  $P$ .
30. Draw a cam to give the following uniform motions to a point: Rise 40 mm in  $90^\circ$ . In upper position for  $75^\circ$ . Fall the same distance in  $120^\circ$ . In lower position for the remaining period of the revolution. Diameter of shaft = 50 mm. Least metal = 20 mm.
31. A circle of 40 mm diameter rolls on the inside of a circle of 90 mm diameter. A point  $P$  lies within the rolling circle at a distance of 15 mm from its centre. Trace the path of the point  $P$  for one revolution of the circle.

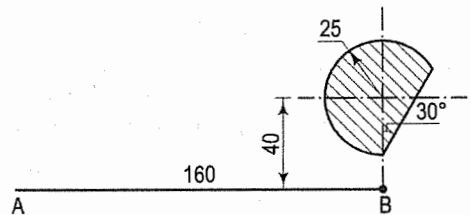


FIG. 6-75

32. The distance between the foci of an ellipse is 100 mm and the minor axis is 65 mm long. Draw the evolute of the ellipse.
33. Draw the evolutes of the two curves, the data of which is given in Ex. 2.
34. Construct the evolute of the hyperbola whose data is given in Ex. 3.
35. Draw the evolute of the cycloid obtained in Ex. 15.
36. Draw the epicycloid and hypocycloid, when the generating circle and the directing circle are of 50 mm and 175 mm diameters respectively. Construct the evolutes of the two curves.
37. In a logarithmic spiral, the shortest radius is 40 mm. The length of adjacent radius vectors enclosing  $30^\circ$  are in the ratio 9.8. Construct one revolution of the spiral. Draw a tangent to the spiral at any point on it.
38. Draw a triangle  $ABC$  with  $AB = 30$  mm,  $AC = 40$  mm and  $\angle BAC = 45^\circ$ .  $B$  and  $C$  are the points on an Archimedian spiral of one convolution of which  $A$  is the pole. Find the initial line and draw the spiral.
39.  $ABC$  is an equilateral triangle of side equal to 70 mm. Trace the loci of the vertices  $A$ ,  $B$  and  $C$  when the circle circumscribing  $ABC$  rolls without slipping along a fixed straight line for one complete revolution.
40. Draw the shape of the cam to give the same motions as in Ex. 30 to a roller of 25 mm diameter.
41. A point is raised by a cam uniformly 25 mm in  $\frac{3}{8}$  of a revolution; kept at the same height for  $\frac{1}{8}$  of the revolution; then allowed to drop through 10 mm height and remain there for  $\frac{1}{8}$  of the revolution; lowered uniformly to its original position in  $\frac{1}{4}$  of the revolution and kept there for the rest of the revolution. Draw the shape of the cam. Diameter of shaft = 40 mm. Least metal = 25 mm.
42. Two pegs  $A$  and  $B$  are fixed on the ground 6 metres apart. A string, 8 metres long has its one end tied to the peg  $A$ , while the other end is passed through a small ring  $C$  and tied to the peg  $B$ . A link  $CD$  of 1 metre length is fixed to the ring  $C$ . The link with the ring slides along the string. Draw the boundary of the area of the ground beyond which  $D$  cannot reach, as  $C$  slides along the string  $AB$ .
43. Fill in the blanks in the following with appropriate words selected from the list given below:
  - (a) When a cone is cut by planes at different angles, the curves of intersection are called \_\_\_\_\_.
  - (b) When the plane makes the same angle with the axis as do the generators, the curve is a \_\_\_\_\_.
  - (c) When the plane is perpendicular to the axis, the curve is a \_\_\_\_\_.
  - (d) When the plane is parallel to the axis, the curve is a \_\_\_\_\_.
  - (e) When the plane makes an angle with the axis greater than what do the generators, the curve is a \_\_\_\_\_.
  - (f) A conic is a locus of a point moving in such a way that the ratio of its distance from the \_\_\_\_\_ and its distance from the \_\_\_\_\_ is always constant. This ratio is called the \_\_\_\_\_. It is \_\_\_\_\_ in case of parabola, \_\_\_\_\_ in case of hyperbola and \_\_\_\_\_ in case of ellipse.

- (g) In a conic the line passing through the fixed point and perpendicular to the fixed line is called the \_\_\_\_\_.
- (h) The vertex is the point at which the \_\_\_\_\_ cuts the \_\_\_\_\_.
- (i) The sum of the distances of any point on the \_\_\_\_\_ from its two foci is always the same and equal to the \_\_\_\_\_.
- (j) The distance of the ends of the \_\_\_\_\_ of an ellipse from the \_\_\_\_\_ is equal to half the \_\_\_\_\_.
- (k) In a \_\_\_\_\_ the product of the distances of any point on it from two fixed lines at right angles to each other is always constant. The fixed lines are called \_\_\_\_\_.
- (l) Curves generated by a fixed point on the circumference of a circle rolling along a fixed line or circle are called \_\_\_\_\_.
- (m) The curve generated by a point on the circumference of a circle, rolling along another circle inside it, is called a \_\_\_\_\_.
- (n) The curve generated by a point on the circumference of a circle rolling along a straight line is called a \_\_\_\_\_.
- (o) The curve generated by a point on the circumference of a circle rolling along another circle outside it, is called \_\_\_\_\_.
- (p) The curve generated by a point fixed to a circle outside its circumference, as it rolls along a straight line is called \_\_\_\_\_.
- (q) The curve generated by a point fixed to a circle inside its circumference, as it rolls along a circle inside it, is called \_\_\_\_\_.
- (r) The curve generated by a point fixed to a circle outside its circumference as it rolls along a circle outside it, is called \_\_\_\_\_.
- (s) The curve traced out by a point on a straight line which rolls, without slipping, along a circle or a polygon is called \_\_\_\_\_.
- (t) The curve traced out by a point moving in a plane in one direction, towards a fixed point while moving around it is called a \_\_\_\_\_.
- (u) The line joining any point on the spiral with the pole is called \_\_\_\_\_.
- (v) In \_\_\_\_\_ the ratio of the lengths of consecutive radius vectors enclosing equal angles is always constant.

List of words for Ex. 43:

- |               |                    |                   |                    |
|---------------|--------------------|-------------------|--------------------|
| 1. Asymptotes | 9. Equal to 1      | 17. Hypotrochoid  | 25. Rectangular    |
| 2. Axis       | 10. Epitrochoid    | 18. Involute      | 26. Smaller than 1 |
| 3. Cycloidal  | 11. Ellipse        | 19. Inferior      | 27. Superior       |
| 4. Conic      | 12. Eccentricity   | 20. Logarithmic   | 28. Spiral         |
| 5. Circle     | 13. Focus          | 21. Minor axis    | 29. Trochoid       |
| 6. Cycloid    | 14. Greater than 1 | 22. Major axes    | 30. Conics         |
| 7. Directrix  | 15. Hyperbola      | 23. Parabola      | 31. Curves         |
| 8. Epicycloid | 16. Hypocycloid    | 24. Radius vector |                    |

Answers to Ex. 43:

- (a) – 30 (b) – 23 (c) – 5 (d) – 15 (e) – 11 (f) – 13, 7, 12, 9, 14 and 26 (g) – 2 (h) – 4 and 2 (i) – 11 and 22 (j) – 21, 13 and 22 (k) – 25, 15 and 1 (l) – 3 and 31 (m) – 16 (n) – 6 (o) – 8 (p) – 27 and 29 (q) – 19 and 17 (r) – 27 and 10 (s) – 18 (t) – 28 (u) – 24 (v) – 20 and 28.

**Miscellaneous problems:**

44.  $ABC$  is an equilateral triangle of side equal to 70 mm. Trace the loci of the vertices  $A$ ,  $B$  and  $C$ , when the circle circumscribing  $ABC$  rolls without slipping along a fixed straight line for one half revolution.
45. The major-axis  $AB$  of an ellipse is 140 mm long with  $P$  as its mid-point. The foci  $F_1$  and  $F_2$  of the ellipse are 48 mm away from the mid-point  $P$ . Draw the ellipse and find the length of the minor axes.
46. Draw an ellipse by 'concentric circles method' and find the length of the minor axis with the help of the following data: (i) Major axes = 100 mm. (ii) Distance between foci 80 mm.
47.  $PQ$  is a diameter of a circle and is 75 mm long. A piece of string is tied tightly round the circumference of the semi-circle starting from  $P$  and finishing at  $Q$ . The end  $Q$  is then untied and the string, always kept taut, is gradually unwound from the circle, until it lies along the tangent at  $P$ . Draw the curve traced by the moving extremity of the string.
48. A half-cone is standing on its half base on the ground with the triangular face parallel to the V.P. An inextensible string is passed round the half-cone from a point on the periphery and brought back to the same point. Find the shortest length of the string. Take the base-circle diameter of the half-cone as 60 mm and height 75 mm.
49. One end of an inelastic thread of 120 mm length is attached to one corner of a regular hexagonal disc having a side of 25 mm. Draw the curve traced out by the other end of the thread when it is completely wound along the periphery of the disc, keeping the thread always tight.
50. A circus man rides a motor-bike inside a globe of 6 metres diameter. The motor-bike has the wheel of 1 metre diameter. Draw the locus of the point on the circumference of the motor-bike wheel for one complete revolution. Adopt suitable scale.
51. A fountain jet discharges water from ground level at an inclination of  $45^\circ$  to the ground. The jet travels a horizontal distance of 7.5 metre from the point of discharge and falls on the ground. Trace the path of the jet. Name the curve.
52. A coin of 35 mm diameter rolls over dining table without slipping. A point on the circumference of the coin is in contact with the table surface in the beginning and after one complete revolution. Draw the curve traced by the point. Draw a tangent and a normal at any point on the curve.
53. Draw a rectangle of 130 mm  $\times$  85 mm. Draw two parabolas in it with their axes bisecting each other.
54. Two concentric discs of 40 mm and 50 mm diameters roll on the horizontal line  $AB$  150 mm long. Both discs start at the same point and roll in the same direction. Plot the curves for the movement of the points lying on their circumferences.
55. Draw a path of the end of string when it is wound on a circle of 40 cm diameter without slipping. The length of the string is 150 cm long. Name the curve and write practical applications.

56. A circle of 60 mm diameter rolls on a horizontal line for a half revolution and then on a vertical line for another half revolution. Draw the curve traced out by a point  $P$  on the circumference of the circle.
57. Draw the path traced out by a point on a circumference of circle when it rolls, without slip, on vertical surface, for the distance equal to the perimeter of the circle of diameter of 40 mm.
58. Draw a helix of pitch equal to 50 mm upon a cylinder of 75 mm diameter and develop the surface of the cylinder. Assume the starting point to be on the vertical centre line in the top view.
59. Draw the projections of a helix having a helix-angle of  $30^\circ$ , on a cylinder of 75 mm diameter.
60. A spiral is made of a wire of rectangular cross-section 25 mm  $\times$  20 mm. Draw two complete turns of the spring. Outside diameter 100 mm, inside diameter 50 mm and pitch 50 mm.
61. Project one complete turn of a helical spring of outside diameter 75 mm and pitch 50 mm, the cross-section of wire being a circle of 20 mm diameter.
62. Draw the projections of three coils of a helical spring of steel wire 20 mm diameter. Outside diameter of the spring 100 mm and pitch 50 mm.
63. Project two complete turns of a triangular thread, outside diameter 125 mm, pitch 25 mm and angle  $60^\circ$ .
64. A screw having triple-start square thread has outside diameter 150 mm, lead 120 mm and pitch 40 mm. Draw its projections.
65. Draw a helix of one convolution upon a cone, diameter of the base 75 mm, axis 100 mm long and pitch 75 mm. Take apex as the starting point for the curve.
66. A point  $P$ , starting from the base-circle of a cone, reaches the apex, while moving around the axis through two complete turns. Assuming the movement of  $P$  towards the apex (measured parallel to the axis) to be uniform with its movement around the axis, draw the projections and the development of the surface of the cone showing the path of  $P$  in each. Diameter of the base of the cone 75 mm; axis 100 mm long.
67. A propeller screw of 20 mm diameter has a helical blade welded on its surface. The diameter of the helix is 80 mm and the pitch is 50 mm. Project two complete turns of the screw with the blade.
68. Two objects  $A$  and  $B$ , 10 m above and 7 m below the ground level respectively, are observed from the top of a tower 35 m high from the ground. Both the objects make an angle of depression of  $45^\circ$  with the horizon. The horizontal distance between  $A$  and  $B$  is 20 m. Draw to scale 1:250, the projections of the objects and the tower and find (a) the true distance between  $A$  and  $B$ , and (b) the angle of depression of another object  $C$  situated on the ground midway between  $A$  and  $B$ .
69. An ant is travelling on a cylindrical surface in a circumferential direction at a uniform angular speed in the clockwise direction as observed from the top and simultaneously advances at a uniform rate in the axial direction. If the diameter of the cylinder is 60 mm and the ant moves 80 mm in the axial direction during one turn, trace the path of the ant and name the curve.

# Chapter

# 7



# LOCI OF POINTS

## 7-0. INTRODUCTION

A machine is made of the various types of mechanisms for transmitting motions. These mechanisms consist of interrelated links moving on the required path. Dimensions (generally lengths) of the links of the mechanism can be determined by plotting the path of required points. This chapter deals with problems of plotting path of points on the links of different mechanisms.

## 7-1. LOCI OF POINTS

A locus (plural loci) is the path of a point which moves in space.

- (i) The locus of a point  $P$  moving in a plane about another point  $O$  in such a way that its distance from it is constant, is a circle of radius equal to  $OP$  (fig. 7-1).
- (ii) The locus of a point  $P$  moving in a plane in such a way that its distance from a fixed line  $AB$  is constant, is a line through  $P$ , parallel to the fixed line (fig. 7-2).
- (iii) When the fixed line is an arc of a circle, the locus will be another arc drawn through  $P$  with the same centre (fig. 7-3).

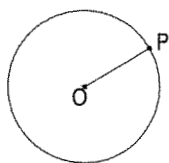


FIG. 7-1

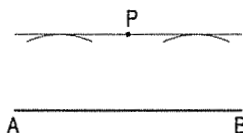


FIG. 7-2

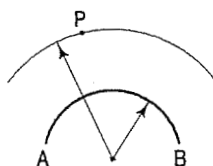


FIG. 7-3

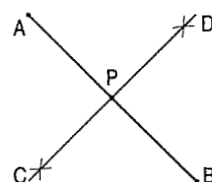


FIG. 7-4

- (iv) The locus of a point equidistant from two fixed points  $A$  and  $B$  in the same plane, is the perpendicular bisector of the line joining the two points (fig. 7-4).
- (v) The locus of a point equidistant from two fixed non-parallel straight lines  $AB$  and  $CD$  will be a straight line bisecting the angle between them (fig. 7-5).

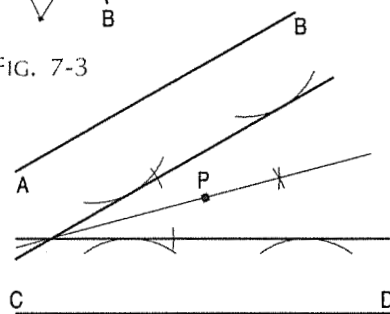


FIG. 7-5

**Problem 7-1.** (fig. 7-6): To draw the locus of a point equidistant from a fixed straight line and a fixed point.

Let  $AB$  be the given line and  $C$  the given point.

- (i) From  $C$ , draw a line  $CD$  perpendicular to  $AB$ . The mid-point  $P$  of  $CD$  is equidistant from  $AB$  and  $C$ , and hence, it lies on the locus.
- (ii) To obtain more points, mark a number of points 1, 2 etc. on  $PC$  and through them, draw lines parallel to  $AB$ .
- (iii) With centre  $C$  and radius  $D1$ , draw an arc cutting the line through 1 at points  $P_1$  and  $P'_1$ .
- (iv) Similarly, obtain more points and through them, draw a smooth curve which will be the required locus.

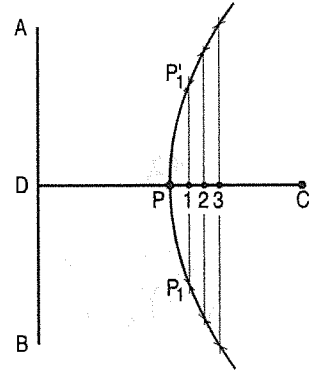


FIG. 7-6

**Problem 7-2.** (fig. 7-7): To draw the locus of a point equidistant from a fixed circle and a fixed point.

A circle with centre  $O$  and a point  $C$  are given.

- (i) Draw a line joining  $O$  and  $C$  and cutting the circle at a point  $A$ . The mid-point  $P$  of the line  $AC$  will lie on the locus.
- (ii) Mark a number of points 1, 2 etc. on  $PC$  and through them, draw arcs with  $O$  as centre.
- (iii) With centre  $C$  and radius equal to  $A1$ , draw an arc cutting the arc through 1 at points  $P_1$  and  $P'_1$ .

Similarly, obtain more points and draw the required curve through them.

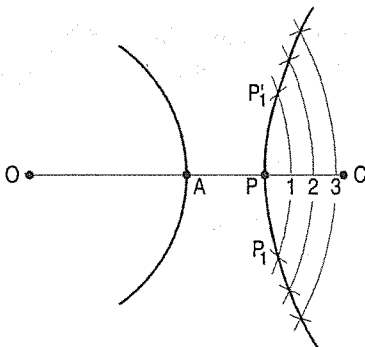


FIG. 7-7

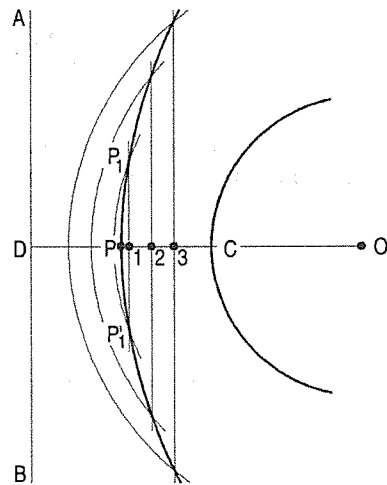


FIG. 7-8

**Problem 7-3.** (fig. 7-8): To draw the locus of a point equidistant from a fixed straight line and a fixed circle.

A line  $AB$  and a circle with centre  $O$  are given.

- (i) From  $O$ , draw a line  $OD$  perpendicular to  $AB$  and cutting the circle at  $C$ .
- (ii) Find the mid-point  $P$  of the line  $DC$ . It will lie on the curve.

- (iii) Mark any point 1 on  $PC$  and through it, draw a line parallel to  $AB$ .
- (iv) With centre  $O$  and radius equal to  $(OC + D1)$ , draw an arc cutting the line through 1 at  $P_1$  and  $P'_1$ .
- (v) Similarly, obtain more points and draw the curve through them.

**Problem 7-4.** (fig. 7-9): To draw the locus of a point equidistant from two given circles. Circles with centres  $A$  and  $B$  are given.

- (i) Draw a line joining  $A$  and  $B$  and cutting the circles at points  $C$  and  $D$ .
- (ii) Find the mid-point  $P$  of the line  $CD$ .
- (iii) Mark any point 1 on  $PD$  and through it, draw an arc with centre  $A$ .
- (iv) With centre  $B$  and radius equal to  $(BD + C1)$ , draw an arc, cutting the arc through 1 at points  $P_1$  and  $P'_1$ .
- (v) Similarly, locate more points and draw the curve through them.

The curves obtained in the above four problems are also the loci of centres of circles which will touch the given line, point or circles as the case may be.

**Problem 7-5.** (fig. 7-9): To draw a circle touching two given circles and a given straight line.

The circles with centres  $A$  and  $B$  and the line  $EF$  are given.

Draw the locus of a point equidistant from one of the circles, say the smaller circle, and the line  $EF$ .

The point of intersection of this curve, with the locus of the point equidistant from the two given circles, viz.  $O$  is the centre of the required circle.

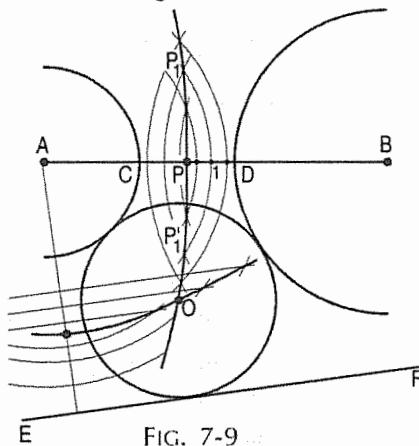


FIG. 7-9

## 7-2. SIMPLE MECHANISMS

In simple mechanisms, it is often necessary to know the paths of points on their moving parts. These are determined by assuming a number of different positions of the moving parts and then locating the corresponding positions of the points.

### 7-2-1. THE SLIDER CRANK MECHANISM

The slider crank mechanism shown diagrammatically in fig. 7-10 is one of the simplest mechanisms and it is used in internal combustion engine, sewing-machine and printing press etc. The end  $A$  of the connecting rod  $AB$  is connected to the crank  $OA$  which rotates about  $O$ . The other end  $B$  is attached to a slider which slides along a straight line. The locus of  $A$  will be a circle and that of the end  $B$  will be a straight line. The locus of any other point, say  $P$ , on the connecting rod will be neither a circle nor a straight line and may be determined:

- (i) by assuming various positions of the crank-end  $A$ ,
- (ii) by obtaining the corresponding positions of the end  $B$  and finally
- (iii) by locating the positions of  $P$  on the lines joining the first two positions.

A smooth curve drawn through the points thus located will be the locus of  $P$ . Such curves are known as *coupler curves*.

**(1) Simple slider crank mechanism:**

This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 15 for the following problem of simple slider crank mechanism.

**Problem 7-6.** (fig. 7-10): In a simple slider crank mechanism, the connecting rod  $AB$  is 750 mm long and the crank  $OA$  is 150 mm long. The end  $B$  moves along a straight line passing through  $O$ . Trace the locus of a point  $P$ , 300 mm from  $A$  along the rod, for one revolution of  $OA$ .

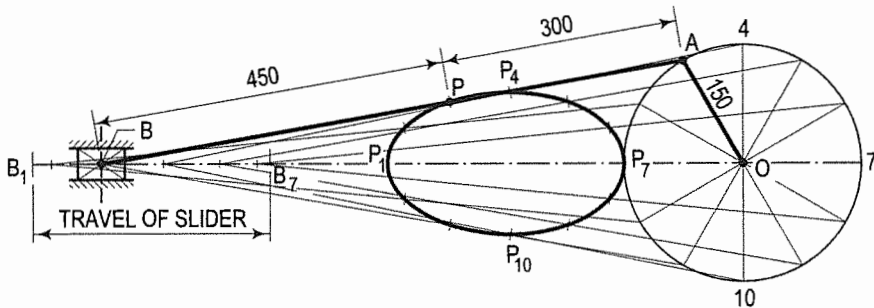


FIG. 7-10

- (i) Divide the circle (path of  $A$ ) into 12 equal parts.
- (ii) With centre 1 and radius  $AB$ , cut the path of  $B$  at a point  $B_1$ .
- (iii) Draw a line joining 1 and  $B_1$ . Again, with centre 1 and radius  $PA$ , cut the line  $1B_1$  at a point  $P_1$ .
- (iv) Obtain other points in similar manner and draw a smooth curve through these points. Then this curve is the locus of the point  $P$ .

Note that the distance  $B_1B_7$  is the travel of the slider and is equal to twice the length of the crank. This distance is known as *stroke length*. But the movement of the slider is not uniform with that of the crank-end  $A$ .

**(2) Offset slider crank mechanism:**

This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 16 for the following problem of offset slider crank mechanism.

**Problem 7-7.** (fig. 7-11): In the offset slider crank mechanism shown in fig. 7-11, the slider-end  $B$  moves in guides along the line  $CD$ , 225 mm below the axis  $O$  of the crank-shaft. Plot the locus of a point  $P$ , 450 mm from  $A$  along  $AB$  and of a point  $Q$  along the extension of the rod, 300 mm beyond  $B$ . Determine also the travel of the slider.

- (i) Divide the circle into 12 equal parts and obtain the positions of the end  $B$  on its path, and of the point  $P$  as shown in problem 7-6.
- (ii) Produce lines  $1B$ ,  $2B$  etc. 300 mm further and mark positions  $Q_1$ ,  $Q_2$  etc. In addition to the above 12 points, it is also necessary to determine the limiting positions of the end  $B$ . They will occur when the connecting rod and the crank are in a straight line.

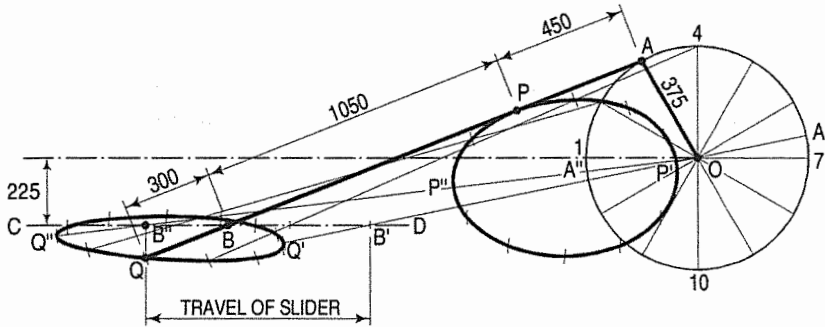


FIG. 7-11

- (iii) These are found by drawing arcs with centre  $O$  and radii (i)  $(AB - AO)$  and (ii)  $(AB + AO)$ , and cutting the path of  $B$  at points  $B'$  and  $B''$ . The travel of the slider is shown by the distance  $B''B'$ . The corresponding positions of the end  $A$  for the limiting positions of  $B$ , viz.  $A'$  and  $A''$  will be on the lines  $B'O$ -produced and  $B''O$  respectively. The positions of  $P$  and  $Q$ , viz.  $P', P''$  and  $Q', Q''$  are obtained as already explained.

**Problem 7-8.** (fig. 7-12): In the mechanism shown in fig. 7-12, the connecting rod is constrained to pass through the guide at  $D$ . Trace the locus of the end  $B$  and of a point  $P$  on  $AB$  for one complete revolution of the crank.  $AB = 1800$  mm,  $AO = 375$  mm and  $AP = 750$  mm.

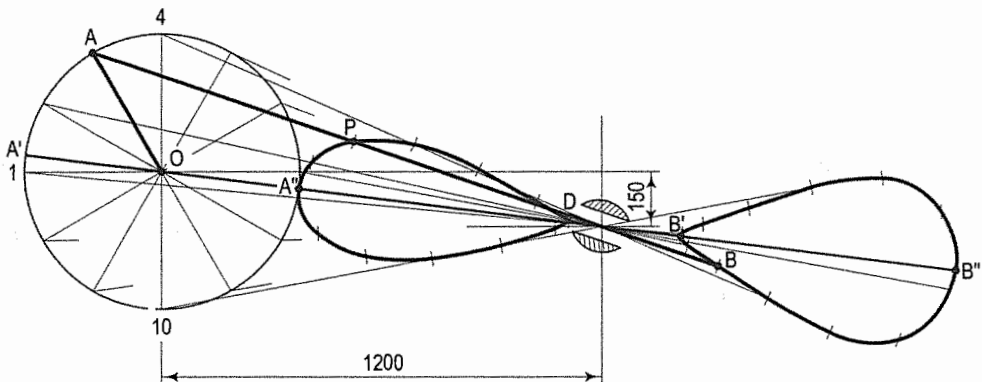


FIG. 7-12

- (i) Divide the circle into 12 equal parts.
- (ii) Draw a line from the point 1, passing through  $D$  and obtain a point  $B_1$  such that  $1B_1 = AB$ .
- (iii) Similarly, locate other positions of the end  $B$  and draw a curve through them. The locus of  $P$  is drawn in the same manner as explained in the previous problem. The limiting positions  $B'$  and  $B''$  (of the end  $B$ ) are found by drawing a line through  $O$  and  $D$  and making  $OB'$  equal to  $(AB - AO)$  and  $OB''$  equal to  $(AB + AO)$ .

Note that  $A'B' = A''B'' = AB$ .

## 7-2-2. A FOUR-BAR MECHANISM

This mechanism consists of four links. It is widely used in locomotive, steering mechanism of the car, pantograph and straight-line mechanisms.

**Problem 7-9.** (fig. 7-13): Two equal cranks  $AB$  and  $CD$  connected by the link  $BD$ , rotate in opposite directions. Draw the locus of a point  $P$  on  $BD$  and of  $Q$  along extension of the rod beyond  $B$  for one revolution of  $AB$ .

$$AB = CD = 450 \text{ mm}; AC = BD = 1500 \text{ mm};$$

$$PD = 300 \text{ mm}; BQ = 300 \text{ mm}.$$

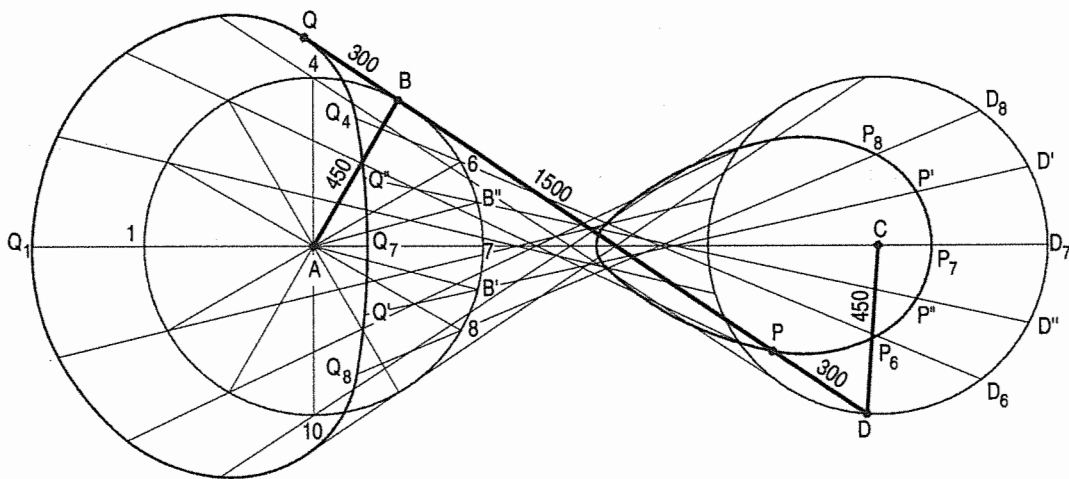


FIG. 7-13

- (i) Divide one of the circles (say path of  $B$ ), into 12 equal parts.
- (ii) Determine the position of  $D$  on its path (the other circle), for every position of  $B$  and find the corresponding positions of  $P$  and  $Q$  for these positions as shown. It will be found that there is a wide gap between the points for positions 6, 7 and 8 of the end  $B$ . A few more positions such as  $AB'$  and  $AB''$  etc. may be taken and points  $P'$ ,  $P''$ ,  $Q'$ ,  $Q''$  etc. may be located.

**Problem 7-10.** (fig. 7-14): Two cranks  $AO$  and  $BQ$  oscillate about  $O$  and  $Q$  respectively. Trace the locus of the mid-point  $P$  of the connecting link  $AB$ .  $AO = 450 \text{ mm}$ ;  $BQ = 675 \text{ mm}$ ;  $AB = 375 \text{ mm}$ .

The limiting positions of the ends  $A$  and  $B$  will be when the link and each of the cranks are in straight lines. They are found as shown below:

- (i) With centre  $Q$  and radius  $(BQ + AB)$ , draw arcs cutting the path of  $A$  at points  $A'$  and  $A''$ .
- (ii) With centre  $O$  and radius  $(AO + AB)$ , draw arcs cutting the path of  $B$  at points  $B'$  and  $B''$ .

Assume  $A$  to be moving downwards. Then  $B$  will move towards  $B'$ . After  $B$  has reached  $B'$ , if  $A$  moves further towards  $A'$ ,  $B$  will begin its return-motion.  $A$  will go upto  $A'$  and then move backwards. The movement will be repeated in a similar manner at  $B''$  and  $A''$  also.

- (i) To draw the locus of  $P$ , mark a number of points on the path of  $A$ .
- (ii) With centre  $A_1$  and radius  $AB$ , draw an arc cutting the path of  $B$  at  $B_1$ .
- (iii) Mark a point  $P_1$  on  $A_1B_1$  such that  $A_1P_1 = AP$ .
- (iv) Similarly, locate other points during the complete oscillation of the crank  $OA$ , from  $A'$  to  $A''$ . It is not necessary to draw the cranks in various positions.

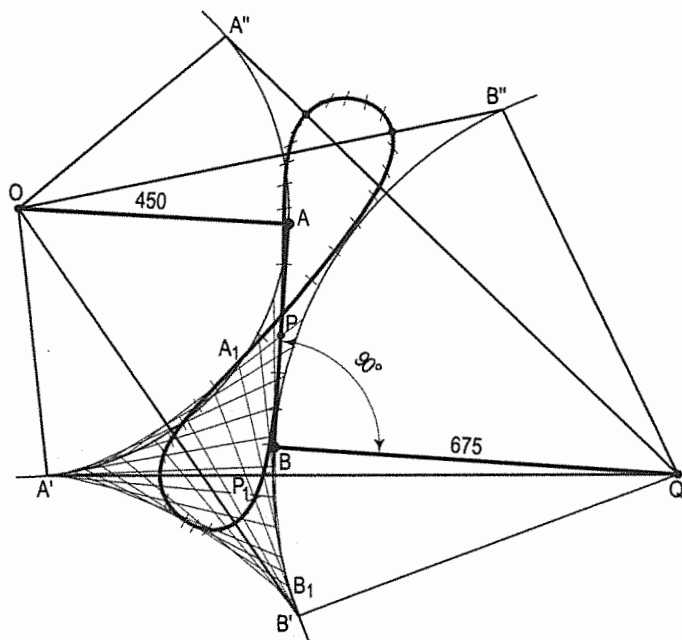


FIG. 7-14

**Problem 7-11.** (fig. 7-15): Two cranks  $AB$  and  $CD$  are connected by a link  $BD$ .  $AB$  rotates about  $A$ , while  $CD$  oscillates about  $C$ . Trace the locus of the mid-point of the link  $BD$  during one complete revolution of the crank  $AB$ .  $AB = 450$  mm,  $CD = 750$  mm,  $BD = 1050$  mm. Distance between  $A$  and  $C$  is equal to  $900$  mm.

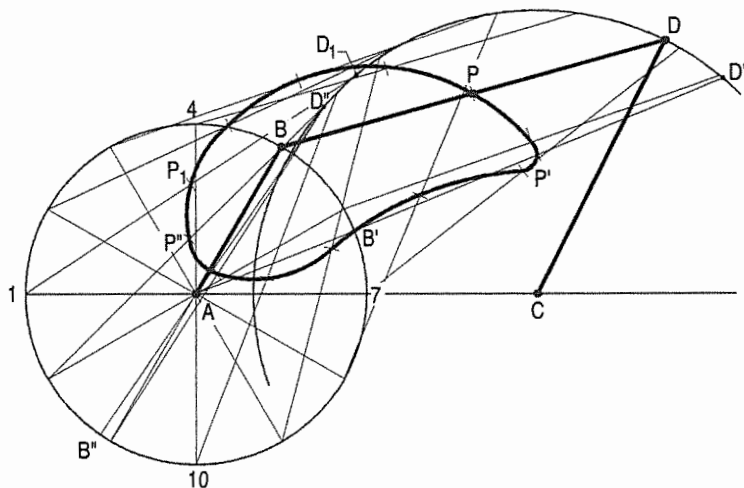


FIG. 7-15

- (i) Divide the circle into 12 equal parts. With centre 1 and radius  $BD$ , cut the path of  $D$  at  $D_1$ .
- (ii) Locate  $P_1$ , the mid-point of  $1D_1$ . Similarly, find other points.
- (iii) In addition, find points  $P'$  and  $P''$  for limiting positions, when  $AD' = (BD + AB)$  and  $AD'' = (BD - AB)$ .



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 17 for the following problem.

**Problem 7-12.** (fig. 7-16): The end  $A$  of a rod  $AB$  rotates about  $O$ , while the end  $B$  slides along a straight line. A crank  $CQ$  oscillates about  $Q$ . Draw the locus of the mid-point  $P$  of the connecting link  $CD$  for one revolution of the crank  $OA$ .  $AB = 1500$  mm,  $CD = 750$  mm,  $OA = 450$  mm and  $CQ = 1200$  mm.

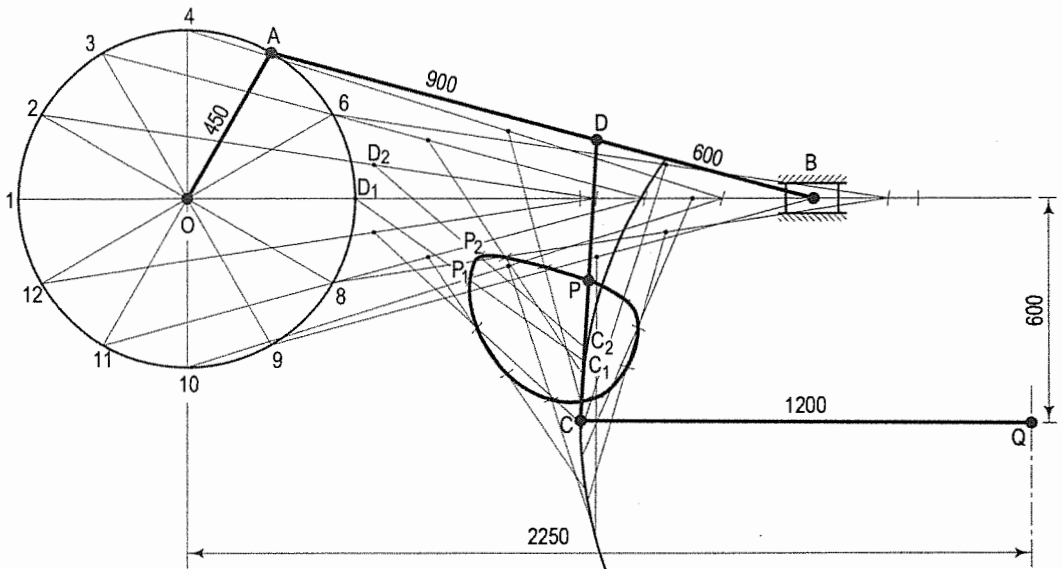


FIG. 7-16

- (i) Determine points  $D_1, D_2$  etc. for various positions of the crank  $OA$ .
- (ii) With centres  $D_1, D_2$  etc. and radius  $CD$ , draw arcs cutting the path of  $C$  at  $C_1, C_2$  etc.
- (iii) Locate the mid-points  $P_1, P_2$  etc. of lines  $D_1C_1, D_2C_2$  etc. and draw the required curve through them.

**Problem 7-13.** (fig. 7-17): A circular disc of 100 mm diameter, revolves clockwise about its centre with uniform angular velocity. A point  $P$  is situated initially at the end  $A$  of the chord  $AB$  (60 mm long), travels along the chord towards the end  $B$ , with uniform velocity. As the disc completes one revolution, the point  $P$  reaches to the end  $B$ . Trace the path of the point  $P$  for one complete revolution of the disc.

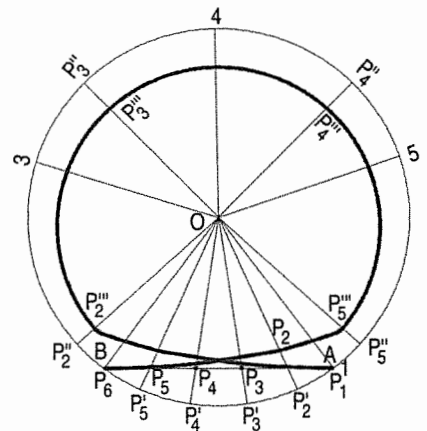


FIG. 7-17

- (i) Draw a circle of 100 mm diameter and divide into 5 equal parts. Mark 1, 2,.....5.
- (ii) Draw a chord AB 60 mm long from the point 1. (The point 1 coincides with the point A.) Divide it into 5 equal parts and number  $P_1, P_2, \dots, P_6$ . (The point  $P_6$  coincides with the point B.)
- (iii) When the line OA moves through one division i.e. arc AB, the point  $P_1$  will move towards B by a distance equal to one division of the chord AB (or  $P_1 P_6$ ).
- (iv) To obtain points systematically, draw arcs with centre O and radii  $OP_2, OP_3, OP_4$  etc. intersecting lines  $OP''_2, OP''_3, OP''_4$  etc. at points  $P'''_2, P'''_3, P'''_4$  etc. respectively. Draw a curve through points  $P_1, P'''_2, \dots, B$ .

**Problem 7-14.** (fig. 7-18): A mechanism as used on cup-board door is in fully open position. Draw the path of end A of the mechanism as the door moves to the fully closed position. Take  $AB = 175$  mm.

- (i) Draw the mechanism to full size scale. Divide  $\angle 120^\circ$  into equal divisions.
- (ii) Mark the successive position of the door. Mark the points  $B_1, B_2$ , etc. at 10 mm away on the perpendicular line to the door position as shown. Draw lines  $B_2O_2, B_3O_2, \dots, B_9O_2$  passing through  $O_2$ .
- (iii) With a centre as  $B_1, B_2, \dots, B_9$  and the radius equal to AB, draw arcs intersecting lines  $B_1O_2, B_2O_2, \dots, B_9O_2$  at points  $A_1, A_2, A_3, \dots, A_9$ .
- (iv) Join the points  $A_1, A_2, \dots, A_9$  by smooth curve.

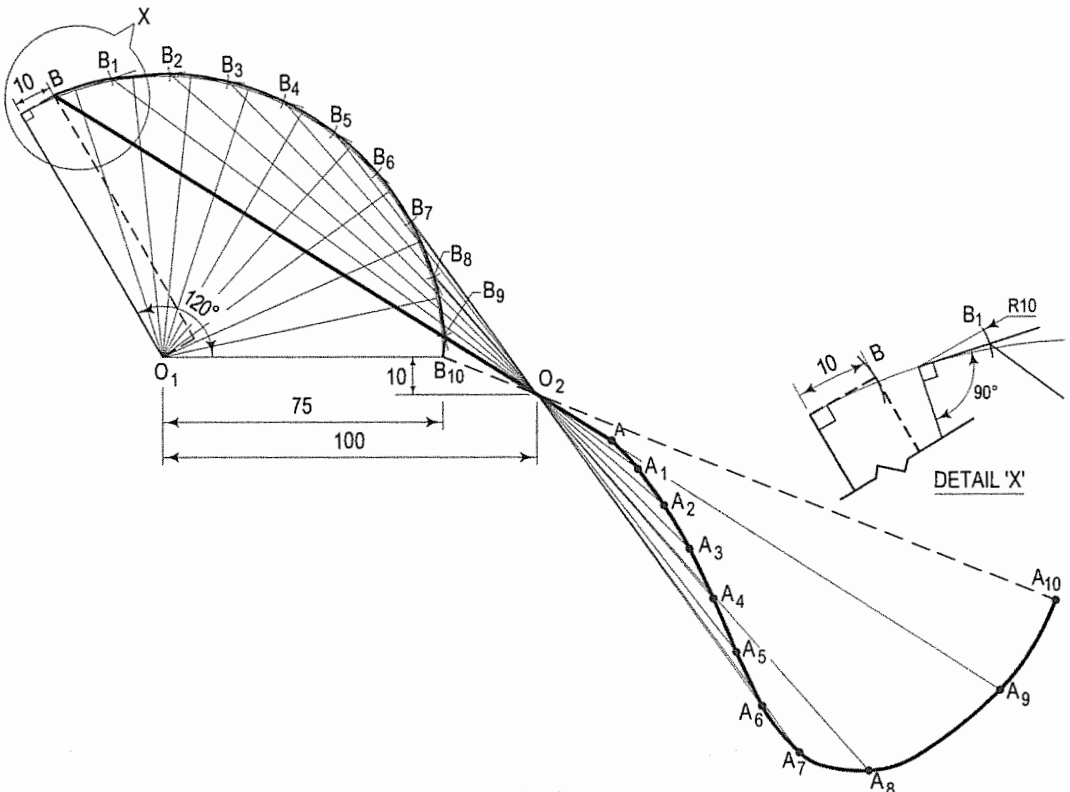


FIG. 7-18

**Problem 7-15.** (fig. 7-19): A link of shaping machine mechanism is shown in fig. 7-19. A sliding block moves towards  $O_4$  along the oscillating link with the uniform motion.

Draw the path of the sliding block when the link  $O_4D$  rotates  $40^\circ$  in anti-clockwise direction. Use following data  $O_4D = 160$  mm,  $O_4B = 113$  mm. The movement of the sliding block along the link = 67 mm.

The construction of path of the sliding block is shown in fig. 7-19.

**Problem 7-16.** (fig. 7-20): A link  $PQ$  100 mm long carries a circular disc of 30 mm diameter having centre  $Q$ . The link  $PQ$  oscillates about the hinged  $P$  from left to right and vic-versa (i.e. right to left) to maximum amplitude of  $45^\circ$  on either side from the vertical.

At the same time the disc rotates uniformly through one revolution in clockwise direction. Draw the path of point on the circumference of the disc.

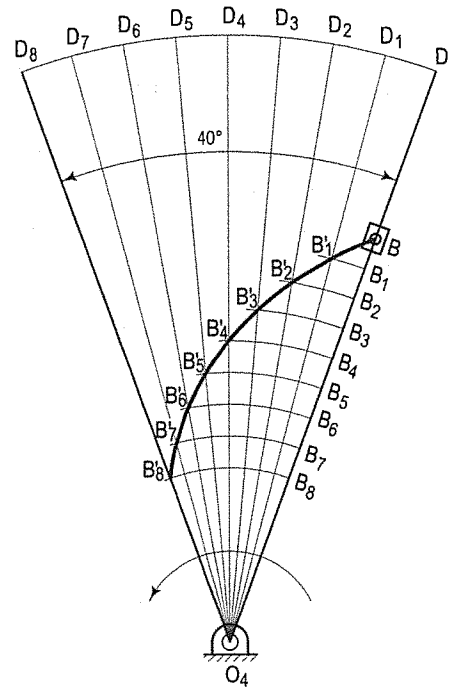


FIG. 7-19

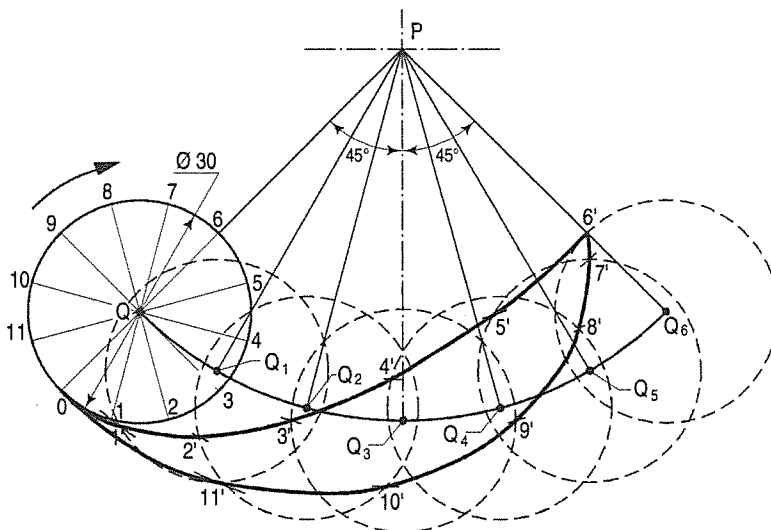


FIG. 7-20

- (i) Mark the point  $P$ . Draw vertical line from  $P$ . Draw the link  $PQ$  at angle of  $45^\circ$  with the vertical on the left side as well as on the right side.
- (ii) Draw circle of radius 15 mm at  $Q$  and divide the circle into twelve equal parts. Number as 0, 1 to 11 from the division of extended  $PQ$  in anticlockwise direction as shown.

- (iii) The disc rotates as well as oscillates with the link  $PQ$  from left to right and then right to left for completing one cycle of oscillation. The angle moved by the link  $PQ$  is  $(2 \times 90 = 180^\circ)$ . Divide half cycle (i.e.  $90^\circ$ ) into six equal parts described by the link. Each division will be of  $15^\circ$ .
- (iv) Now draw the arc joining initial position and left extreme position. Mark the intersection of radial lines drawn from  $P$  and this arc as  $Q, Q_1, Q_2, Q_3, Q_4, Q_5, Q_6$  and for return cycle on the same arc  $Q_7, Q_8, Q_9, Q_{10}, Q_{11}$ .
- (v) When the disc rotates from 0 to 1, the rod moves  $PQ$  to  $PQ_1$ . Therefore the centre of disc moves to  $Q_1, Q_2, Q_3, \dots, Q_{11}$  successive position, the rod (radii) will occupy the position  $PQ_1, PQ_2, \dots, PQ_6$  and  $PQ_7$  to  $PQ_{11}$ .
- (vi)  $Q_1, Q_2, Q_3, Q_4, Q_5, Q_6$  as centre, draw the circle for successive position of the disc. Take point  $O$  on its circumference of the circle for the initial position as shown.
- (vii) Now  $1P$  as radius, draw arc intersecting the circle drawn from centre  $Q_1$ . Name this point as  $1'$ . Similarly  $P$  as centre and radii  $2P, 3P, 4P, 5P, 6P, \dots, 11P$  obtain other the intersection points. Join with all points by the smooth curve.
- (viii) Fig. 7-20 is drawn to half scale.

**Problem 7-17.** (fig. 7-21): A rod  $PQ$  40 mm long rotates about  $R$  with a speed of 60 rpm, while  $Q$  moves along a straight line  $QR$  towards  $R$  with a speed of 108 mm per second (i.e.  $QR = 108$  mm). Draw to full size scale the path traced out by the end  $P$  for one complete revolution of the rod. Assume that initial position of the rod  $PQ$  is along the extension of the line  $RQ$ .

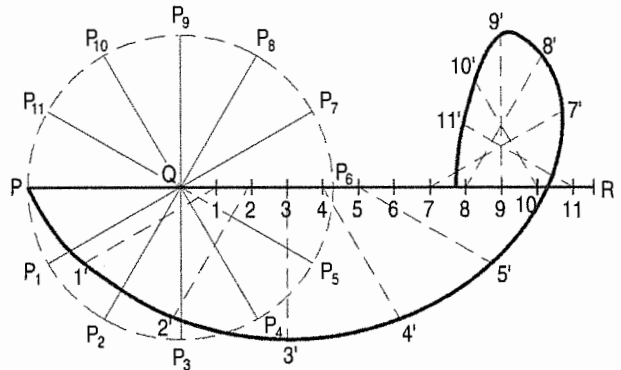


FIG. 7-21

- (i) Draw a line  $PQR$  of length = 148 mm, ( $PQ + QR = 40 + 108$  mm)
- (ii) Now  $Q$  as centre  $PQ = 40$  mm as radius, draw the circle and divide circle into twelve equal parts, label them  $P, P_1, \dots, P_{11}$ . Each part of the circle is covered by the rod in  $\frac{1}{12}$  second.  $\left(\frac{360}{30} = 12^\circ\right)$  Similarly the line  $QR$  is also divide in twelve parts. Mark division as  $Q, 1, 2, \dots, 11, R$ .
- (iii) From 1 draw a parallel line to  $QP_1$ . Now 1 as centre and  $QP_1$  as radius, cut the parallel line at  $1'$ . Similarly from 2, 3, 4, 5,  $\dots, 11, R$ , draw parallel lines to the respective radii of the circle  $QP_2, QP_3, QP_4, \dots, QP_{11}$ .
- (iii) Draw parallel lines from  $P_1, P_2, P_3, \dots, P_{11}$  to  $PR$ . Take radius equal to  $PQ$  and centre as 1, intersect the line drawn parallel to  $PQ$  from  $P_1$ . Similarly with same radius and successive centres say 2, 3,  $\dots, R$ , intersect the line drawn parallel to  $PQ$  from  $P_2, P_3, \dots, R$ .
- (iv) Note that  $QP_6$  will be along the line  $QR$ .
- (v) Join points with the smooth curve.
- (vi) Scale adopted is half.

**Problem 7-18.** (fig. 7-22): The crank  $O_1P_1$  turns about  $O_1$  and the connecting rod  $P_1Q_1$  slides in the same plane on the curved surface of a shaft (with centre  $O_2$ ) of 30 mm diameter. Trace the loci of the point  $Q_1$  and  $R_1$  which is extension of  $P_1Q_1$  at 30 mm from the  $P_1$  when  $O_1P_1$  revolves one revolution. Take  $O_1P_1 = 30$  mm.  $P_1Q_1 = 100$  mm and  $O_1O_2 = 70$  mm. Take scale 10 mm = 1 mm

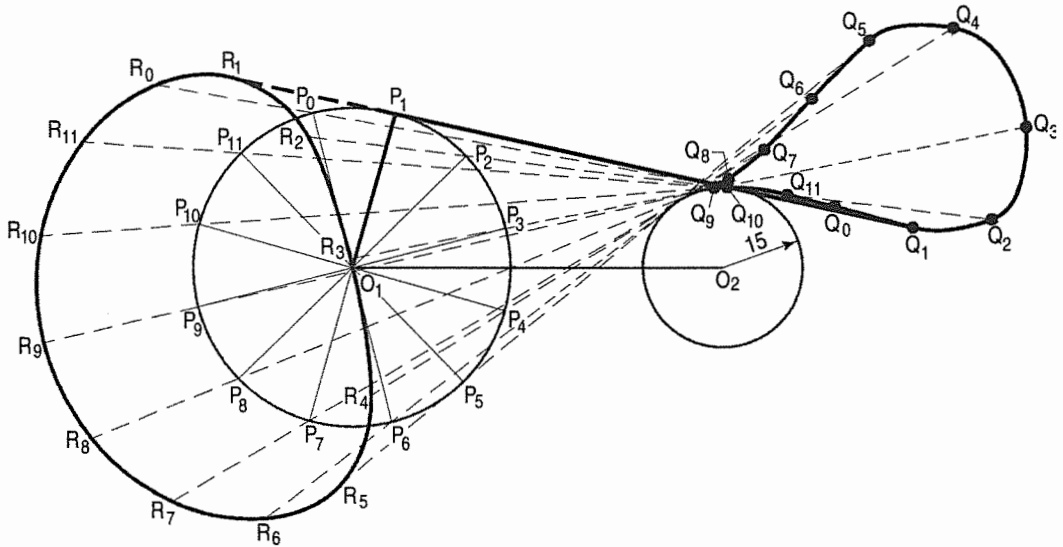


FIG. 7-22

- (i) Draw a circle of 30 mm radius and centre  $O_1$ .
- (ii) Mark  $O_1O_2 = 70$  mm and draw the circle of radius 15 mm and the centre as  $O_2$  the representing shaft.
- (iii) Mark any point  $P_1$  on the circle of  $Q_1P_1$ . Mark line  $P_1Q_1 = 100$  mm making tangent to the circle of the shaft. Extend  $O_1P_1$  to left side to  $R_1$  such that  $P_1R_1 = 30$  mm.
- (iv) Divide the circle of radius  $O_1P$  into twelve divisions and number them  $P_0, P_1, P_2, \dots, P_{11}$ . From  $P_0, P_1, P_2, \dots, P_{11}$  draw tangent on the same side to the circle of the shaft. On each extended tangent  $QP$  mark the points  $P_0R_0, P_1R_1, \dots, P_{11}R_{11}$  as shown in fig. 7-22. Similarly from  $P_0, P_1, \dots, P_{11}$  mark the length 100 mm each tangent showing end  $Q_1, Q_2, Q_3, \dots, Q_{11}$ .
- (v) Join all  $R$  points and  $Q$  points by the smooth curve.

**Problem 7-19.** (fig. 7-23): A mechanism  $O_1A$  rotates clockwise direction about fixed point  $O_1$ . ( $O_1AB$ ) is an offset slider crank mechanism. Link  $O_2E$  oscillates about fixed point  $O_2$ .  $AB$  and  $CE$  are connecting links. Draw the loci of the points  $C$  and  $D$  for one complete revolution of the link ( $O_1A$ ).

- (i) First mark the points  $O_1$  and  $O_2$  2000 mm apart to some suitable scale.
- (ii)  $O_1$  and  $O_2$  as centres, draw the circle showing path of point  $A$  and point  $E$ . Mark the points  $A$  and  $E$  above and below  $O_1O_2$ .
- (iii) Draw a line parallel to  $O_1O_2$  at a distance of 1700 mm above as shown.  $A$  as centre and  $AB$  as radius, mark arc intersecting to parallel line drawn at  $B$ . Mark points  $C$  and  $D$  with given distances on  $AB$  and  $CE$  respectively.

- (iv) Divide the circle of  $O_1 A$  into twelve parts. Number them as  $A_1, A_2, A_3, \dots, A_{11}$ . Now  $A_1$  as centre and  $AB$  distance as radius, draw intersection on the horizontal line  $MN$  at  $B_1$ . From  $B_1$  mark the distance of  $CB$  on  $A_1 B_1$  from  $B_1$  at 1400 mm. This is new position of  $C$  call as  $C_1$ .  $C$  as centre  $CE$  as radius, cut the arc at  $E_2$  drawn from  $O_2$ .  $O_2 E$  as radius, mark the point  $D$  at given distance from  $E$  (680 mm). From  $A_2, A_3, \dots, A_{11}$  taking  $AB$  as radius draw arcs intersecting the path  $MN$  at  $B_2, B_3, \dots, B_{11}$ . Join  $A_1 B_1, A_2 B_2$  upto  $A_{11} B_{11}$ . On these lines mark  $B_1 C_1 = B_2 C_2 = B_3 C_3 = B_{11} C_{11}$ . Join the points  $C, C_1, C_2, \dots, C_{11}$  by smooth curve. Similarly from  $C_1, C_2, C_3, \dots, C_{11}$ , draw arc of radius  $CE$  intersecting points  $E_1, E_2, E_3, \dots, E_{11}$ . Join  $C_1 E_1, C_2 E_2, C_3 E_3, \dots, C_{11} E_{11}$ . Along the lines  $C_1 E_1, C_2 E_2, C_3 E_3, \dots, C_{11} E_{11}$ , mark  $DE$  respectively. Let these points be  $D_1, D_2, D_3, \dots, D_{11}$ . Join the points by smooth curve.

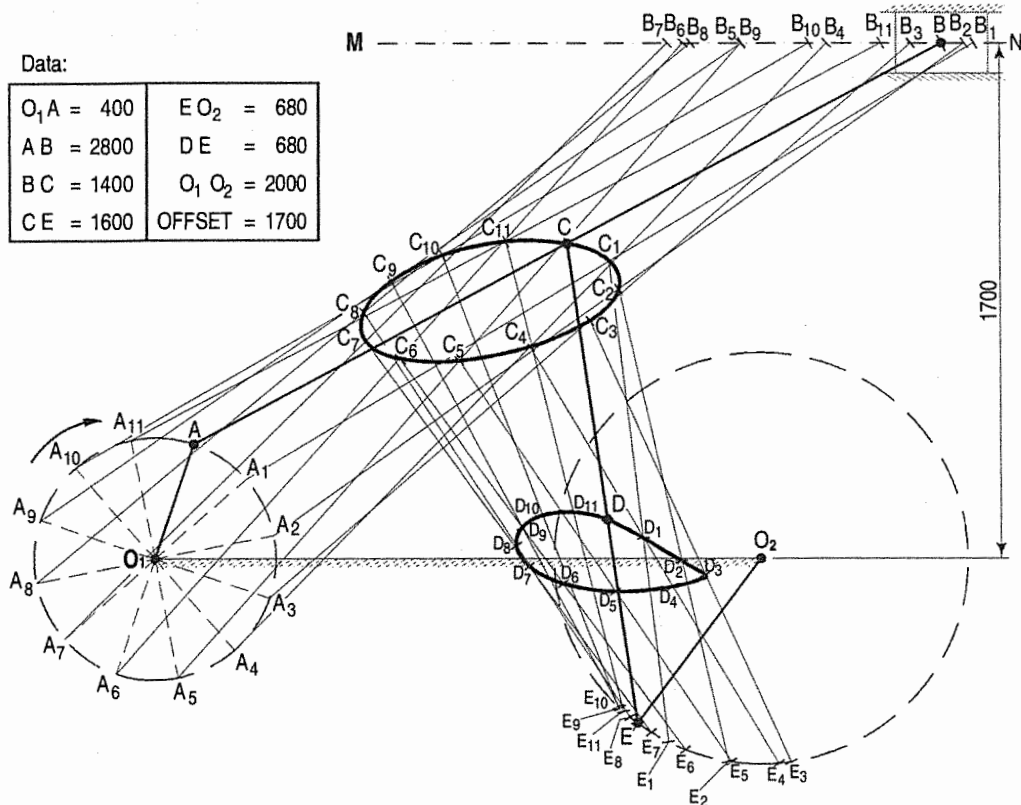


FIG. 7-23

**Problem 7-20.** (fig. 7-24):  $PQ$  and  $PR_1$  are fastened rigidly at  $P$  at angle of  $90^\circ$  to each other. The ends  $P$  and  $R$  are sliding in the guides vertically and horizontally respectively. Draw the loci of the points  $Q$  and mid-point of the link  $PR$  when the link  $PR$  moves from vertically to the horizontal. Name the curve trace by the mid-point. Take  $PQ = 40$  mm and  $PR = 70$  mm.

- (i) Draw two links  $PQ$  and  $PR_1$  of 40 mm and 70 mm respectively at right angle.

- (ii) Make slotted link of 10 mm of width and at  $P$  and  $R$ . Construct sliders of the convenient length at  $P$  and  $R_1$ .
- (iii) Mark the mid point of  $PR_1$ , say  $S_1$ .
- (iv) Assume that the slider  $P$  moves vertically downward while  $R_1$  will moves horizontally.
- (v) Divide vertical line  $PR_1$  into equal division and number them as 1, 2, 3, 4.....14,  $R_1$  as shown ( $P$ , 1 are coinciding).
- (vi) Now 1 as centre and  $PR_1$  as radius, draw arc to cut the horizontal line passing through  $R_1$ . Let us call this point as  $R_1$ . Join  $P1R_1$ ,  $P1$  as centre mark the mid-point of  $P1R_1$ , say this point  $S_1$ , similarly from the point 2, 3..... 14, draw arc cutting the horizontal line and number them respectively  $R_2, R_3..... R_{14}$ . Join all points  $Q$  and  $S$  respective respectively. Mark on these lines the mid-points  $S_2, S_3, S_4..... S_{14}$ . Join them with smooth curve. The curve is quarter portion of ellipse. The mechanism is known as *trammel mechanism*.
- (vii) The path of  $Q$  is simple to trace as shown in the fig. 7-24.

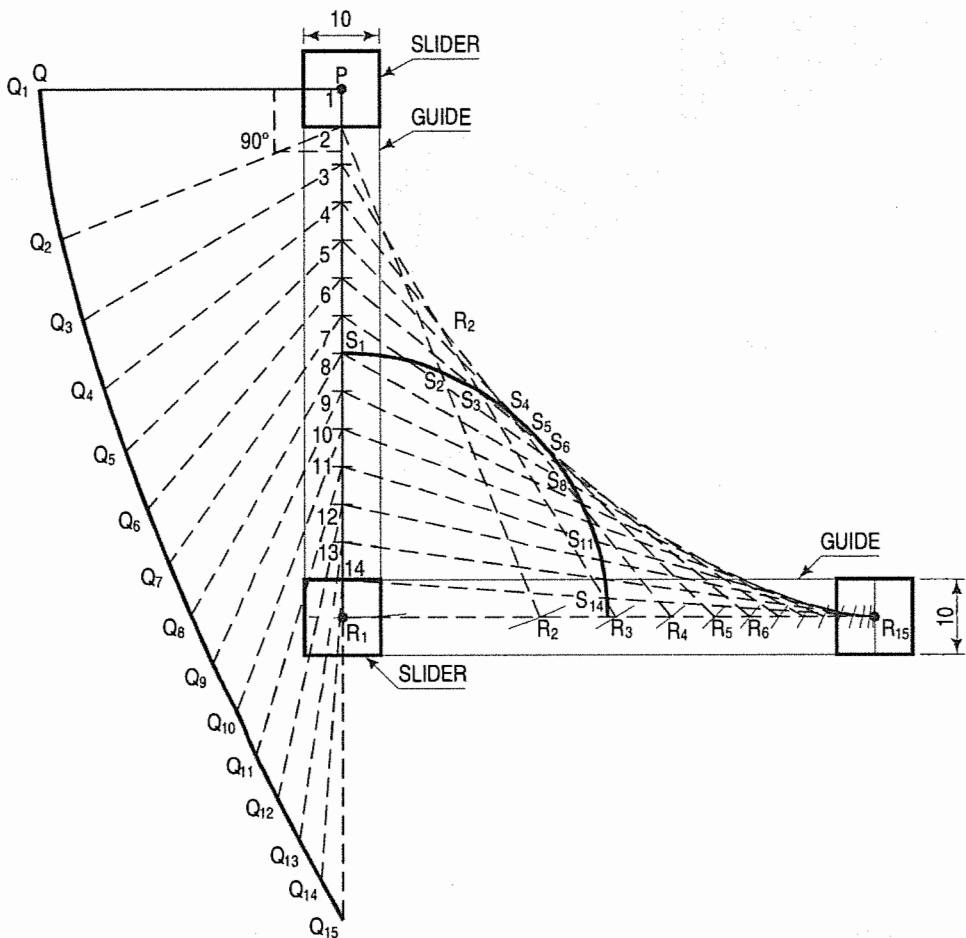


FIG. 7-24

**Problem 7-21.** The rod  $PQ$  as shown in fig. 7-25 is hinged (pinned) to the crank  $AO$  at  $A$ .  $OA$  rotates about  $O$  and the rod  $PQ$  is constrained to pass through the point  $R$ . Draw the loci of the ends  $P$  and  $Q$  for one complete revolution of  $OA$ .  $PQ = 1100$  mm,  $OA = 220$  mm,  $AP = 300$  mm and  $OR = 500$  mm.

Scale  $100$  mm =  $10$  mm.

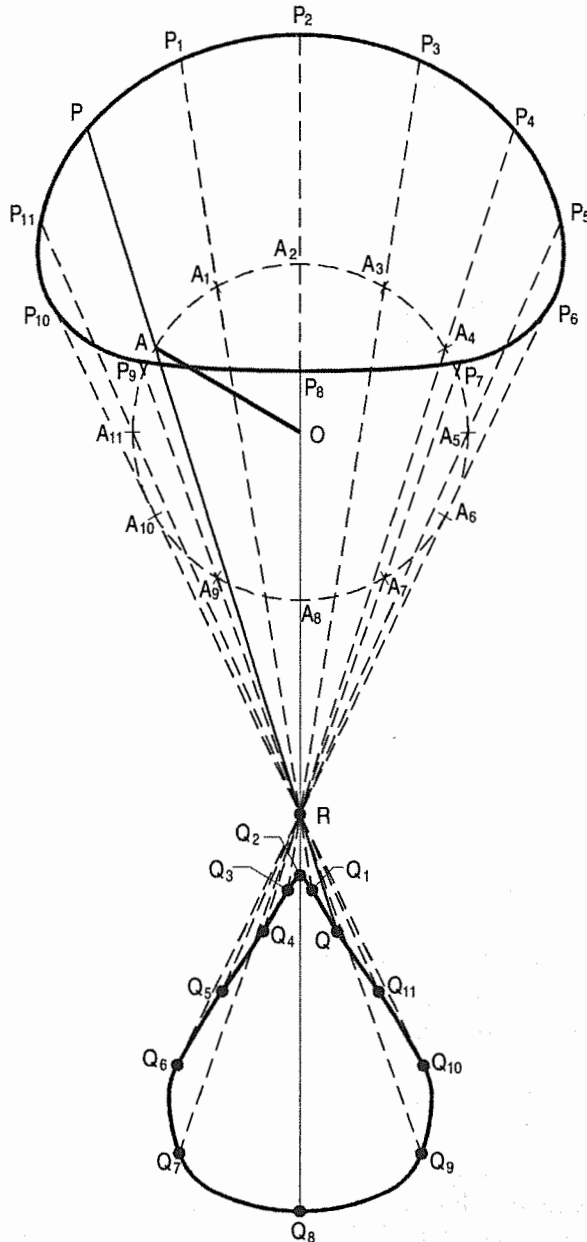


FIG. 7-25

- (i) Draw a mechanism to scale  $100$  mm =  $10$  mm for given dimensions.

- (ii) Divide circle described by the crank  $OA$  into 12 equal divisions. Say these division  $A, A_1, A_2, \dots, A_{11}$ .
- (iii) From each points of  $A_1, A_2, \dots, A_{11}$ , draw lines passing through  $R$  and mark length  $AP$  and  $RQ$  on respective lines. Say these points are  $P, P_1, P_2, \dots, P_{11}$  and  $Q, Q_1, \dots, Q_{11}$ . Join respective points by smooth curves.

## EXERCISES 7

- $P, Q$  and  $R$  are the centres of three circles of diameters 75 mm, 45 mm and 30 mm respectively.  $PQ = 95$  mm,  $QR = 50$  mm and  $PR = 75$  mm. Draw a circle touching the three circles.
- In a slider-crank mechanism, the crank  $OA$  is 450 mm long, and the connecting rod  $AB$ , 1050 mm long. Plot the locus of (i) the mid-point  $P$  of  $AB$ , and (ii) a point 600 mm from  $A$  on  $BA$  extended, for one revolution of the crank.
- The end  $P$  of a 100 mm long line  $PQ$  (fig. 7-26) slides vertically downwards. The end  $Q$  moves along the line  $AB$  towards  $A$  and then back to  $B$ . Plot the locus of the point  $O$  on  $PQ$  and 40 mm from  $P$ .

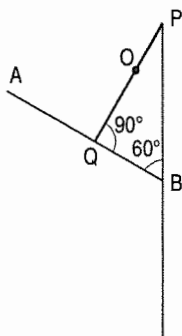


FIG. 7-26

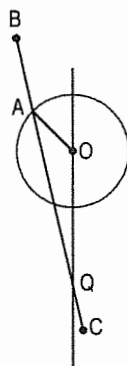


FIG. 7-27

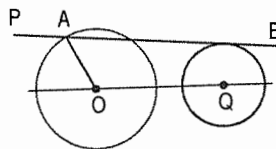


FIG. 7-28

- The rod  $BC$  (fig. 7-27) is attached to the crank  $AO$  at  $A$ .  $OA$  rotates about  $O$  and the rod  $BC$  is constrained to pass through the point  $Q$ . Draw the loci of the ends  $B$  and  $C$  for one complete revolution of  $OA$ .  $BC = 1200$  mm,  $OA = 225$  mm,  $AB = 300$  mm and  $OQ = 525$  mm.
- The cranks  $OA$  (fig. 7-28) rotates about  $O$  and the connecting rod  $AB$  slides in the same plane on the curved surface of a shaft (with centre  $Q$ ) of 450 mm diameter. Trace the locus of (i) the end  $B$  and (ii) the point  $P$  beyond  $AB$  and 300 mm from  $A$  for one revolution of  $OA$ .  $OA = 375$  mm,  $AB = 1200$  mm and  $OQ = 700$  mm.
- $PO$  is a rod 50 mm long. It rotates about its end  $O$  with a speed of one revolution per second, while  $O$  moves along a straight line  $OB$  towards  $B$  with a speed of 120 mm per second. Draw to full-size scale the path traced out by the end  $P$  for one complete revolution of the rod. In the starting position assume the rod  $PO$  to be lying along the extension of the line  $BO$ . Show all construction lines.

7. Two equal cranks  $AB$  and  $CD$  (fig. 7-29) rotate in opposite directions about  $A$  and  $C$  and are connected by the rod  $BD$ . Plot the locus of the end  $P$  of the link  $PQ$  attached at right angles to  $BD$  at its mid-point  $Q$  for one complete revolution of the cranks.  $AB = 300$  mm;  $BD = AC = 1050$  mm;  $PQ = 225$  mm.

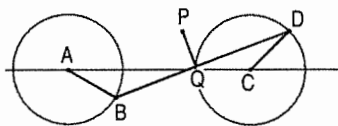


FIG. 7-29

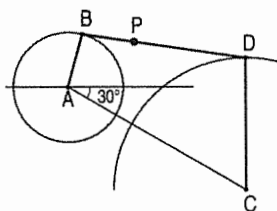


FIG. 7-30

8. Two cranks  $AB$  and  $CD$  (fig. 7-30) are connected by a link  $BD$ . The end  $B$  moves round the circumference of the circle with centre  $A$ , while the end  $D$  oscillates on an arc about  $C$  as centre. Plot the locus of the point  $P$  on  $BD$ , 450 mm from  $B$ , for one complete revolution of  $AB$ .  $AB = 450$  mm,  $CD = 1050$  mm,  $BD = 1350$  mm and  $AC = 1650$  mm.
9. Two equal links  $AB$  and  $CD$  (fig. 7-31) connected by a rod  $BD$ , oscillate about their ends  $A$  and  $C$ . Plot the locus of (i) the mid-point  $P$  of  $BD$  and (ii) the point  $Q$  on  $BD$ .  $AB = CD = 1200$  mm;  $BD = 900$  mm;  $BQ = 225$  mm.

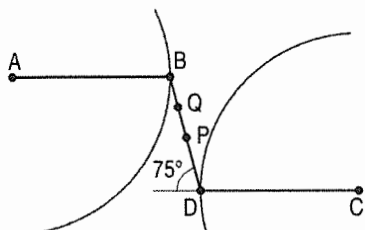


FIG. 7-31

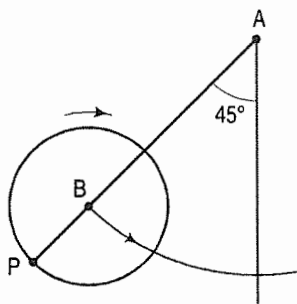


FIG. 7-32

10. The link  $AB$  (fig. 7-32) is 120 mm long and carries a circular disc of 40 mm radius. The end  $A$  is hinged while the disc can revolve about its centre  $B$ . The link turns uniformly to the right through  $90^\circ$  and at the same time the disc revolves uniformly in clockwise  $\curvearrowright$  direction through one complete revolution. Plot the locus of the point  $P$  situated on the circumference of the disc.
11. A thin rod  $AB$ , 100 mm long revolves uniformly about its centre  $O$ . During its one revolution a point  $P$  moves along  $AB$  at uniform speed from  $A$  to  $B$ . Draw the locus of  $P$ .
12. Two cranks  $AB$  and  $CD$  are connected by a link  $BD$ .  $AB$  rotates about  $A$ , while  $CB$  oscillates about  $C$ . Trace the locus of the mid-point  $P$  of link  $BD$  during one complete revolution of the link  $AB$ . Assume that the points  $B$  and  $D$  are opposite sides of  $AC$ .  $AB = 450$  mm,  $CD = 750$  mm,  $DB = 1050$  mm,  $AC = 900$  mm.  $\angle CAB = 120^\circ$ .

13. In a slider-crank mechanism, the crank  $O_2P$  is 600 mm long and the connecting rod  $PQ$ , 1400 mm long. Plot the path of a point 600 mm from  $P$  on  $QP$  extended for one revolution of the crank.
14. A link  $AB$  120 mm long rotates about fixed pivot  $A$  in an anti-clockwise direction. An ant is situated at 20 mm from the pivot point moves towards the end  $B$  with uniform velocity, while the link has rotated through  $\frac{1}{6}$  of a revolution. Trace the path of an ant.

## Chapter

## 8



# ORTHOGRAPHIC PROJECTION

### 8-0. INTRODUCTION

*Practical solid geometry or descriptive geometry* deals with the representation of points, lines, planes and solids on a flat surface (such as a sheet of paper), in such a manner that their relative positions and true forms can be accurately determined.

### 8-1. PRINCIPLE OF PROJECTION

If straight lines are drawn from various points on the contour of an object to meet a plane, the object is said to be projected on that plane. The figure formed by joining, in correct sequence, the points at which these lines meet the plane, is called the *projection* of the object. The lines from the object to the plane are called *projectors*.

### 8-2. METHODS OF PROJECTION

In engineering drawing following *four* methods of projection are commonly used, they are:

- |                             |                             |
|-----------------------------|-----------------------------|
| (1) Orthographic projection | (3) Oblique projection      |
| (2) Isometric projection    | (4) Perspective projection. |

In the above methods (2), (3) and (4) represent the object by a pictorial view as eyes see it. In these methods of projection a three dimensional object is represented on a projection plane by one view only. While in the orthographic projection an object is represented by two or three views on the mutual perpendicular projection planes. Each projection view represents two dimensions of an object. For the complete description of the three dimensional object at least *two* or *three* views are required.

### 8-3. ORTHOGRAPHIC PROJECTION

When the projectors are parallel to each other and also perpendicular to the plane, the projection is called *orthographic projection*.

**Step 1:** Imagine that a person looks at the block [fig. 8-1(i)] from a theoretically infinite distance, so that the rays of sight from his eyes are parallel to one another and perpendicular to the front surface *F*. The view of this block will be the shaded figure, showing the front surface of the object in its true shape and proportion.

**Step 2:** If these rays of sight are extended further to meet perpendicularly a vertical plane (marked V.P.) set up behind the block.

**Step 3:** The points at which they meet the plane are joined in proper sequence, the resulting figure (marked E) will also be exactly similar to the front surface and this is known as an *elevation* or *front-view*. This figure is the projection of the block. The lines from the block to the plane are the projectors. As the projectors are perpendicular to the plane on which the projection is obtained, it is the orthographic projection. The projection is shown separately in fig. 8-1(ii). It shows only two dimensions of the block viz. the height  $H$  and the width  $W$ . It does not show the thickness. Thus, we find that only one projection is insufficient for complete description of the block.

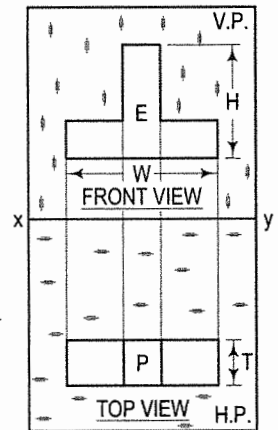
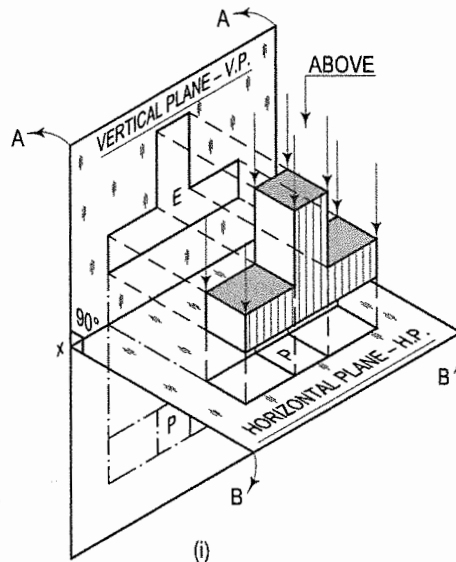
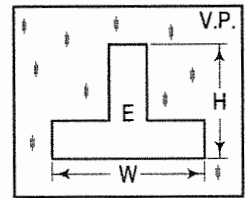
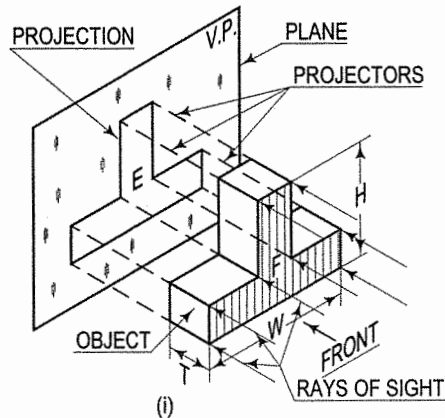


FIG. 8-2

Let us further assume that another plane marked H.P. (horizontal plane) [fig. 8-2(i)] is hinged at right angles to the first plane, so that the block is in front of the V.P. and above the H.P. The projection on the H.P. (figure P) shows the top surfaces of the block. If a person looks at the block from above, he will obtain the same view as the figure P and is known as a *plan* or *top-view*. It shows the width  $W$  and the thickness  $T$  of the block. It however does not show the height of the block.

One of the planes is now rotated or turned around on the hinges so that it lies in extension of the other plane. This can be done in two ways:

- (i) by turning the V.P. in direction of arrows A
- (ii) by turning the H.P. in direction of arrows B.

The H.P. when turned and brought in line with the V.P. is shown by dashed lines. The two projections can now be drawn on a flat sheet of paper, in correct relationship with each other, as shown in fig. 8-2(ii).

When studied together, they supply all information regarding the shape and the size of the block. Any solid may thus be represented by means of orthographic projections or orthographic views.

#### 8-4. PLANES OF PROJECTION

The two planes employed for the purpose of orthographic projections are called *reference planes* or *principal planes of projection*. They intersect each other at right angles. The *vertical plane* of projection (in front of the observer) is usually denoted by the letters V.P. It is often called the *frontal plane* and denoted by the letters F.P.

The other plane is the *horizontal plane* of projection known as the H.P. The line in which they intersect is termed the *reference line* and is denoted by the letters xy. The projection on the V.P. is called the *front view* or the *elevation* of the object. The projection on the H.P. is called the *top view* or the *plan*.

#### 8-5. FOUR QUADRANTS

When the planes of projection are extended beyond the line of intersection, they form four quadrants or dihedral angles which may be numbered as in fig. 8-3. The object may be situated in any one of the quadrants, its position relative to the planes being described as "above or below the H.P." and "in front of or behind the V.P."

The planes are assumed to be transparent. The projections are obtained by drawing perpendiculars from the object to the planes, i.e. by looking from the front and from above. They are then shown on a flat surface by rotating one of the planes as already explained. *It should be remembered that the first and the third quadrants are always opened out while rotating the planes.*

The positions of the views with respect to the reference line will change according to the quadrant in which the object may be situated. This has been explained in detail in the next chapter.

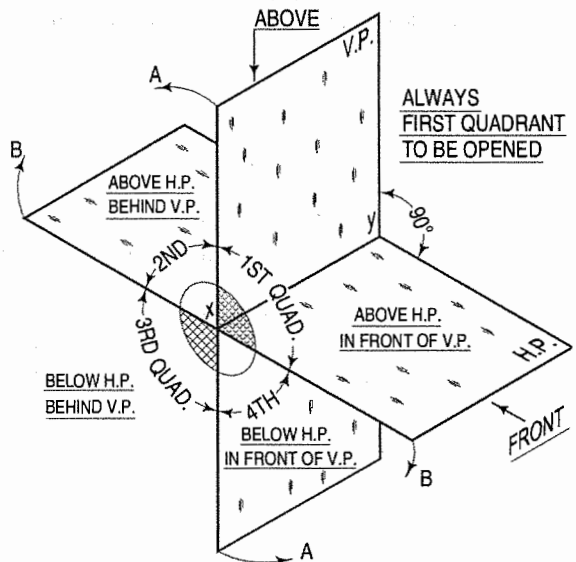


FIG. 8-3

#### 8-6. FIRST-ANGLE PROJECTION

We have assumed the object to be situated in front of the V.P. and above the H.P. i.e. in the first quadrant and then projected it on these planes. This method of projection is known as *first-angle projection method*. The object lies between the observer and the plane of projection. In this method, when the views are drawn in their relative positions, the top view comes below the front view. In other words, the view seen from above is placed on the other side of (i.e. below) the front view. Each projection shows the view of that surface (of the object) which is remote from the plane on which it is projected and which is nearest to the observer.

TABLE 8-1  
DIFFERENCE BETWEEN FIRST-ANGLE PROJECTION METHOD  
AND THIRD-ANGLE PROJECTION METHOD

| No. | First-angle projection method                                                                                                                                                                                                            | Third-angle projection method                                                                                                                                                                                         |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | The object is kept in the <i>first quadrant</i> .                                                                                                                                                                                        | The object is assumed to be kept in the <i>third quadrant</i> .                                                                                                                                                       |
| 2.  | The object lies between the observer and the plane of projection.                                                                                                                                                                        | The plane of projection lies between the observer and the object.                                                                                                                                                     |
| 3.  | The plane of projection is assumed to be non-transparent.                                                                                                                                                                                | The plane of projection is assumed to be <i>transparent</i> .                                                                                                                                                         |
| 4.  | In this method, when the views are drawn in their relative positions, the <i>plan</i> comes <i>below</i> the <i>elevation</i> , the view of the object as observed from the <i>left-side</i> is drawn to the <i>right of elevation</i> . | In this method, when the views are drawn in their relative positions, the <i>plan</i> , comes <i>above</i> the <i>elevation</i> , <i>left hand side view</i> is drawn to the <i>left hand side of the elevation</i> . |
| 5.  | This method of projection is now recommended by the "Bureau of Indian Standards" from 1991.                                                                                                                                              | This method of projection is used in U.S.A. and also in other countries.                                                                                                                                              |

8-7. THIRD-ANGLE PROJECTION

In this method of projection, the object is assumed to be situated in the third quadrant [fig. 8-4(i)]. The planes of projection are assumed to be transparent. They lie between the object and the observer. When the observer views the object from the front, the rays of sight intersect the V.P. The figure formed by joining the points of intersection in correct sequence is the front view of the object. The top view is obtained in a similar manner by looking from above. When the two planes are brought in line with each other, the views will be seen as shown in fig. 8-4(ii). The top view in this case comes above the front view.

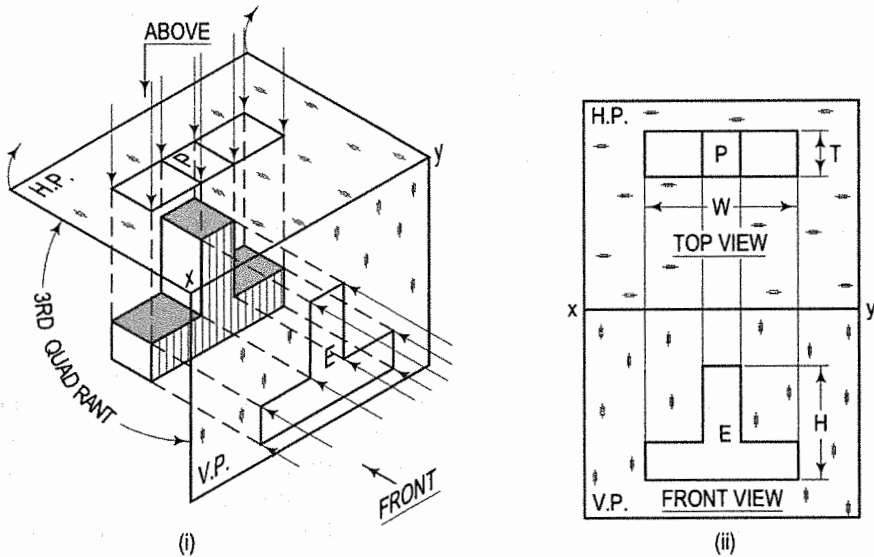


FIG. 8-4

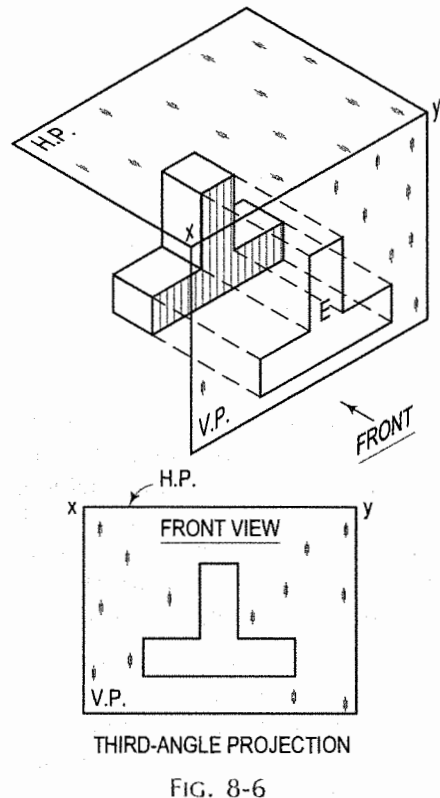
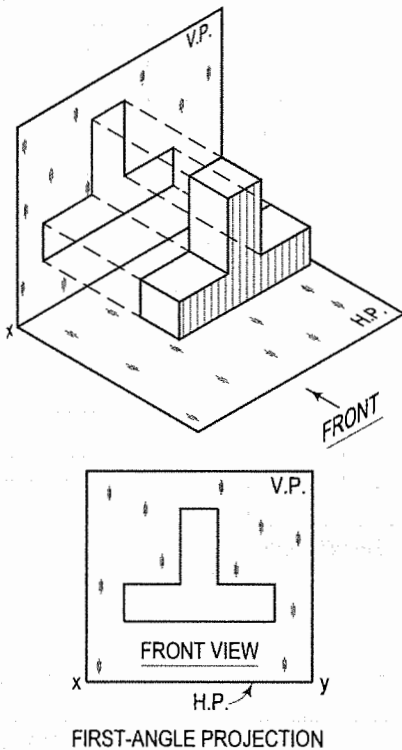
In other words, the view seen from above the object is placed on the same side of (i.e. above) the front view.

Each projection shows the view of that surface (of the object) which is nearest to the plane on which it is projected.

On comparison, it is quite evident that the views obtained by the two methods of projection are completely identical in shape, size and all other details. The difference lies in their relative positions only.

## 8-8. REFERENCE LINE

Studying the projections independently, it can be seen that while considering the front view (fig. 8-5 and fig. 8-6), which is the view as seen from the front, the H.P. coincides with the line  $xy$ . In other words,  $xy$  represents the H.P.



Similarly, while considering the top view (fig. 8-7 and fig 8-8), which is the view obtained by looking from above, the same line  $xy$  represents the V.P. Hence, when the two projections are drawn in correct relationship with each other (fig. 8-9),  $xy$  represents both the H.P. and the V.P. This line  $xy$  is called the *reference line*. The squares or rectangles for individual planes are thus unnecessary and are therefore discarded.

Further, in first-angle projection method, the H.P. is always assumed to be so placed as to coincide with the ground on or above which the object is situated. Hence, in this method, the line  $xy$  is also the line for the ground.

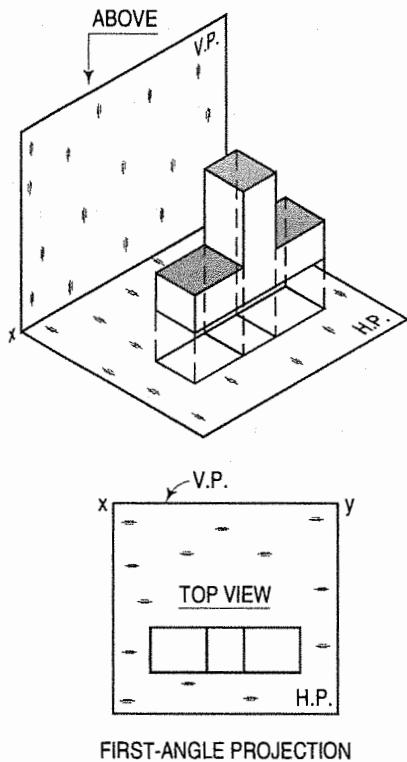


FIG. 8-7

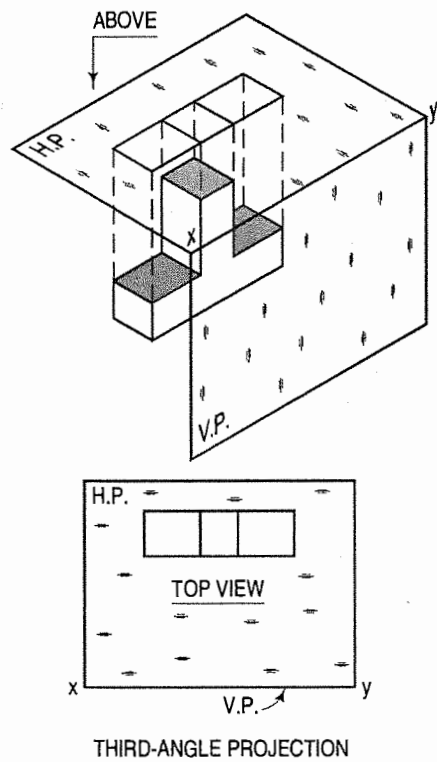


FIG. 8-8

In third-angle projection method, the H.P. is assumed to be placed above the object. The object may be situated on or above the ground. Hence, in this method, the line  $xy$  does not represent the ground. The line for the ground, denoted by letters GL, may be drawn parallel to  $xy$  and below the front view [fig. 8-9(ii)].

In brief, when an object is situated on the ground, in first-angle projection method, the bottom of its front view will coincide with  $xy$ ; in third-angle projection method, it will coincide with GL, while  $xy$  will be above the front view and parallel to Ground line.

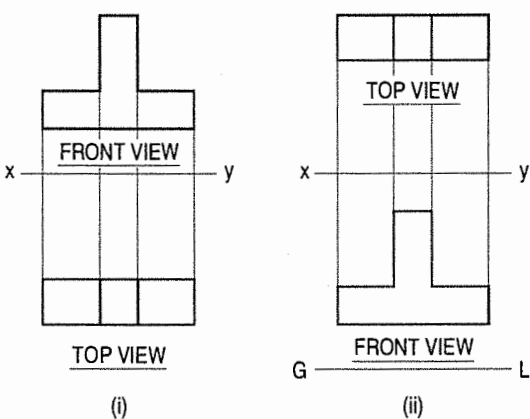
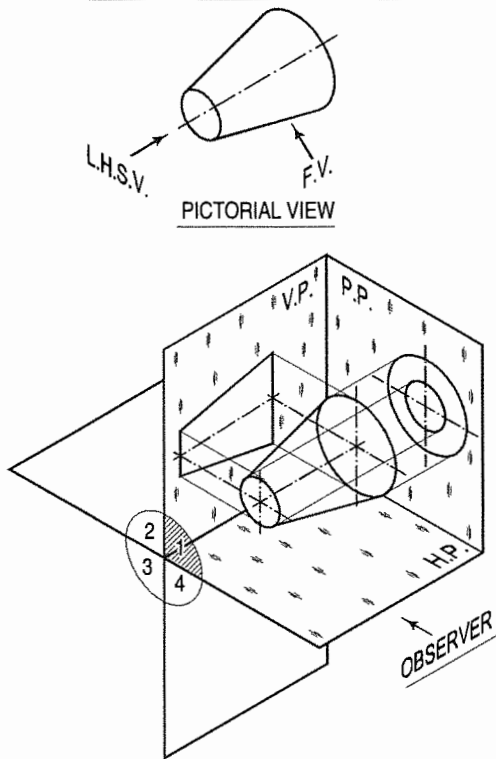


FIG. 8-9

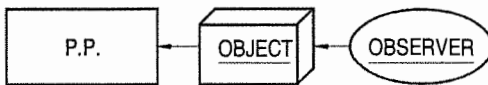
**Symbols for methods of projection:** For every drawing it is absolutely essential to indicate the method of projection adopted. This is done by means of a symbolic figure drawn within the title block on the drawing sheet.

The symbolic figure for the first-angle projection method is shown in fig. 8-10, while that for the third-angle projection method is shown in fig. 8-11 which are self explanatory. These symbolic figures are actually the projections of a frustum of cone of convenient dimensions according to the size of drawing.

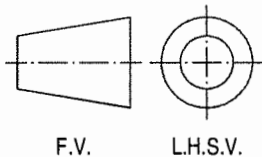
FIRST ANGLE PROJECTION METHOD



FIRST ANGLE PROJECTION



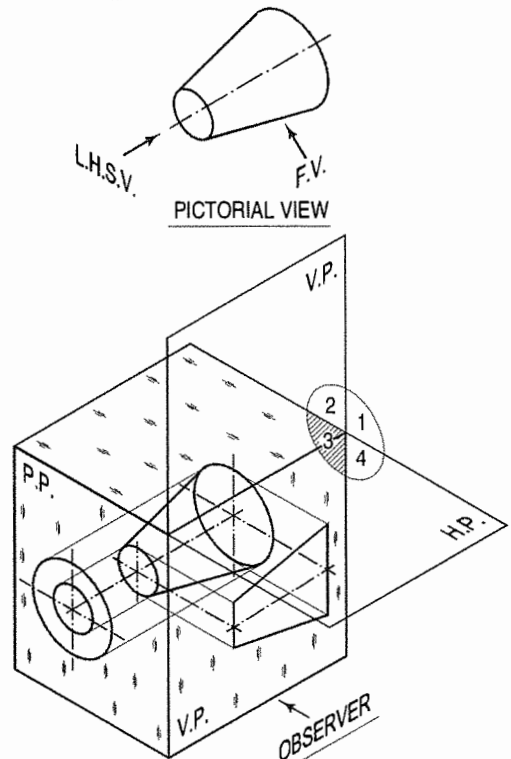
RELATION BETWEEN OBSERVER, OBJECT AND P.P.



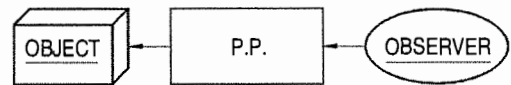
IDENTIFYING GRAPHICAL SYMBOL OF  
FIRST ANGLE PROJECTION

FIG. 8-10

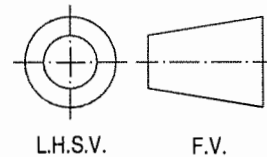
THIRD ANGLE PROJECTION METHOD



THIRD ANGLE PROJECTION



RELATION BETWEEN OBSERVER, OBJECT AND P.P.



IDENTIFYING GRAPHICAL SYMBOL OF  
THIRD ANGLE PROJECTION

FIG. 8-11

**Six views of an Object:** There are *three* important elements of this projection system, namely

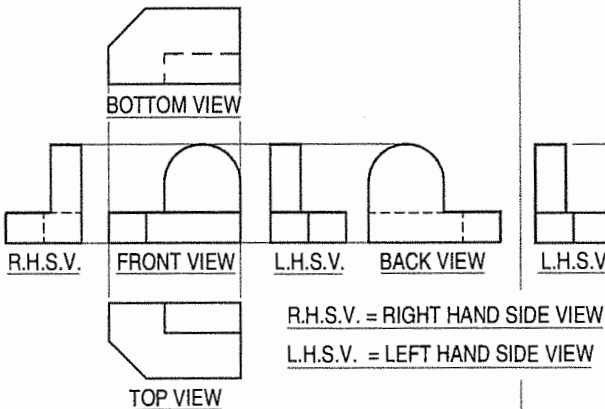
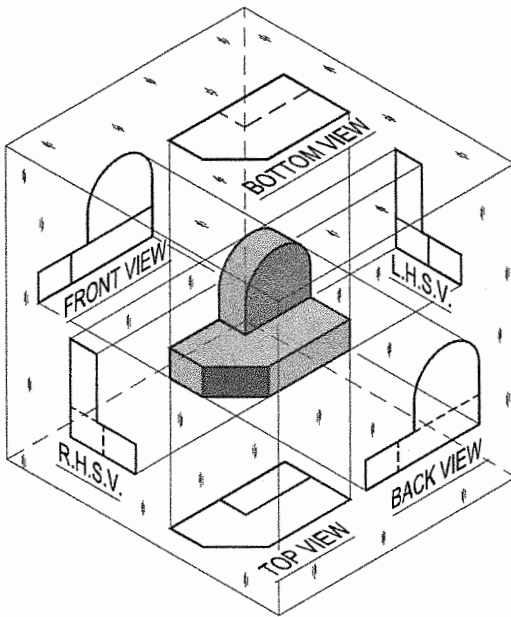
- (a) an object
- (b) plane of projection
- (c) an observer.

Very often, two views are not sufficient to describe an object completely. The planes of projection being imaginary, following six views are obtained:

- |                         |                          |
|-------------------------|--------------------------|
| (1) Front view          | (4) Right hand side view |
| (2) Top view            | (5) Back view            |
| (3) Left hand side view | (6) Bottom view          |

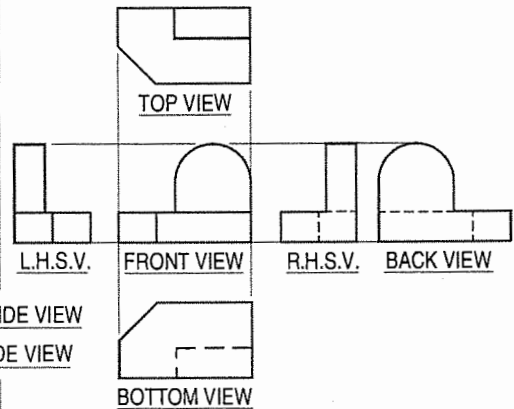
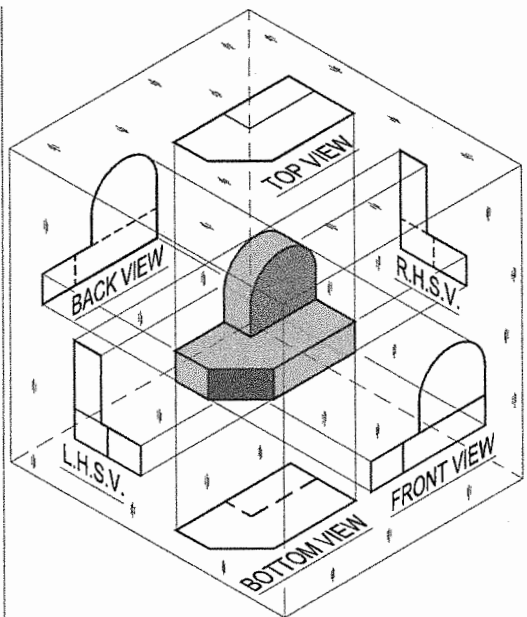
These projections are shown projected on the respective planes, placed by the methods of first-angle projection and third-angle projection as shown in fig. 8-12 and fig. 8-13 respectively.

Ordinarily, two views — the front view and top view are shown. Two other views i.e. L.H.S.V. or R.H.S.V. may be required to describe an object completely. Only in exceptional cases, when an object is of a very complex nature, five or six views may be found necessary.



FIRST ANGLE PROJECTION

FIG. 8-12



THIRD ANGLE PROJECTION

FIG. 8-13

## 8-9. B.I.S. CODE OF PRACTICE



The method of first-angle projection is the British standard practice. The third-angle projection is the standard practice followed in America and in the continent of Europe.

In our country, the first-angle projection method was formerly in use. The Indian Standards Institution (I.S.I.) now Bureau of Indian Standards (B.I.S.), in its earlier versions of Indian Standard (IS:696) 'Code of Practice for General Engineering Drawing' published in 1955 and revised in 1960 had recommended the use of third-angle projection method.

In the second revised version of this standard published in December 1973, the committee responsible for its preparation left the option of selecting first-angle or third-angle projection method to the users.

The committee again reviewed the position and finally recommended revised SP:46-1988 and SP:46-2003 for implementation of first-angle method of projection in our country, by replacing earlier IS:696 drawing standard.

Persons engaged in engineering profession may come across drawings from industries and organizations following any one method. It is therefore necessary for them to be perfectly conversant with both the methods.

*In this book, the method of first-angle projection has been generally followed. Third-angle projection method is also adequately treated in the form of illustrative problems and set exercises.*

**Conventions employed:** In this book, actual points, ends of lines, corners of solids etc., in space are denoted by capital letters *A, B, C* etc. Their *top* views are marked by corresponding small letters *a, b, c*, etc., their *front* views by small letters with dashes *a', b', c'*, their side views by *a<sub>1</sub>, b<sub>1</sub>, c<sub>1</sub>*, and their auxiliary views by *a'<sub>1</sub>, b'<sub>1</sub>, c'<sub>1</sub>*, etc. In pictorial views, the projectors from the points in space to the planes are shown by dashed lines.

The lines from the projections to the reference line *xy* (which are also called projectors, though they are the projections of the projectors) are shown as dash and dot lines. In orthographic views, the projectors and other construction lines are shown continuous, but thinner than the lines for actual projections.

## 8-10. TYPICAL PROBLEMS



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 18 for the following problem.

**Problem 8-1.** (fig. 8-14): A *L-shaped* solid object having dimensions of length (*L*), width (*W*) and height (*H*) in the fig. 8-14. Assuming that this object is lying in the first quadrant. Draw its front view, top view and side view.

When the given object is in the first quadrant, its front view appears in the imaginary vertical plane V.P. behind it while its top view appears in an imaginary horizontal plane, H.P. below. The side view appears to the right or left of the front view depending on from which side the object is being viewed.

- (i) Mark the visible corners of the given block as shown

**Drawing front orthogonal view:**

Assume that you are viewing the object in the direction of the arrow towards the imaginary V.P. What you will see is a rectangle of height  $H$  and width  $W$  on V.P. This will be the front view. To draw this view:

- (ii) Draw a reference line  $xy$ , which represents the intersecting line of the planes V.P. and H.P. Draw a rectangle as shown  $W$  and  $H$ , above  $xy$  make sure that the width is parallel to the line  $xy$ . The rectangle is the front orthogonal view of the object.
- (iii) Draw a line parallel to and thickness of  $h$  to the line 1-2. The rectangle 1-2-4-3 is the front view of the horizontal  $L$ -shaped stem of the object.

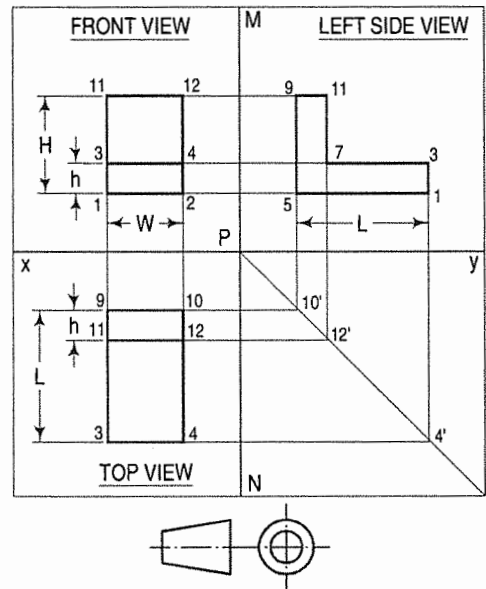
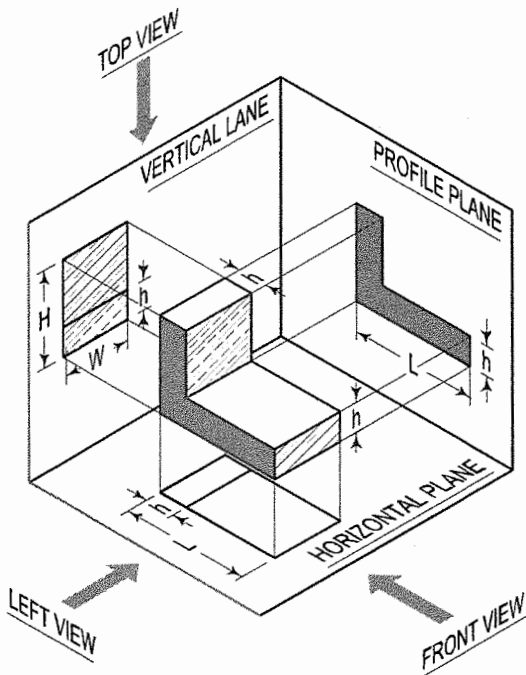


FIG. 8-14

**Drawing top orthogonal view:**

Now, if you look the object from the top, you will see a rectangle of Length  $L$  and width  $W$  on the horizontal plane. This is the top view of plan of the object. To draw this view:

- (iv) Draw vertical projectors from 1 and 2 and extend them beyond the line  $xy$ . Draw a line 9-10 below and parallel to the reference line  $xy$ . Draw the lines 9-3 and 10-4 equal to the length  $L$  of the object. Join line 3-4. The rectangle 9-10-3-4 is the top view of the object.
- (v) Draw a line 11-12 parallel to and below 9-10 of thickness  $h$ . The rectangle 9-11-12-10 is the view of the vertical stem of the object.

**Drawing the side orthogonal view:**

Now if you look at the object from the left side, what you will see is an  $L$ -shaped image having a length of  $L$  and height of  $H$  on the auxiliary plane,  $AP$ . This view appears adjacent and to the right of front view. To draw this view:

- (vi) Draw a reference vertical line  $MN$  at right angles to  $xy$  cutting it at  $P$ . From  $P$  draw a construction line at  $45^\circ$  in the fourth quadrant.
- (vii) Project lines from the points 10, 12 and 4 of the top view to meet this inclined line at  $10'$ ,  $12'$  and  $4'$ .
- (viii) Project lines from points 2, 4 and 12 from the front view parallel to line  $xy$ . From points  $10'$ ,  $12'$  and  $4'$  project lines vertically upwards to meet these horizontal projections.
- (ix) Join points 5-1-3-7-11-9. This will be the side view of the object.
- (x) Finally draw the symbol of first angle projection at the right bottom corner of the drawing.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 19 for the following problem.

**Problem 8-2.** (fig. 8-15 and fig. 8-16): A pictorial view of a machine bracket is shown in the figure. Draw using the first angle projection method front view, top view and right end side view.

Assume that you are viewing the object in the direction of the arrow towards the imaginary V.P. What will you see? It is a rectangle of height 70 mm and width 100 mm on V.P. This will be the front view. To draw this view:

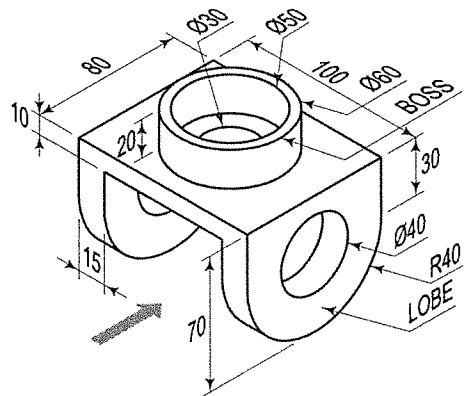


FIG. 8-15

- (i) Draw a Reference line  $xy$ , which represents the intersecting line of the planes V.P. and H.P.
- (ii) Draw rectangular block of size  $100 \times 70$  and the thickness is 15 mm each on the two parallel portions of the slot.
- (iii) Now on the top of this rectangular block draw a rectangle of size  $60 \text{ mm} \times 20 \text{ mm}$ . This rectangle represents a boss of  $\phi 60$  on the rectangular block.
- (iv) Draw a hole of  $\phi 50$  in the above boss. Note carefully that the lines are dashed lines as they are invisible from the front side. The  $\phi 30$  hole is in the rectangular block.

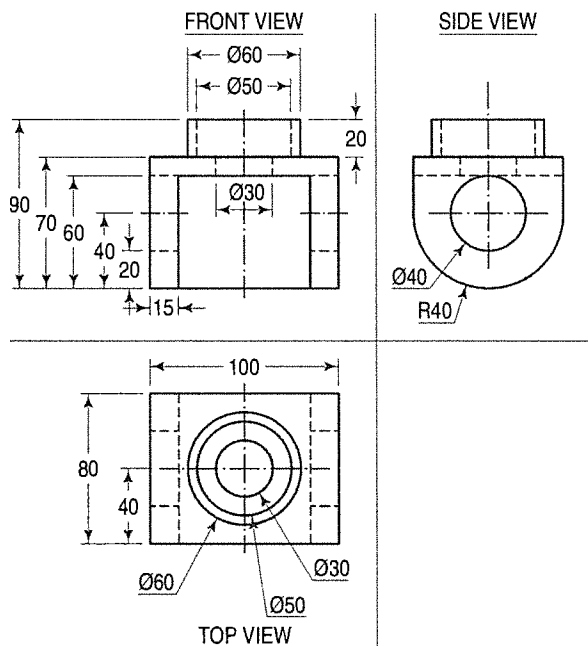


FIG. 8-16

- (v) Draw the two holes on the parallel edges of the rectangular block. These holes are represented by dotted lines in the front view as they are hidden. This completes the front view construction.
- (vi) To draw the top view, project all the details from front view.
- (vii) Draw a vertical centre line from the centre of the rectangular block.
- (viii) The top view of the rectangular block appears as a rectangle of size  $100 \times 80$ .
- (ix) Draw the holes in the centre of the rectangle. These holes are drawn taking the common centre point on vertical centre line.
- (x) The two holes of  $\phi 40$  on the lobes are not visible, so they are projected as dashed lines. This completes the top view.
- (xi) To draw the side view, draw projectors from all the points in front view and top view to side view. Join the intersection points. This completes the side view.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 20 for the following problem.

**Problem 8-3.** Fig. 8-17 shows the pictorial view of the object. Draw the front view, top view and left hand side view.

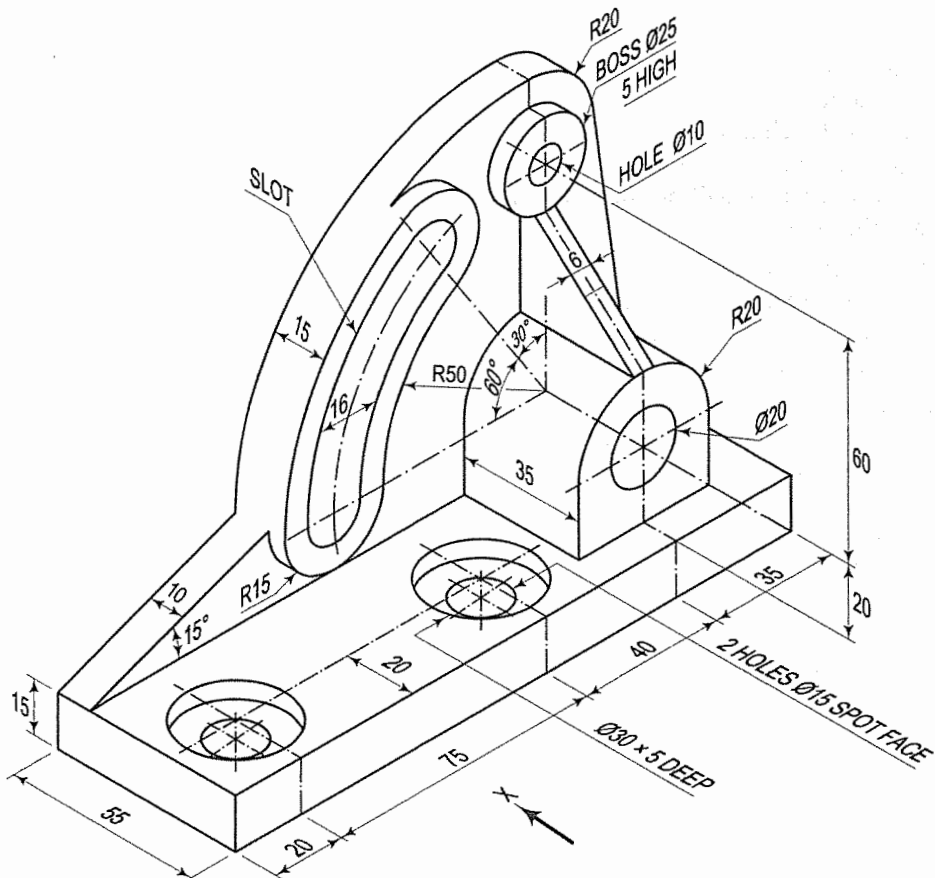
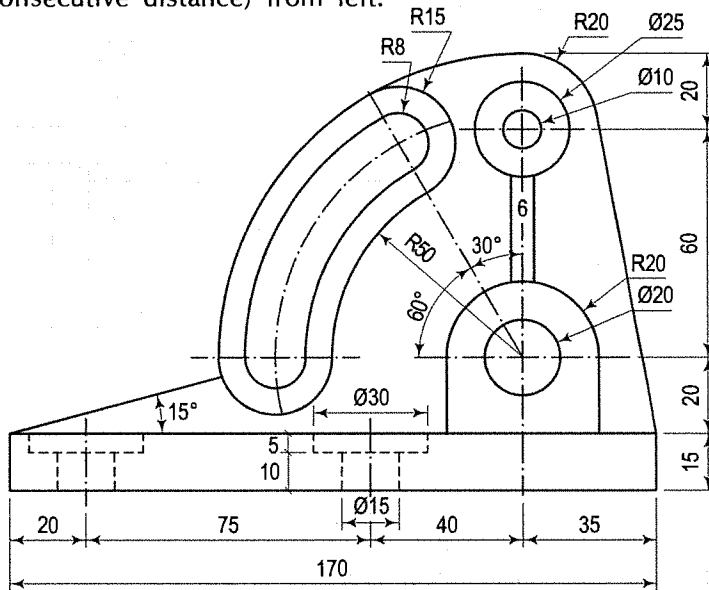


FIG. 8-17

- (i) To draw F.V. in the direction of X, L.H.S.V. and T.V.
- (ii) As the L.H.S.V. is to be drawn, fix the position of  $xy$  at centre of page and  $x_1y_1$  to the left side as L.H.S.V. will come to the right of F.V.
- (iii) Draw the rectangles for the F.V. ( $170 \times 115$ ), project it down to locate T.V. ( $170 \times 55$ ) and take the projections of F.V. to left to draw L.H.S.V., complete the rectangle of L.H.S.V. by taking the projections from T.V.
- (iv) First we will start with F.V. (fig. 8-18).
  - (a) Draw the rectangle for base plate  $170 \times 15$
  - (b) Draw vertical centre lines at 20 mm and 75 mm and 40 mm (consecutive distance) from left.

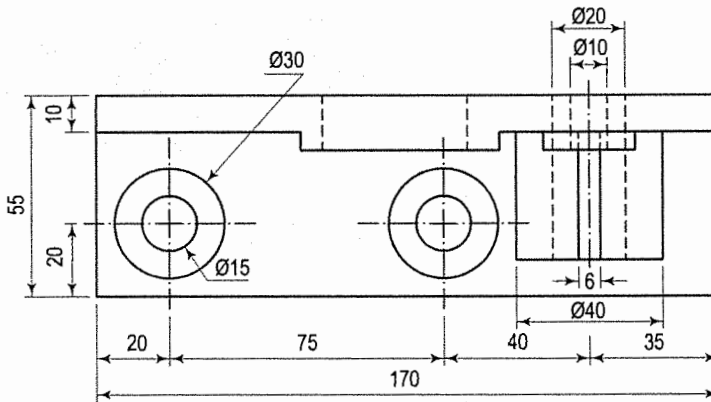


Front view

FIG. 8-18

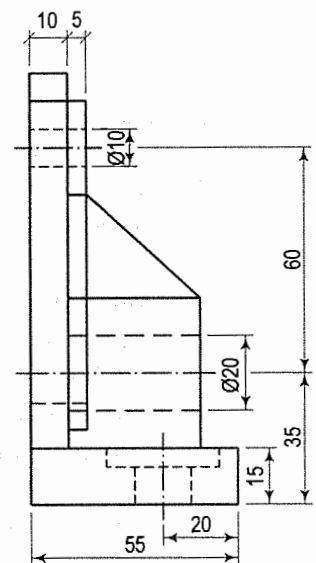
- (c) Draw hidden lines for 5 mm depth for two  $\phi 30$  spot face holes, draw dotted lines for 2 holes of  $\phi 15$  (Inside these holes).
- (d) Draw horizontal centre line at 35 mm from base, in semi-circle.
- (e) Referring third centre line and this horizontal line draw the circle of 10 mm radius and semicircle of 20 mm radius.
- (f) Draw vertical lines from end of this semi-circle. Till it touches to base.
- (g) Draw another horizontal centre line at 60 mm above previous horizontal centre line.
- (h) With this as centre, draw circles of  $\phi 10$  and  $\phi 25$ .
- (i) Also draw the arc of radius 20 mm, join this to base with inclined line to represent rib.
- (j) Draw two lines at 3 mm to both the sides, vertical centre lines of the circles to represent the rib.
- (k) From third centre point, draw centre line at  $30^\circ$ .
- (l) Draw arcs of 50 mm, 73 mm, 57 mm, 80 mm.

- (m) Draw an arc of centre line pattern of 65 mm radius. (Draw centre line for an arc)
- (n) Taking intersection of this as reference and horizontal centre line draw semi-circle of radius 15 mm and 8 mm.
- (o) Also draw same arcs, where this line intersects with  $30^\circ$  inclined line.
- (p) Erase unwanted arcs if any.
- (q) Draw inclined line at  $15^\circ$  from left end of block.
- (r) Show the required dimensions.
- (s) Now project all the required line to draw T.V.
- (v) Now draw the top view (fig. 8-19).



Top view  
FIG. 8-19

- (a) Complete the rectangle of the top view ( $170 \times 55$ )
- (b) Project the centre lines. (Vertical at 20, 75 and 40 consecutive distances from left and at 20 above base)
- (c) Draw the two circles of  $\varnothing 30$  and  $\varnothing 15$  to represent spot holes.
- (d) Draw projection of inclined rib 10 thick and circular slot 15 thick.
- (e) Draw hidden lines for inner circular slot.
- (f) Draw rectangle for semicircular projection from F.V. projections
- (g) Draw hidden lines for hole of  $\varnothing 20$  in it.
- (h) Show the projections of boss holes.
- (i) Draw hidden lines for hole of  $\varnothing 10$ .
- (j) Show the required dimensions.
- (vi) Take projections from F.V. and T.V. to complete L.H.S.V. (fig. 8-20).



L.H.S.V.  
FIG. 8-20

- (a) Complete the base rectangle of  $55 \text{ mm} \times 15 \text{ mm}$ . To represent the base.

- (b) Complete the rectangle at left side in  $L$  shape of 10 mm thick.
- (c) Show the projections of projected semi-circle.
- (d) Show hidden lines of hole in it.
- (e) Show projection of projected slot and show hidden lines in it.
- (f) Draw projection of boss and show hole in it by dotted line.
- (g) Show inclined line for the rib.
- (h) Draw front view, top view and L.H.S.V. in their relative position.

## EXERCISES 8



1. Define orthographic projection. Describe briefly the method of obtaining an orthographic projection of an object.
2. Write short notes on:  
Reference planes; Reference line; Projector; Front view; Ground line.
3. Sketch neatly the symbols used for indicating the method of projection adopted in a drawing. State where this symbol is drawn on a drawing sheet.
4. Explain briefly how the reference line represents both the principal planes of projection.
5. Explain clearly the difference between the first-angle projection method and the third-angle projection method.
6. Fill-up the blanks in the following with appropriate words selected from the list of words given below:
  - (a) In \_\_\_\_\_ projection, the \_\_\_\_\_ are perpendicular to the \_\_\_\_\_ of projection.
  - (b) In first-angle projection method,
    - (i) the \_\_\_\_\_ comes between the \_\_\_\_\_ and the \_\_\_\_\_.
    - (ii) the \_\_\_\_\_ view is always \_\_\_\_\_ the \_\_\_\_\_ view.
  - (c) In third-angle projection method,
    - (i) the \_\_\_\_\_ comes between the \_\_\_\_\_ and the \_\_\_\_\_.
    - (ii) the \_\_\_\_\_ view is always \_\_\_\_\_ the \_\_\_\_\_ view.

List of words for Exercise (6):

- |          |                 |               |
|----------|-----------------|---------------|
| 1. Above | 5. Object       | 9. Projectors |
| 2. Below | 6. Orthographic | 10. Plane     |
| 3. Front | 7. Observer     | 11. Side      |
| 4. Left  | 8. Right        | 12. Top.      |

Answer to Exercise (6):

- (a) – 6, 9 and 10,
  - (b) (i) – 5, 7 and 10,
  - (b) (ii) – 12, 2 and 3,
  - (c) (i) – 10, 5 and 7,
  - (c) (ii) – 12, 1 and 3.
7. Why second and fourth quadrants are not used in practice?
  8. What is the convention of representing first-angle projection method?

9. The pictorial view of different types of objects are shown in fig. 8-21. Sketch, looking from arrow, elevation, plan and end-view using first-angle projection method.

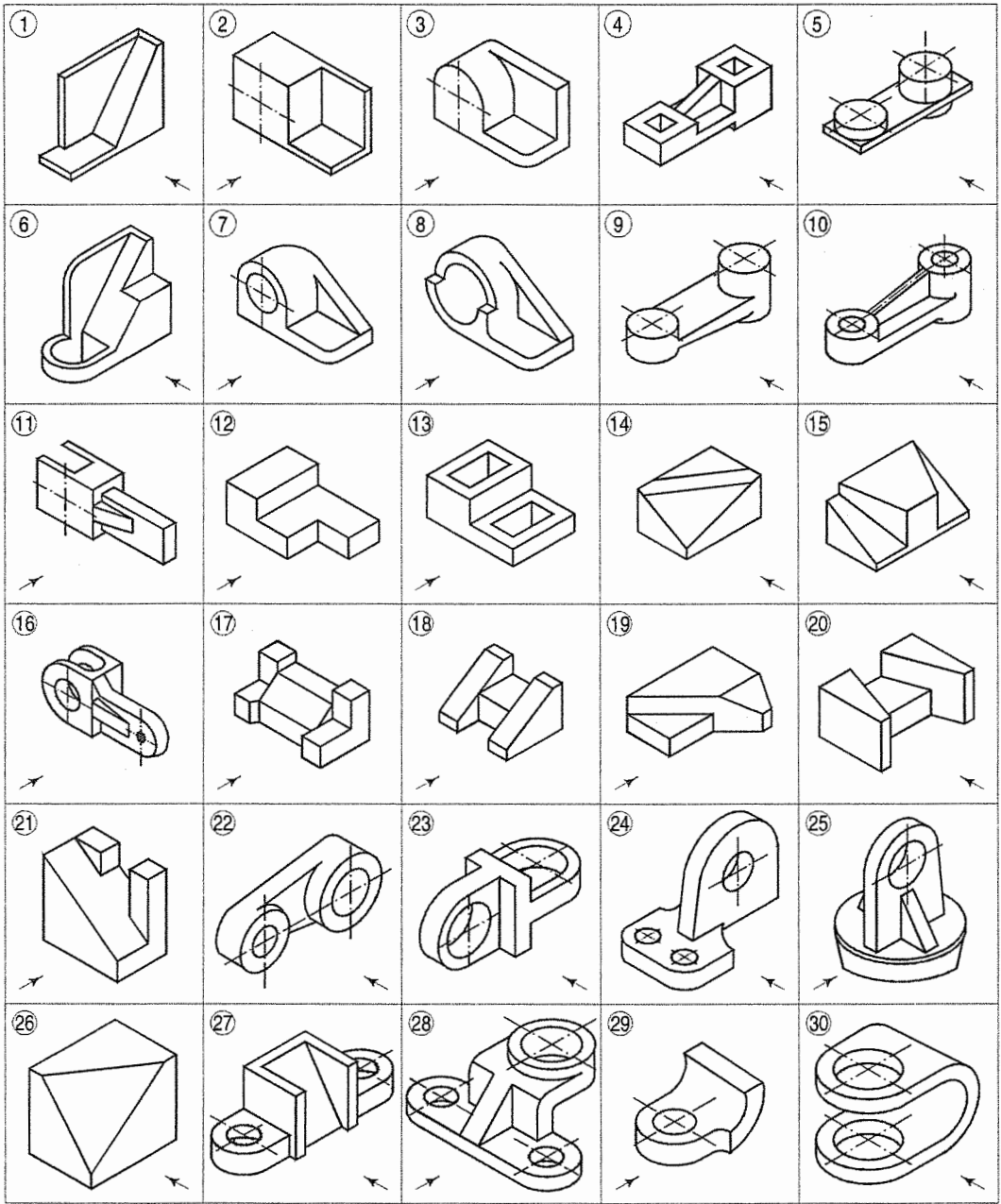


FIG. 8-21

[Draw orthographic projections of each object and then compare your answer with the solution given in fig. 8-26.]

10. What dimensions of an object are given by
- Front view or elevation?
  - Plan or top view?
  - Left-hand side view and right-hand side view?
11. Fig. 8-22 and fig. 8-23 show the orthographic projections of the objects in the first-angle projection method.  
Draw them in the third-angle projection method.

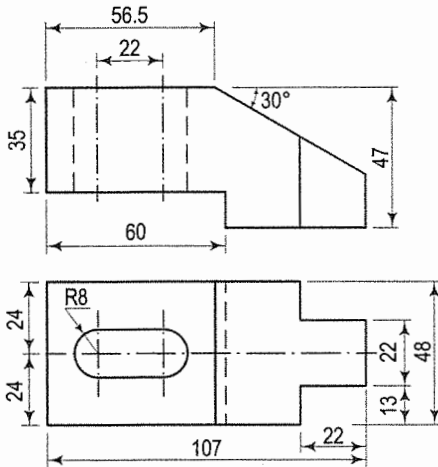


FIG. 8-22

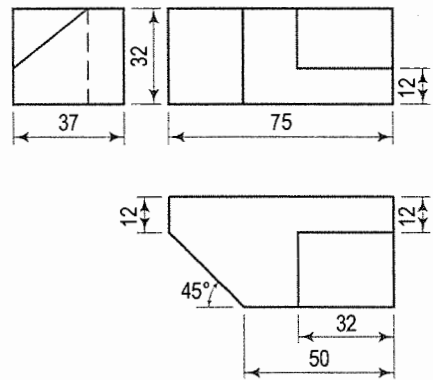


FIG. 8-23

12. Fig. 8-24 and fig. 8-25 show the orthographic projections of the objects in the third-angle projection method.  
Redraw them in the first-angle projection method.

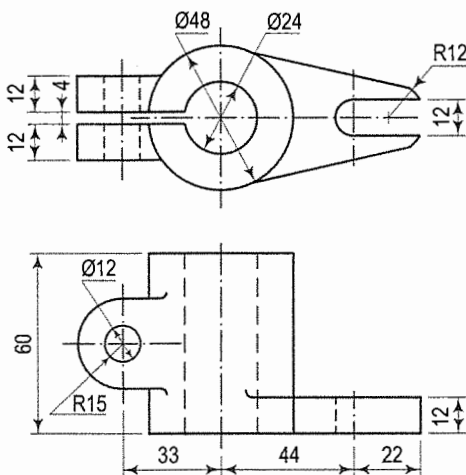


FIG. 8-24

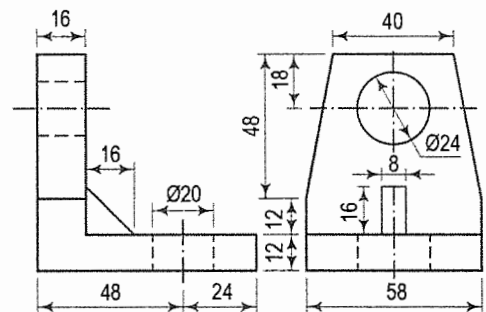
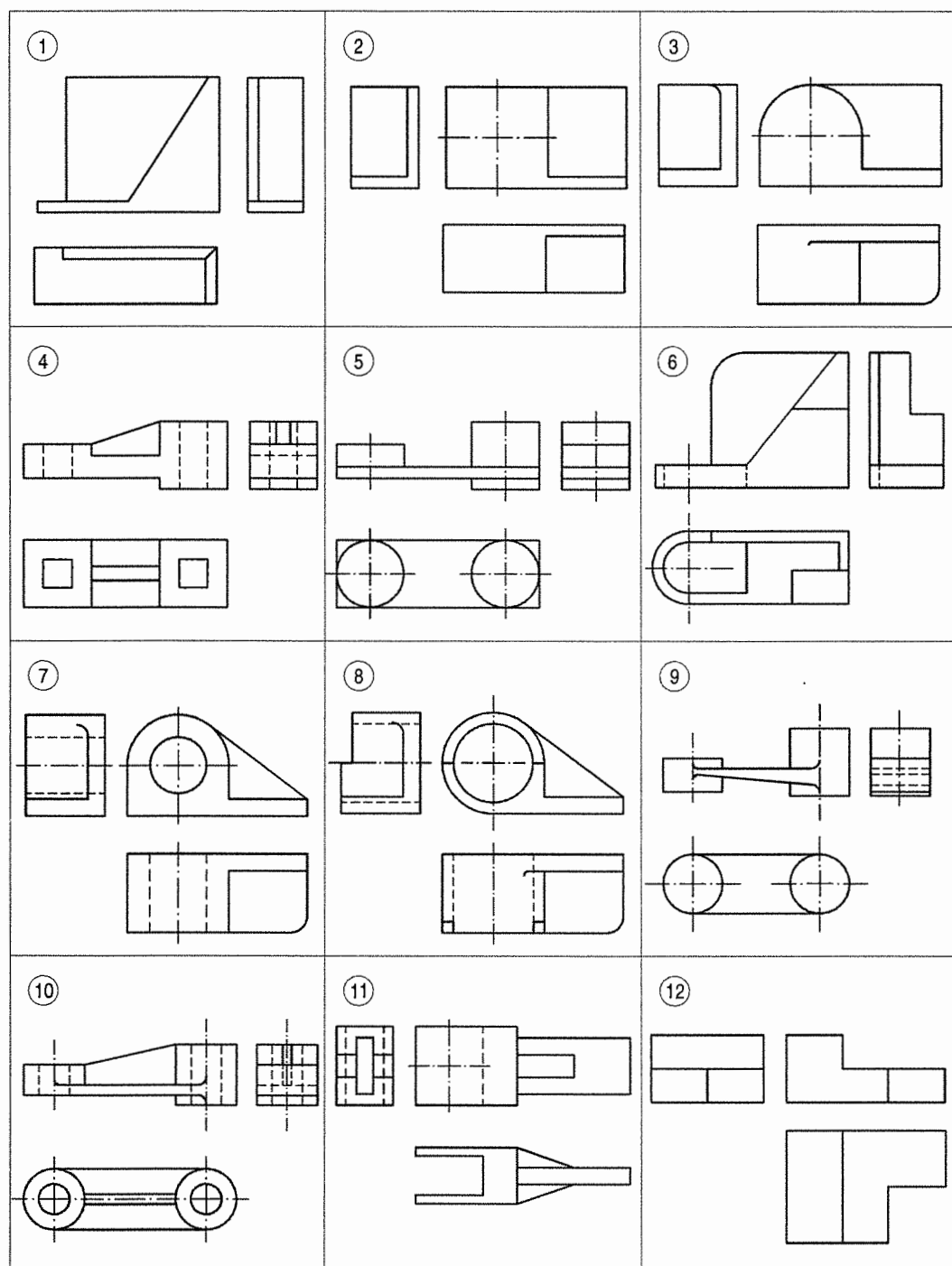


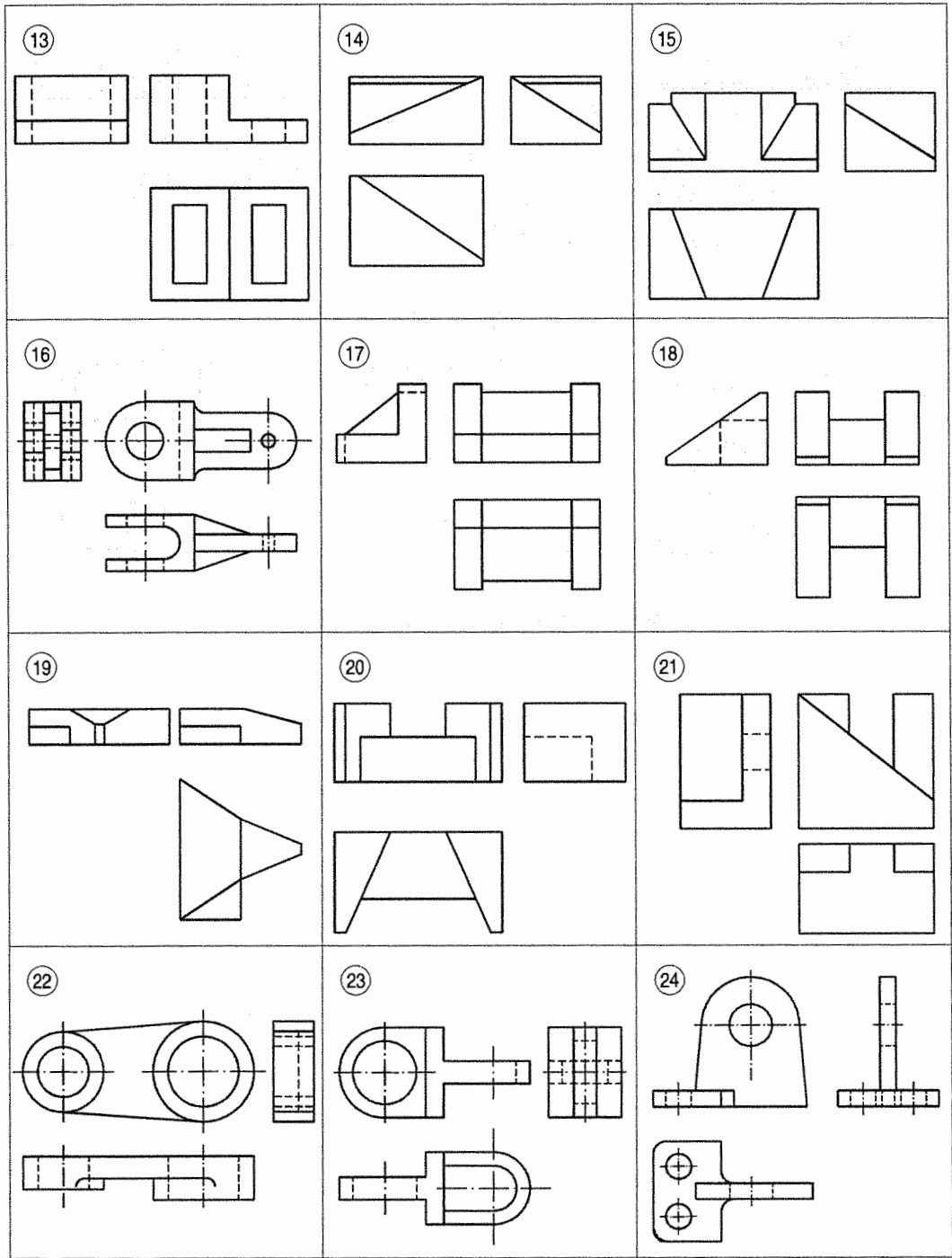
FIG. 8-25

13. What are the important elements of the projection system? Draw any simple object and obtain its six views, projected on imaginary planes by the methods of first angle projection and third angle projection.



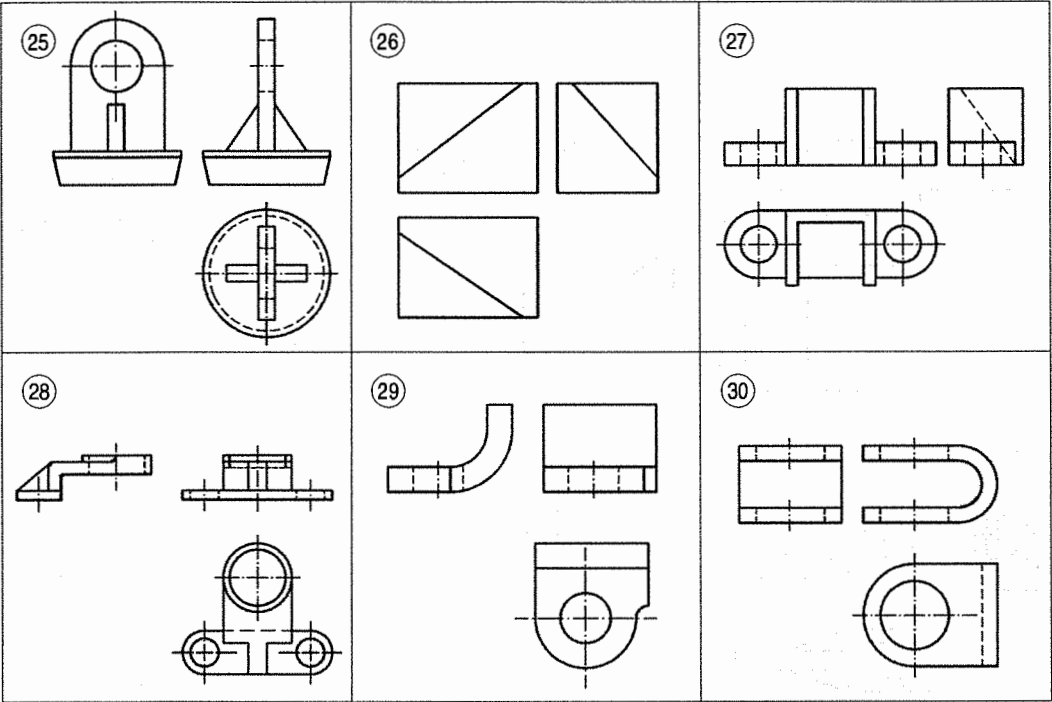
[Answer to Exercise (9), fig. 8-21] (Continued...)

FIG. 8-26



[Answer to Exercise (9), fig. 8-21] (Continued...)

FIG. 8-26



[Answer to Exercise (9), fig. 8-21]  
FIG. 8-26

# Chapter

# 9



# PROJECTIONS OF POINTS

## 9-0. INTRODUCTION

A point may be situated, in space, in any one of the four quadrants formed by the two principal planes of projection or may lie in any one or both of them. Its projections are obtained by extending projectors perpendicular to the planes.

One of the planes is then rotated so that the first and third quadrants are opened out. The projections are shown on a flat surface in their respective positions either above or below or in  $xy$ .



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 21 for the projections of points.

## 9-1. A POINT IS SITUATED IN THE FIRST QUADRANT

The pictorial view [fig. 9-1(i)] shows a point  $A$  situated above the H.P. and in front of the V.P., i.e. in the first quadrant.  $a'$  is its front view and  $a$  the top view. After rotation of the plane, these projections will be seen as shown in fig. 9-1(ii).

The front view  $a'$  is above  $xy$  and the top view  $a$  below it. The line joining  $a'$  and  $a$  (which also is called a projector), intersects  $xy$  at right angles at a point  $o$ . It is quite evident from the pictorial view that  $a'o = Aa$ , i.e. the distance of the front view from  $xy$  = the distance of  $A$  from the H.P. viz.  $h$ . Similarly,  $ao = Aa'$ , i.e. the distance of the top view from  $xy$  = the distance of  $A$  from the V.P. viz.  $d$ .

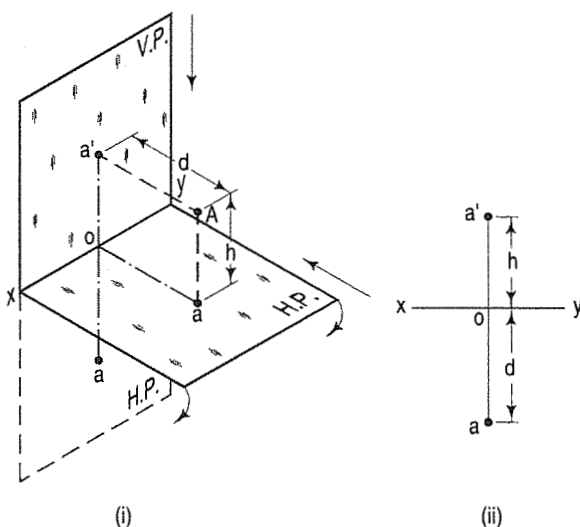


FIG. 9-1

### 9-2. A POINT IS SITUATED IN THE SECOND QUADRANT

A point  $B$  (fig. 9-2) is above the H.P. and behind the V.P., i.e. in the second quadrant.  $b'$  is the front view and  $b$  the top view.

When the planes are rotated, both the views are seen above  $xy$ . Note that  $b'o = Bb$  and  $bo = Bb'$ .

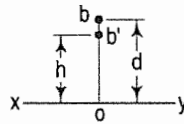
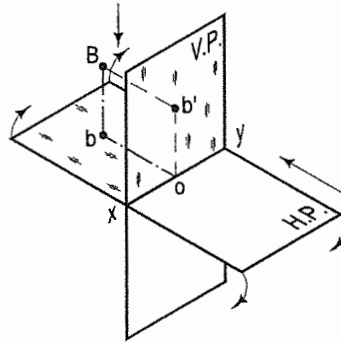


FIG. 9-2

### 9-3. A POINT IS SITUATED IN THE THIRD QUADRANT

A point  $C$  (fig. 9-3) is below the H.P. and behind the V.P., i.e. in the third quadrant. Its front view  $c'$  is below  $xy$  and the top view  $c$  above  $xy$ . Also  $c'o = Cc$  and  $co = Cc'$ .

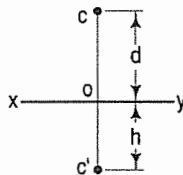
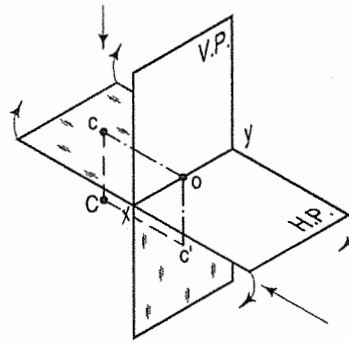


FIG. 9-3

### 9-4. A POINT IS SITUATED IN THE FOURTH QUADRANT

A point  $E$  (fig. 9-4) is below the H.P. and in front of the V.P., i.e. in the fourth quadrant. Both its projections are below  $xy$ , and  $e'o = Ee$  and  $eo = Ee'$ .

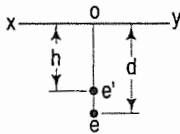
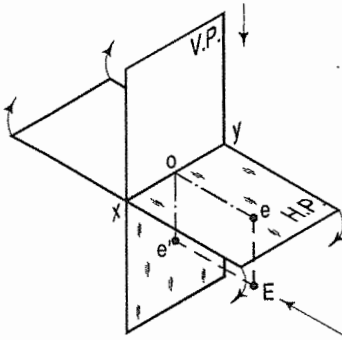


FIG. 9-4

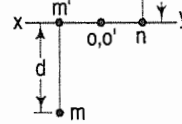
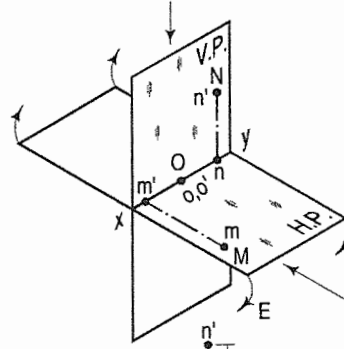


FIG. 9-5

Referring to fig. 9-5, we see that,

- (i) A point  $M$  is in the H.P. and in front of the V.P. Its front view  $m'$  is in  $xy$  and the top view  $m$  below it.
- (ii) A point  $N$  is in the V.P. and above the H.P. Its top view  $n$  is in  $xy$  and the front view  $n'$  above it.
- (iii) A point  $O$  is in both the H.P. and the V.P. Its projection  $o$  and  $o'$  coincide with each other in  $xy$ .

### 9-5. GENERAL CONCLUSIONS

- (i) The line joining the top view and the front view of a point is always perpendicular to  $xy$ . It is called a *projector*.
- (ii) When a point is above the H.P., its front view is above  $xy$ ; when it is below the H.P., the front view is below  $xy$ . The distance of a point from the H.P. is shown by the length of the projector from its front view to  $xy$ , e.g.  $a'o$ ,  $b'o$  etc.
- (iii) When a point is in front of the V.P., its top view is below  $xy$ ; when it is behind the V.P., the top view is above  $xy$ . The distance of a point from the V.P. is shown by the length of the projector from its top view to  $xy$ , e.g.  $ao$ ,  $bo$  etc.
- (iv) When a point is in a reference plane, its projection on the other reference plane is in  $xy$ .

**Problem 9-1.** (fig. 9-1): A point  $A$  is 25 mm above the H.P. and 30 mm in front of the V.P. Draw its projections.

- (i) Draw the reference line  $xy$  [fig. 9-1(ii)].

- (ii) Through any point  $o$  in it, draw a perpendicular.

As the point is above the H.P. and in front of the V.P. its front view will be above  $xy$  and the top view below  $xy$ .

- (iii) On the perpendicular, mark a point  $a'$  above  $xy$ , such that  $a'o = 25$  mm. Similarly, mark a point  $a$  below  $xy$ , so that  $ao = 30$  mm.  $a'$  and  $a$  are the required projections.

**Problem 9-2.** (fig. 9-6): A point  $A$  is 20 mm below the H.P. and 30 mm behind the V.P. Draw its projections.

As the point is below the H.P. and behind the V.P., its front view will be below  $xy$  and the top view above  $xy$ .

Draw the projections as explained in problem 9-1 and as shown in fig. 9-6.

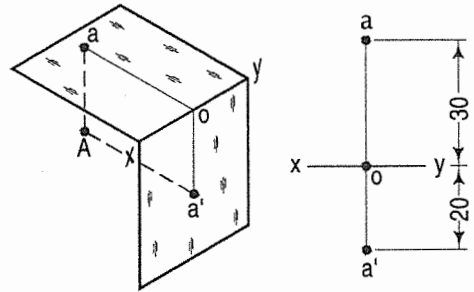


FIG. 9-6

**Problem 9-3.** (fig. 9-7): A point  $P$  is in the first quadrant. Its shortest distance from the intersection point of H.P., V.P. and Auxiliary vertical plane, perpendicular to the H.P. and V.P. is 70 mm and it is equidistant from principal planes (H.P. and V.P.). Draw the projections of the point and determine its distance from the H.P. and V.P.

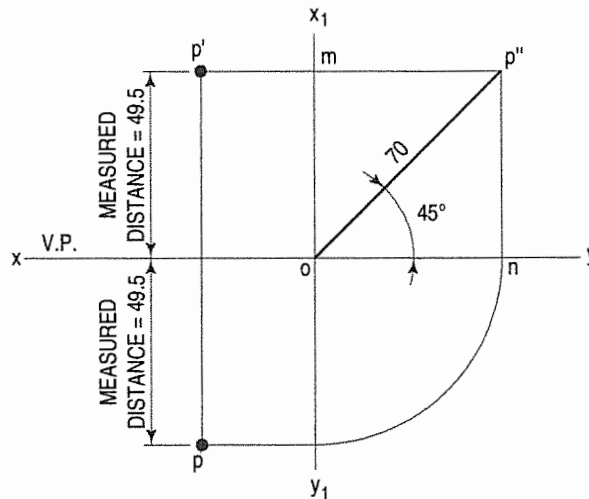


FIG. 9-7

*Note:*  $O$  represents intersection of H.P., V.P. and A.V.P.

- (i) Draw  $xy$  and  $x_1 y_1$  perpendicular reference lines.
- (ii)  $O$  represents intersection of H.P., V.P. and A.V.P.
- (iii) Draw from  $O$  a line inclined at  $45^\circ$  of 70 mm length.
- (iv) Project from  $P''$  on  $xy$  line and  $x_1 y_1$ . The projections are  $n$  and  $m$  respectively as shown in figure. From  $O$  draw arc intersecting  $x_1 y_1$ .
- (v) Draw a parallel line at convenient distance from  $x_1 y_1$ . Extend  $P''m$  to intersect a parallel line at  $p'$  and  $p$  as shown.
- (vi) Measure distance from  $xy$  line, which is nearly 49.4974 mm say 49.5 mm.

**Projections on auxiliary plane:** Sometime projections of object on the principal (H.P. and V.P.) are insufficient. In such situation, another projection plane perpendicular to the principal planes is taken. This plane is known as auxiliary plane. The projection on the auxiliary plane is known as side view or side elevation. Refer fig. 9-8.

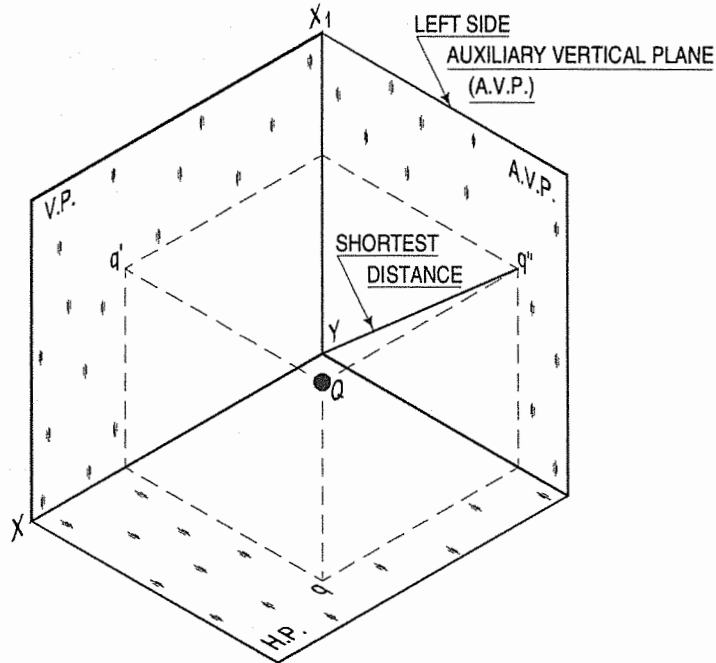


FIG. 9-8

The A.V.P. can be also taken right side also. For more details on projection on auxiliary plane, refer chapter 11.

## EXERCISES 9

- Draw the projections of the following points on the same ground line, keeping the projectors 25 mm apart.  
 A, in the H.P. and 20 mm behind the V.P.  
 B, 40 mm above the H.P. and 25 mm in front of the V.P.  
 C, in the V.P. and 40 mm above the H.P.  
 D, 25 mm below the H.P. and 25 mm behind the V.P.  
 E, 15 mm above the H.P. and 50 mm behind the V.P.  
 F, 40 mm below the H.P. and 25 mm in front of the V.P.  
 G, in both the H.P. and the V.P.
- A point *P* is 50 mm from both the reference planes. Draw its projections in all possible positions.
- State the quadrants in which the following points are situated:
  - A point *P*; its top view is 40 mm above *xy*; the front view, 20 mm below the top view.
  - A point *Q*, its projections coincide with each other 40 mm below *xy*.

4. A point  $P$  is 15 mm above the H.P. and 20 mm in front of the V.P. Another point  $Q$  is 25 mm behind the V.P. and 40 mm below the H.P. Draw projections of  $P$  and  $Q$  keeping the distance between their projectors equal to 90 mm. Draw straight lines joining (i) their top views and (ii) their front views.
5. Projections of various points are given in fig. 9-9. State the position of each point with respect to the planes of projection, giving the distances in centimetres.

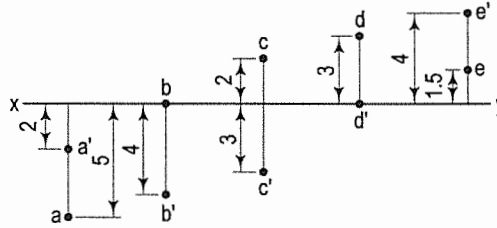


FIG. 9-9

6. Two points  $A$  and  $B$  are in the H.P. The point  $A$  is 30 mm in front of the V.P., while  $B$  is behind the V.P. The distance between their projectors is 75 mm and the line joining their top views makes an angle of  $45^\circ$  with  $xy$ . Find the distance of the point  $B$  from the V.P.
7. A point  $P$  is 20 mm below H.P. and lies in the third quadrant. Its shortest distance from  $xy$  is 40 mm. Draw its projections.
8. A point  $A$  is situated in the first quadrant. Its shortest distance from the intersection point of H.P., V.P. and auxiliary plane is 60 mm and it is equidistant from the principal planes. Draw the projections of the point and determine its distance from the principal planes.
9. A point 30 mm above  $xy$  line is the plan-view of two points  $P$  and  $Q$ . The elevation of  $P$  is 45 mm above the H.P. while that of the point  $Q$  is 35 mm below the H.P. Draw the projections of the points and state their position with reference to the principal planes and the quadrant in which they lie.
10. A point  $Q$  is situated in first quadrant. It is 40 mm above H.P. and 30 mm in front of V.P. Draw its projections and find its shortest distance from the intersection of H.P., V.P. and auxiliary plane.

## Chapter 10



# PROJECTIONS OF STRAIGHT LINES

### 10-0. INTRODUCTION

A straight line is the shortest distance between two points. Hence, the projections of a straight line may be drawn by joining the respective projections of its ends which are points.

The position of a straight line may also be described with respect to the two reference planes. It may be:

1. Parallel to one or both the planes.
2. Contained by one or both the planes.
3. Perpendicular to one of the planes.
4. Inclined to one plane and parallel to the other.
5. Inclined to both the planes.
6. Projections of lines inclined to both the planes.
7. Line contained by a plane perpendicular to both the reference planes.
8. True length of a straight line and its inclinations with the reference planes.
9. Traces of a line.
10. Methods of determining traces of a line.
11. Traces of a line, the projections of which are perpendicular to  $xy$ .
12. Positions of traces of a line.

### 10-1. LINE PARALLEL TO ONE OR BOTH THE PLANES

(FIG. 10-1)

- (a) Line  $AB$  is parallel to the H.P.

$a$  and  $b$  are the top views of the ends  $A$  and  $B$  respectively. It can be clearly seen that the figure  $ABba$  is a rectangle. Hence, the top view  $ab$  is equal to  $AB$ .

$a'b'$  is the front view of  $AB$  and is parallel to  $xy$ .

- (b) Line  $CD$  is parallel to the V.P.

The line  $c'd'$  is the front view and is equal to  $CD$ ; the top view  $cd$  is parallel to  $xy$ .

- (c) Line  $EF$  is parallel to the H.P. and the V.P.

$ef$  is the top view and  $e'f'$  is the front view; both are equal to  $EF$  and parallel to  $xy$ .

Hence, when a line is parallel to a plane, its projection on that plane is equal to its true length; while its projection on the other plane is parallel to the reference line.

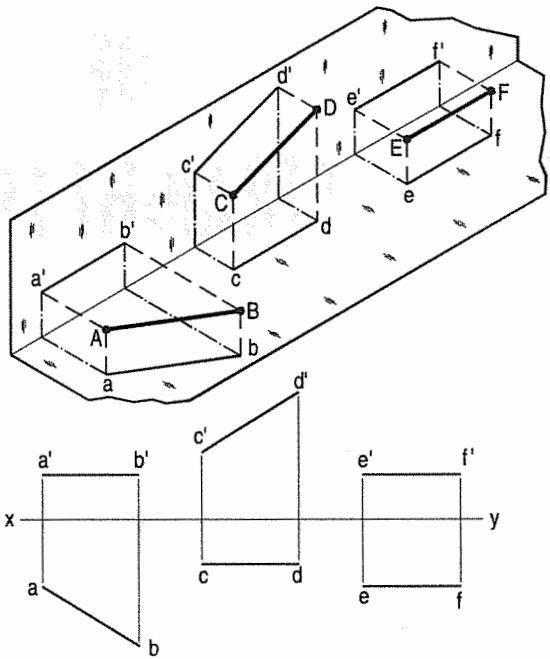


FIG. 10-1

**10-2. LINE CONTAINED BY ONE OR BOTH THE PLANES**

(FIG. 10-2)

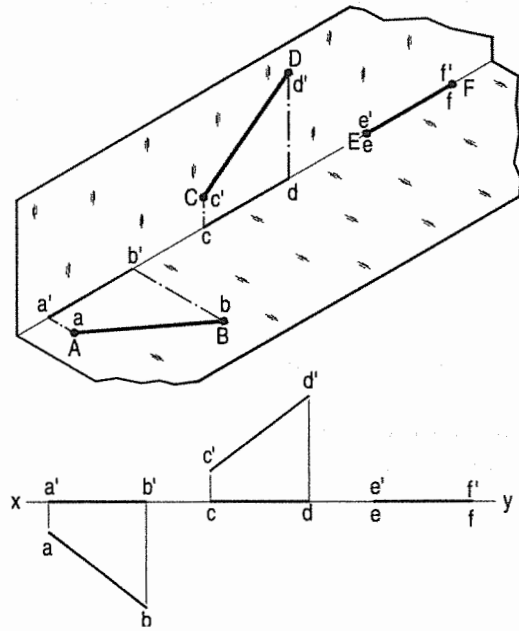


FIG. 10-2

Line  $AB$  is in the H.P. Its top view  $ab$  is equal to  $AB$ ; its front view  $a' b'$  is in  $xy$ .

Line  $CD$  is in the V.P. Its front view  $c'd'$  is equal to  $CD$ ; its top view  $cd$  is in  $xy$ .

Line  $EF$  is in both the planes. Its front view  $e' f'$  and the top view  $ef$  coincide with each other in  $xy$ .

Hence, when a line is contained by a plane, its projection on that plane is equal to its true length; while its projection on the other plane is in the reference line.

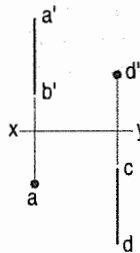
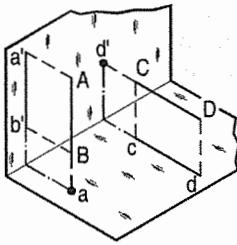
### 10-3. LINE PERPENDICULAR TO ONE OF THE PLANES

(FIG. 10-3)

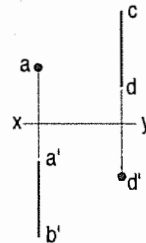


This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 22 for the line perpendicular to one of the planes.

When a line is perpendicular to one reference plane, it will be parallel to the other.



(FIRST-ANGLE PROJECTION)



(THIRD-ANGLE PROJECTION)

FIG. 10-3

- Line  $AB$  is perpendicular to the H.P. The top views of its ends coincide in the point  $a$ . Hence, the top view of the line  $AB$  is the point  $a$ . Its front view  $a' b'$  is equal to  $AB$  and perpendicular to  $xy$ .
- Line  $CD$  is perpendicular to the V.P. The point  $d'$  is its front view and the line  $cd$  is the top view.  $cd$  is equal to  $CD$  and perpendicular to  $xy$ .

Hence, when a line is perpendicular to a plane its projection on that plane is a point; while its projection on the other plane is a line equal to its true length and perpendicular to the reference line.

In first-angle projection method, when top views of two or more points coincide, the point which is comparatively farther away from  $xy$  in the front view will be visible; and when their front views coincide, that which is farther away from  $xy$  in the top view will be visible.

In third-angle projection method, it is just the reverse. When top views of two or more points coincide the point which is comparatively nearer  $xy$  in the front view will be visible; and when their front views coincide, the point which is nearer  $xy$  in the top view will be visible.

## 10-4. LINE INCLINED TO ONE PLANE AND PARALLEL TO THE OTHER



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 23 for the line inclined to one plane and parallel to the other.

*The inclination of a line to a plane is the angle which the line makes with its projection on that plane.*

- (a) Line  $PQ_1$  [fig. 10-4(i)] is inclined at an angle  $\theta$  to the H.P. and is parallel to the V.P. The inclination is shown by the angle  $\theta$  which  $PQ_1$  makes with its own projection on the H.P., viz. the top view  $pq_1$ .

The projections [fig. 10-4(ii)] may be drawn by first assuming the line to be parallel to both the H.P. and the V.P. Its front view  $p'q'$  and the top view  $pq$  will both be parallel to  $xy$  and equal to the true length. When the line is turned about the end  $P$  to the position  $PQ_1$  so that it makes the angle  $\theta$  with the H.P. while remaining parallel to the V.P., in the front view the point  $q'$  will move along an arc drawn with  $p'$  as centre and  $p'q'$  as radius to a point  $q'_1$  so that  $p'q'_1$  makes the angle  $\theta$  with  $xy$ . In the top view,  $q$  will move towards  $p$  along  $pq$  to a point  $q_1$  on the projector through  $q'_1$ .  $p'q'_1$  and  $pq_1$  are the front view and the top view respectively of the line  $PQ_1$ .

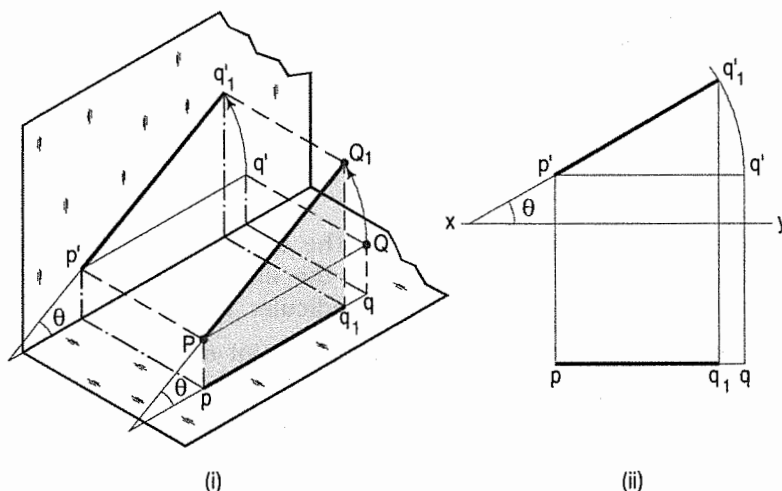


FIG. 10-4

- (b) Line  $RS_1$  [fig. 10-5(i)] is inclined at an angle  $\phi$  to the V.P. and is parallel to the H.P. The inclination is shown by the angle  $\phi$  which  $RS_1$  makes with its projection on the V.P., viz. the front view  $r's'_1$ . Assuming the line to be parallel to both the H.P. and the V.P., its projections  $r's'$  and  $rs$  are drawn parallel to  $xy$  and equal to its true length [fig. 10-5(ii)].

When the line is turned about its end  $R$  to the position  $RS_1$  so that it makes the angle  $\phi$  with the V.P. while remaining parallel to the H.P., in

the top view the point  $s$  will move along an arc drawn with  $r$  as centre and  $rs$  as radius to a point  $s_1$  so that  $rs_1$  makes the angle  $\phi$  with  $xy$ . In the front view, the point  $s'$  will move towards  $r'$  along the line  $r's'$  to a point  $s'_1$  on the projector through  $s_1$ .  $rs_1$  and  $r's'_1$  are the projections of the line  $RS_1$ .

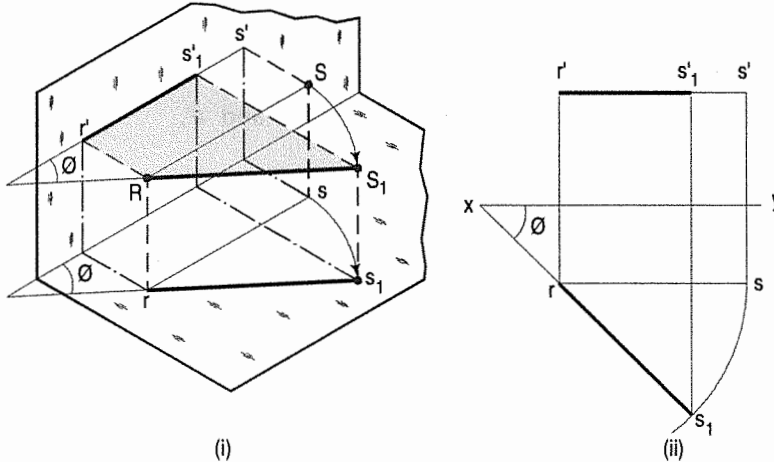


FIG. 10-5

Therefore, when the line is inclined to the H.P. and parallel to the V.P., its top view is shorter than its true length, but parallel to  $xy$ ; its front view is equal to its true length and is inclined to  $xy$  at its true inclination with the H.P. And when the line is inclined to the V.P. and parallel to the H.P., its front view is shorter than its true length but parallel to  $xy$ ; its top view is equal to its true length and is inclined to  $xy$  at its true inclination with the V.P.

Hence, when a line is inclined to one plane and parallel to the other, its projection on the plane to which it is inclined, is a line shorter than its true length but parallel to the reference line. Its projection on the plane to which it is parallel, is a line equal to its true length and inclined to the reference line at its true inclination.

In other words, the inclination of a line with the H.P. is seen in the front view and that with the V.P. is seen in the top view.

**Problem 10-1.** (fig. 10-6): A line  $PQ$ , 90 mm long, is in the H.P. and makes an angle of  $30^\circ$  with the V.P. Its end  $P$  is 25 mm in front of the V.P. Draw its projections.

As the line is in the H.P., its top view will show the true length and the true inclination with the V.P. Its front view will be in  $xy$ .

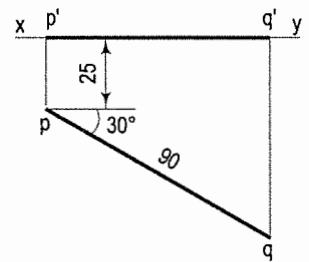


FIG. 10-6

- (i) Mark a point  $p$ , the top view 25 mm below  $xy$ . Draw a line  $pq$  equal to 90 mm and inclined at  $30^\circ$  to  $xy$ .
- (ii) Project  $p$  to  $p'$  and  $q$  to  $q'$  on  $xy$ .  
 $pq$  and  $p'q'$  are the required top view and front view respectively.

**Problem 10-2.** (fig. 10-7): The length of the top view of a line parallel to the V.P. and inclined at  $45^\circ$  to the H.P. is 50 mm. One end of the line is 12 mm above the H.P. and 25 mm in front of the V.P. Draw the projections of the line and determine its true length.

As the line is parallel to the V.P., its top view will be parallel to  $xy$  and the front view will show its true length and the true inclination with the H.P.

- (i) Mark  $a$ , the top view, 25 mm below  $xy$  and  $a'$ , the front view, 12 mm above  $xy$ .
- (ii) Draw the top view  $ab$  50 mm long and parallel to  $xy$  and draw a projector through  $b$ .
- (iii) From  $a'$  draw a line making  $45^\circ$  angle with  $xy$  and cutting the projector through  $b$  at  $b'$ . Then  $a'b'$  is the front view and also the true length of the line.

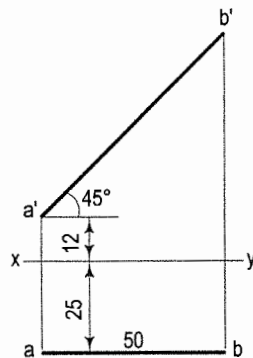


FIG. 10-7

**Problem 10-3.** (fig. 10-8): The front view of a 75 mm long line measures 55 mm. The line is parallel to the H.P. and one of its ends is in the V.P. and 25 mm above the H.P. Draw the projections of the line and determine its inclination with the V.P.

As the line is parallel to the H.P., its front view will be parallel to  $xy$ .

- (i) Mark  $a$ , the top view of one end in  $xy$ , and  $a'$ , its front view, 25 mm above  $xy$ .
- (ii) Draw the front view  $a'b'$ , 55 mm long and parallel to  $xy$ . With  $a$  as centre and radius equal to 75 mm, draw an arc cutting the projector through  $b'$  at  $b$ . Join  $a$  with  $b$ .  $ab$  is the top view of the line. Its inclination with  $xy$ , viz.  $\theta$  is the inclination of the line with the V.P.

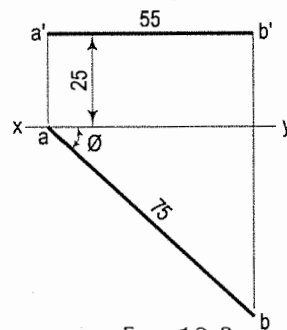


FIG. 10-8

## EXERCISES 10(a)

1. Draw the projections of a 75 mm long straight line, in the following positions:
  - (a) (i) Parallel to both the H.P. and the V.P. and 25 mm from each.
  - (ii) Parallel to and 30 mm above the H.P. and in the V.P.
  - (iii) Parallel to and 40 mm in front of the V.P. and in the H.P.
  - (b) (i) Perpendicular to the H.P., 20 mm in front of the V.P. and its one end 15 mm above the H.P.
  - (ii) Perpendicular to the V.P., 25 mm above the H.P. and its one end in the V.P.
  - (iii) Perpendicular to the H.P., in the V.P. and its one end in the H.P.
  - (c) (i) Inclined at  $45^\circ$  to the V.P., in the H.P. and its one end in the V.P.
  - (ii) Inclined at  $30^\circ$  to the H.P. and its one end 20 mm above it; parallel to and 30 mm in front of the V.P.
  - (iii) Inclined at  $60^\circ$  to the V.P. and its one end 15 mm in front of it; parallel to and 25 mm above the H.P.

2. A 100 mm long line is parallel to and 40 mm above the H.P. Its two ends are 25 mm and 50 mm in front of the V.P. respectively. Draw its projections and find its inclination with the V.P.
3. A 90 mm long line is parallel to and 25 mm in front of the V.P. Its one end is in the H.P. while the other is 50 mm above the H.P. Draw its projections and find its inclination with the H.P.
4. The top view of a 75 mm long line measures 55 mm. The line is in the V.P., its one end being 25 mm above the H.P. Draw its projections.
5. The front view of a line, inclined at  $30^\circ$  to the V.P. is 65 mm long. Draw the projections of the line, when it is parallel to and 40 mm above the H.P., its one end being 30 mm in front of the V.P.
6. A vertical line  $AB$ , 75 mm long, has its end  $A$  in the H.P. and 25 mm in front of the V.P. A line  $AC$ , 100 mm long, is in the H.P. and parallel to the V.P. Draw the projections of the line joining  $B$  and  $C$ , and determine its inclination with the H.P.
7. Two pegs fixed on a wall are 4.5 metres apart. The distance between the pegs measured parallel to the floor is 3.6 metres. If one peg is 1.5 metres above the floor, find the height of the second peg and the inclination of the line joining the two pegs, with the floor.
8. Draw the projections of the lines in Exercises 1 to 6, assuming them to be in the third quadrant, taking the given positions to be below the H.P. instead of above the H.P., and behind the V.P., instead of in front of the V.P.

### 10-5. LINE INCLINED TO BOTH THE PLANES



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 24 for the line inclined to both the planes.

- (a) A line  $AB$  (fig. 10-9) is inclined at  $\theta$  to the H.P. and is parallel to the V.P. The end  $A$  is in the H.P.  $AB$  is shown as the hypotenuse of a right-angled triangle, making the angle  $\theta$  with the base.

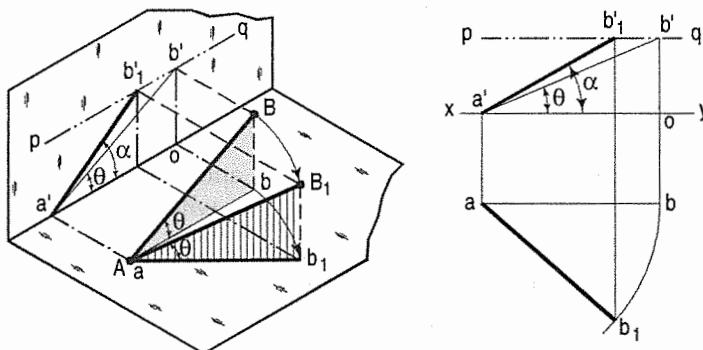


FIG. 10-9

The top view  $ab$  is shorter than  $AB$  and parallel to  $xy$ . The front view  $a'b'$  is equal to  $AB$  and makes the angle  $\theta$  with  $xy$ .

Keeping the end  $A$  fixed and the angle  $\theta$  with the H.P. constant, if the end  $B$  is moved to any position, say  $B_1$ , the line becomes inclined to the V.P. also.

In the top view,  $b$  will move along an arc, drawn with  $a$  as centre and  $ab$  as radius, to a position  $b_1$ . The new top view  $ab_1$  is equal to  $ab$  but shorter than  $AB$ .

In the front view,  $b'$  will move to a point  $b'_1$  keeping its distance from  $xy$  constant and equal to  $b'o$ ; i.e. it will move along the line  $pq$ , drawn through  $b'$  and parallel to  $xy$ . This line  $pq$  is the locus or path of the end  $B$  in the front view.  $b'_1$  will lie on the projector through  $b_1$ . The new front view  $a'b'_1$  is shorter than  $a'b'$  (i.e.  $AB$ ) and makes an angle  $\alpha$  with  $xy$ .  $\alpha$  is greater than  $\theta$ .

Thus, it can be seen that as long as the inclination  $\theta$  of  $AB$  with the H.P. is constant, even when it is inclined to the V.P.

- (i) its length in the top view, viz.  $ab$  remains constant; and
  - (ii) the distance between the paths of its ends in the front view, viz.  $b'o$  remains constant.
- (b) The same line  $AB$  (fig. 10-10) is inclined at  $\phi$  to the V.P. and is parallel to the H.P. Its end  $A$  is in the V.P.  $AB$  is shown as the hypotenuse of a right-angled triangle making the angle  $\phi$  with the base.

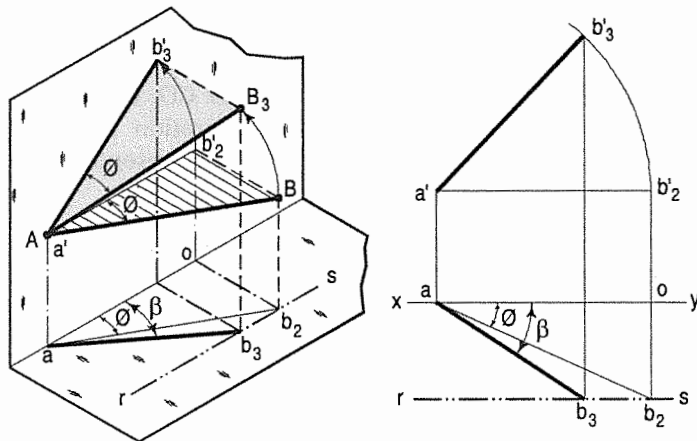


FIG. 10-10

The front view  $a'b'_2$  is shorter than  $AB$  and parallel to  $xy$ . The top view  $ab_2$  is equal to  $AB$  and makes an angle  $\phi$  with  $xy$ .

Keeping the end  $A$  fixed and the angle  $\phi$  with the V.P. constant, if  $B$  is moved to any position, say  $B_3$ , the line will become inclined to the H.P. also.

In the front view,  $b'_2$ , will move along the arc, drawn with  $a'$  as centre and  $a'b'_2$  as radius, to a position  $b'_3$ . The new front view  $a'b'_3$  is equal to  $a'b'_2$  but is shorter than  $AB$ .

In the top view,  $b_2$  will move to a point  $b_3$  along the line  $rs$ , drawn through  $b_2$  and parallel to  $xy$ , thus keeping its distance from the path of  $a$ , viz.  $b_2o$  constant.  $rs$  is the locus or path of the end  $B$  in the top view. The point  $b_3$  lies on the projector through  $b'_3$ . The new top view  $ab_3$  is shorter than  $ab_2$  (i.e.  $AB$ ) and makes an angle  $\beta$  with  $xy$ .  $\beta$  is greater than  $\theta$ .

Here also we find that, as long as the inclination of  $AB$  with the V.P. does not change, even when it becomes inclined to the H.P.

- (i) its length in the front view, viz.  $a'b'_2$  remains constant; and
- (ii) the distance between the paths of its ends in the top view, viz.  $b_2o$  remains constant.

Hence, when a line is inclined to both the planes, its projections are shorter than the true length and inclined to  $xy$  at angles greater than the true inclinations. These angles viz.  $\alpha$  and  $\beta$  are called apparent angles of inclination.

## 10-6. PROJECTIONS OF LINES INCLINED TO BOTH THE PLANES



From Art. 10-5(a) above, we find that as long as the inclination of  $AB$  with the H.P. is constant

- (i) its length in the top view, viz.  $ab$  remains constant, and
- (ii) in the front view, the distance between the loci of its ends, viz.  $b'o$  remains constant.

In other words if

- (i) its length in the top view is equal to  $ab$ , and
- (ii) the distance between the paths of its ends in the front view is equal to  $b'o$ , the inclination of  $AB$  with the H.P. will be equal to  $\theta$ .

Similarly, from Art. 10-5(b) above, we find that as long as the inclination of  $AB$  with the V.P. is constant

- (i) its length in the front view, viz.  $a'b'_2$  remains constant, and
- (ii) in the top view, the distance between the loci of its ends, viz.  $b_2o$  remains constant.

The reverse of this is also true, viz.

- (i) if its length in the front view is equal to  $a'b'_2$ , and
- (ii) the distance between the paths of its ends in the top view is equal to  $b_2o$ , the inclination of  $AB$  with the V.P. will be equal to  $\theta$ .

Combining the above two findings, we conclude that when  $AB$  is inclined at  $\theta$  to the H.P. and at  $\theta$  to the V.P.

- (i) its lengths in the top view and the front view will be equal to  $ab_2$  and  $a'b'_2$  respectively, and
- (ii) the distances between the paths of its ends in the front view and the top view will be equal to  $b'_2o$  and  $b_2o$  respectively.

The two lengths when arranged with their ends in their respective paths and in projections with each other will be the projections of the line  $AB$ , as illustrated in problem 10-4.

**Problem 10-4.** Given the line  $AB$ , its inclinations  $\theta$  with the H.P. and  $\phi$  with the V.P. and the position of one end  $A$ . To draw its projections.

Mark the front view  $a'$  and the top view  $a$  according to the given position of  $A$  (fig. 10-12).

Let us first determine the lengths of  $AB$  in the top view and the front view and the paths of its ends in the front view and the top view.

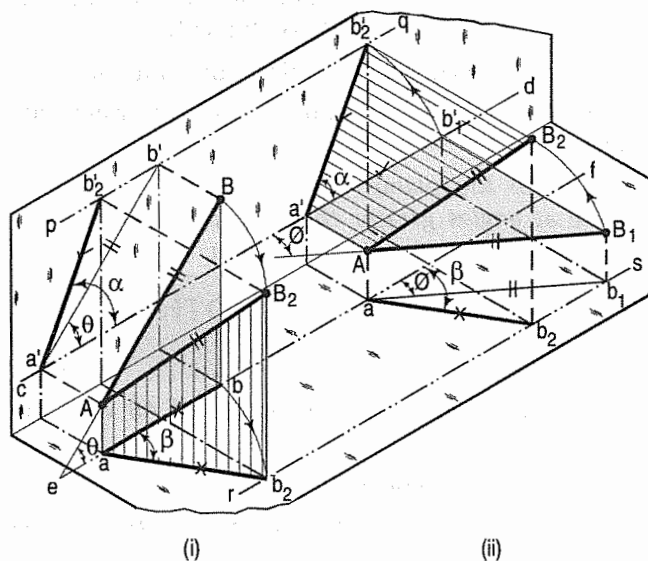


FIG. 10-11

- (2) Again, assume  $AB_1$  (equal to  $AB$ ) to be parallel to the H.P. and inclined at  $\phi$  to the V.P. In the pictorial view [fig. 10-11(ii)],  $AB_1$  is shown as a side of the trapezoid  $AB_1b'_1a'$ . Draw the top view  $ab_1$  equal to  $AB$  [fig. 10-12(ii)] and inclined at  $\phi$  to  $xy$ . Project the front view  $a'b'_1$  parallel to  $xy$ . Through  $a$  and  $b_1$ , draw lines  $ef$  and  $rs$  respectively parallel to  $xy$ .  $a'b'_1$  is the length of  $AB$  in the front view and,  $ef$  and  $rs$  are the paths of  $A$  and  $B$  respectively in the top view.

We may now arrange

- (i)  $ab$  (the length in the top view) between its paths  $ef$  and  $rs$ , and
- (ii)  $a'b'_1$  (the length in the front view) between the paths  $cd$  and  $pq$ , keeping them in projection with each other, in one of the following two ways:

- (a) In case (1) [fig. 10-11(i)], if the side  $Bb$  is turned about  $Aa$ , so that  $b$  comes on the path  $rs$ , the line  $AB$  will become inclined at  $\phi$  to the V.P. Therefore, with  $a$  as centre [fig. 10-12(i)] and radius equal to  $ab$ , draw an arc cutting  $rs$  at a point  $b_2$ . Project  $b_2$  to  $b'_2$  on the path  $pq$ .

- (1) Assume  $AB$  to be parallel to the V.P. and inclined at  $\theta$  to the H.P.  $AB$  is shown in the pictorial view as a side of the trapezoid  $ABba$  [fig. 10-11(i)]. Draw the front view  $a'b'$  equal to  $AB$  [fig. 10-12(i)] and inclined at  $\theta$  to  $xy$ . Project the top view  $ab$  parallel to  $xy$ . Through  $a'$  and  $b'$ , draw lines  $cd$  and  $pq$  respectively parallel to  $xy$ .  $ab$  is the length of  $AB$  in the top view and,  $cd$  and  $pq$  are the paths of  $A$  and  $B$  respectively in the front view.

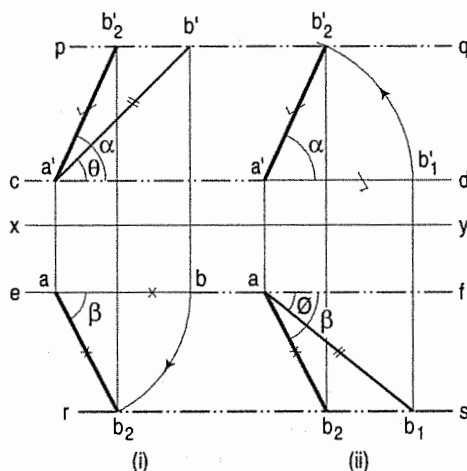


FIG. 10-12

Draw lines joining  $a$  with  $b_2$ , and  $a'$  with  $b'_2$ .  $ab_2$  and  $a'b'_2$  are the required projections. Check that  $a'b'_2 = a'b'_1$ .

- (b) Similarly, in case (2) [fig. 10-11(ii)], if the side  $B_1b'_1$  is turned about  $Aa'$  till  $b'_1$  is on the path  $pq$ , the line  $AB_1$  will become inclined at  $\theta$  to the H.P. Hence, with  $a'$  as centre [fig. 10-12(ii)] and radius equal to  $a'b'_1$ , draw an arc cutting  $pq$  at a point  $b'_2$ . Project  $b'_2$  to  $b_2$  in the top view on the path  $rs$ .

Draw lines joining  $a$  with  $b_2$ , and  $a'$  with  $b'_2$ .  $ab_2$  and  $a'b'_2$  are the required projections. Check that  $ab_2 = ab$ .

Fig. 10-13 shows (in pictorial and orthographic views) the projections obtained with both the above steps combined in one figure and as described below.

First, determine

- (i) the length  $ab$  in the top view and the path  $pq$  in the front view and
- (ii) the length  $a'b'_1$  in the front view and the path  $rs$  in the top view.

Then, with  $a$  as centre and radius equal to  $ab$ , draw an arc cutting  $rs$  at a point  $b_2$ . With  $a'$  as centre and radius equal to  $a'b'_1$ , draw an arc cutting  $pq$  at a point  $b'_2$ .

Draw lines joining  $a$  with  $b_2$  and  $a'$  with  $b'_2$ .  $ab_2$  and  $a'b'_2$  are the required projections. Check that  $b_2$  and  $b'_2$  lie on the same projector.

It is quite evident from the figure that the apparent angles of inclination  $\alpha$  and  $\beta$  are greater than the true inclinations  $\theta$  and  $\phi$  respectively.

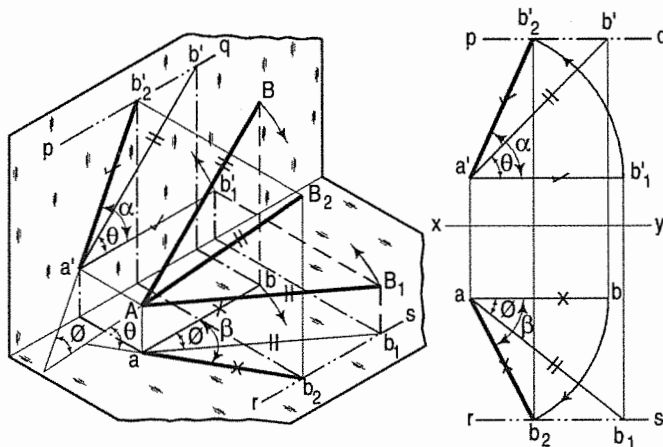


FIG. 10-13

## 10-7. LINE CONTAINED BY A PLANE PERPENDICULAR TO BOTH THE REFERENCE PLANES

As the two reference planes are at right angles to each other, the sum total of the inclinations of a line with the two planes, viz.  $\theta$  and  $\phi$  can never be more than  $90^\circ$ . When  $\theta + \phi = 90^\circ$ , the line will be contained by a third plane called the profile plane, perpendicular to both the H.P. and the V.P.

A line  $EF$  (fig. 10-14), is inclined at  $\theta$  to the H.P. and at  $\phi$  [equal to  $(90^\circ - \theta)$ ] to the V.P. The line is thus contained by the profile plane marked P.P.

The front view  $e'f'$  and the top view  $ef$  are both perpendicular to  $xy$  and shorter than  $EF$ .

Therefore, when a line is inclined to both the reference planes and contained by a plane perpendicular to them, i.e. when the sum of its inclinations with the H.P. and the V.P. is  $90^\circ$ , its projections are perpendicular to  $xy$  and shorter than the true length.

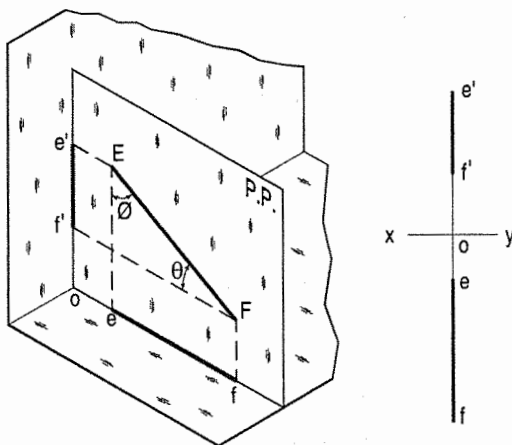


FIG. 10-14

### 10-8. TRUE LENGTH OF A STRAIGHT LINE AND ITS INCLINATIONS WITH THE REFERENCE PLANES

When projections of a line are given, its true length and inclinations with the planes are determined by the application of the following rule:

*When a line is parallel to a plane, its projection on that plane will show its true length and the true inclination with the other plane.*

The line may be made parallel to a plane, and its true length obtained by any one of the following three methods:

#### Method I:

Making each view parallel to the reference line and projecting the other view from it. This is the exact reversal of the processes adopted in Art. 10-5 for obtaining the projections.

#### Method II:

Rotating the line about its projections till it lies in the H.P. or in the V.P.

#### Method III:

Projecting the views on auxiliary planes parallel to each view.

(This method will be dealt with in chapter 11).

The following problem shows the application of the first two methods and problem 10-29 and problem 10-31 show application of third method.

**Problem 10-5.** *The top view  $ab$  and the front view  $a'b'$  of a line  $AB$  are given. To determine its true length and the inclinations with the H.P. and the V.P.*

#### Method I:

Fig. 10-15(i) shows  $AB$  the line,  $a'b'$  its front view and  $ab$  its top view. If the trapezoid  $ABba$  is turned about  $Aa$  as axis, so that  $AB$  becomes parallel to the V.P., in the top view,  $b$  will move along an arc drawn with centre  $a$  and radius equal to  $ab$ , to  $b_1$ , so that  $ab_1$  is parallel to  $xy$ . In the front view,  $b'$  will move along its locus  $pq$ , to a point  $b'_1$  on the projector through  $b_1$ .

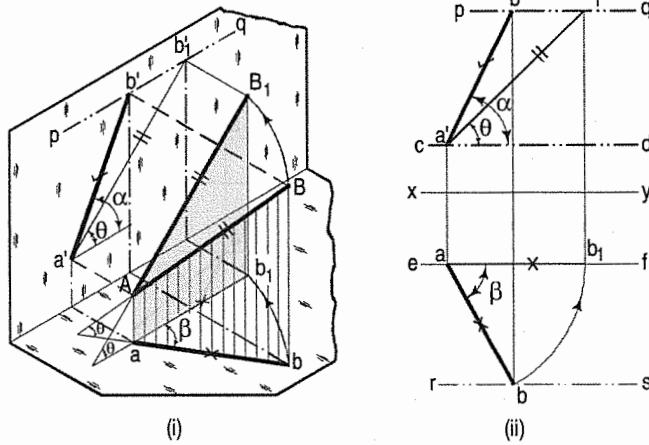


FIG. 10-15

- (i) Therefore, with centre  $a$  and radius equal to  $ab$  [fig. 10-15(ii)], draw an arc to cut  $ef$  at  $b_1$ .
- (ii) Draw a projector through  $b_1$  to cut  $pq$  (the path of  $b'$ ) at  $b'_1$ .
- (iii) Draw the line  $a'b'_1$  which is the true length of  $AB$ . The angle  $\theta$ , which it makes with  $xy$  is the inclination of  $AB$  with the H.P.

Again, in fig. 10-16(i)  $AB$  is shown as a side of a trapezoid  $ABB'a'$ . If the trapezoid is turned about  $Aa'$  as axis so that  $AB$  is parallel to the H.P., the new top view will show its true length and true inclination with the V.P.

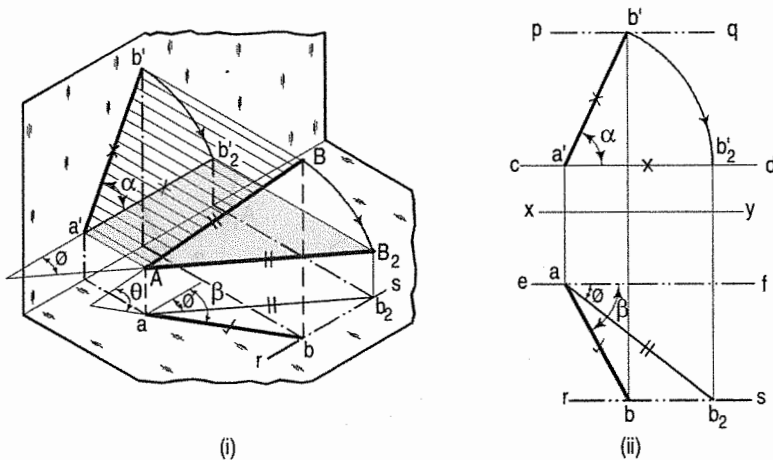


FIG. 10-16

- (i) With  $a'$  as centre and radius equal to  $a'b'$  [fig. 10-16(ii)], draw an arc to cut  $cd$  at  $b'_2$ .
- (ii) Draw a projector through  $b'_2$  to cut  $rs$  (the path of  $b$ ) at  $b_2$ .
- (iii) Draw the line  $ab_2$ , which is the true length of  $AB$ . The angle  $\theta$  which it makes with  $xy$  is the inclination of  $AB$  with the V.P.

Fig. 10-17(i) shows the above two steps combined in one figure.

The same results will be obtained by keeping the end  $B$  fixed and turning the end  $A$  [fig. 10-17(ii)], as explained below.

- (i) With centre  $b$  and radius equal to  $ba$ , draw an arc cutting  $rs$  at  $a_1$  (thus making  $ba$  parallel to  $xy$ ).
- (ii) Project  $a_1$  to  $a'_1$  on  $cd$  (the path of  $a'$ ).  $a'_1b'$  is the true length and  $\theta$  is the true inclination of  $AB$  with the H.P.
- (iii) Similarly, with centre  $b'$  and radius equal to  $b'a'$ , draw an arc cutting  $pq$  at  $a'_2$ .
- (iv) Project  $a'_2$  to  $a_2$  on  $ef$  (the path of  $a$ ).  $a_2b$  is the true length and  $\phi$  is the true inclination of  $AB$  with the V.P.

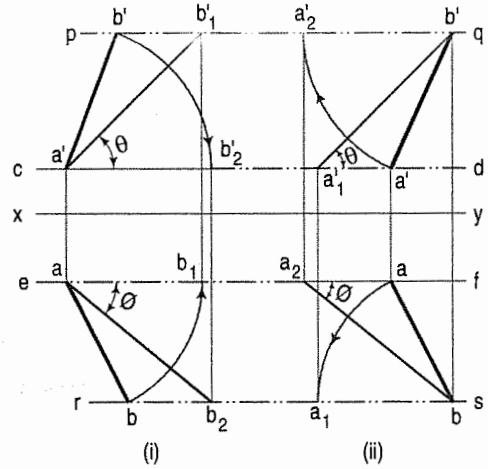


FIG. 10-17

### Method II:

Referring to the pictorial view in fig. 10-18(i) we find that  $AB$  is the line,  $ab$  its top view and  $a'b'$  its front view.

In the trapezoid  $ABb'a'$  (i)  $a'A$  and  $b'B$  are both perpendicular to  $a'b'$  and are respectively equal to  $ao_1$  and  $bo_2$  (the distances of  $a$  and  $b$  from  $xy$  in the top view), and (ii) the angle between  $AB$  and  $a'b'$  is the angle of inclination  $\phi$  of  $AB$  with the V.P.

Assume that this trapezoid is rotated about  $a'b'$ , till it lies in the V.P.

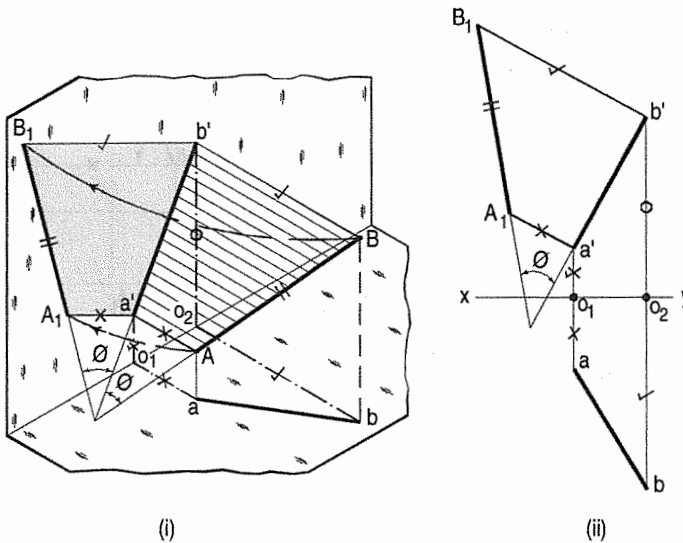


FIG. 10-18

In the orthographic view [fig. 10-18(ii)], this trapezoid is obtained by drawing perpendiculars to  $a'b'$ , viz.  $a'A_1$  (equal to  $ao_1$ ) and  $b'B_1$  (equal to  $bo_2$ ) and then joining  $A_1$  with  $B_1$ . The line  $A_1B_1$  is the true length of  $AB$  and its inclination  $\phi$  with  $a'b'$  is the inclination of  $AB$  with the V.P.

Similarly, in trapezoid  $ABba$  in fig. 10-19(i),  $AB$  is the line and  $ab$  its top view.  $Aa$  and  $Bb$  are both perpendicular to  $ab$  and are respectively equal to  $a'o_1$  and  $b'o_2$  (the distances of  $a'$  and  $b'$  from  $xy$  in the front view). The angle  $\theta$  between  $AB$  and  $ab$  is the inclination of  $AB$  with the H.P.

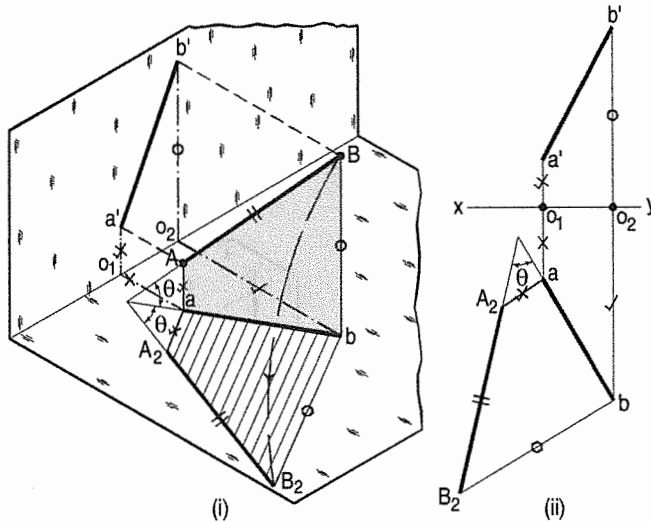


FIG. 10-19

This figure may now be assumed to be rotated about  $ab$  as axis, so that it lies in the H.P.

In the orthographic view [fig. 10-19(ii)], this trapezoid is obtained by erecting perpendiculars to  $ab$ , viz.  $aA_2$  equal to  $a'o_1$  and  $bB_2$  equal to  $b'o_2$  and joining  $A_2$  with  $B_2$ . The line  $A_2B_2$  is the true length of  $AB$  and its inclination  $\theta$  with  $ab$  is the inclination of  $AB$  with the H.P.

Note: The perpendiculars on  $ab$  or  $a'b'$  can also be drawn on its other side assuming the trapezoid to be rotated in the opposite direction.

## 10-9. TRACES OF A LINE



When a line is inclined to a plane, it will meet that plane, produced if necessary. The point in which the line or line-produced meets the plane is called its *trace*.

The point of intersection of the line with the H.P. is called the *horizontal trace*, usually denoted as H.T. and that with the V.P. is called the *vertical trace* or V.T.

Refer to fig. 10-20.

- (i) A line  $AB$  is parallel to the H.P. and the V.P. It has no trace.
- (ii) A line  $CD$  is inclined to the H.P. and parallel to the V.P. It has only the H.T. but no V.T.
- (iii) A line  $EF$  is inclined to the V.P. and parallel to the H.P. It has only the V.T. but no H.T.

Thus, when a line is parallel to a plane it has no trace upon that plane.

Refer to fig. 10-21.

- (i) A line  $PQ$  is perpendicular to the H.P. Its H.T. coincides with its top view which is a point. It has no V.T.

- (ii) A line  $RS$  is perpendicular to the V.P. Its V.T. coincides with its front view which is a point. It has no H.T.

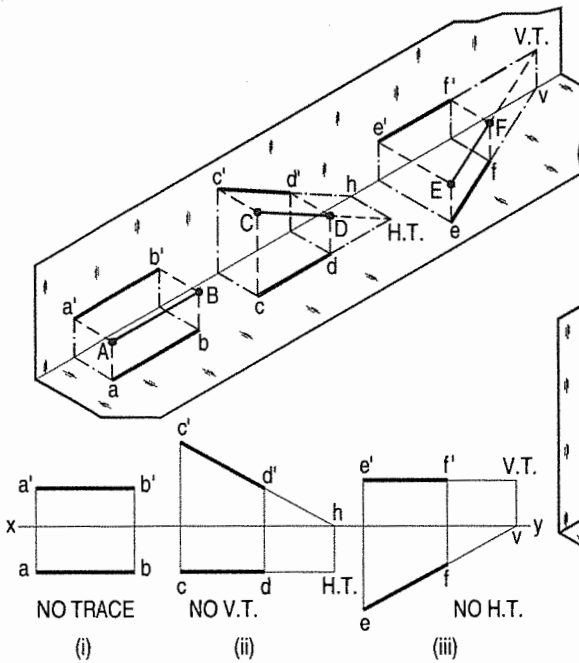


FIG. 10-20

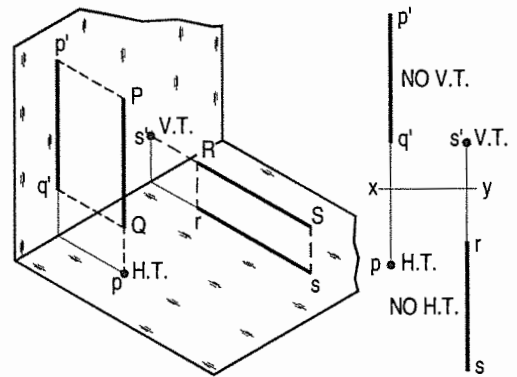


FIG. 10-21

Hence, when a line is perpendicular to a plane, its trace on that plane coincides with its projection on that plane. It has no trace on the other plane.

Refer to fig. 10-22.

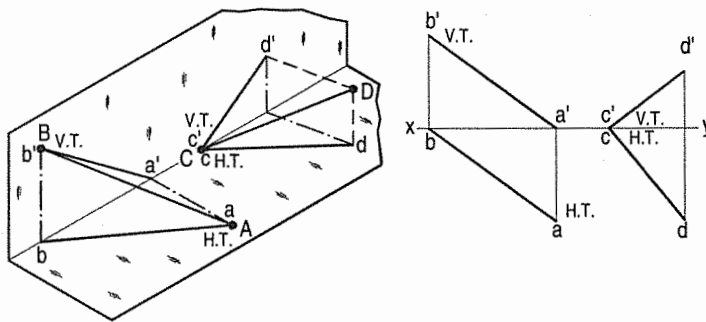


FIG. 10-22

- (i) A line  $AB$  has its end  $A$  in the H.P. and the end  $B$  in the V.P. Its H.T. coincides with  $a$  the top view of  $A$  and the V.T. coincides with  $b'$  the front view of  $B$ .
- (ii) A line  $CD$  has its end  $C$  in both the H.P. and the V.P. Its H.T. and V.T. coincide with  $c$  and  $c'$  (the projections of  $C$ ) in  $xy$ .

Hence, when a line has an end in a plane, its trace upon that plane coincides with the projection of that end on that plane.

## 10-10. METHODS OF DETERMINING TRACES OF A LINE

### Method I:

Fig. 10-23(i) shows a line  $AB$  inclined to both the reference planes. Its end  $A$  is in the H.P. and  $B$  is in the V.P.

$a'b'$  and  $ab$  are the front view and the top view respectively [fig. 10-23(ii)].

The H.T. of the line is on the projector through  $a'$  and coincides with  $a$ . The V.T. is on the projector through  $b$  and coincides with  $b'$ .

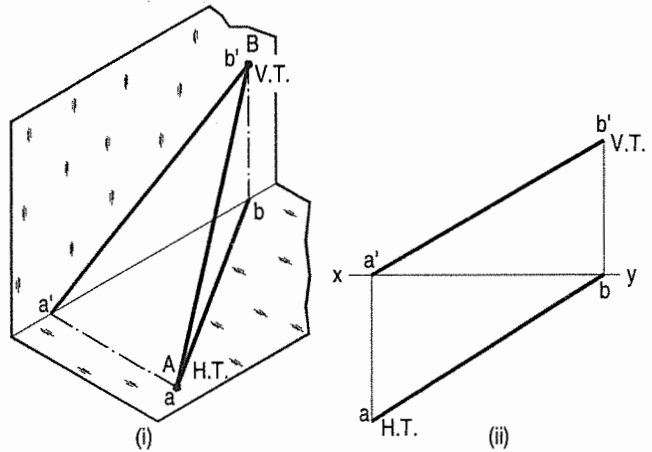


FIG. 10-23

Let us now assume that  $AB$  is shortened from both its ends, its inclination with the planes remaining constant. The H.T. and V.T. of the new line  $CD$  are still the same as can be seen clearly in fig. 10-24(i).

$c'd'$  and  $cd$  are the projections of  $CD$  [fig. 10-24(ii)]. Its traces may be determined as described below.

- (i) Produce the front view  $c'd'$  to meet  $xy$  at a point  $h$ .
- (ii) Through  $h$ , draw a projector to meet the top view  $cd$ -produced, at the H.T. of the line.

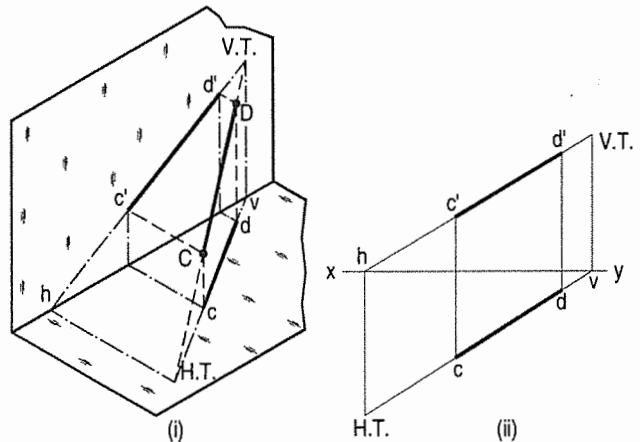


FIG. 10-24

- (iii) Similarly, produce the top view  $cd$  to meet  $xy$  at a point  $v$ .
- (iv) Through  $v$ , draw a projector to meet the front view  $c'd'$ -produced, at the V.T. of the line.

### Method II:

$c'd'$  and  $cd$  are the projections of the line  $CD$  [fig. 10-25(ii)]. Determine the true length  $C_1D_1$  from the front view  $c'd'$  by trapezoid method. The point of intersection between  $c'd'$ -produced and  $C_1D_1$ -produced is the V.T. of the line.

Similarly, determine the true length  $C_2D_2$  from the top view  $cd$ . Produce them to intersect at the H.T. of the line.

The above is quite evident from the pictorial view shown in fig. 10-25(i).

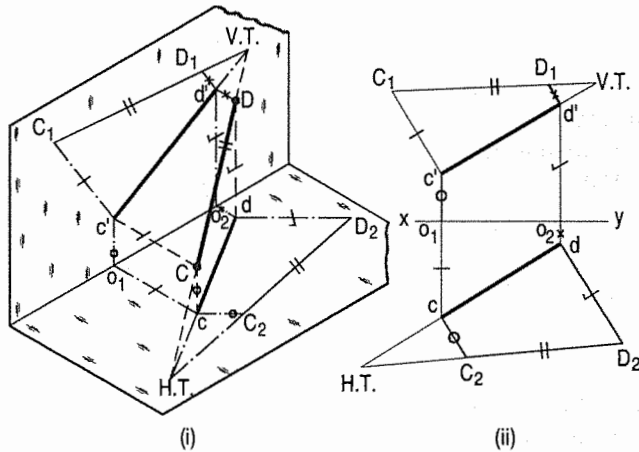


FIG. 10-25

### 10-11. TRACES OF A LINE, THE PROJECTIONS OF WHICH ARE PERPENDICULAR TO $xy$

When the projections of a line are perpendicular to  $xy$ , i.e. when the sum of its inclinations with the two principal planes of projection is  $90^\circ$ , it is not possible to find the traces by the first method. Method II must, therefore, be adopted as shown in fig. 10-26.

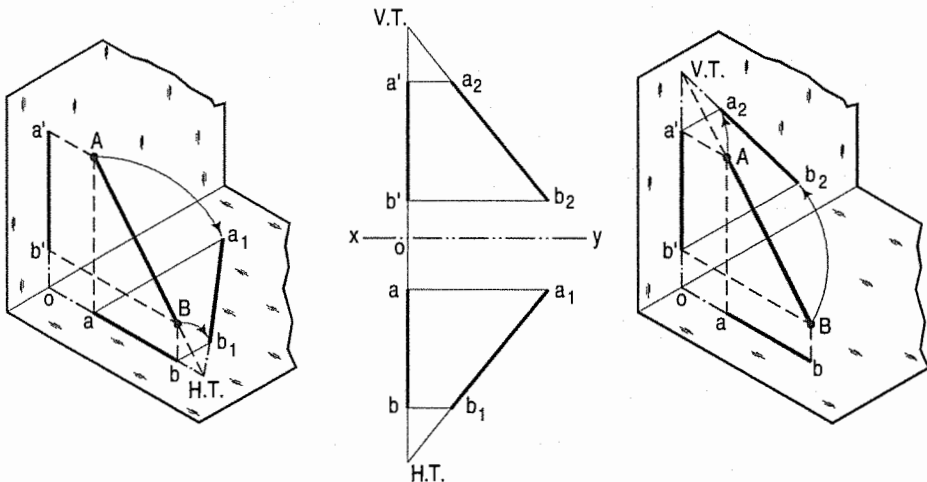


FIG. 10-26

### 10-12. POSITIONS OF TRACES OF A LINE

Although the line may be situated in the third quadrant, its both traces may be above or below  $xy$ , as shown in problem 10-6 and in fig. 10-27 and fig. 10-28. When a line intersects a plane, its traces on that plane will be contained by its projection on that plane as shown in problem 10-7.

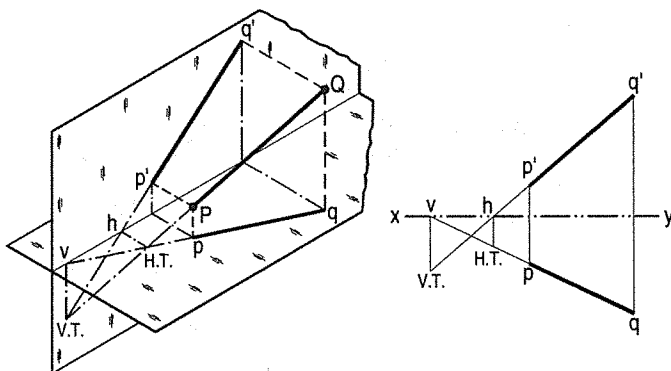


FIG. 10-27

**Problem 10-6.** Projections of a line  $PQ$  are given. Determine the positions of its traces.

Let  $pq$  and  $p'q'$  be the projections of  $PQ$  (fig. 10-27 and fig. 10-28).

- Produce the top view  $pq$  to meet  $xy$  at  $v$ . Draw a projector through  $v$  to meet the front view  $p'q'$ -produced at the V.T.
- Through  $h$ , the point of intersection between  $p'q'$ -produced and  $xy$ , draw a projector to meet the top view  $pq$ -produced at the H.T.

Note that in fig. 10-27, the traces are below  $xy$  while in fig. 10-28 they are above it.

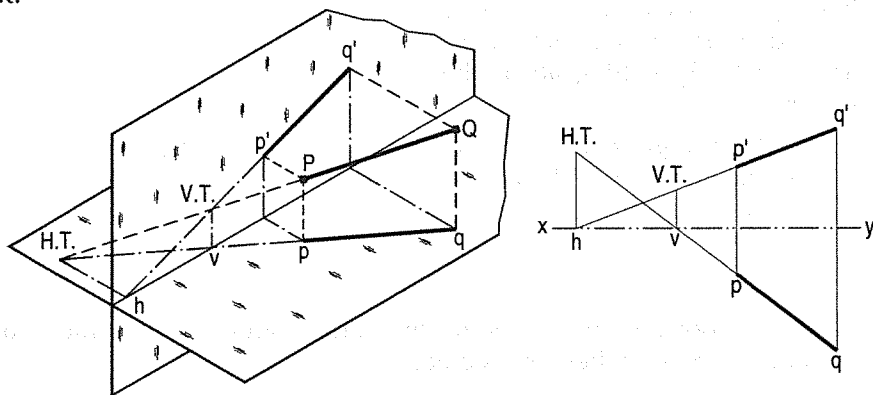


FIG. 10-28

**Problem 10-7.** A point  $A$  is 50 mm below the H.P. and 12 mm behind the V.P. A point  $B$  is 10 mm above the H.P. and 25 mm in front of the V.P. The distance between the projectors of  $A$  and  $B$  is 40 mm. Determine the traces of the line joining  $A$  and  $B$ .

Draw the projections  $ab$  and  $a'b'$  of the line  $AB$ .

**Method I:** (fig. 10-29):

- Through  $v$ , the point of intersection between  $ab$  and  $xy$ , draw a projector to meet  $a'b'$  at the V.T. of the line.
- Similarly, through  $h$ , the point of intersection between  $a'b'$  and  $xy$ , draw a projector to cut  $ab$  at the H.T. of the line.

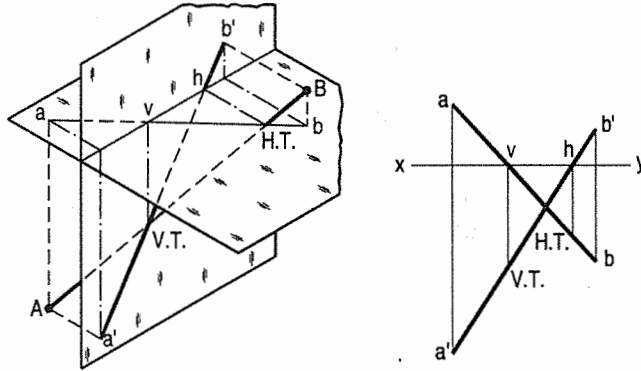


FIG. 10-29

**Method II:** (fig. 10-30):

At the ends  $a'$  and  $b'$ , draw perpendiculars to  $a'b'$ , viz.  $a'A_1$  equal to  $ao_1$  and  $b'B_1$  equal to  $bo_2$  on its opposite sides (as  $a$  and  $b$  are on opposite sides of  $xy$ ).

Draw the line  $A_1B_1$  intersecting  $a'b'$  at the V.T. of the line.

Similarly, at the ends  $a$  and  $b$ , draw perpendiculars to  $ab$ , viz.  $aA_2$  equal to  $a'o_1$  and  $bB_2$  equal to  $b'o_2$ , on its opposite sides (as  $a'$  and  $b'$  are on opposite sides of  $xy$ ). Join  $A_2$  with  $B_2$  cutting  $ab$  at the H.T. of the line.

Note that  $A_1B_1 = A_2B_2 = AB$  and that  $\theta$  and  $\phi$  are the inclinations of  $AB$  with the H.P. and the V.P. respectively.

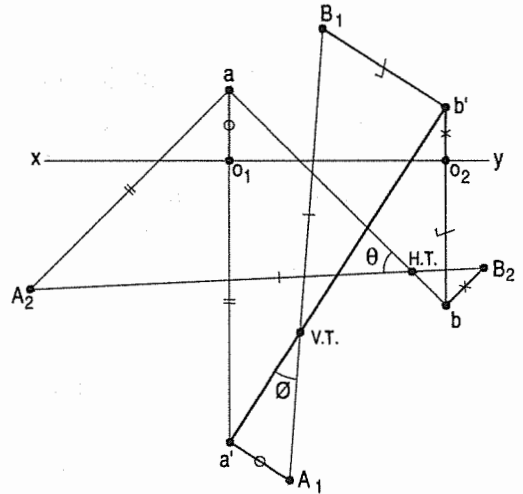


FIG. 10-30

### 10-13. ADDITIONAL ILLUSTRATIVE PROBLEMS

In the following problems, the ends of the lines should be assumed to be in the first quadrant, unless otherwise stated.

**Problem 10-8.** (fig. 10-31): A line  $AB$ , 50 mm long, has its end  $A$  in both the H.P. and the V.P. It is inclined at  $30^\circ$  to the H.P. and at  $45^\circ$  to the V.P. Draw its projections.

As the end  $A$  is in both the planes, its top view and the front view will coincide in  $xy$ .

- (i) Assuming  $AB$  to be parallel to the V.P. and inclined at  $\theta$  (equal to  $30^\circ$ ) to the H.P., draw its front view  $ab'$  (equal to  $AB$ ) and project the top view  $ab$ .

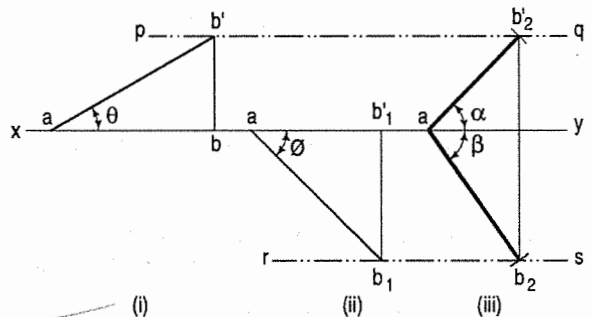


FIG. 10-31

- (ii) Again assuming  $AB$  to be parallel to the H.P. and inclined at  $\theta$  (equal to  $45^\circ$ ) to the V.P., draw its top view  $ab_1$  (equal to  $AB$ ). Project the front view  $ab'_1$ .

$ab$  and  $ab'_1$  are the lengths of  $AB$  in the top view and the front view respectively, and  $pq$  and  $rs$  are the loci of the end  $B$  in the front view and the top view respectively.

- (iii) With  $a$  as centre and radius equal to  $ab'_1$ , draw an arc cutting  $pq$  in  $b'_2$ . With the same centre and radius equal to  $ab$ , draw an arc cutting  $rs$  in  $b_2$ .

Draw lines joining  $a$  with  $b'_2$  and  $b_2$ .  $ab'_2$  and  $ab_2$  are the required projections.

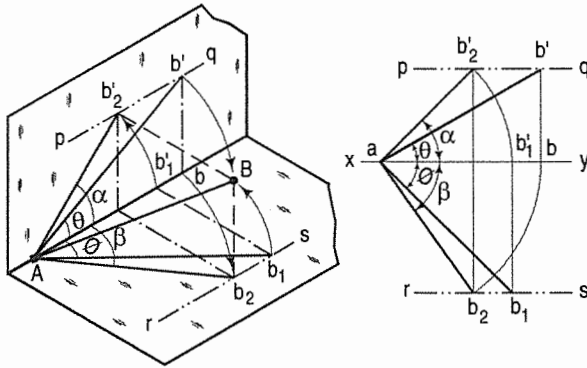


FIG. 10-32

Fig. 10-32 shows in pictorial and orthographic views, the solution obtained with all the above steps combined in one figure only.

**Problem 10-9.** (fig. 10-33): A line  $PQ$  75 mm long, has its end  $P$  in the V.P. and the end  $Q$  in the H.P. The line is inclined at  $30^\circ$  to the H.P. and at  $60^\circ$  to the V.P. Draw its projections.

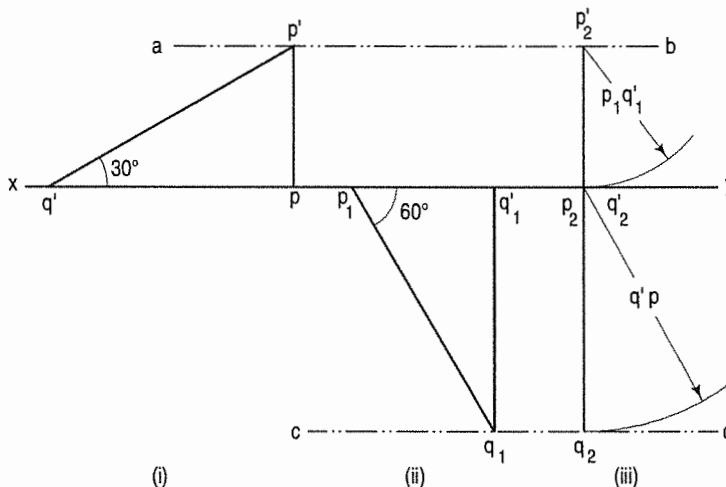


FIG. 10-33

The top view of  $P$  and the front view of  $Q$  will be in  $xy$ . As shown in the previous problem, determine

- (i) the length of  $PQ$  in the top view, viz.  $q'p$  and the path  $ab$  of the end  $P$  in the front view;
- (ii) the length  $p_1q'_1$  in the front view and the path  $cd$  of the end  $Q$  in the top view.
- (iii) Mark any point  $p_2$  (the top view of  $P$ ) in  $xy$  and project its front view  $p'_2$  in  $ab$ .
- (iv) With  $p'_2$  as centre and radius equal to  $p_1q'_1$ , draw an arc cutting  $xy$  in  $q'_2$ . It coincides with  $p_2$ .
- (v) With  $p_2$  as centre and radius equal to  $q'p$ , draw an arc cutting  $cd$  in  $q_2$ .  $p_2q_2$  and  $p'_2q'_2$  are the required projections. They lie in a line perpendicular to  $xy$  because the sum of the two inclinations is equal to  $90^\circ$ .

**Problem 10-10.** (fig. 10-34): A line  $PQ$  100 mm long, is inclined at  $30^\circ$  to the H.P. and at  $45^\circ$  to the V.P. Its mid-point is in the V.P. and 20 mm above the H.P. Draw its projections, if its end  $P$  is in the third quadrant and  $Q$  in the first quadrant.

The front view and the top view of  $P$  will be below and above  $xy$  respectively, while those of  $Q$  will be above and below  $xy$  respectively.

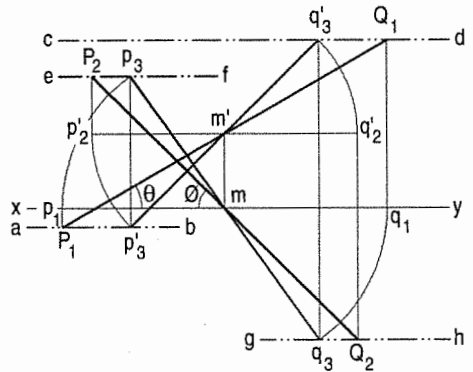


FIG. 10-34

- (i) Mark  $m$ , the top view of the mid-point in  $xy$  and project its front view  $m'$ , 20 mm above  $xy$ .
- (ii) Through  $m'$ , draw a line making an angle  $\theta$  (equal to  $30^\circ$ ) with  $xy$  and with the same point as centre and radius equal to  $\frac{1}{2} PQ$ , cut it at  $P_1$  below  $xy$  and at  $Q_1$  above  $xy$ . Project  $P_1Q_1$  to  $p_1q_1$  on  $xy$ .  $p_1q_1$  is the length of  $PQ$  in the top view.  $ab$  and  $cd$  are the paths of  $P$  and  $Q$  respectively in the front view.
- (iii) Similarly, through  $m$ , draw a line making angle  $\phi$  (equal to  $45^\circ$ ) with  $xy$  and cut it with the same radius at  $P_2$  above  $xy$  and at  $Q_2$  below it.
- (iv) Project  $P_2Q_2$  to  $p'_2q'_2$  on the horizontal line through  $m'$ .  $p'_2q'_2$  is the length of  $PQ$  in the front view and  $ef$  and  $gh$  are the paths of  $P$  and  $Q$  respectively in the top view.
- (v) With  $m$  as centre and radius equal to  $mp_1$  or  $mq_1$ , draw arcs cutting  $ef$  at  $p_3$  and  $gh$  at  $q_3$ . With  $m'$  as centre and radius equal to  $m'p'_2$  or  $m'q'_2$ , draw arcs cutting  $ab$  at  $p'_3$  and  $cd$  at  $q'_3$ .  $p_3q_3$  and  $p'_3q'_3$  are the required projections.

**Problem 10-11.** (fig. 10-35): The top view of a 75 mm long line  $AB$  measures 65 mm, while the length of its front view is 50 mm. Its one end  $A$  is in the H.P. and 12 mm in front of the V.P. Draw the projections of  $AB$  and determine its inclinations with the H.P. and the V.P.

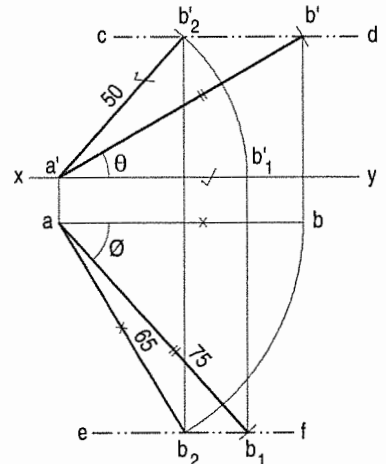


FIG. 10-35

- (i) Mark the front view  $a'$  and the top view  $a$  of the given end  $A$ .

- (ii) Assuming  $AB$  to be parallel to the V.P., draw a line  $ab$  equal to 65 mm and parallel to  $xy$ . With  $a'$  as centre and radius equal to 75 mm, draw an arc cutting the projector through  $b$  at  $b'$ . The line  $cd$  through  $b'$  and parallel to  $xy$ , is the locus of  $B$  in the front view and  $\theta$  is the inclination of  $AB$  with the H.P.
- (iii) Similarly, draw a line  $a'b'_1$  in  $xy$  and equal to 50 mm. With  $a$  as centre and radius equal to  $AB$ , draw an arc cutting the projector through  $b'_1$  at  $b_1$ .  $ef$  is the locus of  $B$  in the top view and  $\phi$  is the inclination of  $AB$  with the V.P.
- (iv) With  $a'$  as centre and radius equal to  $a'b'_1$ , draw an arc cutting  $cd$  in  $b'_2$ . With  $a$  as centre and radius equal to  $ab$ , draw an arc cutting  $ef$  in  $b_2$ .  $a'b'_2$  and  $ab_2$  are the required projections.

**Problem 10-12.** (fig. 10-36): A line  $AB$ , 65 mm long, has its end  $A$  20 mm above the H.P. and 25 mm in front of the V.P. The end  $B$  is 40 mm above the H.P. and 65 mm in front of the V.P. Draw the projections of  $AB$  and show its inclinations with the H.P. and the V.P.

- (i) As per given positions, draw the loci  $cd$  and  $gh$  of the end  $A$ , and  $ef$  and  $jk$  of the end  $B$  in the front view and the top view respectively.
- (ii) Mark any point  $a$  (the top view of  $A$ ) in  $gh$  and project it to  $a'$  on  $cd$ . With  $a'$  as centre and radius equal to 65 mm, draw an arc cutting  $ef$  in  $b'$ . Join  $a'$  with  $b'$ .  $\theta$ , the inclination of  $a'b'$  with  $xy$ , is the inclination of  $AB$  with the H.P. Project  $b'$  to  $b$  on  $gh$ .  $ab$  is the length of  $AB$  in the top view.
- (iii) With  $a$  as centre and radius equal to 65 mm, draw an arc cutting  $jk$  in  $b_1$ . Join  $a$  with  $b_1$ .  $\phi$ , the inclination of  $ab_1$

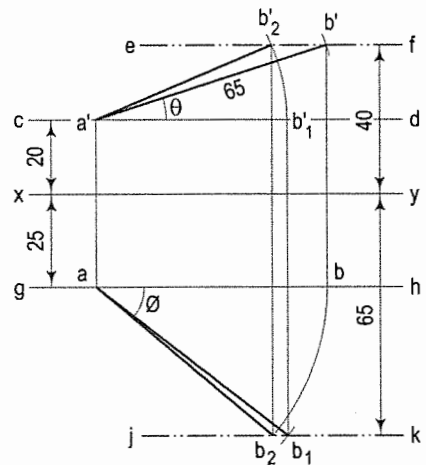


FIG. 10-36

with  $xy$ , is the inclination of  $AB$  with the V.P. Project  $b_1$  to  $b'_1$  on  $cd$ .  $a'b'_1$  is the length of  $AB$  in the front view.

Arrange  $ab$  and  $a'b'_1$  between their respective paths as shown.  $a'b'_2$  and  $ab_2$  are the required projections of  $AB$ .

**Problem 10-13.** (fig. 10-37 and fig. 10-38): The projectors of the ends of a line  $AB$  are 50 mm apart. The end  $A$  is 20 mm above the H.P. and 30 mm in front of the V.P. The end  $B$  is 10 mm below the H.P. and 40 mm behind the V.P. Determine the true length and traces of  $AB$ , and its inclinations with the two planes.

Draw two projectors 50 mm apart. On one projector, mark the top view  $a$  and the front view  $a'$  of the end  $A$ . On the other, mark the top view  $b$  and the front view  $b'$  of the end  $B$ , as per given distances.  $ab$  and  $a'b'$  are the projections of  $AB$ .

Determine the true length, traces and inclinations by any one of the following two methods:

**Method I:**

By making the line parallel to a plane (fig. 10-37):

- (i) Keeping  $a$  fixed, turn  $ab$  to a position  $ab_1$ , thus making it parallel to  $xy$ . Project  $b_1$  to  $b'_1$  on the locus of  $b'$ .  $a'b'_1$  is the true length of  $AB$  and  $\theta$  is its true inclination with the H.P.
- (ii) Similarly, turn  $a'b'$  to the position  $a'_1b'$  and project  $a'_1$  to  $a_1$  on the path of  $a$  (because the end  $a$  has been moved).  $a_1b'$  is the true length of  $AB$  and  $\phi$  is its inclination with the V.P.

Traces:

- (i) Through  $v$  the point of intersection of the top view  $ab$  with  $xy$ , draw a projector to cut  $a'b'$  at the V.T.
- (ii) Through  $h$  the point of intersection of the front view  $a'b'$  with  $xy$ , draw a projector to cut  $ab$  at the H.T. of the line.

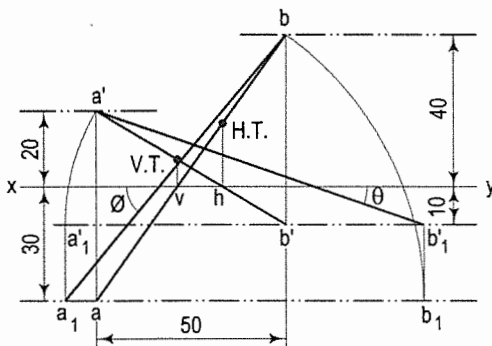


FIG. 10-37

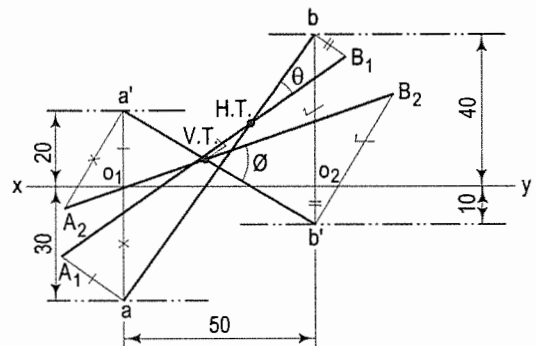


FIG. 10-38

**Method II:**

By rotating the line about its projections till it lies in H.P. or V.P. (fig. 10-38):

- (i) At the ends  $a$  and  $b$  of the top view  $ab$ , draw perpendiculars to  $ab$ , viz.  $aA_1$  equal to  $a'o_1$  and  $bB_1$  equal to  $b'o_2$ , on opposite sides of it (because  $a'$  and  $b'$  are on opposite sides of  $xy$ ).  $A_1B_1$  is the true length of  $AB$ .  $\theta$  (its inclination with  $ab$ ) is the inclination of  $AB$  with the H.P. and the point at which  $A_1B_1$  intersects  $ab$  is the H.T. of  $AB$ .
- (ii) Similarly, at the ends  $a'$  and  $b'$  of the front view  $a'b'$ , draw perpendiculars to  $a'b'$ , viz.  $a'A_2$  equal to  $ao_1$  and  $b'B_2$  equal to  $bo_2$ , on opposite sides of it.  $A_2B_2$  is the true length of  $AB$ .  $\phi$  (its inclination with  $a'b'$ ) is the inclination of  $AB$  with the V.P. and the point at which  $A_2B_2$  intersects  $a'b'$  is the V.T. of  $AB$ .

**Problem 10-14.** (fig. 10-39): A line  $AB$ , 90 mm long, is inclined at  $45^\circ$  to the H.P. and its top view makes an angle of  $60^\circ$  with the V.P. The end  $A$  is in the H.P. and 12 mm in front of the V.P. Draw its front view and find its true inclination with the V.P.

- (i) Mark  $a$  and  $a'$ , the projections of the end  $A$ .

- (ii) Assuming  $AB$  to be parallel to the V.P. and inclined at  $45^\circ$  to the H.P., draw its front view  $a'b'$  equal to  $AB$  and making an angle of  $45^\circ$  with  $xy$ . Project  $b'$  to  $b$  so that  $ab$  the top view is parallel to  $xy$ . Keeping the end  $a$  fixed, turn the top view  $ab$  to a position  $ab_1$  so that it makes an angle of  $60^\circ$  with  $xy$ . Project  $b_1$  to  $b'_1$  on the locus of  $b'$ . Join  $a'$  with  $b'_1$ .  $a'b'_1$  is the front view of  $AB$ .
- (iii) To find the true inclination with the V.P., draw an arc with  $a$  as centre and radius equal to  $AB$ , cutting the locus of  $b_1$  in  $b_2$ . Join  $a$  with  $b_2$ .  $\phi$  is the true inclination of  $AB$  with the V.P.

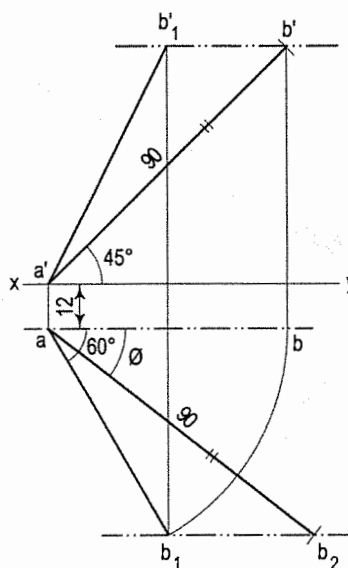
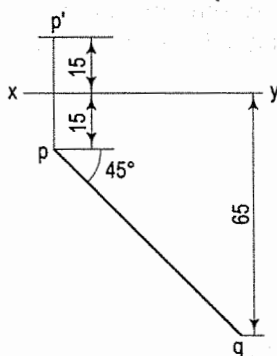
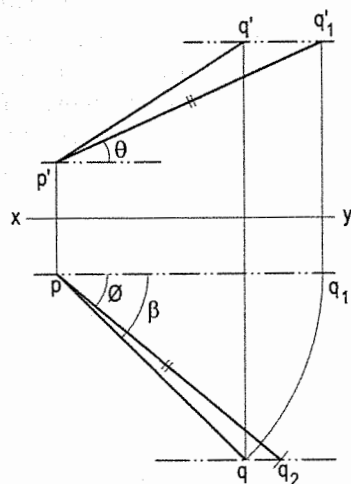


FIG. 10-39



(i)



(ii)

FIG. 10-40

**Problem 10-15.** (fig. 10-40): Incomplete projections of a line  $PQ$ , inclined at  $30^\circ$  to the H.P. are given in fig. 10-40(i). Complete the projections and determine the true length of  $PQ$  and its inclination with the V.P.

- Turn the top view  $pq$  [fig. 10-40(ii)] to a position  $pq_1$ , so that it is parallel to  $xy$ . Through  $p'$ , draw a line making an angle  $\theta$  (equal to  $30^\circ$ ) with  $xy$  and cutting the projector through  $q_1$  at  $q'_1$ .  $p'q'_1$  is the front view of  $PQ$ .
- Through  $q'_1$ , draw a line parallel to  $xy$  and cutting the projector through  $q$  at  $q'$ .  $p'q'$  is the front view of  $PQ$ .
- With  $p$  as centre and radius equal to  $p'q'_1$ , draw an arc cutting the locus of  $q$  at  $q_2$ .
- Join  $p$  with  $q_2$ .  $\phi$  is the inclination of  $PQ$  with the V.P.

**Problem 10-16.** (fig. 10-41): The end  $A$  of a line  $AB$  is 25 mm behind the V.P. and is below the H.P. The end  $B$  is 12 mm in front of the V.P. and is above the H.P. The distance between the projectors is 65 mm. The line is inclined at  $40^\circ$  to the H.P. and its H.T. is 20 mm behind the V.P. Draw the projections of the line and determine its true length and the V.T.

Draw the top view  $ab$  and mark the H.T. on it, 20 mm above  $xy$ .

We have seen that the line representing the true length obtained by the trapezoid method, intersects the top view or the top view-produced, at the H.T. at an angle equal to the true inclination of the line with the V.P.

- (i) Hence, at the ends  $a$  and  $b$ , draw perpendiculars to  $ab$  on its opposite sides (as one end is below the H.P. and the other end above it). Through the H.T., draw a line making angle  $\theta$  (equal to  $40^\circ$ ) with  $ab$  and cutting the perpendiculars at  $A_1$  and  $B_1$ , as shown.  $A_1B_1$  is the true length of  $AB$ .  $aA_1$  and  $bB_1$  are the distances of the ends  $A$  and  $B$  respectively from the H.P.

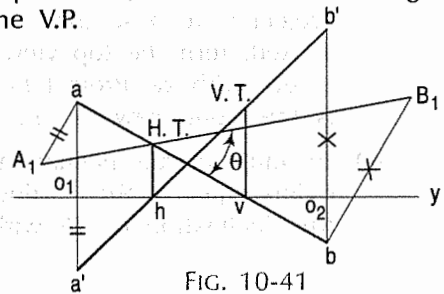


FIG. 10-41

- (ii) Project  $a$  and  $b$  to  $a'$  and  $b'$ , making  $a'o_1$  equal to  $aA_1$  and  $b'o_2$  equal to  $bB_1$ .  $a'b'$  is the front view of  $AB$ . Through  $v$ , the point of intersection between  $ab$  and  $xy$ , draw a projector cutting  $a'b'$  at the V.T. of the line.

**Problem 10-17.** (fig. 10-42): A line  $AB$ , 90 mm long, is inclined at  $30^\circ$  to the H.P. Its end  $A$  is 12 mm above the H.P. and 20 mm in front of the V.P. Its front view measures 65 mm. Draw the top view of  $AB$  and determine its inclination with the V.P.

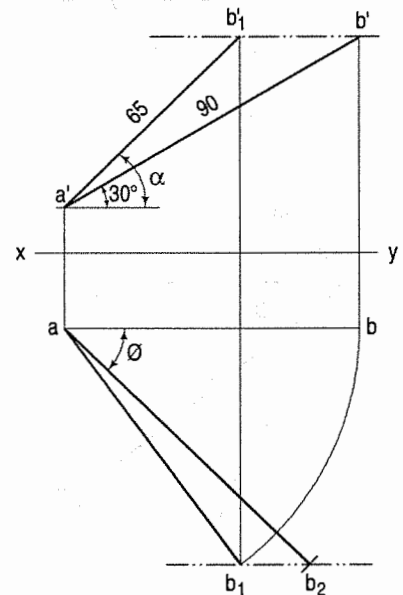


FIG. 10-42

- (i) Mark  $a$  and  $a'$  the projections of the end  $A$ . Through  $a'$ , draw a line  $ab'$  90 mm long and making an angle of  $30^\circ$  with  $xy$ .
- (ii) With  $a'$  as centre and radius equal to 65 mm, draw an arc cutting the path of  $b'$  at  $b'_1$ .  $a'b'_1$  is the front view of  $AB$ .
- (iii) Project  $b'$  to  $b$ , so that  $ab$  is parallel to  $xy$ .  $ab$  is the length of  $AB$  in the top view.
- (iv) With  $a$  as centre and radius equal to  $ab$ , draw an arc cutting the projector through  $b'_1$  at  $b_1$ . Join  $a$  with  $b_1$ .  $ab_1$  is the required top view.

Determine  $\theta$  as described in problem 10-14.

**Problem 10-18.** (fig. 10-43): The ends of a line  $PQ$  are on the same projector. The end  $P$  is 30 mm below the H.P. and 12 mm behind the V.P. The end  $Q$  is 55 mm above the H.P. and 45 mm in front of the V.P. Determine the true length and traces of  $PQ$  and its inclinations with the two planes.

**Note:** When the ends of a line are on the same projector or sum of angles of inclinations of a line with  $xy$  is  $90^\circ$ , use Method II only.

Draw the projections  $pq$  and  $p'q'$  as per given positions of the ends  $P$  and  $Q$ . They will partly coincide with each other.

- (i) At the ends  $p$  and  $q$  of the top view  $pq$ , erect perpendiculars, viz.  $pP_1$  equal to  $p'o$ , and  $qQ_1$  equal to  $q'o$  and on opposite sides of  $pq$ .  $P_1Q_1$  is the true length of  $PQ$ .  $\theta$  is the inclination of  $PQ$  with the H.P. and the point of intersection between  $P_1Q_1$  and  $pq$  is the H.T. of  $PQ$ .

- (ii) Similarly, draw perpendiculars to  $p'q'$ , viz.  $p'P_2$  equal to  $po$  and  $q'Q_2$  equal to  $qo$  and on opposite sides of  $p'q'$ .  $P_2Q_2$  is the true length.  $\theta$  is the true inclination of  $PQ$  with the V.P. and the point where  $P_2Q_2$  cuts  $p'q'$  is the V.T. of  $PQ$ .

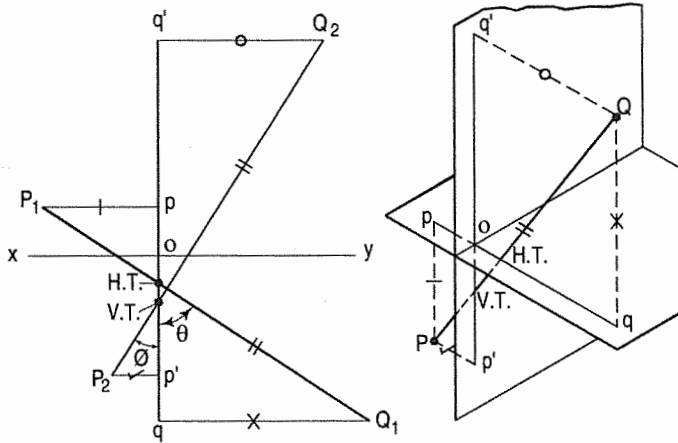


FIG. 10-43

**Problem 10-19.** (fig. 10-44): A line  $AB$ , inclined at  $40^\circ$  to the V.P., has its ends 50 mm and 20 mm above the H.P. The length of its front view is 65 mm and its V.T. is 10 mm above the H.P. Determine the true length of  $AB$ , its inclination with the H.P. and its H.T.

- (i) Draw the front view  $a'b'$  as per given positions of  $A$  and  $B$  and the given length.
- (ii) Draw a line parallel to and 10 mm above  $xy$ . This line will contain the V.T. Produce  $a'b'$  to cut this line at the V.T. Draw a projector through V.T. to  $v$  on  $xy$ .
- (iii) Assuming  $a'$  V.T. to be the front view of a line which makes  $40^\circ$  angle with the V.P. and whose one end  $v$  is in the V.P., let us determine its true length.
- (iv) Keeping V.T. fixed, turn the end  $a'$  to  $a_1$  so that the line becomes parallel to  $xy$ . Through  $v$ , draw a line making an angle of  $40^\circ$  with  $xy$  and cutting the projector through  $a_1$  at  $a_1$ . The line through  $a_1$ , drawn parallel to  $xy$ , is the locus of  $A$  in the top view. Project  $a'$  to  $a$  on this line.  $av$  is the top view of the line, whose front view is  $a'$  V.T. and whose true length is equal to  $a_1v$ .
- (v) But  $a'b'$  is the given front view of  $AB$ . Therefore, project  $b'$  to  $b$  on  $av$ .  $ab$  is the top view of  $AB$ . Obtain the inclination  $\theta$  with the H.P. by making the top view  $ab$  parallel to  $xy$ , as shown.

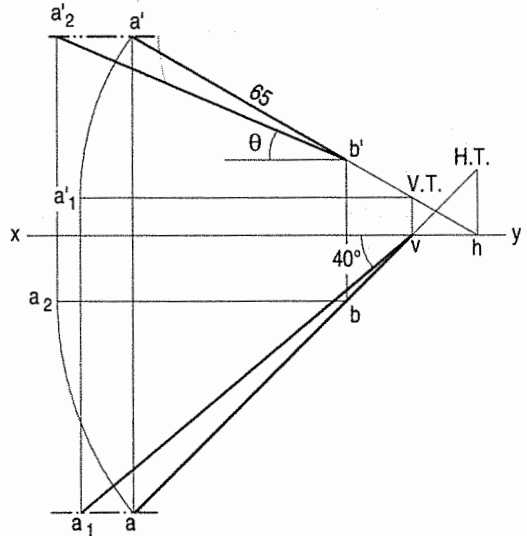


FIG. 10-44

Produce  $a'b'$  to meet  $xy$  at  $h$ . Draw a projector through  $h$  to cut  $ab$ -produced, at the H.T. of the line.

**Problem 10-20.** (fig. 10-45): The front view  $a'b'$  and the H.T. of a line  $AB$ , inclined at  $23^\circ$  to the H.P. are given in fig. 10-45(i). Determine the true length of  $AB$ , its inclination with the V.P. and its V.T.

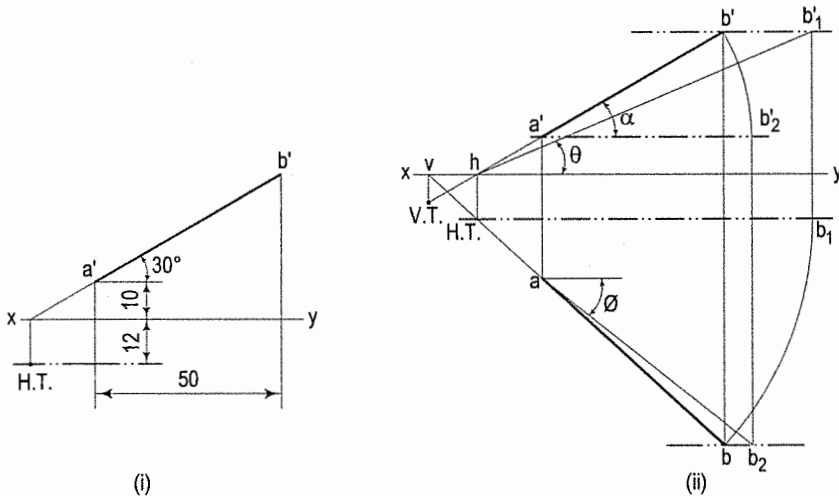


FIG. 10-45

Consider that  $hb'$  is the front view of a line inclined at  $23^\circ$  to the H.P. and the top view of whose one end is in H.T.

- (i) Through  $h$  [fig. 10-45(ii)], draw a line making an angle of  $\theta = 23^\circ$  with  $xy$  and cutting the locus of  $B$  in the front view in  $b'_1$ .  $hb'_1$  is the true length of the line whose length in the top view is H.T.  $b_1$ .
- (ii) With H.T. as centre and radius equal to H.T.  $b_1$ , draw an arc cutting the projector through  $b'$  at  $b$ . H.T.  $b$  is the top view and  $hb'$  is the front view of a line which contains  $AB$ .
- (iii) Therefore, through  $a'$ , draw a projector cutting H.T.  $b$  at  $a$ .  $ab$  is the top view of  $AB$ .

Obtain the true length  $ab_2$  (of  $AB$ ) and its inclination  $\phi$  with the V.P. by making  $a'b'$  parallel to  $xy$ .

- (iv) Produce  $ba$  to meet  $xy$  in  $v$ . Draw a projector through  $v$  to cut  $b'a'$ -produced, at the V.T. of the line.

**Problem 10-21.** (fig. 10-46): A tripod stand rests on the floor. One of its legs is 150 mm long and makes an angle of  $70^\circ$  with the floor. The other two legs are 163 mm and 175 mm long respectively. The upper ends of the legs are attached to the corners of a horizontal equilateral triangular frame of 50 mm side, one side of which is parallel to the V.P. In the top view, the legs appear as lines  $120^\circ$  apart, which if produced, would meet in a point. Draw the projections of the tripod and determine the angle which each of the other two legs makes with the floor. Assume the thickness of the frame and of the legs to be equal to that of the line.

- (i) At any point  $P$  on  $xy$ , draw a line  $PA$ , 150 mm long and making  $70^\circ$  angle with  $xy$ .  $h$  is the height of the tripod and  $PA_1$  is the length of the leg in the top view.

- (ii) Draw an equilateral triangle  $abc$  of 50 mm side with one side parallel to and below  $xy$ . Project the front view  $a'b'c'$  at the height  $h$  above  $xy$ . Determine the lengths of the other two legs in the top view as described below.
- (iii) With  $b'$  as centre and radius equal to 163 mm, draw an arc cutting  $xy$  in  $q'_1$ . Similarly with  $a'$  and  $c'$  as centre and radius equal to 150 mm and 175 mm, draw an arc cutting  $xy$  in  $p'_1$  and  $r'_1$  respectively  $ap_1$ ,  $bq_1$  and  $cr_1$  are the lengths of the three legs in the top view and  $\alpha$  and  $\beta$  respectively are their inclinations with the floor (H.P.).

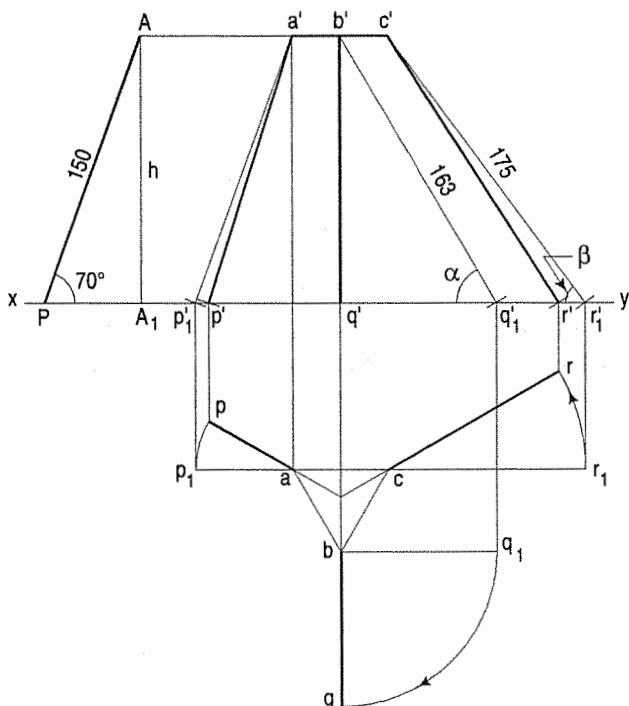


FIG. 10-46

- (iv) The legs in the top view are to be inclined at  $120^\circ$  to each other and to meet at a point, if produced. Therefore, draw lines bisecting the angles of the triangle, making  $ap$  equal to  $PA_1$ ,  $bq$  equal to  $bq_1$  and  $cr$  equal to  $cr_1$ , thus completing the top view.
- (v) Project  $p, q$  and  $r$  to  $p', q'$  and  $r'$  respectively on  $xy$ . Complete the front view by drawing lines  $a'p'$ ,  $b'q'$  and  $c'r'$ .

**Problem 10-22.** (fig. 10-47): A straight road going uphill from a point A, due east to another point B, is 4 km long and has a slope of  $15^\circ$ . Another straight road from B, due  $30^\circ$  east of north, to a point C is also 4 km long but is on ground level. Determine the length and slope of the straight road joining the points A and C. Scale, 10 mm = 0.4 km.

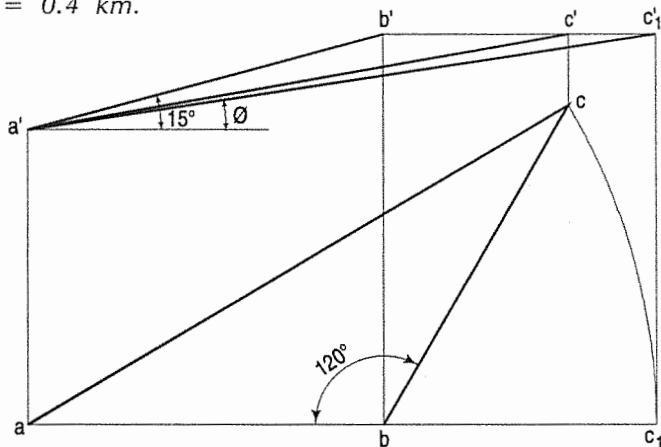


FIG. 10-47

- (i) Mark any point  $a'$ . Draw a line  $a'b'$ , 100 mm long, to the right of  $a'$  and inclined upwards at  $15^\circ$  to the horizontal (to represent the road from A to B). Project its top view  $ab$  keeping it horizontal.
- (ii) As the road from B to C is on ground level, the top view  $bc$  will be equal to 100 mm and inclined at  $(90^\circ + 30^\circ)$  i.e.  $120^\circ$  to  $ab$ .
- (iii) From  $b$ , draw a line  $bc$ , 100 mm long and making  $120^\circ$  angle with  $ab$ . Project  $c$  to  $c'$  making  $b'c'$  horizontal.  $a'c'$  and  $ac$  are the front view and the top view respectively of the road from A to C.

Determine the true length  $a'c'_1$  and the angle  $\theta$  as shown, which are respectively the length and slope of the road from A to C.

**Problem 10-23.** (fig. 10-48): Two lines AB and AC make an angle of  $120^\circ$  between them in their front view and top view. AB is parallel to both the H.P. and the V.P. Determine the real angle between AB and AC.

Draw any line  $b'a'$  parallel to and above  $xy$ , and another line  $a'c'$  of any length making  $120^\circ$  angle with  $b'a'$ . Join  $b'$  with  $c'$ .

- (i) Project the top view  $ba$  parallel to  $xy$  and the top view  $ac$ , making  $120^\circ$  angle with  $ba$ . Join  $b$  with  $c$ .  $b'a'$  or  $ba$  is the true length of AB. Determine the true lengths of AC and BC, viz.  $a'c'_1$  and  $b'c'_2$ , as shown.

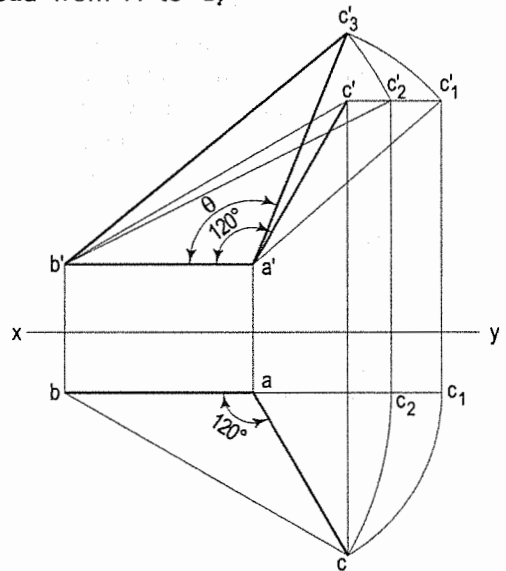


FIG. 10-48

- (ii) Draw a triangle  $a'b'c'_3$  making  $a'c'_3$  equal to  $a'c'_1$  and  $b'c'_3$  equal to  $b'c'_2$ .  $\angle b'a'c'_3$  is the real angle between AB and AC.

**Problem 10-24.** (fig. 10-49): An object O is placed 1.2 m above the ground and in the centre of a room  $4.2 \text{ m} \times 3.6 \text{ m} \times 3.6 \text{ m}$  high. Determine graphically its distance from one of the corners between the roof and two adjacent walls. Scale,  $10 \text{ mm} = 0.5 \text{ m}$ .

- (i) Draw the front view (of the room)  $a'b'c'd'$  as seen from the front of, say 3.6 m wall.  $a'b'$  is the width of the room and  $a'd'$  is the height. The front view  $o'$  of the object will be seen 1.2 m above the mid-point of  $a'b'$ .  $c'$  and  $d'$  are the top corners of the room.  $o'c'$  is the front view of the line joining the object with a top corner.

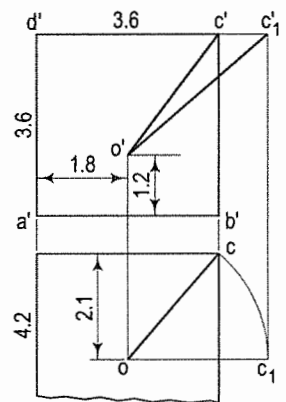


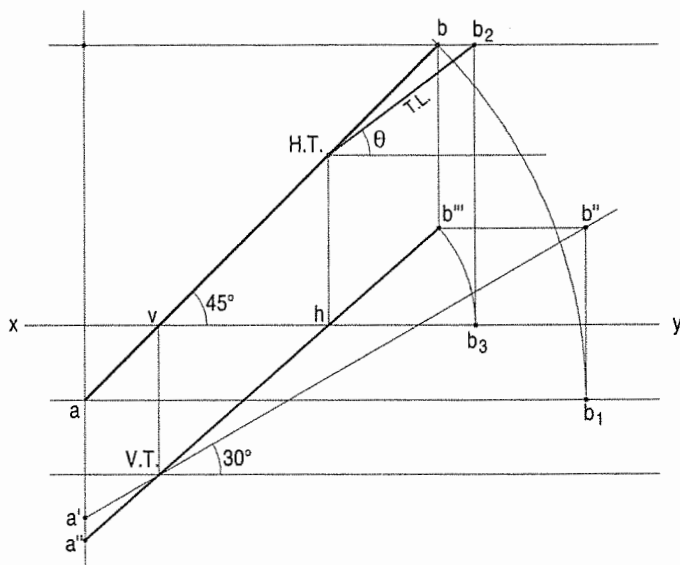
FIG. 10-49

- (ii) Draw the top view of the room. It will be a rectangle of sides equal to 3.6 m and 4.2 m. The top view  $o$  of the object will be in the centre of the rectangle.  $oc$  is the top view of the line joining the object with the top corner.

Determine the true length  $o'c'_1$ , which will show the distance of the object from one of the top corners of the room.

**Problem 10-25.** The straight line  $AB$  is inclined at  $30^\circ$  to  $H.P.$ , while its top view at  $45^\circ$  to a line  $xy$ . The end  $A$  is 20 mm in front of the  $V.P.$  and it is below the  $H.P.$  The end  $B$  is 75 mm behind the  $V.P.$  and it is above the  $H.P.$  Draw the projections of the line when its  $V.T.$  is 40 mm below. Find the true length of the portion of the straight line which is in the second quadrant and locate its  $H.T.$

Refer to fig. 10-50.



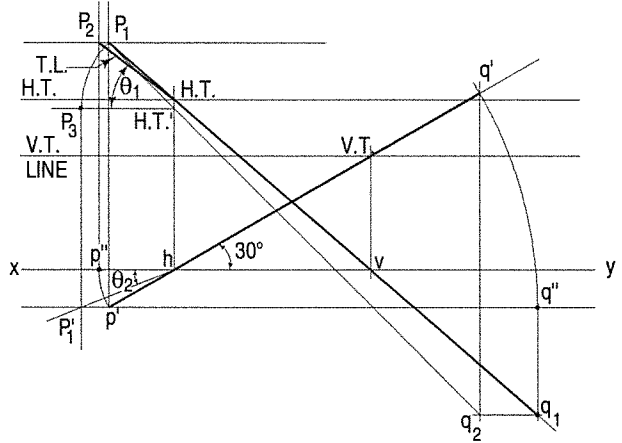
T.L. = 50 mm.  $\angle V.P.$ ,  $\theta = 37^\circ$

FIG. 10-50

- (i) Mark the points  $a$  (top view of  $A$ ) and  $b$  (top view of  $B$ ) at the distances of 20 mm and 75 mm below and above  $xy$  respectively.
- (ii) Through the point  $a$ , draw a line at  $45^\circ$  intersecting  $xy$  and the path of  $b$  at  $v$  and  $b$  respectively as shown.
- (iii) Construct a line containing  $V.T.$  40 mm below  $xy$ . Draw perpendicular from  $v$  to the line  $V.T.$
- (iv) With  $a$  as centre and radius equal to  $ab$ , draw an arc which intersects at  $b_1$  a line drawn from  $a$  parallel to  $xy$ . From  $V.T.$  draw a line at  $30^\circ$  intersecting the projector of  $b_1$  at  $b''$ . From  $b''$ , draw  $b''b'''$  parallel to  $xy$  to intersect projector of  $b$  at  $b'''$ . Join  $V.T. b'''$ . Produce it to meet the projector from  $a$  at  $a''$ .  $a''b'''$  is the required projection.  $a''b'''$  intersects line  $xy$  at  $h$ . From  $h$  draw the perpendicular to meet  $ab$ . The intersection point represents  $H.T.$
- (v) With  $h$  as centre and radius equal to  $hb'''$ , draw an arc intersecting at  $b_3$ . Draw projector from  $b_3$  to cut the path of  $b$  at  $b_2$ . Join  $H.T. b_2$ . Measure the angle  $H.T. b_2$  with  $xy$ . This is an angle made by the line with the  $V.P.$

**Problem 10-26.** (fig. 10-51.): The front view of a line  $PQ$  makes an angle of  $30^\circ$  with  $xy$ . The  $H.T.$  of the line is 45 mm behind the  $V.P.$  While its  $V.T.$  is 30 mm above the  $H.P.$  The end  $P$  of the line is 10 mm below the  $H.P.$  and the end  $Q$  is in the first quadrant. The line is 150 mm long. Draw the projections of the line and determine the true-length of the portion of the line which is in the second quadrant. Also find the angle of the line with the  $H.P.$  and  $V.P.$

- (i) Draw lines containing H.T. and V.T. at 45 mm and 30 mm above  $xy$  respectively. Mark point  $p'$  at 10 mm below line  $xy$ . Draw  $p'q'$  at  $30^\circ$  from  $p'$  intersecting  $xy$  at  $h$  and line V.T. at V.T. From  $h$ , draw perpendicular to line H.T. to locate H.T.

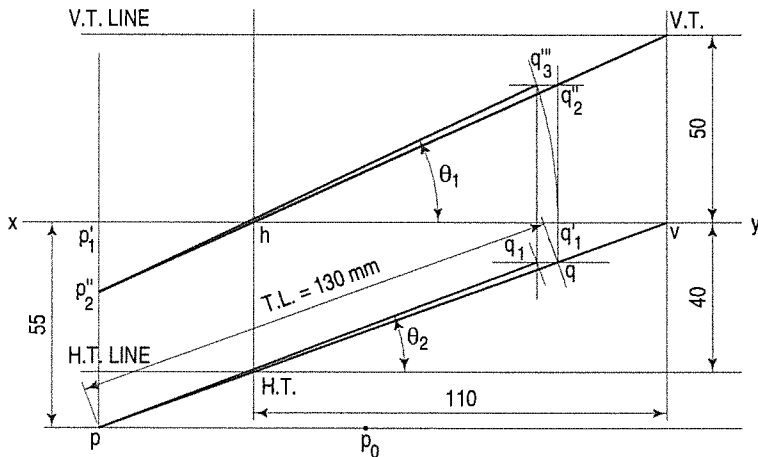


T.L. = 25 mm.  $\angle V.P., \theta_1 = 37^\circ$   $\angle V.P., \theta_2 = 22^\circ$

FIG. 10-51

- (ii) Draw perpendicular from V.T. to intersect  $xy$  at  $v$ . Join  $v$  H.T. and produce to intersect the projector of  $p'$  at  $p_1$ . Draw  $p_1 q_1$  of 150 mm representing true length of the line in top view. From  $p'$ , draw line parallel to  $xy$  representing front view of the line  $PQ$ . Draw projector from  $q_1$  to cut the line drawn from  $p'$  at  $q'$ .
- (iii) Keeping  $p'$  fixed, turn  $p'q'$  such that it cuts the line drawn from  $p'$  at  $q'$ . From  $q_1$  draw line parallel to  $xy$  which intersects the vertical projector drawn from  $q'$  at  $q_2$ . Join  $P_1 q_2$ . This is the required projection.
- (iv) Keeping  $h$  fixed, rotate  $hp'$  and make it parallel to  $xy$ . From  $P''$  draw projector intersecting the path of  $p_1$  at  $p_2$ . H.T.  $p_2$  is true-length of the line. Similarly keeping H.T.' fixed, turn H.T.'  $p_1$  to make it parallel to  $xy$  as shown. From  $P_3$ , draw projector to intersect the horizontal line drawn from  $P'P_1$ . Measure angle  $xhp'_1$ . This is the required angle with H.P.

**Problem 10-27.** (fig. 10-52): The end  $P$  of a line  $PQ$  130 mm long, is 55 mm in front of the V.P. The H.T. of the line is 40 mm in front of the V.P. and the V.T. is 50 mm above the H.P. The distance between H.T. and V.T. is 110 mm. Draw the projections of the line  $PQ$  and determine its angles with the H.P. and the V.P.



$\angle H.P., \theta_1 = 25.5^\circ$ ;  $\angle V.P., \theta_2 = 21^\circ$

FIG. 10-52

- (i) Mark  $p$  below  $xy$  at a distance of 55 mm. Draw lines containing H.T. and V.T. at 40 mm below and 50 mm above  $xy$  respectively. Construct projectors through H.T. and V.T. 110 mm apart intersecting  $xy$  at  $h$  and  $v$  respectively.
- (ii) Draw perpendiculars from  $h$  and  $v$  intersecting the lines containing H.T. and V.T. Join H.T.  $v$  and produce to cut the line drawn from point  $P$  as shown. Join  $h$  V.T. and extend further to intersect the projector drawn from  $P$  at  $P''_2$ .
- (iii) Mark true length 130 mm on H.T.v. Let it be  $pq$ . Draw projector from  $q$  cutting  $xy$  at  $q'_1$ . With  $p'_1$  as centre and radius equal to  $p'_1q'_1$ , draw an arc cutting a horizontal line drawn from  $q''_2$  at  $q'''_3$ . Join  $pq_1$  and  $p''_2q'''_3$ , are the required projections.  $\theta_1 = 25.5^\circ$  and  $\theta_2 = 21^\circ$  are the measured angles.

**Problem 10-28.** (fig. 10-53): The distance between the end projectors of a line  $AB$  is 70 mm and the projectors through the traces are 110 mm apart. The end of a line is 10 mm above H.P. If the top view and the front view of the line make  $30^\circ$  and  $60^\circ$  with  $xy$  line respectively, draw the projections of the line and determine

- (i) the traces, (ii) the angles with the H.P. and the V.P., (iii) the true length of the line.

Assume that the line is in the first quadrant.

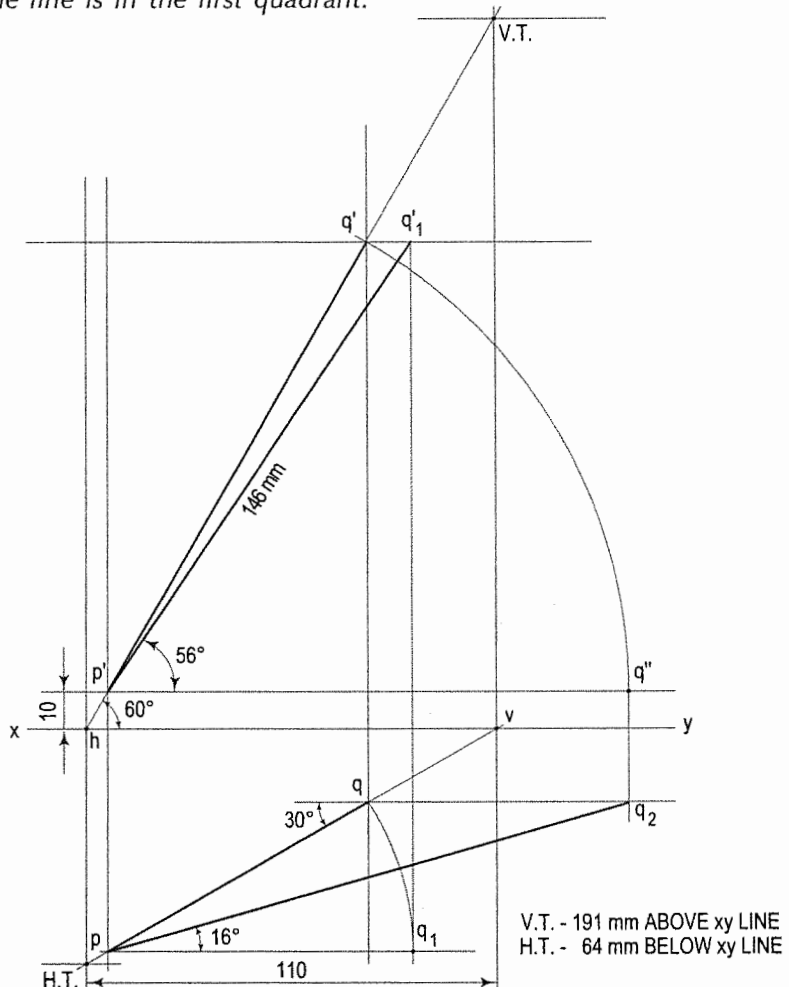


FIG. 10-53

- (i) Draw two vertical lines 110 mm apart representing projectors through the traces of the line. Mark intersection of these projectors  $h$  and  $v$  on  $xy$  as shown.
- (ii) From  $v$  and  $h$ , draw lines at  $30^\circ$  and  $60^\circ$ .
- (iii) Mark  $p'$ , 10 mm above  $xy$ . Draw two vertical projectors at 70 mm apart keeping equal distance from the projectors through traces.  $p'q'$  and  $pq$  represent front view and top view of the line as shown.
- (iv) Keeping  $p$  fixed, turn  $pq$  to position  $pq_1$ . From  $q_1$ , draw a vertical projector intersecting the path of  $q'$  at  $q'_1$ . Join  $p'q'_1$ . This is the true length of the line. Measure angle  $q''p'q'_1$  with the horizontal line as shown. This is the angle made by the line with the H.P. Similarly rotate  $p'q'$  making it parallel to  $xy$  as shown.

Draw a vertical projector from  $q''$  to intersect the path of  $q$  at  $q_2$ . Measure the angle  $q_1pq_2$  with the horizontal line. This is the angle made by the line with V.P.

*Note:* Problem 10-29 and problem 10-31 are solved by using auxiliary plane method.

**Problem 10-29.** Two pipes  $PQ$  and  $RS$  seem to intersect at  $a'$  and  $a$  in front view and top view as shown in fig. 10-54. The point  $A$  is 400 mm above H.P. and 300 mm in front of a wall.

Neglecting the thickness of the pipes, determine the clearance between the pipes.

Refer to fig. 10-54.

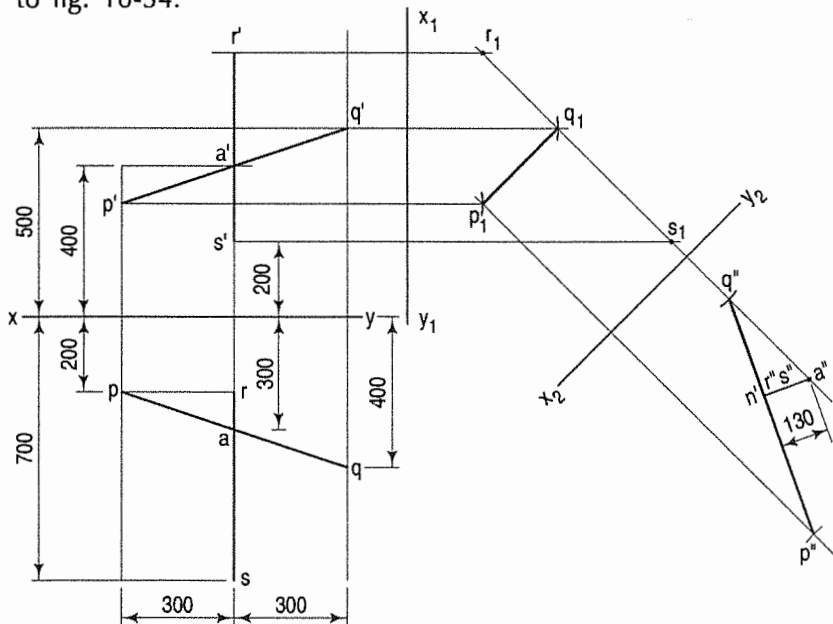


FIG. 10-54

- (i) Draw the projections of pipes treating them as lines.
- (ii) Draw  $x_1y_1$  perpendicular to the line  $xy$  and project auxiliary top view. Note that the distances of  $p_1$ ,  $q_1$ ,  $r_1$  and  $s_1$  from  $x_1y_1$  are equal to the distances of  $p$ ,  $q$ ,  $r$  and  $s$  from  $xy$ .

(iii) Mark an auxiliary reference line  $x_2y_2$  perpendicular to the line  $r_1s_1$  for obtaining the point view of the line. Draw an auxiliary front view as shown. Note that the distances of  $p''$ ,  $q''$ ,  $r''$  and  $s''$  from  $x_2y_2$  are equal to the distances of  $p'$ ,  $q'$ ,  $r'$  and  $s'$  from  $x_1y_1$ .

(iv)  $a''n'$  represents clearance between two pipes which is approximately 130 mm.

**Problem 10-30.** The end projectors of line AB are 22 mm apart. A is 12 mm in front of the V.P. and 12 mm above the H.P. The point B 6 mm in front of the V.P. and 40 mm above the H.P. Locate the H.T. and the V.T. of the line and also determine its inclinations with the V.P. and the H.P.

If the line AB is shifted to II, III and IV quadrants as shown in fig. 10-55 (assume that the distances of A and B from the projection-planes are same as the first quadrant), draw the projections of line and locate the traces.

The solution of first part of problem is shown in pictorial view. For the second part of the problem, the locations of the line in the respective quadrants are shown in fig. 10-55.

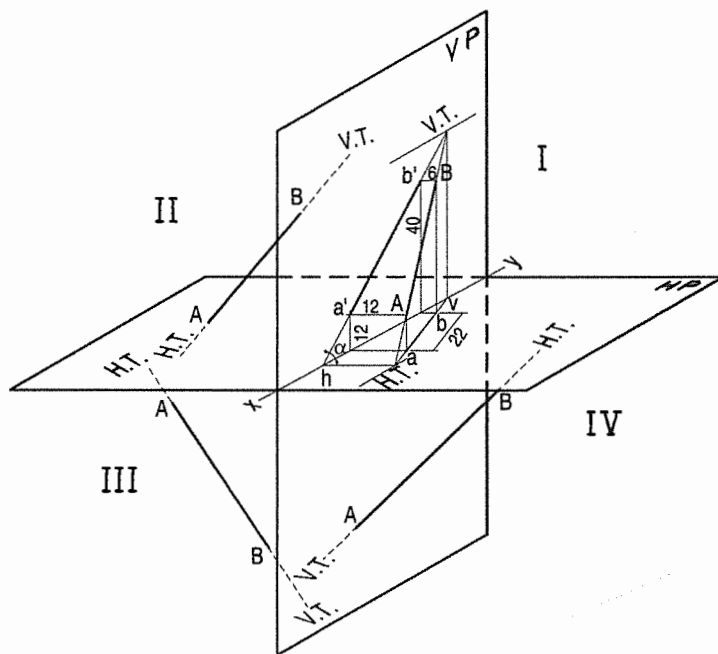


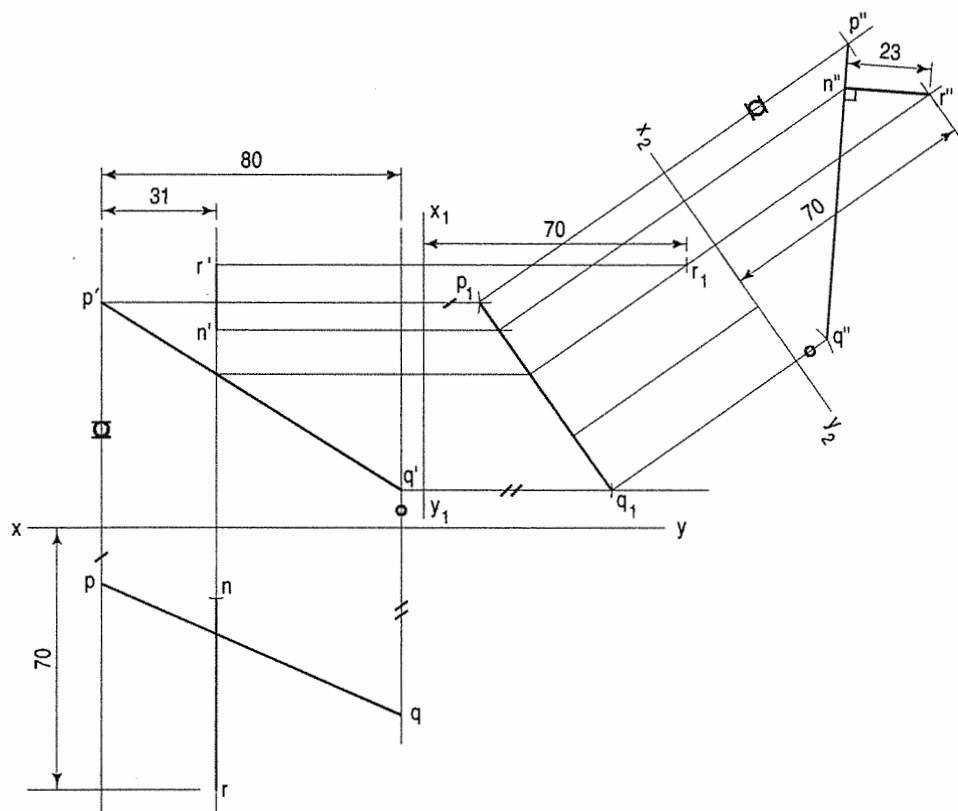
FIG. 10-55

Students are advised to draw the orthographic projections of the line for the respective quadrants.

**Problem 10-31.** The end projectors of line PQ are 80 mm apart. The end P is 15 mm in front of the V.P. and 60 mm above the H.P. While Q is 50 mm in front of the V.P. and 10 mm above the H.P. A point R is situated on the projector at a distance of 31 mm from the projector through P measured towards the projector of Q. The point R is 70 mm in front of the V.P. and above the H.P. A perpendicular is drawn from R on PQ. Draw its projections.

Refer to fig. 10-56.

- (i) Draw two end projectors of the line  $PQ$  at 80 mm apart.
- (ii) Mark the front view  $p'$  and the top view  $p$  at 60 mm and 15 mm from  $xy$  on the end projector of  $P$ . Similarly mark  $q'$  and  $q$  at 10 mm and 50 mm from  $xy$  on the end projector of  $Q$ .
- (iii) Draw a vertical line at 31 mm away from  $p'p$  towards  $q'q$ . Mark the position of  $R$  in the top view  $r$  and the front view  $r'$  at 70 mm from  $xy$  as shown.
- (iv) Draw  $x_1y_1$  perpendicular to  $xy$  as shown. Draw projectors from  $p'$ ,  $q'$  and  $r'$  on  $x_1y_1$ . Transfer the distances 15 mm, 50 mm and 70 mm of  $p$ ,  $q$  and  $r$  from  $xy$  to the new top view  $p_1$ ,  $q_1$  and  $r_1$  from  $x_1y_1$ .
- (v) Draw another reference line  $x_2y_2$  for the new front view parallel to  $p_1q_1$ . Transfer the distances 60 mm, 10 mm and 70 mm of  $p'$ ,  $q'$  and  $r'$  from  $xy$  to the new front view  $p''$ ,  $q''$  and  $r''$  from  $x_2y_2$ .
- (vi) From  $r''$ , draw perpendicular to  $p''q''$  intersecting at  $n''$  as shown which is measured as 23 mm.



$$n''r'' = 23 \text{ mm}$$

FIG. 10-56

**Problem 10-32.** The end projectors of a line  $PQ$  are 65 mm apart.  $P$  is 25 mm behind the V.P. and 30 mm below the H.P. The point  $Q$  is 40 mm above the H.P. and 15 mm in front of the V.P. Find the third point  $C$  in the H.P. and in front of the V.P. such that its distance from a point  $P$  is 45 mm and that from  $Q$  is 60 mm. Determine inclinations of  $PQ$  with the H.P. and the V.P.

Refer to fig. 10-57.

- (i) Draw the end projectors of the line  $PQ$  65 mm apart. Mark the projection of ends  $P$  and  $Q$  according to given distances.  $p'q'$  and  $pq$  are the front view and the top view respectively.
- (ii) Mark the front view of point  $c$  in the line  $xy$  because it is lying in the H.P. Extend the projection line from  $c'$  to  $c$  on the top view  $pq$ .
- (iii) Keeping  $c$  fixed, turn  $cp$  to  $cp_2$  making parallel to  $xy$ .
- (iv) Project  $p_2$  to  $p''$ . Join  $c'p''$ . This is the true distance of the line  $cp$ . Similarly turn  $cq$  to  $cq_2$  making it parallel to  $xy$ . Project  $q_2$  to  $q''$  join  $c'q''$ . This represents the true distance of the line  $cq$ .
- (v) Measure angles  $\theta$  and  $\phi$  as shown.

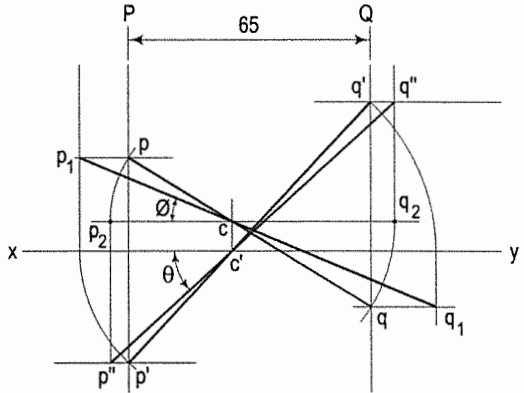
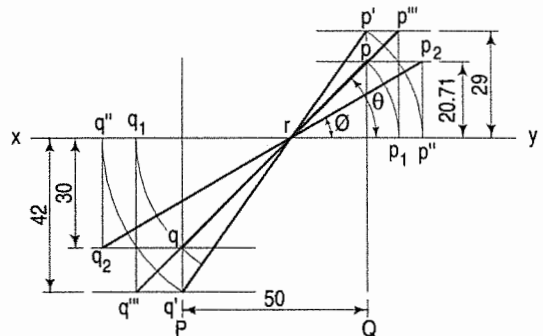


FIG. 10-57

**Problem 10-33.** (fig. 10-58): The distance between the end-projectors of a line  $PQ$  is 50 mm. A point  $P$  is 29 mm above H.P. and 20.71 mm behind V.P. While a point  $Q$  is 42 mm below H.P. and 30 mm in front of V.P.

Draw the projections of the line and determine the true length and the true inclinations of line with H.P. and V.P.

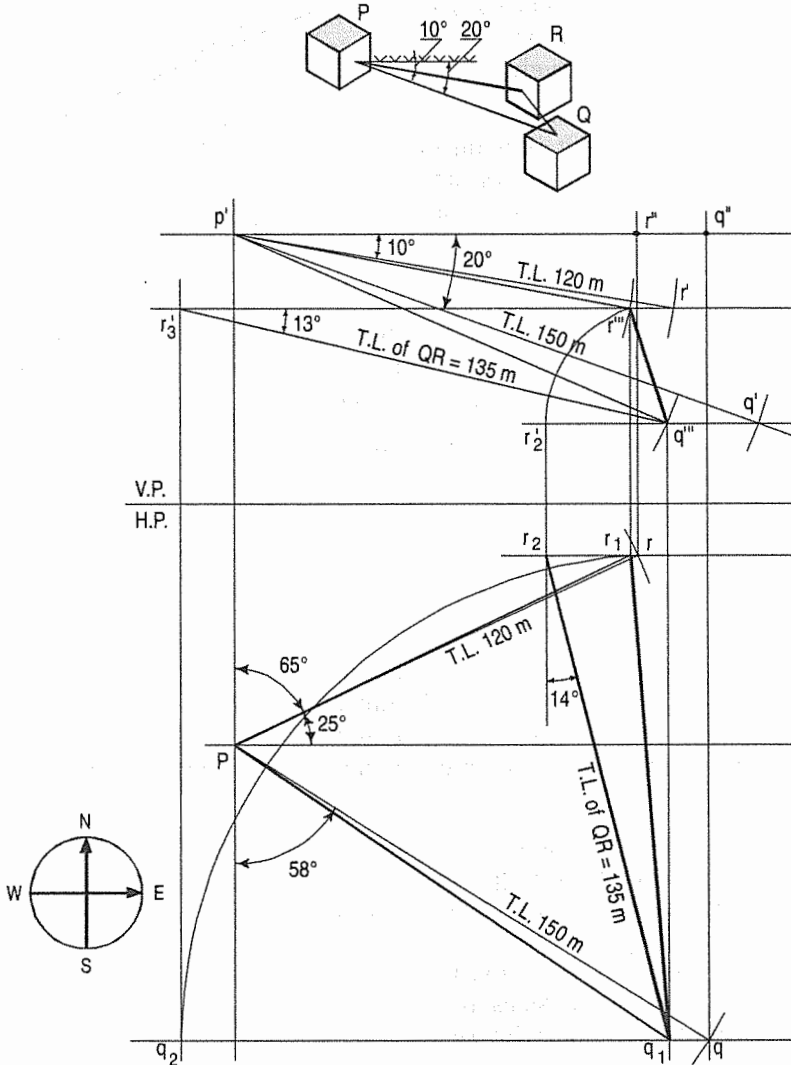
- (i) Draw the end-projectors 50 mm apart. Mark  $p'$  and  $p$ , the front view and the top view of the end  $P$  at 29 mm and 20.71 mm respectively. Similarly mark  $q$  and  $q'$  at 42 mm and 30 mm on the end-projector  $Q$  as shown. Join  $p'q'$  and  $pq$ . They are intersecting  $xy$  at  $r$ . Mark paths of  $p'$ ,  $q'$ ,  $p$  and  $q$  parallel to  $xy$ .
- (ii) With centre  $r$  and radius  $rp'$ , draw an arc intersecting  $xy$  at  $p''$ . Through  $p''$ , draw projector cutting the path of  $p$  at  $p_2$ . Similarly with the same centre and radius  $rq'$ , draw an arc intersecting  $xy$  at  $q''$ . Through  $q''$ , draw projector cutting the path of  $q$  at  $q_2$ . Join  $p_2q_2$  which represents true length. Measure  $\phi$  angle made by  $p_2q_2$  with  $xy$  at  $r$ . This is an angle made by the line with V.P.
- (iii) Similarly centre as  $r$  and radii  $rp$  and  $rq$ , draw arcs intersecting  $xy$  at  $p_1$  and  $q_1$  respectively. Through  $p_1$  and  $q_1$  draw projectors to intersect the paths of  $p'$  and  $q'$  at  $p'''$  and  $q'''$ . Join  $p'''q'''$ . Measure  $\theta$  angle made by it with  $xy$ . This is an angle by line with H.P.



$\angle \theta = 45^\circ$ ;  $\angle \phi = 30^\circ$ ,  
true length = 100 mm

FIG. 10-58

**Problem 10-34.** (fig. 10-59): Two pipes emerge from a common tank. The pipe PQ is 150 metre long and bears S 58° E on a downward slope of 20°. The pipe PR is 120 metre long and bears N 65° E on a downward slope of 10°. Determine the length of pipe required to connect Q and R. Take scale 1 mm = 1 m.



T.L. of QR = 135 m. S 13° E at downward slope of 13°

FIG. 10-59

- (i) Mark position of  $P$  in the top view and the front view as shown.
- (ii) At  $P$ , draw a line  $pq$  150 mm at 58° with the vertical measuring anti-clockwise as the angle is required to measure from south to east. Similarly draw a line  $pr$  120 mm at 25° measured from east to north.
- (iii) Draw the horizontal lines from  $q$  and  $r$ .
- (iv) From point  $p'$ , draw a line  $p'q'$  150 mm and  $p'r'$  120 mm at 20° and 10° with the horizontal lines. Draw horizontal lines from  $r'$  and  $q'$ .

- (v) Draw a projector from  $q$  to intersect the horizontal line drawn from  $p'$  at  $q''$ .  $p'q''$  is the front view of  $PQ$ . With  $p'$  as centre and radius equal to  $p'q''$ , draw an arc intersecting the path of  $q'$  at  $q'''$ . Join  $p'q'''$  and draw a projector from  $q'''$  cutting the path of  $q$  at  $q_1$ . Then  $p'q'''$  and  $pq_1$  are required projection. Similarly obtain the projection of  $p'r'''$  and  $pr_1$  for the line  $PR$  as shown. Join  $r'''q'''$  and  $r_1q_1$ . With centre  $q'''$  and a radius  $q'''r'''$ , draw an arc intersecting the path of  $q'''$  at  $r'_2$ . Draw a projector from  $r'_2$  cutting the path of  $r$  at  $r_2$ . Join  $q_1r_2$  and measure its true length and angle.
- (vi) With centre  $q_1$  and radius  $q_1r_1$ , draw an arc intersecting the path of  $q$  at  $q_2$ . Draw a projector from  $q_2$  cutting the path of  $r'$  at  $r'_3$ . Join  $q'''r'_3$  and measure its true length and angle.
- (vii)  $QR = 135$  m is the measured true length and  $S 13^\circ E$  at downward slope of  $13^\circ$  is the measured angle.

**Note:** Depression or front view angles are seen in front view while bearing angles are seen in top view.

**Problem 10-35.** (fig. 10-60): The projectors of two points  $P$  and  $Q$  are 70 mm apart. The point  $P$  is 25 mm behind the V.P. and 30 mm below the H.P. The point  $Q$  is 40 mm above the H.P. and 15 mm in front of the V.P. Find the third point  $S$  which is in the H.P. and in front of the V.P. such that its distance from point  $P$  is 90 mm and that from  $Q$  is 60 mm.

- (i) Draw  $xy$  line.
- (ii) Draw the projectors of  $P$  and  $Q$  70 mm apart.
- (iii) Mark on the projector of the point  $p$  the front view and top view of the point  $p$  at 25 mm and 30 mm from  $xy$  respectively, say  $p'$  and  $p$ . Similarly on the projector of the point  $Q$ , mark the front view  $q'$  and the top view  $q$  for given distance from the  $xy$ .
- (iv) Join  $p'q'$  and  $pq$ . They are the front view and the top view of the line  $PQ$ .

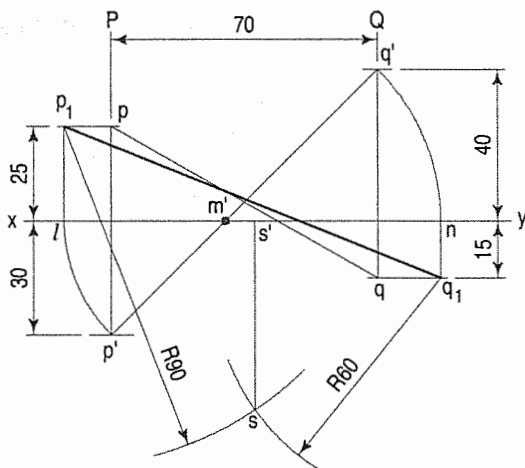


FIG. 10-60

- (v) The front view  $p'q'$  intersects  $xy$  line at  $m'$ . Taking  $m'$  as centre,  $m'p'$  and  $m'q'$  as radii rotate so that  $p'q'$  becomes parallel to the  $xy$  line or intersect  $xy$  line at  $l$  and  $n$ .
- (vi) Draw the projectors from  $l$  and  $n$  to intersect path of the point  $p$  and the point  $q$  at  $p_1$  and  $q_1$ . Join  $p_1$  and  $q_1$ , it represents true length of the line  $PQ$ .
- (vii) Now  $p_1$  as centre and 90 mm radius, draw the arc. Take  $q_1$  as centre and 60 mm as radius, draw the another arc so that it intersects previous arc at  $s$ . From  $s$  draw the projector to intersect  $xy$  line at  $s'$ , which is front view of  $s$ .

**Problem 10-36.** (fig. 10-61): Two unequal lines  $PQ$  and  $PR$  meeting at  $P$  makes an angle of  $130^\circ$  between them in their front view and top view. Line  $PQ$  is parallel and 6 mm away from both the principal planes. Assume the front view length of  $PQ$  and  $PR$  50 mm and 60 mm respectively. Determine real angle between them.

- (i) Mark reference line  $xy$ . Draw at convenient distance above and below two parallel lines to the  $xy$  representing the front view and top view of  $PQ$  as shown. At the point  $P'$  construct angle of  $130^\circ$  with the length of  $p'q'$  and  $p'r'$  of 50 mm and 60 mm respectively.
- (ii) Draw projector from  $r'$  to intersect the line at angle of  $130^\circ$  at  $p$  in the top view of  $pq$ .
- (iii) Make  $pr$  and  $qr$  parallel to  $xy$  line and project above  $xy$  to intersect the path of  $r'$  at  $r''$  and  $r'''$  respectively. Join  $p'r'''$  and  $q'r''$ . They are true length of the line  $PR$  and the line  $QR$ .
- (iv) Construct triangle with sides  $PQ = 50$  mm,  $PR = p'r'''$  and  $QR = q'r''$  as shown. Measure angle  $\angle QPR$  equal and it is  $120^\circ$  approximately.

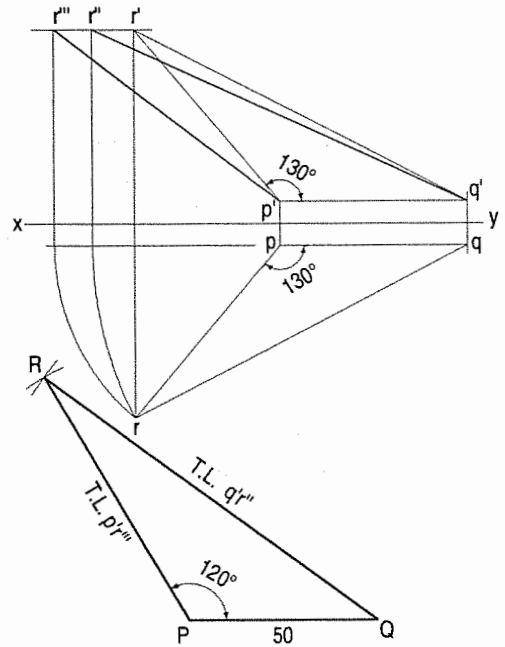


FIG. 10-61

**Problem 10-37.** (fig. 10-62): The distance between end projectors of a straight line  $AB$  is 80 mm. The point  $A$  is 15 mm below the H.P. and 20 mm in front of the V.P.  $B$  is 60 mm behind the V.P. Draw projections of the line if it is inclined at  $45^\circ$  to V.P. Determine also true length and inclination with the H.P.

- (i) Draw  $xy$  line and two vertical parallel lines 80 mm apart showing the end projectors of the line  $AB$ .
- (ii) Mark the point  $a'$  and the point  $a$  15 mm and 20 mm below  $xy$  line on the projector of  $A$ .
- (iii) Mark the point  $b$  above  $xy$  line at distance of 60 mm on the projector of  $B$ . Join the point  $a$  and the point  $b$ .
- (iv) If we measure angle of  $ab$  with  $xy$  line it is  $45^\circ$  which is also inclination of the line  $AB$  with V.P.
- (v) Therefore  $ab$  shows true length. From  $a'$  draw parallel line to  $xy$  to intersect the projector of  $B$  at  $b'$ . It is front view of the line  $AB$ . Measure angle of  $a'b$  with  $xy$  line. It is  $43^\circ$  with  $xy$  line.

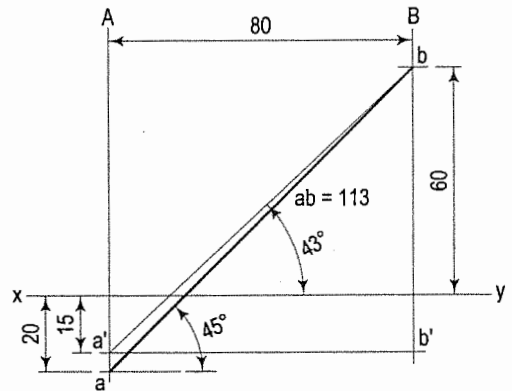


FIG. 10-62

**Problem 10-38.** (fig.10-63): Two mangoes on a tree are respectively 2.0 m and 3.5 m above the ground and 1.5 m and 2.0 m away from 0.2 m thick compound wall, but on the opposite sides of it. The distance between the mangoes, measured along the ground and parallel to the wall is 2.7 m. Determine the real distance between the mangoes. Take scale 1 m = 10 mm.

- (i) Pictorial view is shown for understanding purpose.
- (ii) Draw reference line (ground line)  $xy$ . Mark two parallel lines as end-projectors at 2.7 m (27 mm) apart.
- (iii) Let  $P$  be mango behind the wall and  $Q$  be in front of the wall.
- (iv) Mark  $P'$  and  $P$  along projector  $P$  to given distances. Similarly mark  $q'$  and  $q$  for given distances on the projector  $Q$ .
- (v) Join the point  $p$  and the point  $q$ , the point  $p'$  and the point  $q'$ . Then  $pq$  and  $p'q'$  are projections of  $PQ$ .
- (vi) Rotates  $pq$  taking  $q$  as centre, make it parallel to the ground line  $xy$ , intersecting at the point  $P_1$ . Draw the projector from  $p_1$ . Draw line parallel to the ground line  $xy$  from  $p'$ , intersecting at  $p''$ .
- (vii) Join  $p''q'$ . The line  $p''q'$  is true distance between mangoes  $P$  and  $Q$ . It is approximately 4.8 m.

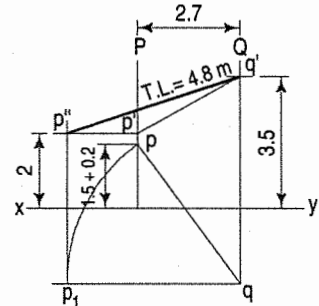
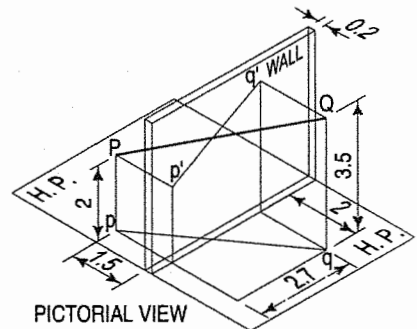


FIG. 10-63

**Problem 10-39.** (fig. 10-64): The front view of straight line  $AB$  is 60 mm long and is inclined at  $60^\circ$  to the reference line  $xy$ . The end point  $A$  is 15 mm above H.P. and 20 mm in front of V.P. Draw the projections of a line  $AB$  if it is inclined at  $45^\circ$  to the V.P. and is situated in the first quadrant (Dihedral angle). Determine its true length, and inclination with the H.P.

See fig. 10-64 which is self explanatory.

**Problem 10-40.** (fig. 10-65): A room 6 m  $\times$  5 m  $\times$  4 m high has a light-bracket above the centre of the longer wall and 1 m below the ceiling. The light bulb is 0.3 m away from the wall. The switch for the light is on an adjacent wall, 1.5 m above the floor and 1 m from the other longer wall. Determine graphically the shortest distance between the bulb and the switch.

- (i) Draw to scale 1 m = 10 mm front view and a top view of the room.
- (ii) Mark mid point of longer wall (i.e. 6 m), say 6 mm.

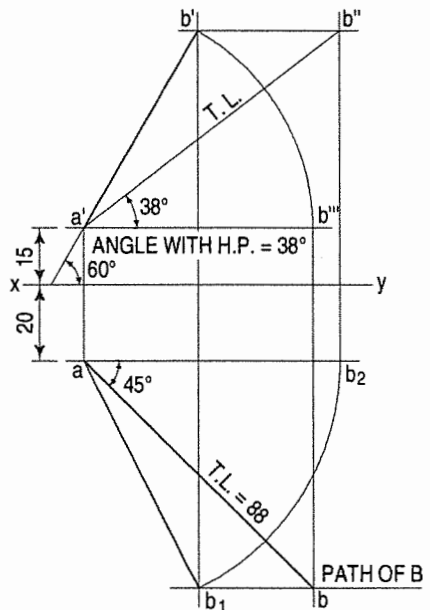


FIG. 10-64

- (iii) Mark the point in the V.P. at distance 1 m (= 10 mm) from midpoint  $m$ . It is the front view of the light bulb, say  $b'$ .
- (iv) From  $xy$  line at 0.3 m (i.e. 3 mm) on the projector passing through  $b'$  mark the point  $b$  (top view).
- (v) Consider adjacent wall right side. Mark  $s'$  front view of the switch at 1.5 m (i.e. 15 mm) above  $xy$  line and on the same line mark  $s$  (top view of switch) at 4 m away from  $xy$ .
- (vi) Join  $b's'$  and  $bs$  which are the front view and the top view of the line.
- (vii)  $s$  as centre and  $bs$  as radius, draw the arc to intersect a parallel line passing through  $s$  at  $b_1$ . From  $b_1$ , draw projector to intersect a parallel line to  $xy$  drawn from  $b'$  at  $b''$ . Join  $s'b''$ .
- (viii) Line  $s'b''$  shows shortest distance between the switch and the bulb. Here it is approximately 5 m.

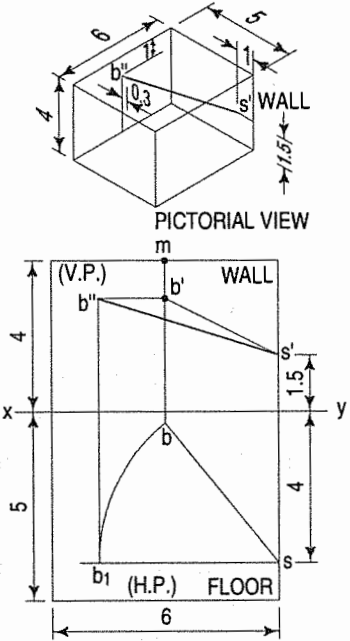


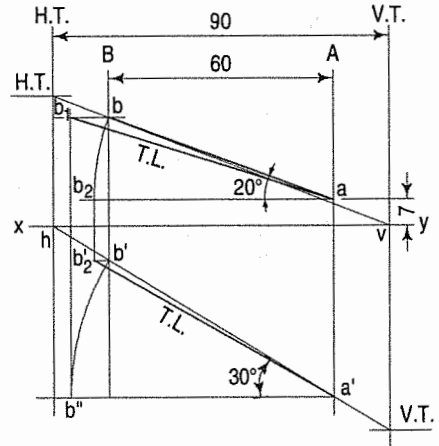
FIG. 10-65

**Problem 10-41.** (fig. 10-66): The distance between end projectors of a line  $AB$  are 60 mm apart, while the projectors passing from H.T. and V.T. are 90 mm apart. The H.T. is 35 mm behind the V.P., the V.T. is 55 mm below the H.P. The point  $A$  is 7 mm behind the V.P. Find graphically true length of the line and inclinations with the H.P. and the V.P.

See fig. 10-66 which is self-explanatory.

**Problem 10-42.** (fig. 10-67): A line  $CD$  is inclined at  $30^\circ$  to the H.P. and it is in the first quadrant. The end  $C$  is 15 mm above the H.P. while the end  $D$  is in the V.P. The mid point  $M$  of the line is 40 mm above H.P. The distance between the end projectors of the line is 70 mm. Draw the projections of the line  $CD$  and the mid point  $M$ . Determine graphically the length of front view and top view and true length of the line. Also determine inclination of the line with the V.P.

- (i) Draw  $xy$  line.
- (ii) Draw two projectors 70 mm, apart.
- (iii) On the projector of  $C$ , mark  $c'$  at 15 mm above  $xy$  line.
- (iv) Draw a line parallel to  $xy$  at 40 mm, to represent the path of mid-point  $M$ .
- (v) From  $c'$  draw a  $30^\circ$  inclined line  $t$  cut the path of mid-point at  $m'$ .  $c'm'$  is half true length. With  $m'$  as centre and radius equal to  $c'm'$ , draw an arc cutting the  $30^\circ$  inclined line at  $d'$ .  $c'd'$  is true length. From  $d'$ , draw a line parallel to  $xy$ , to represent path of  $D$  in front view.



True length = 74;  
Angle with H.P. =  $30^\circ$ ;  
Angle with V.P. =  $20^\circ$

FIG. 10-66